

(45) **Date of Patent:** Sep. 16, 2025

- (Continued)

- (56) **References Cited**

- U.S. PATENT DOCUMENTS

- | | | | | | |
|-----------|------|--------|--------------|-------------|--------|
| 6,930,417 | B2 * | 8/2005 | Kaneko | H02K 11/33 | 310/58 |
| 8,151,914 | B2 * | 4/2012 | Horie | H01M 10/658 | 290/16 |

- (Continued)

- FOREIGN PATENT DOCUMENTS

- JP 2017-070140 A 4/2017

- Primary Examiner* — Terrance L Kenerly

- Assistant Examiner — Theodore L Perkins

- (74) *Attorney, Agent, or Firm* — Oliff PLC

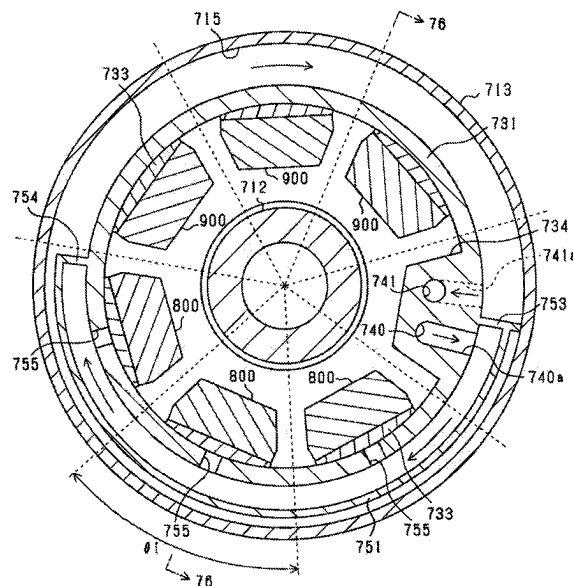
- (57) **ABSTRACT**

- A rotating electric machine includes a housing member including a cylindrical portion. First and second electrical components are arranged in the circumferential direction along the cylindrical portion in a radial direction of the cylindrical portion. A coolant passage between an inlet portion and an outlet portion is formed in the cylindrical portion. The housing member includes a partition wall portion partitioning a portion of the coolant passage from the inlet portion to a specific position between the inlet portion and the outlet portion in the circumferential direction as an upstream-side path. The partition wall portion is separated in the radial direction from a wall portion of the cylindrical portion. The first electrical component is arranged on an inner side in the radial direction relative to the upstream-side path, and the second electrical component is arranged on the inner side in the radial direction relative to a downstream-side path.

- 12 Claims, 67 Drawing Sheets**

- (Continued)

- (52) **U.S. Cl.**
CPC ***H02K 11/33*** (2016.01); ***H02K 5/203***
(2021.01); ***H02K 15/12*** (2013.01); ***H02K***
1/2786 (2013.01); ***H02P 27/08*** (2013.01)



(51) **Int. Cl.***H02K 5/20* (2006.01)*H02K 15/12* (2025.01)*H02P 27/08* (2006.01)(58) **Field of Classification Search**

USPC 310/52

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

2010/0165680 A1 * 7/2010 Yoshinaga B60L 50/51
363/141
2010/0289365 A1 * 11/2010 Bando H02K 16/00
310/156.01
2016/0056683 A1 * 2/2016 Nakanishi H02K 5/18
310/54
2016/0226346 A1 * 8/2016 Roberts H02K 9/00
2019/0229584 A1 7/2019 Shinozaki et al.

* cited by examiner

FIG. 1

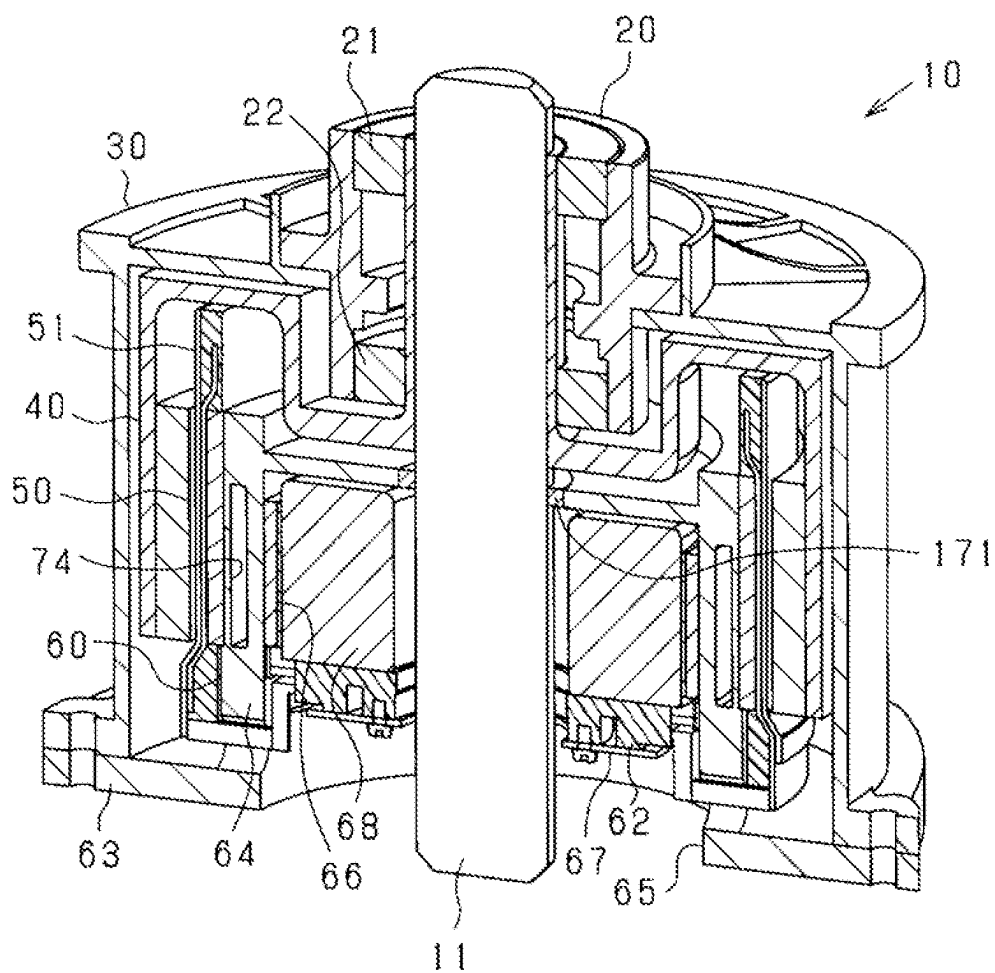


FIG.2

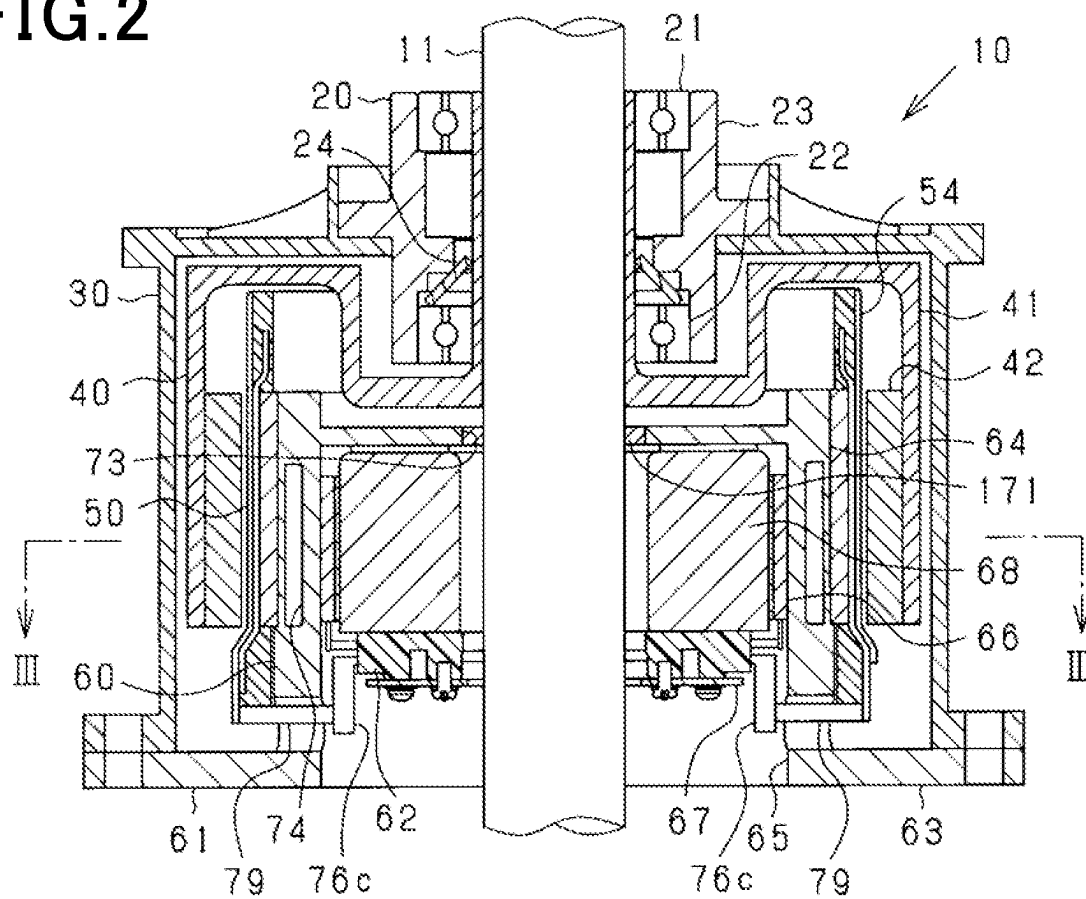


FIG.3

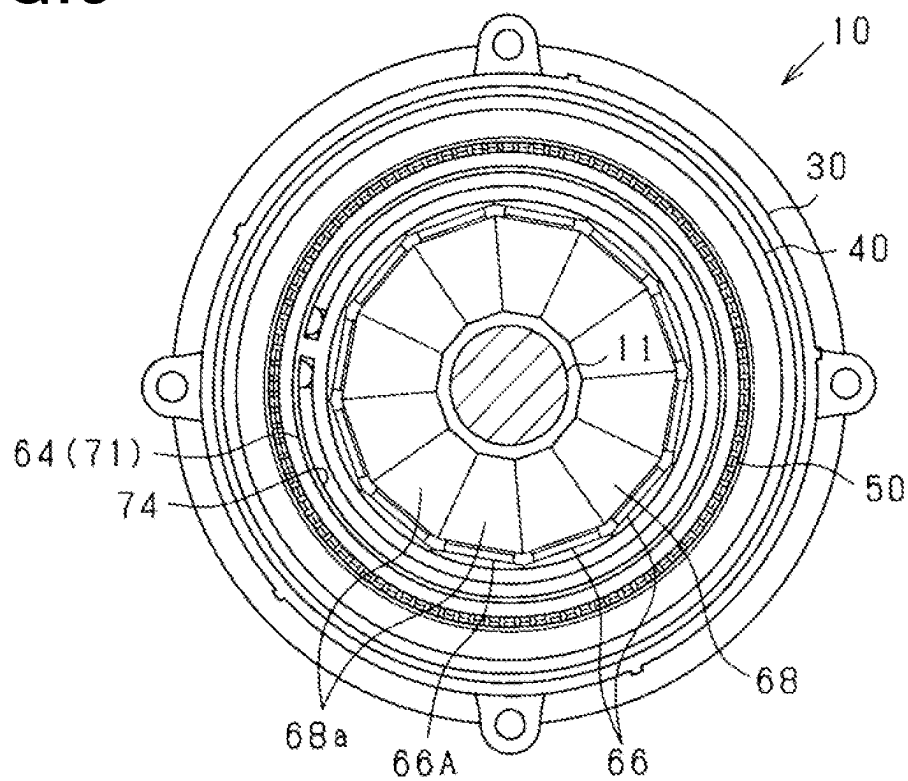


FIG. 4

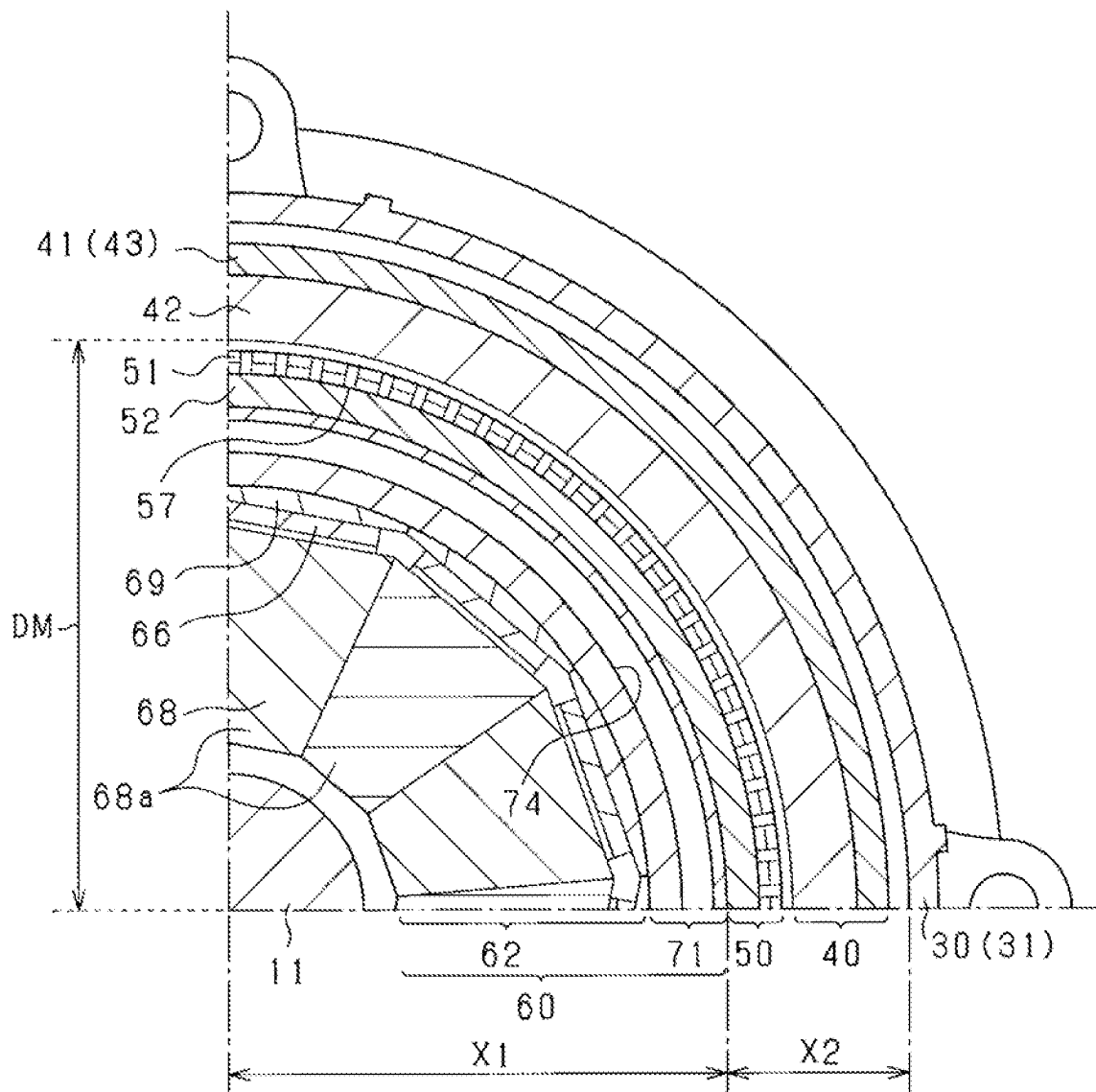


FIG. 5

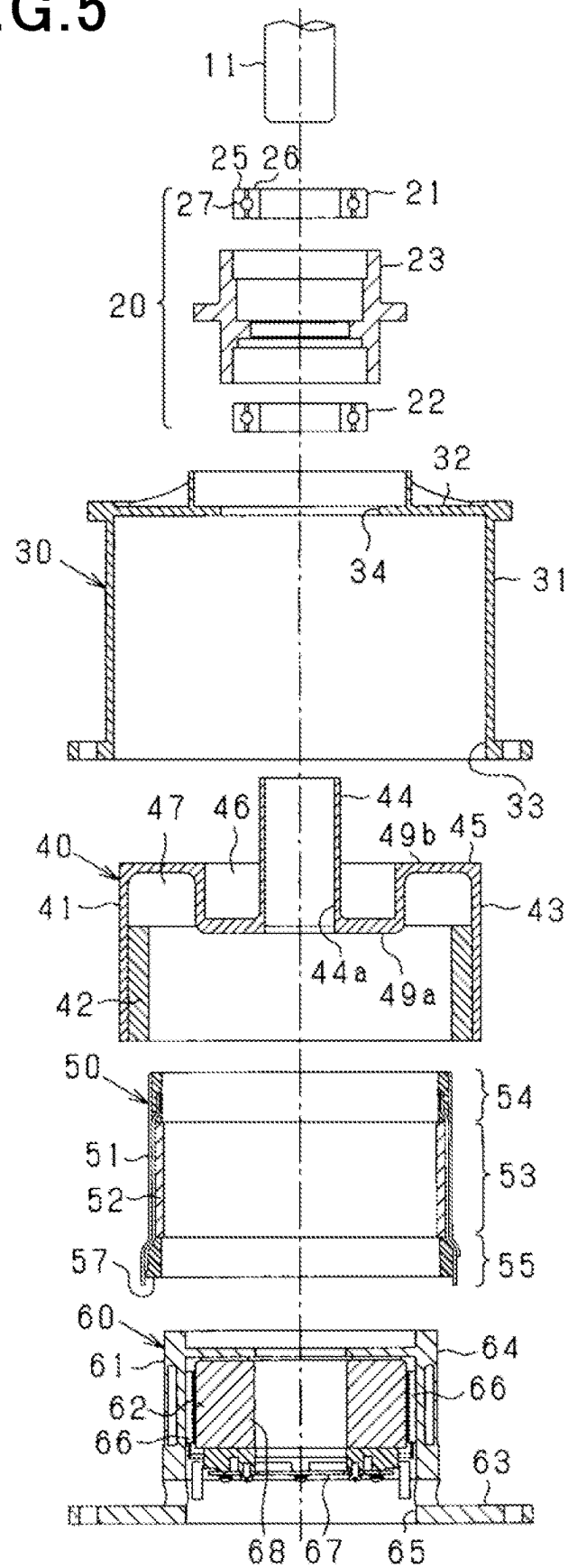


FIG. 6

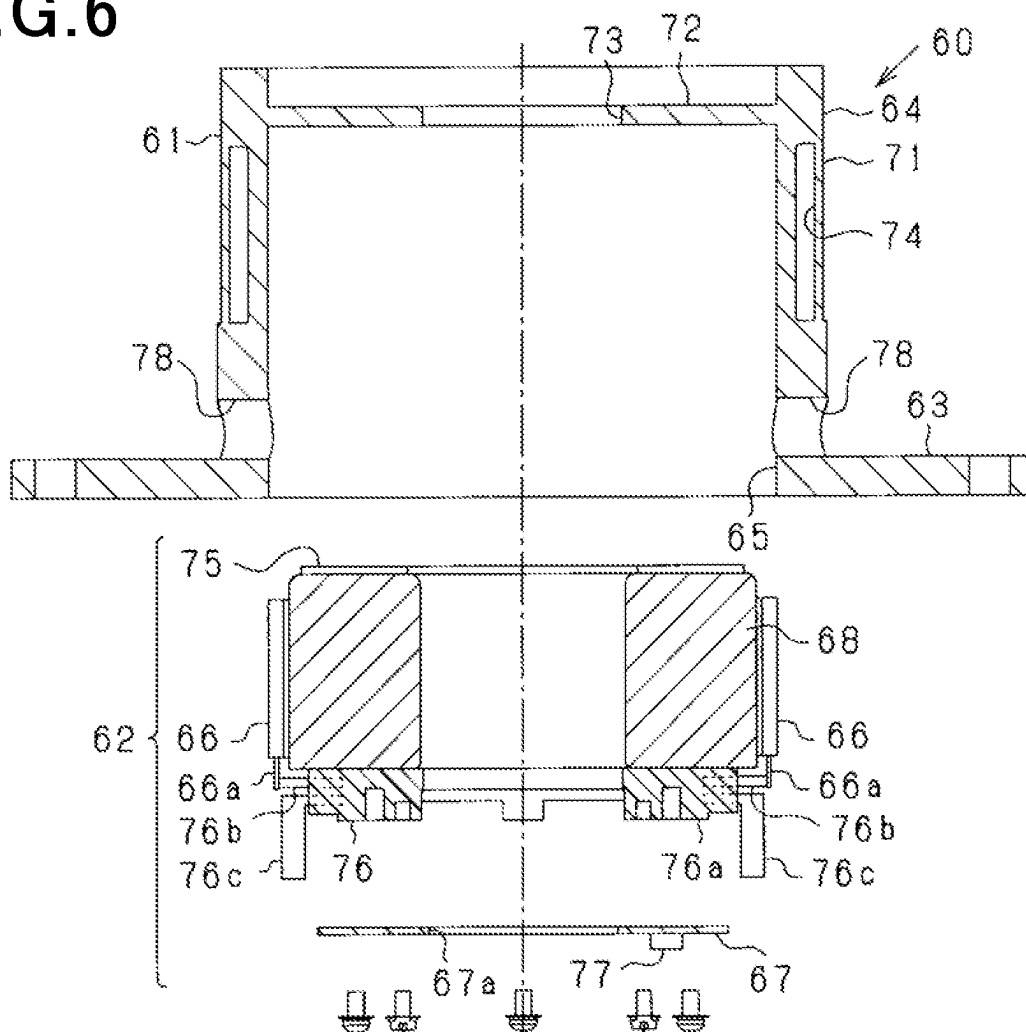


FIG. 7

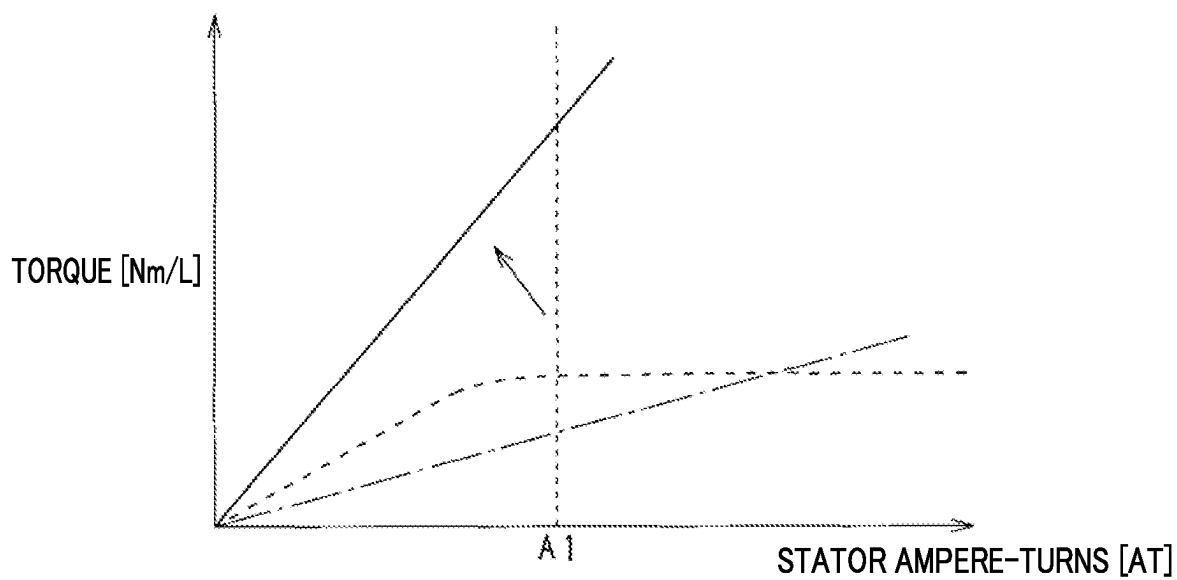


FIG. 8

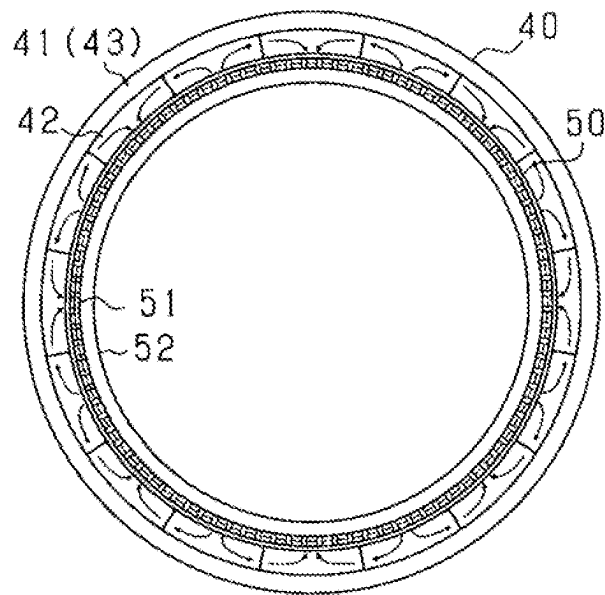


FIG. 9

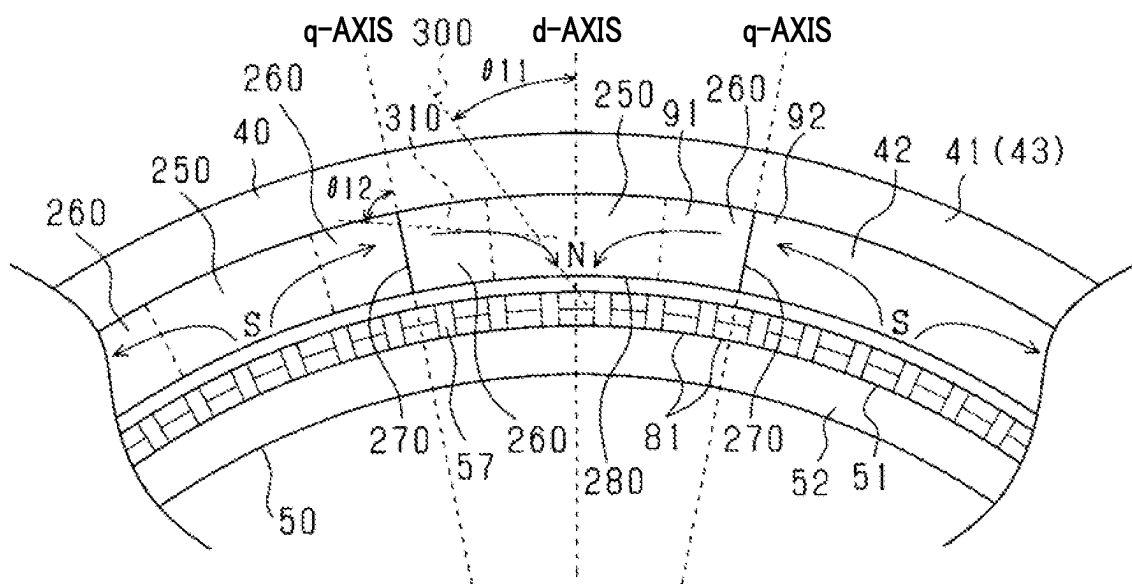


FIG.10

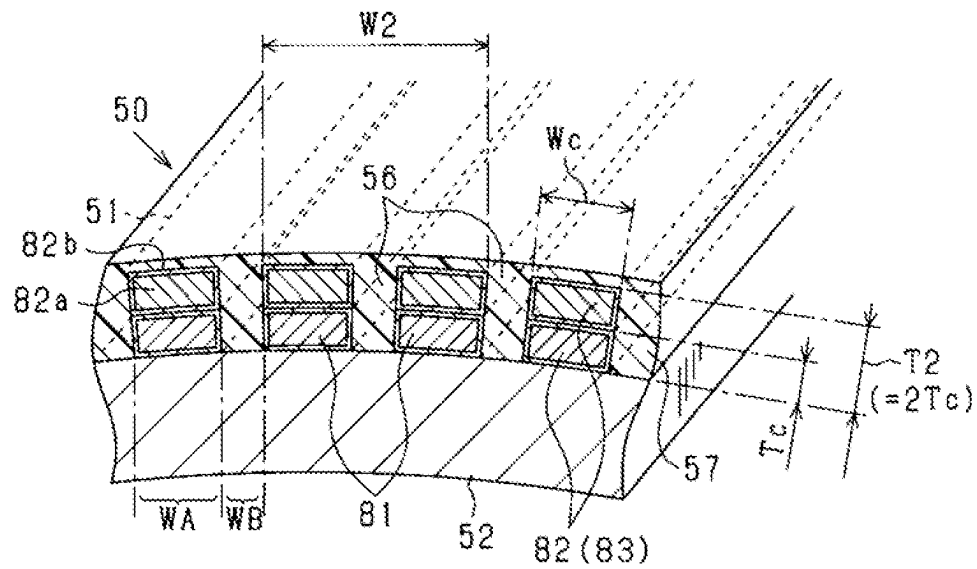


FIG.11

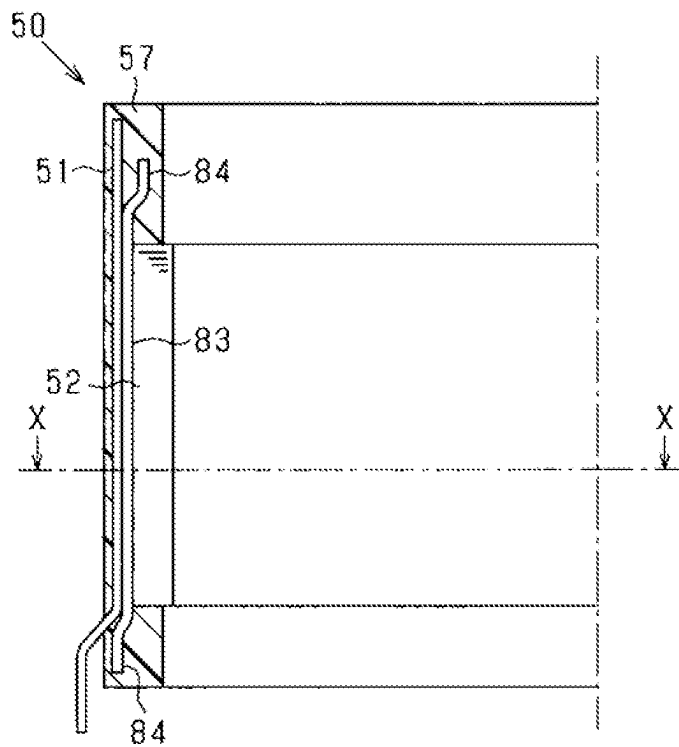


FIG. 12

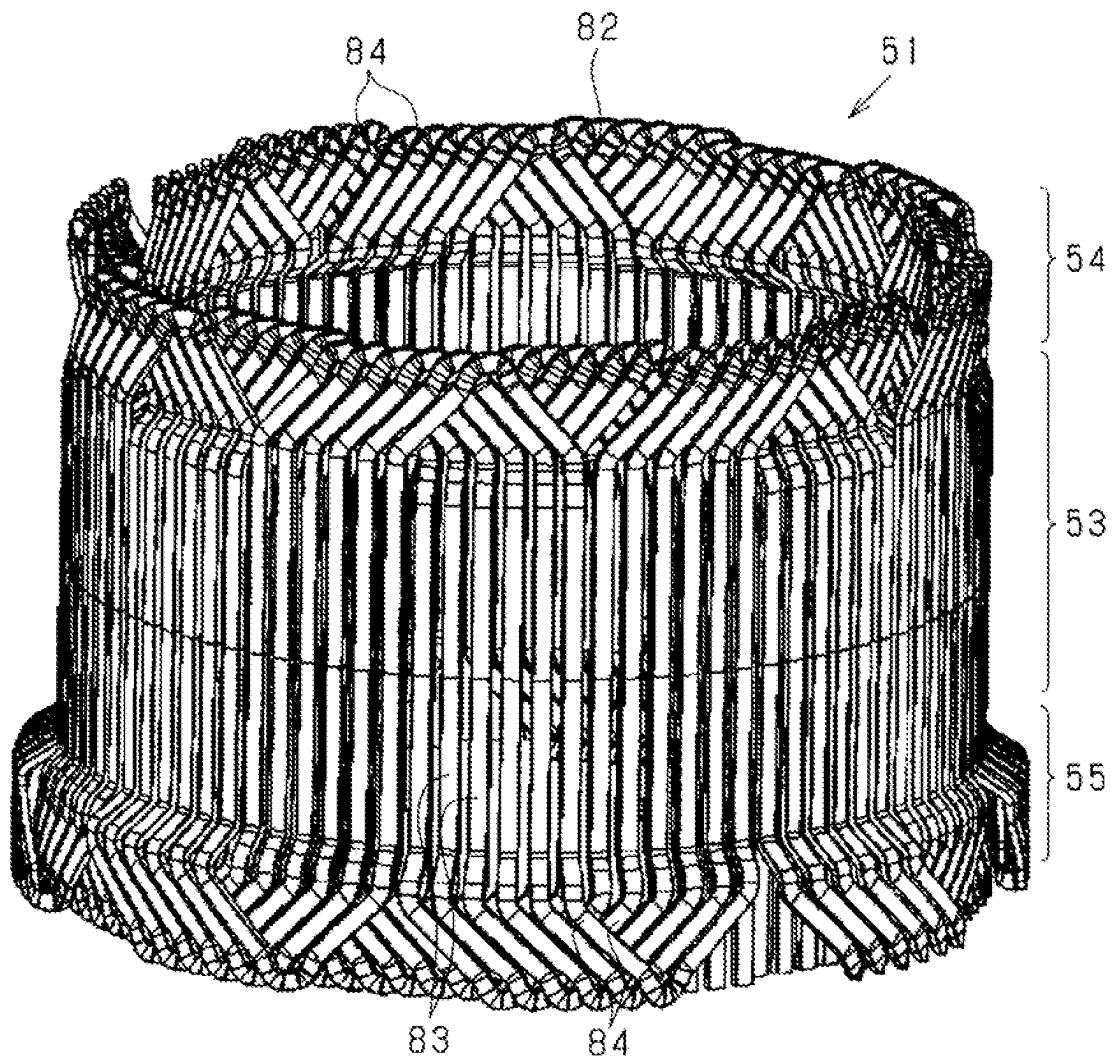


FIG. 13

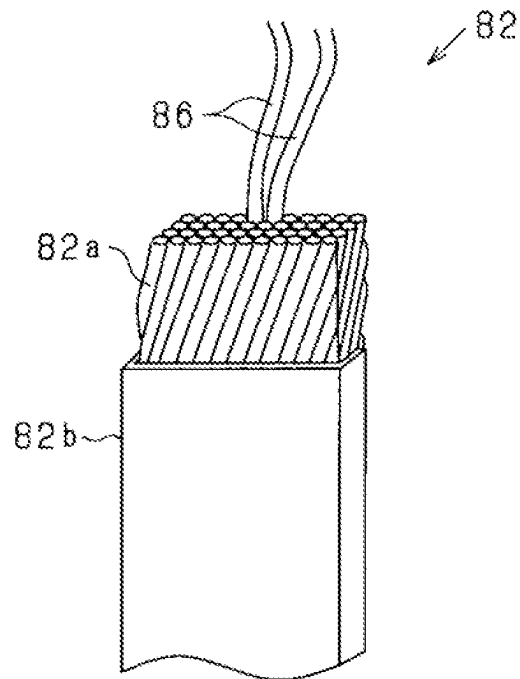


FIG. 14

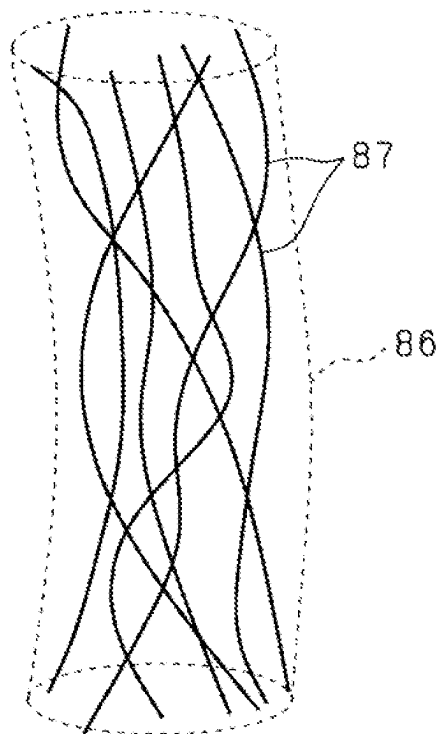


FIG. 15

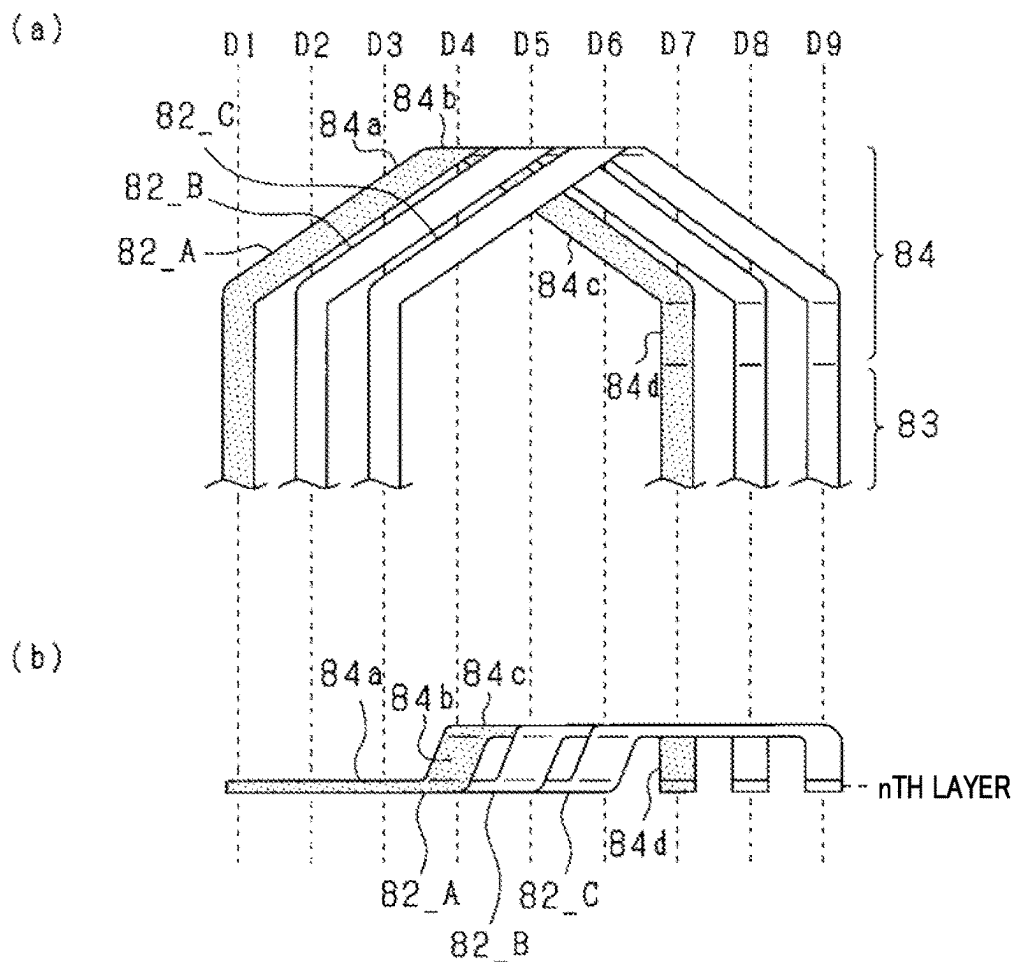


FIG. 16

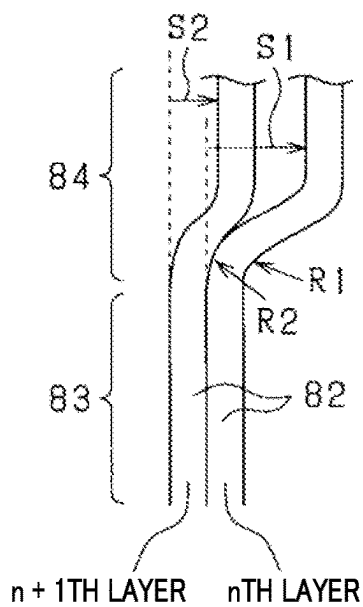


FIG.17

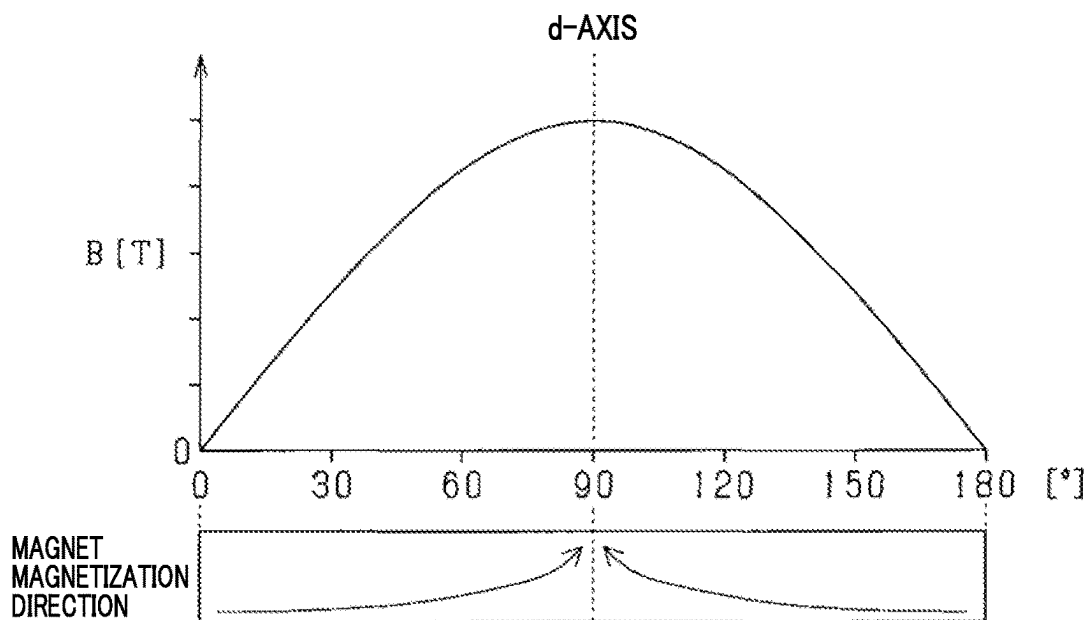


FIG.18

COMPARATIVE EXAMPLE

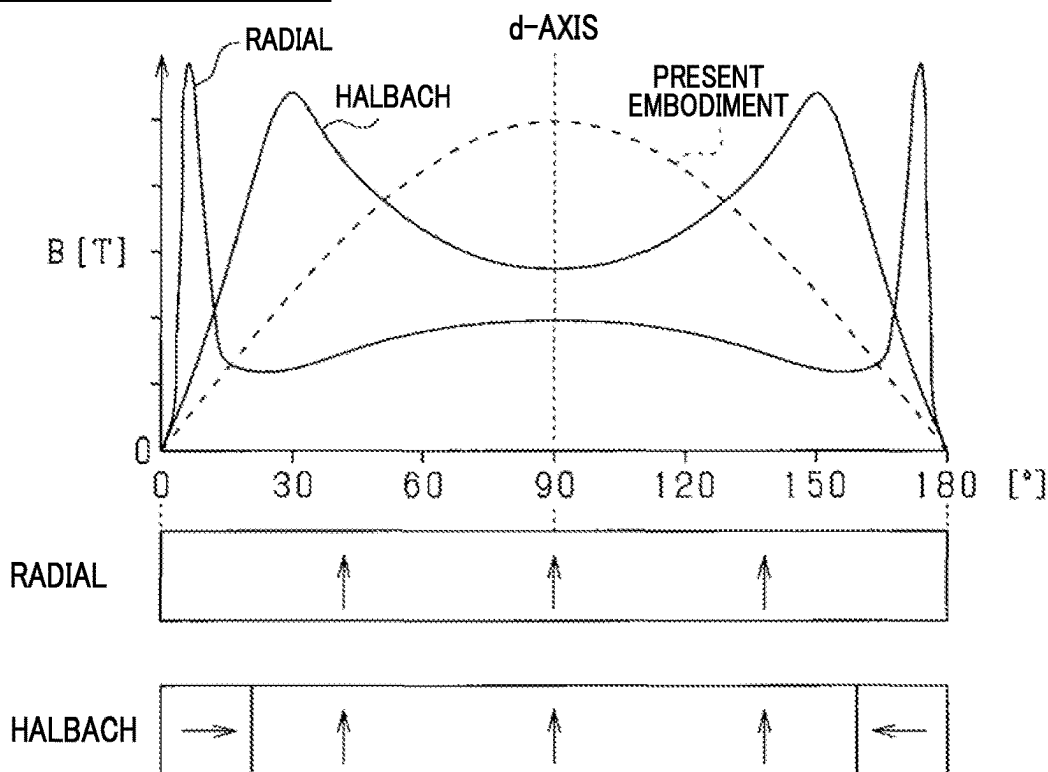


FIG.19

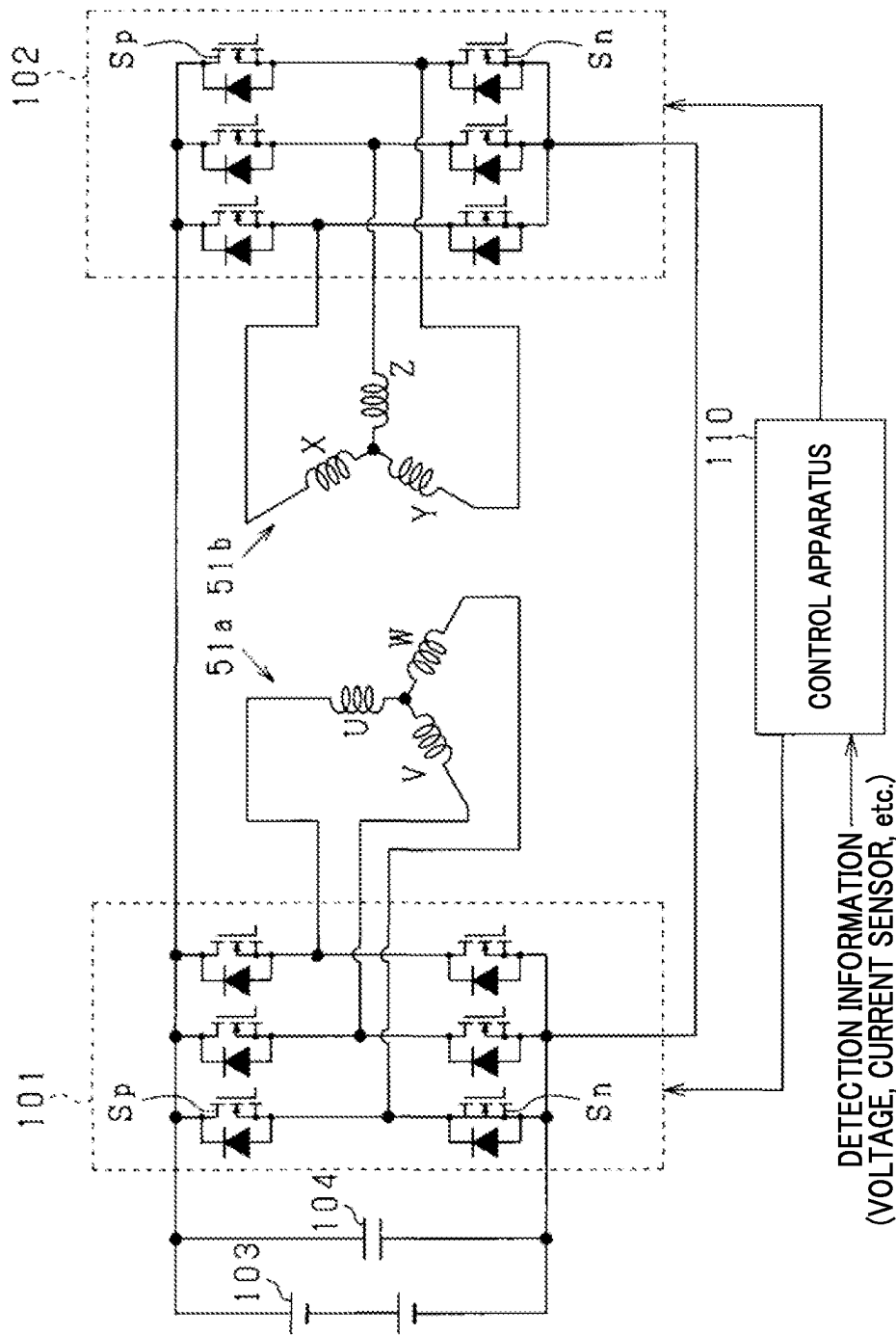


FIG. 20

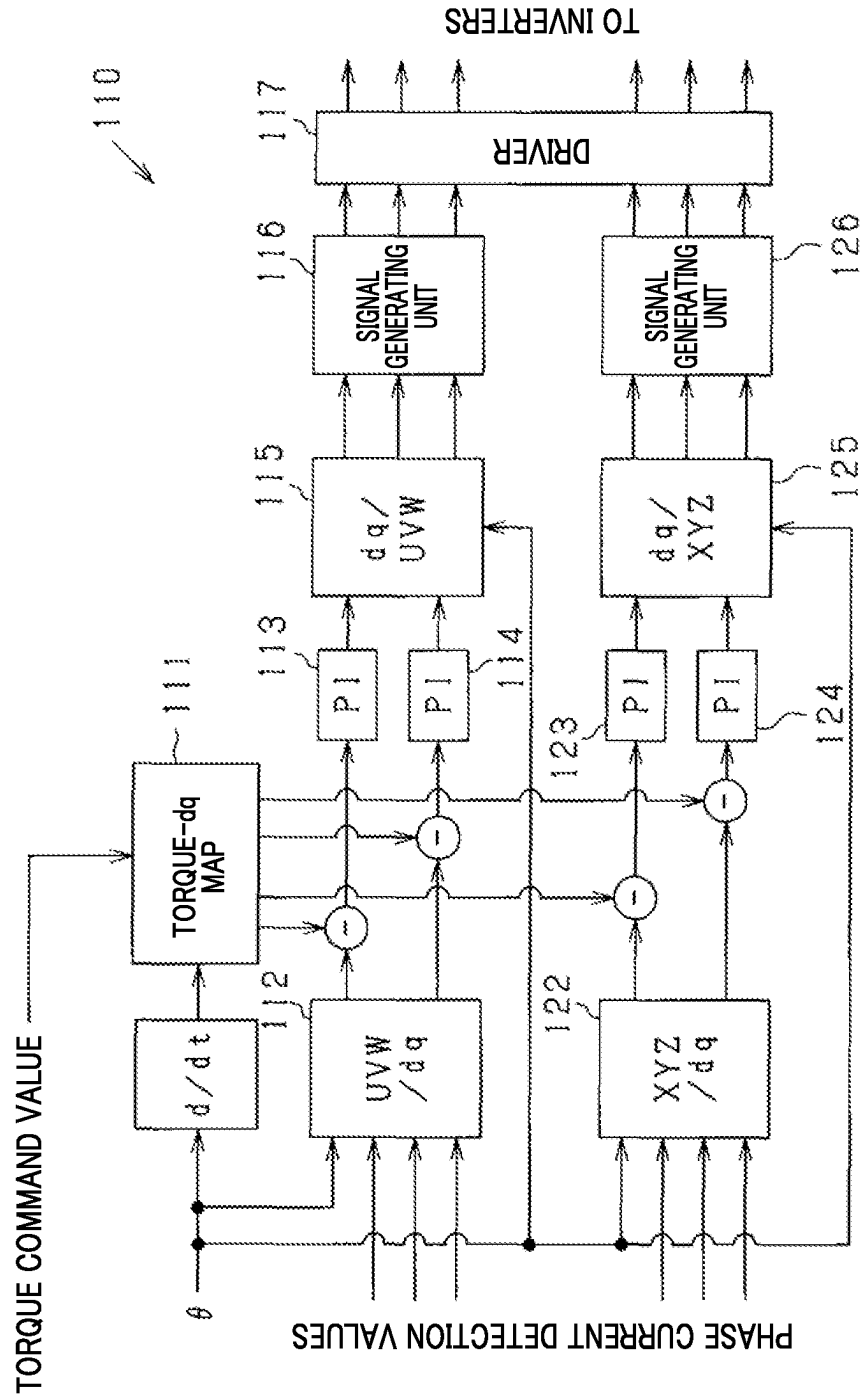


FIG. 21

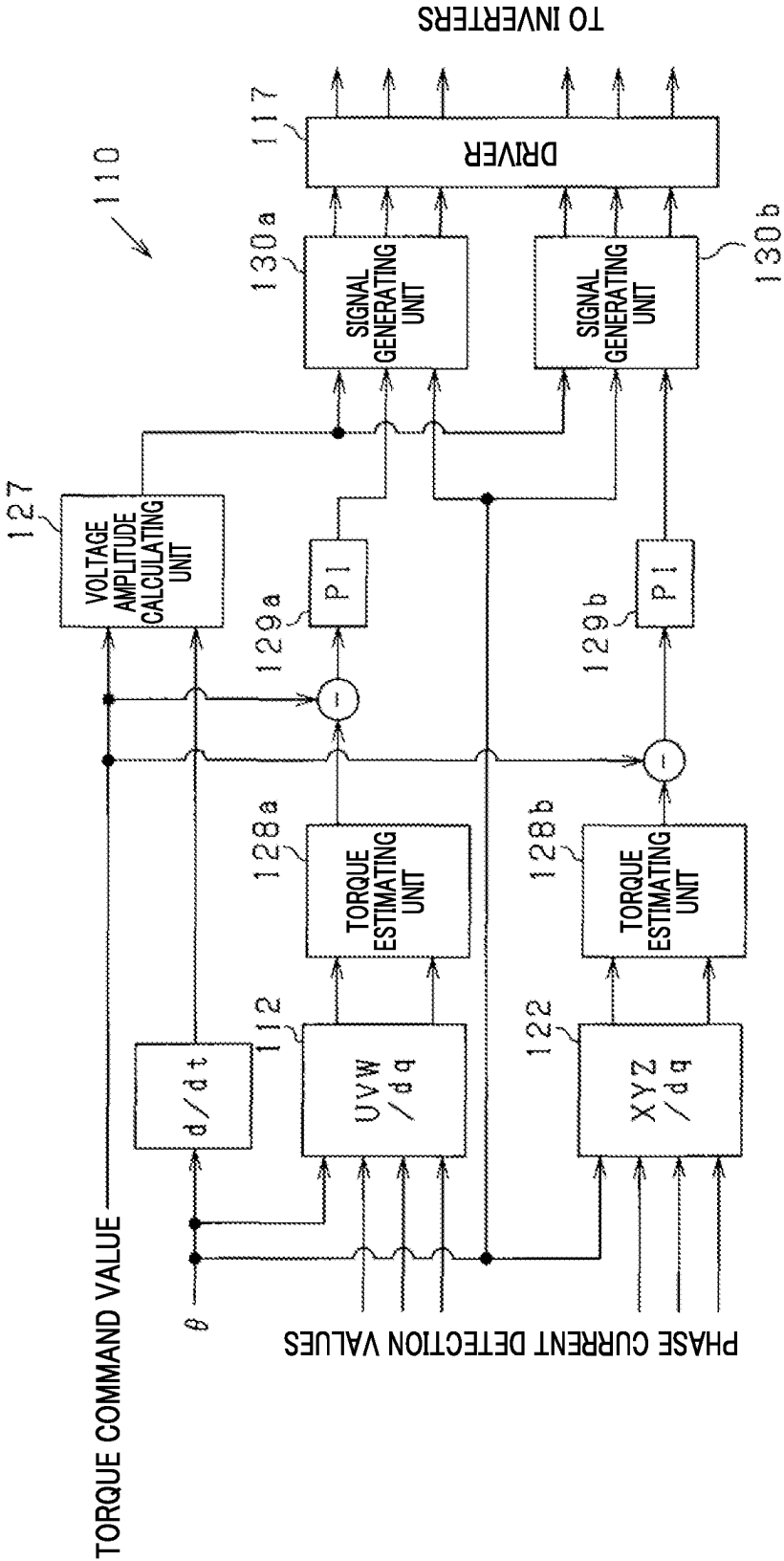


FIG.22

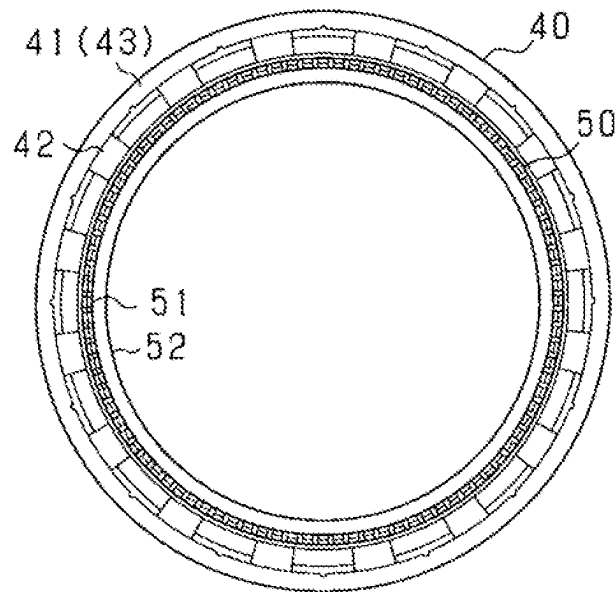


FIG.23

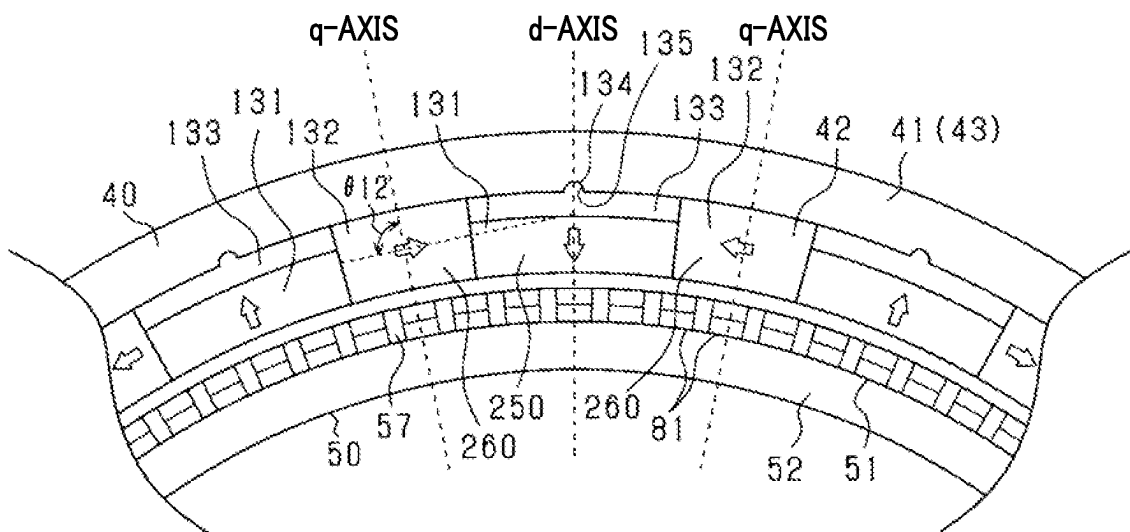
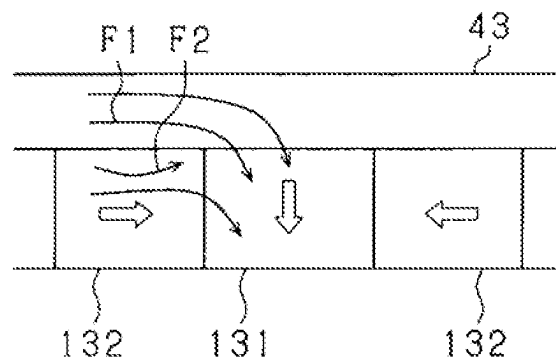


FIG. 24

(a)



(b)

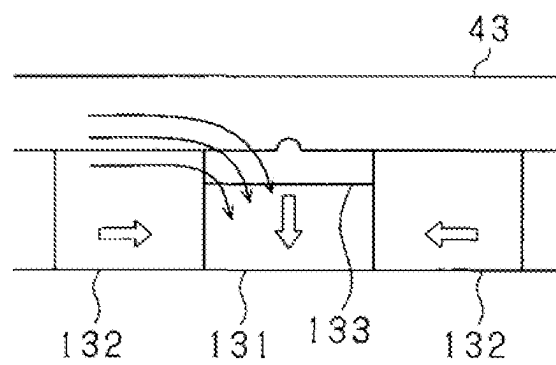


FIG.25

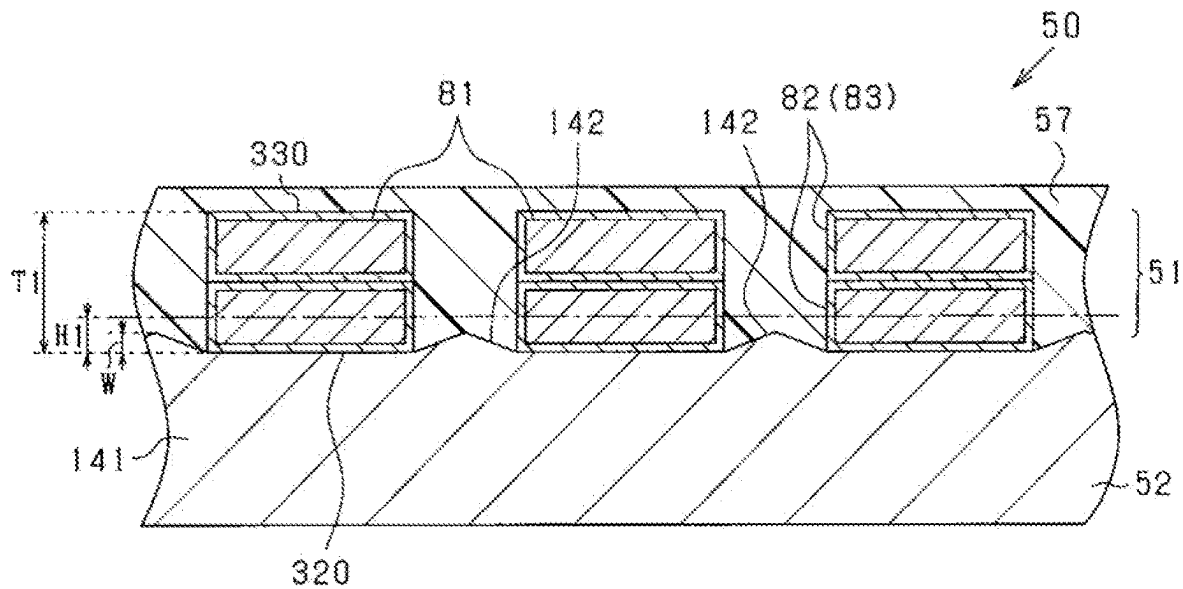


FIG.26

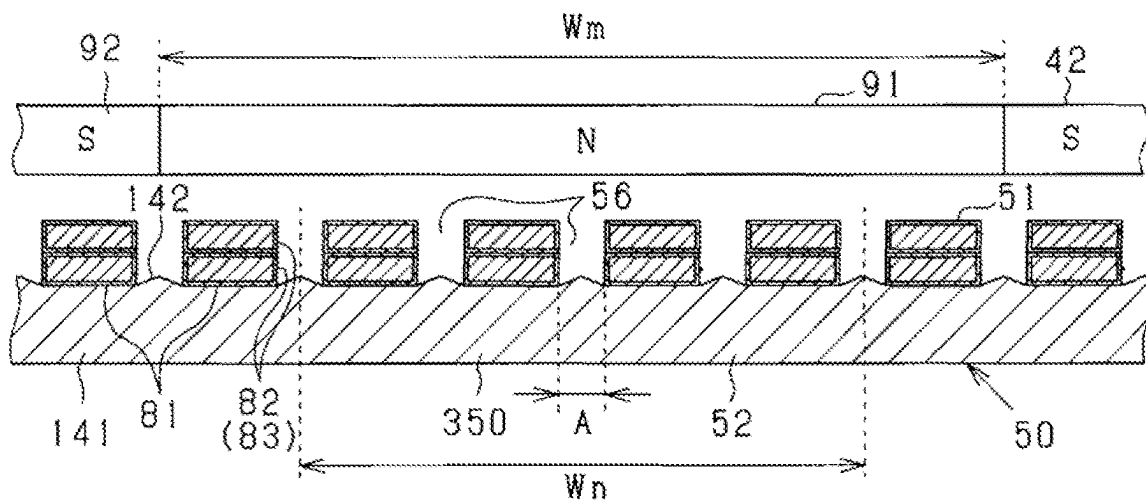


FIG.27

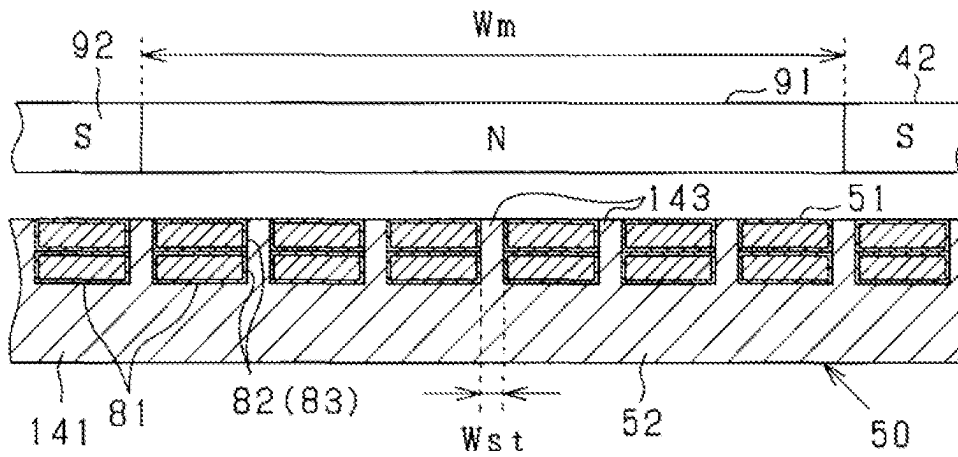


FIG.28

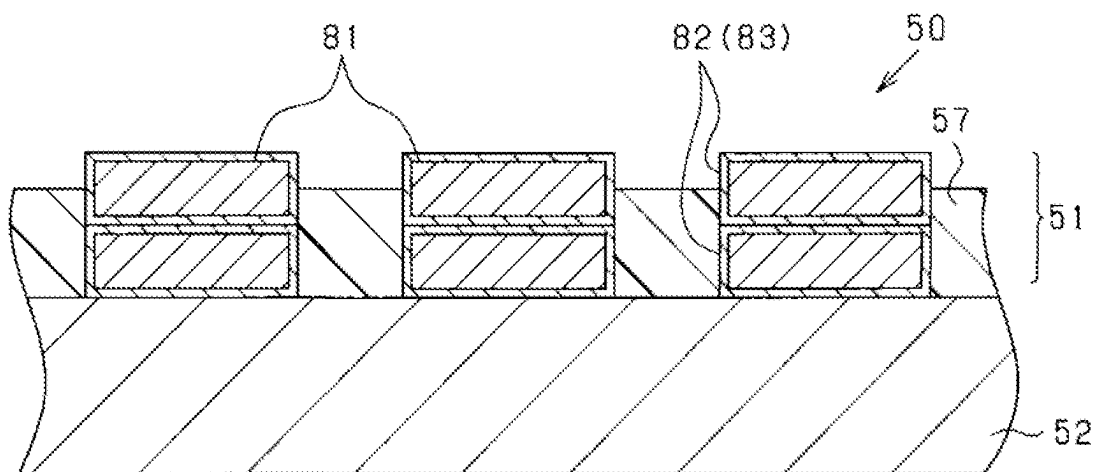


FIG.29

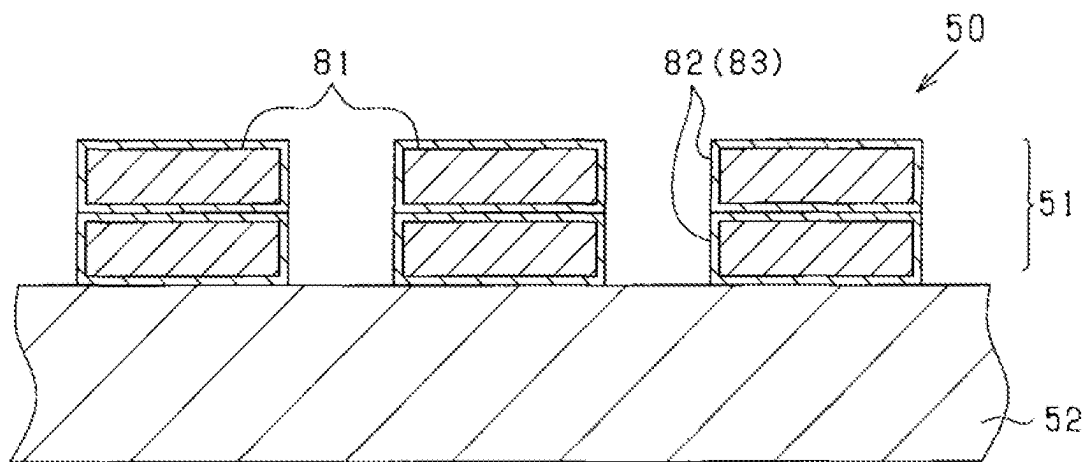


FIG.30

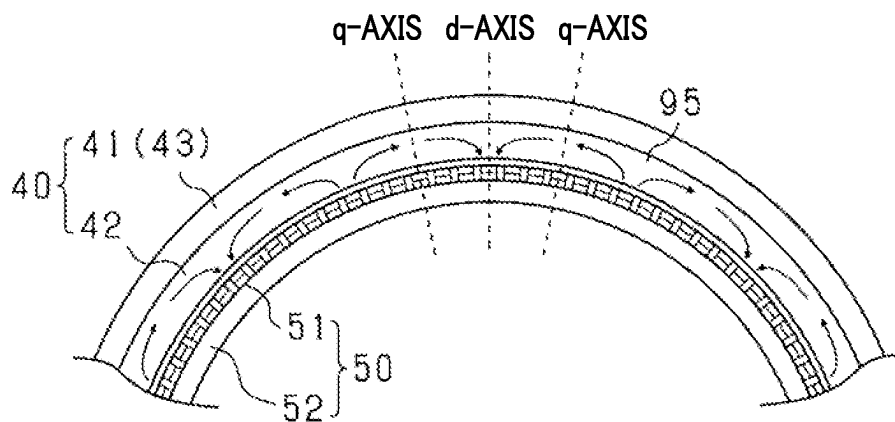


FIG.31

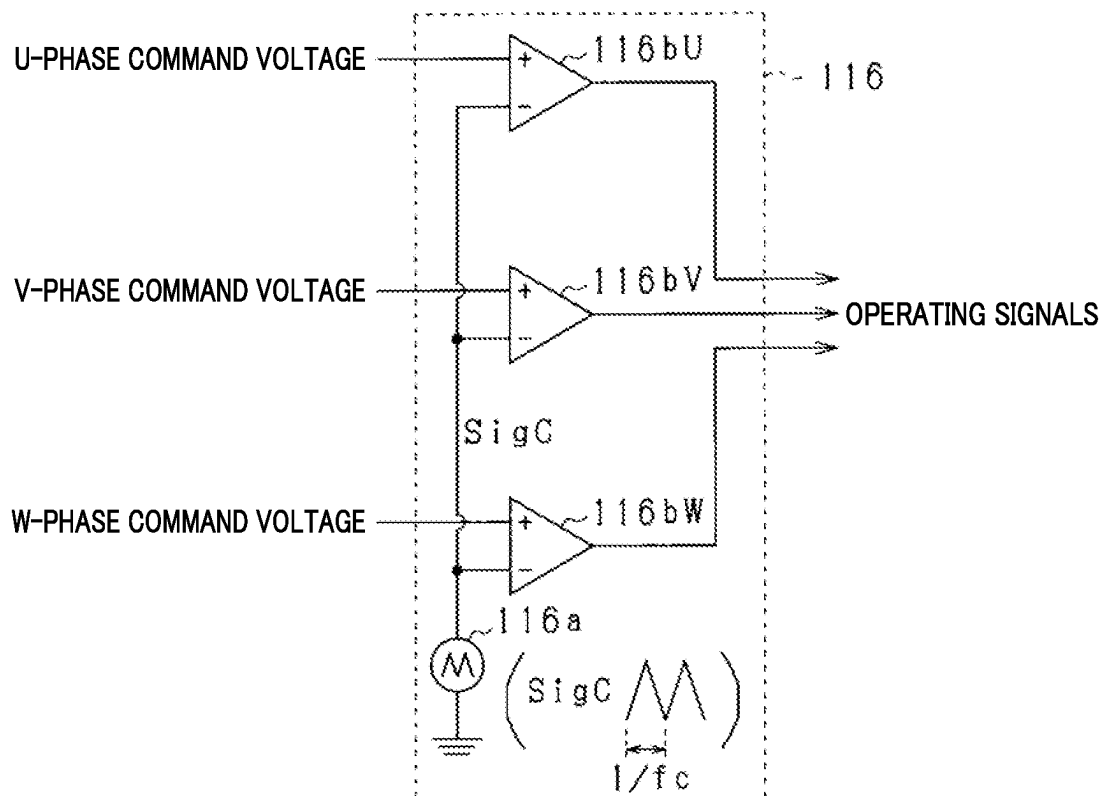


FIG.32

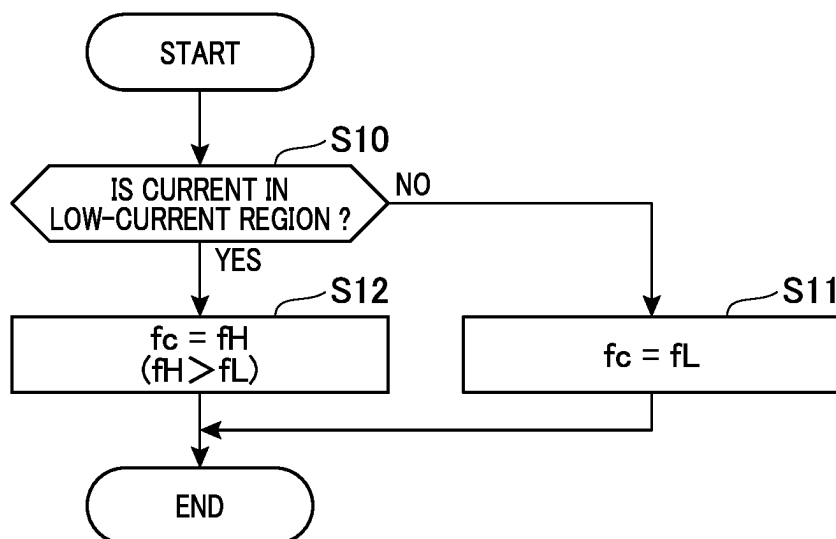


FIG. 33

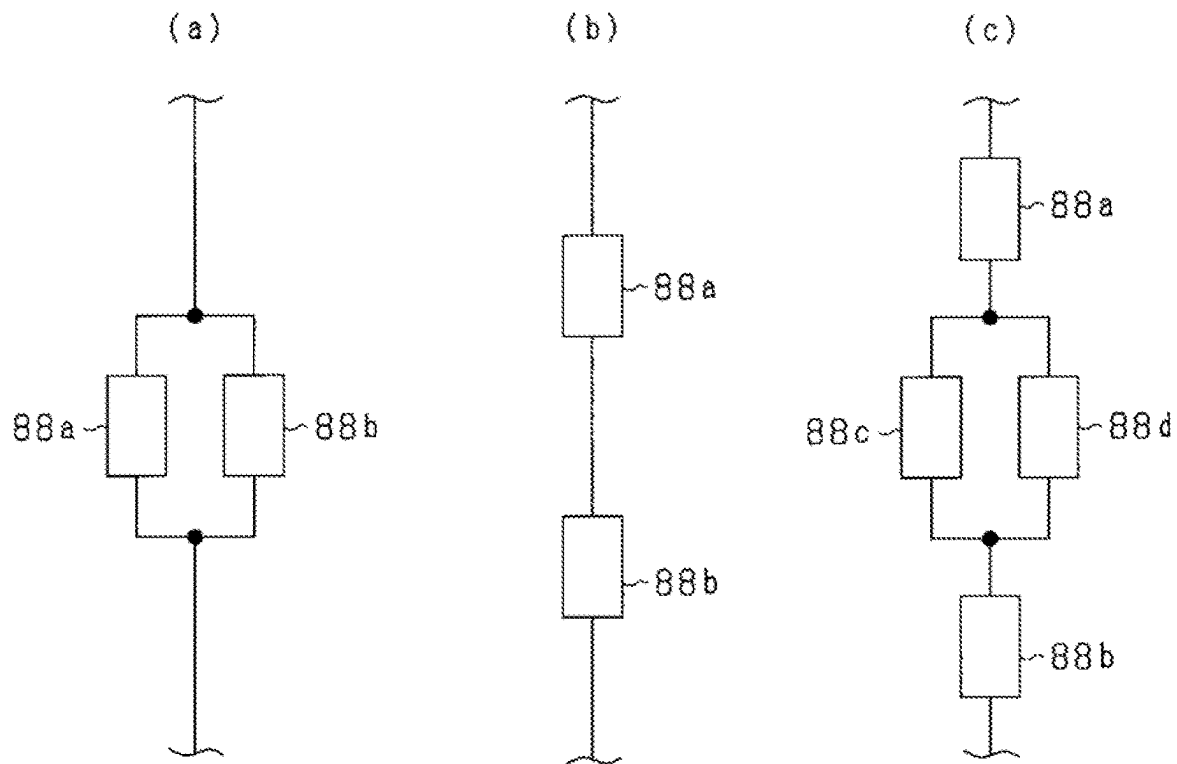


FIG. 34

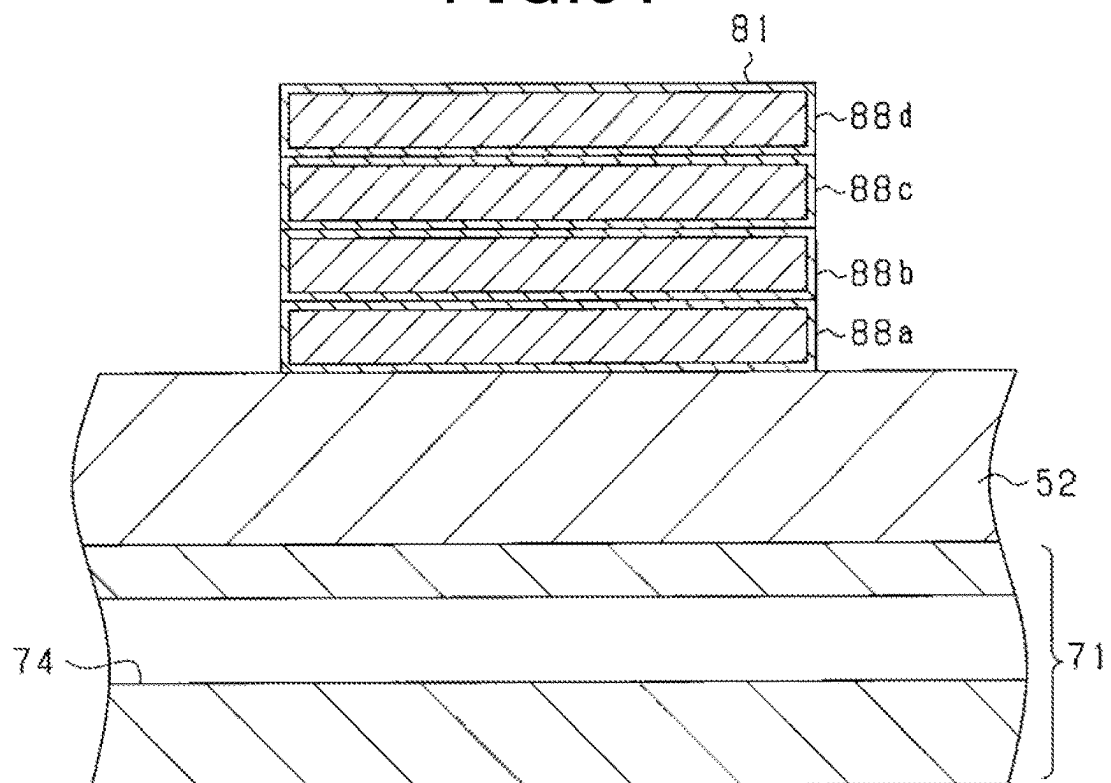


FIG.35

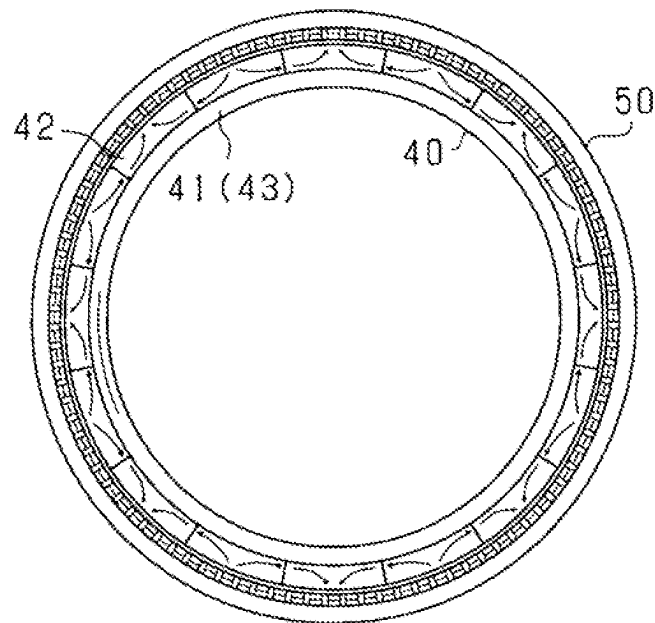


FIG.36

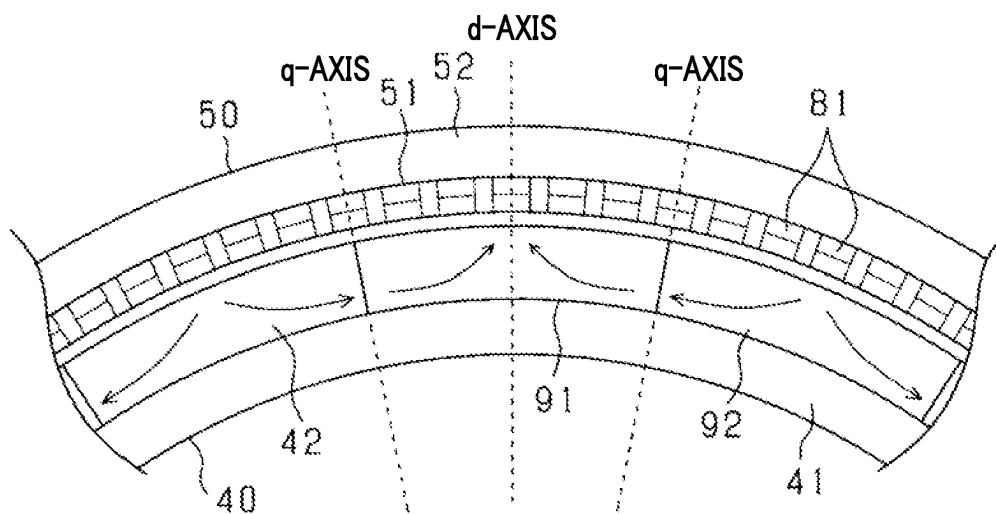


FIG.37

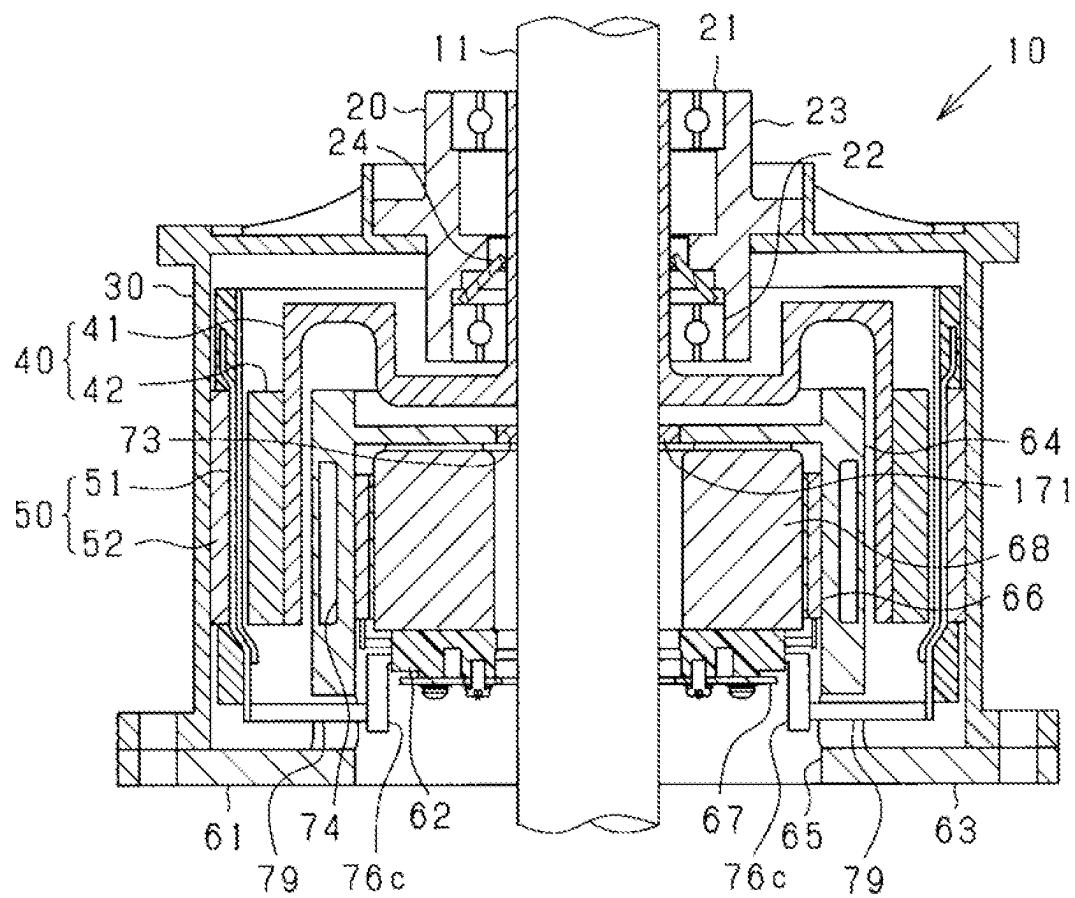


FIG. 38

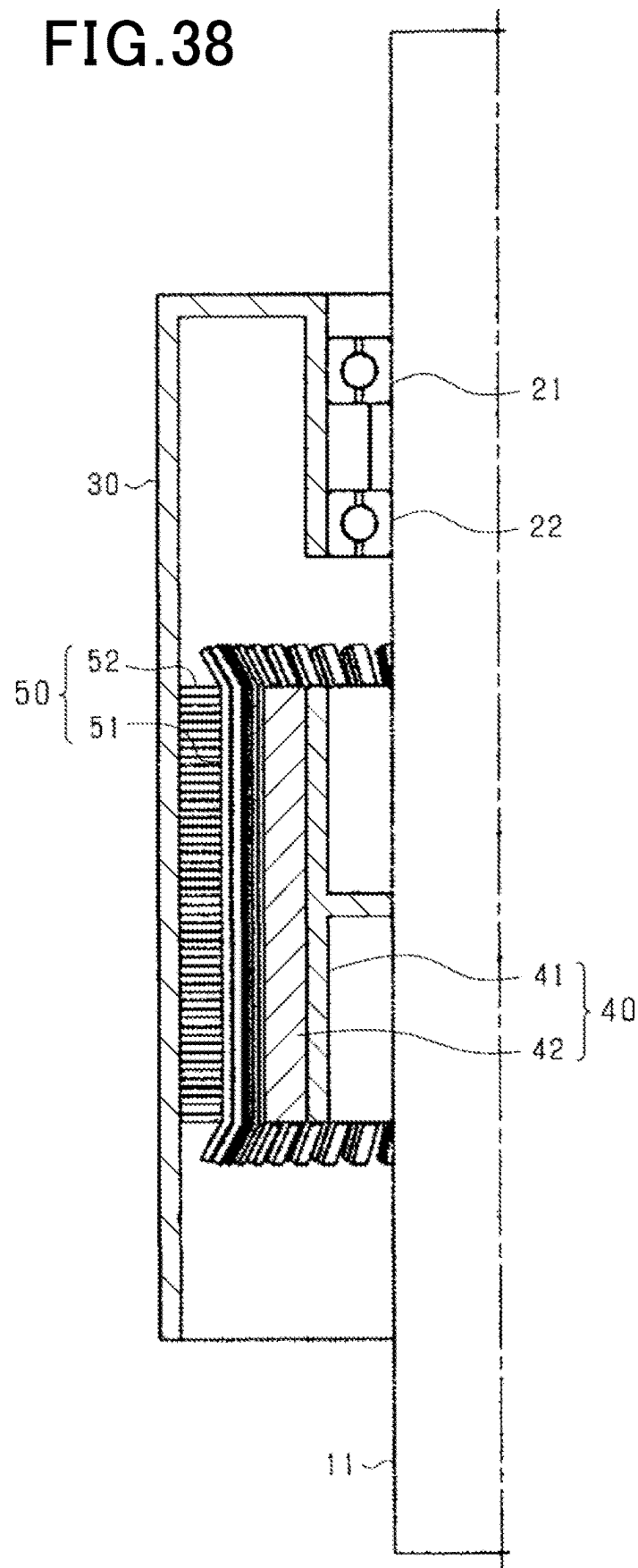


FIG. 39

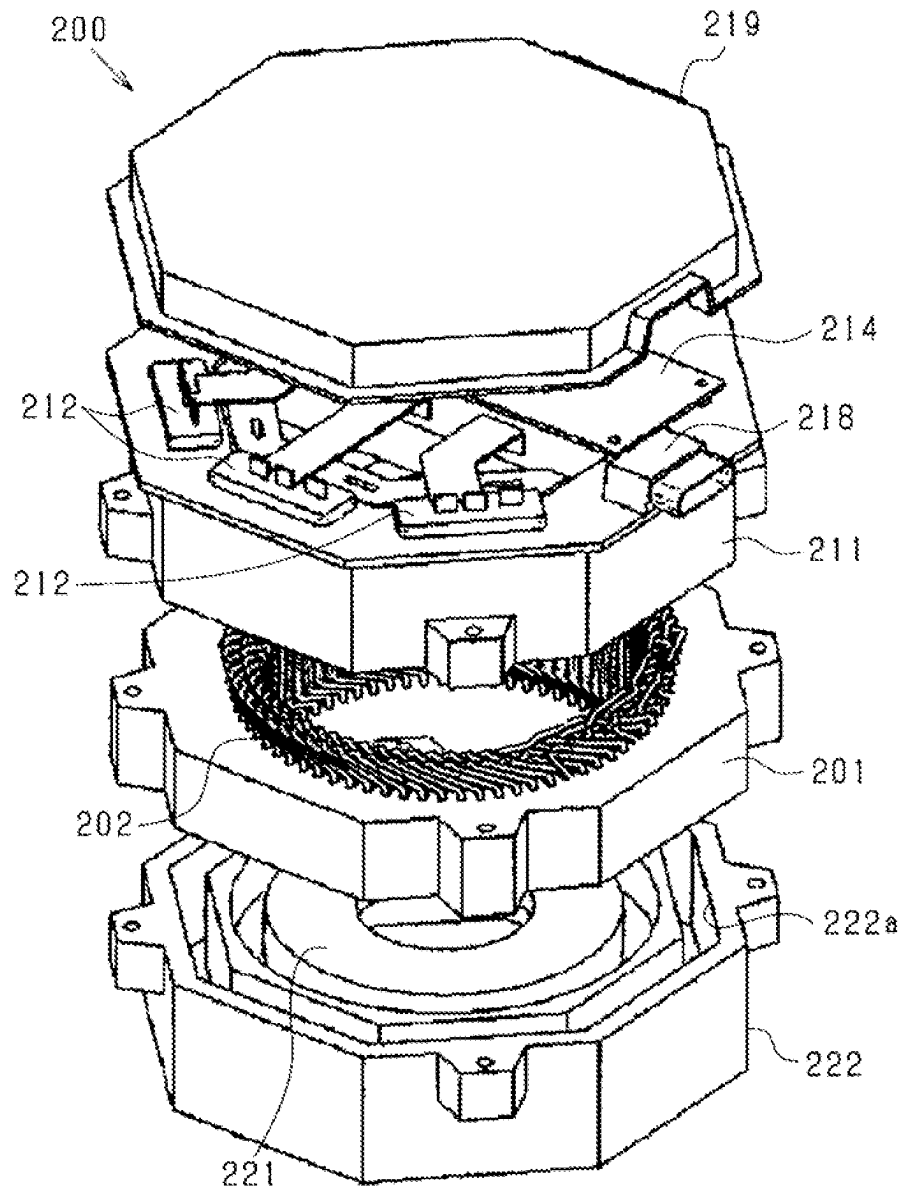


FIG. 41

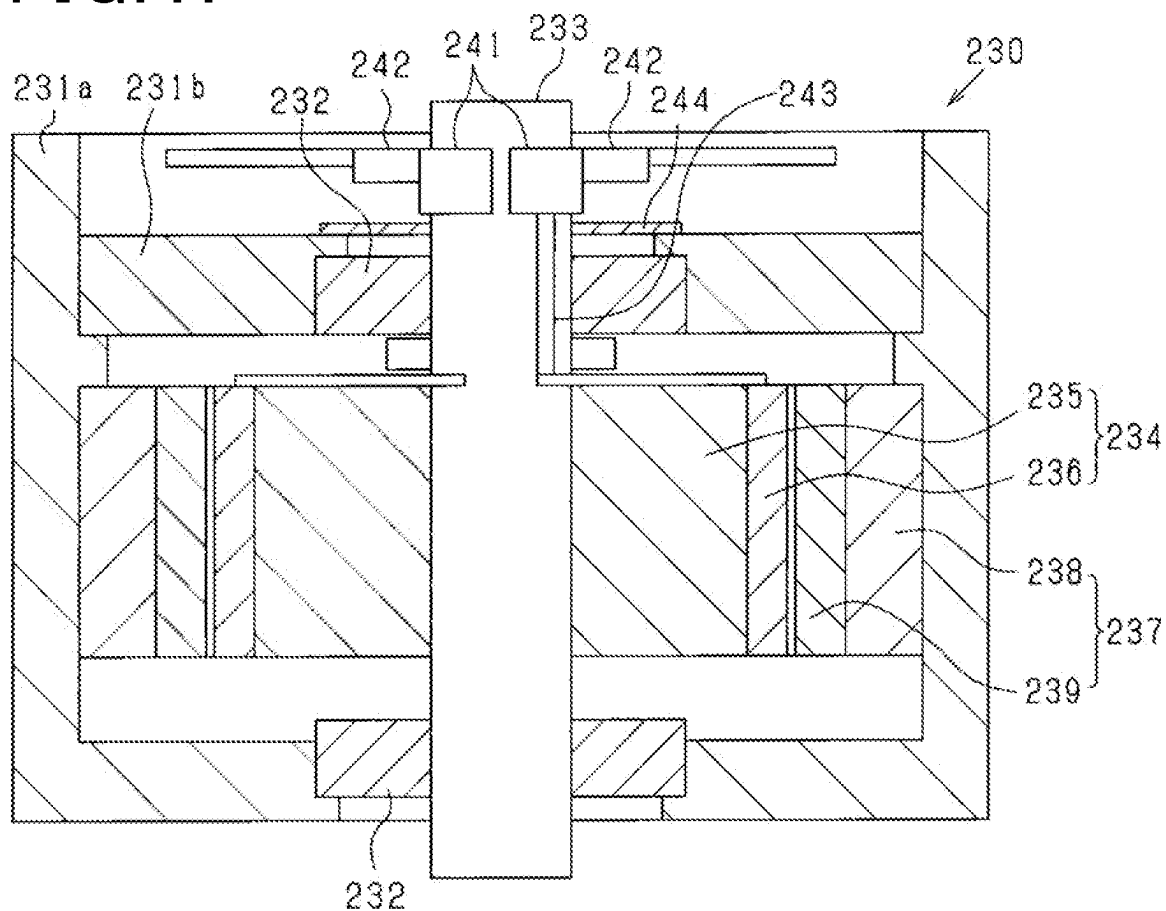


FIG. 42

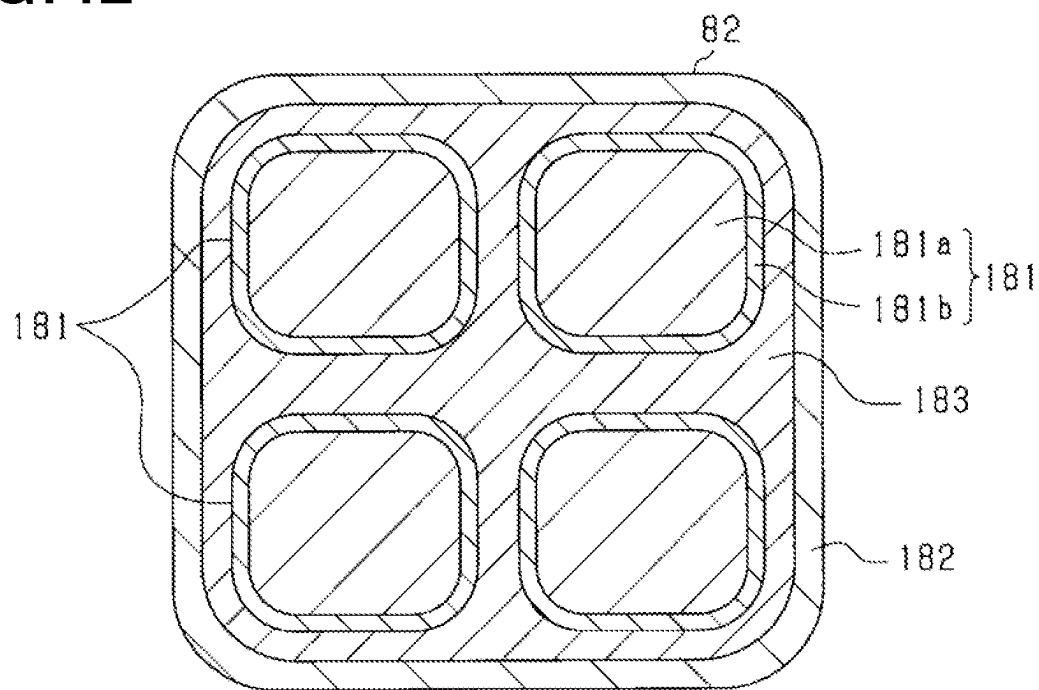


FIG. 43

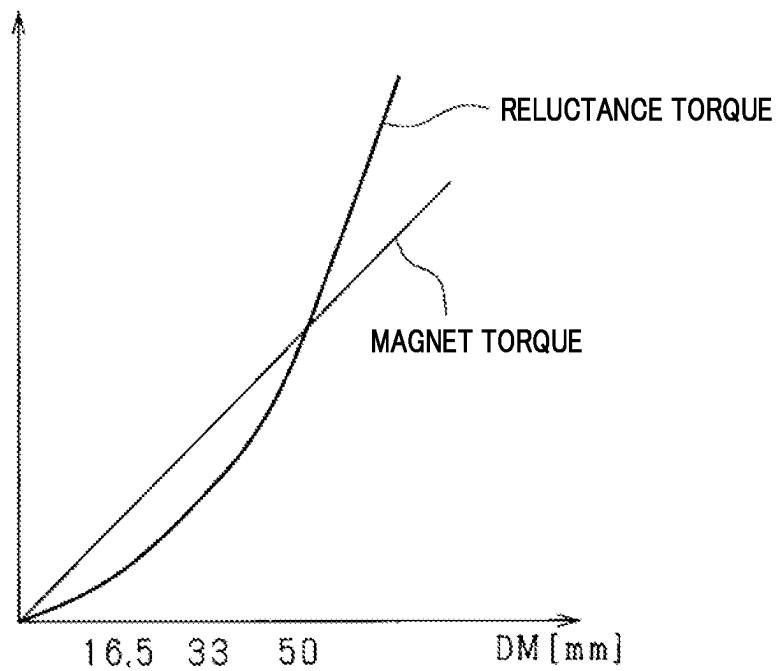


FIG. 44

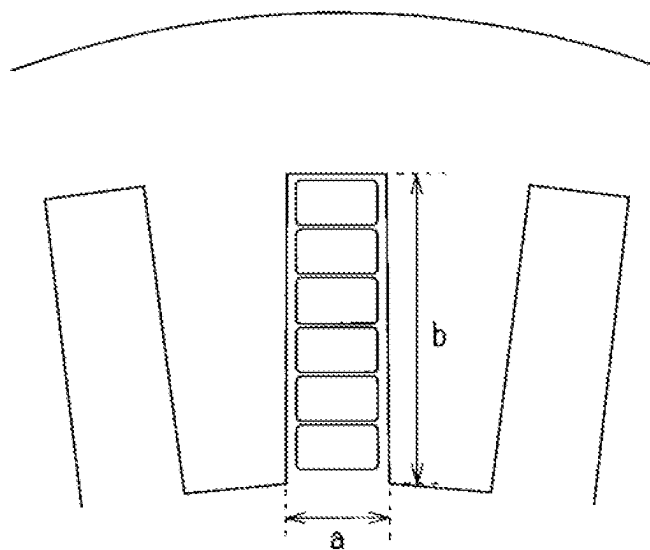


FIG.45

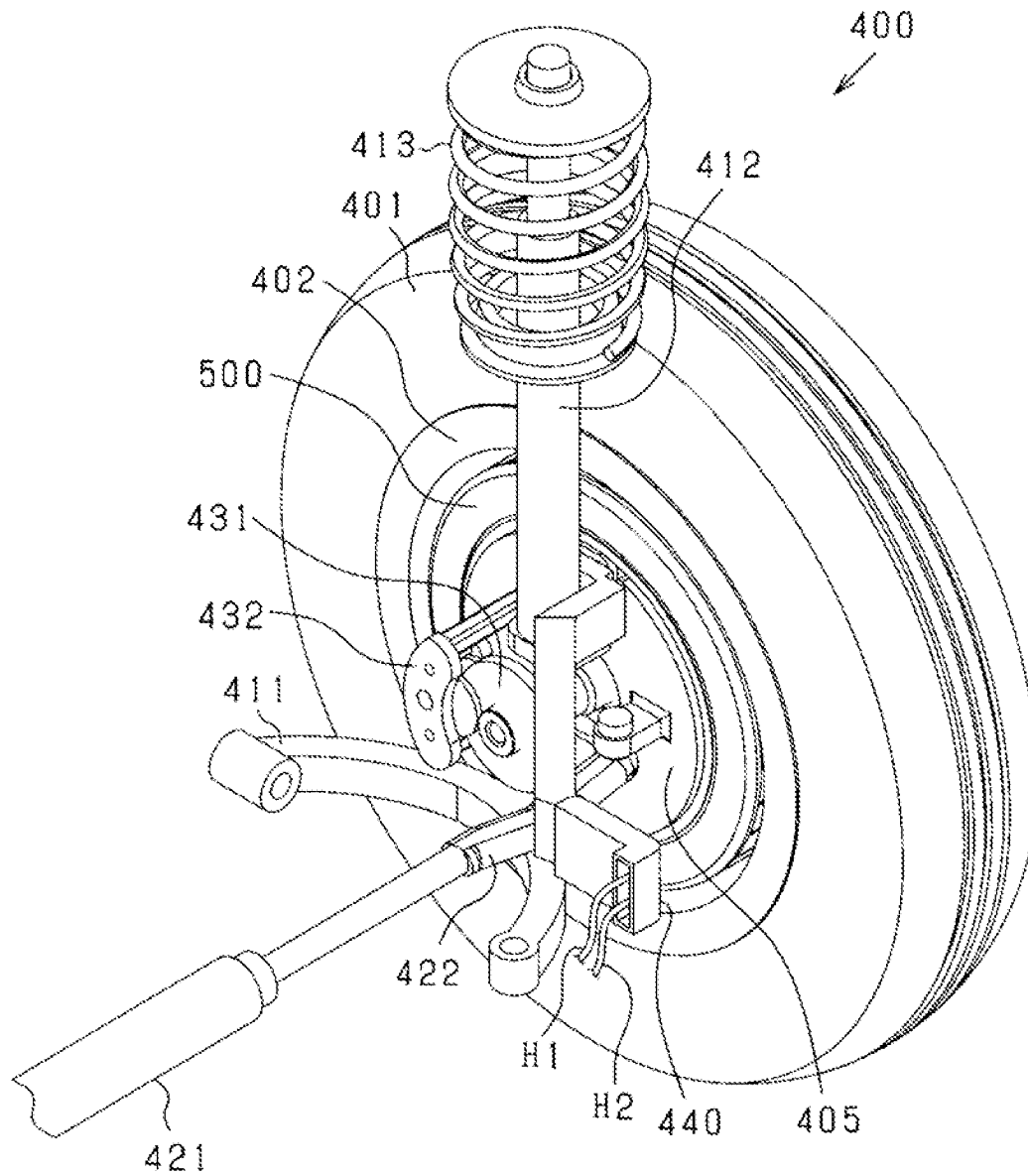
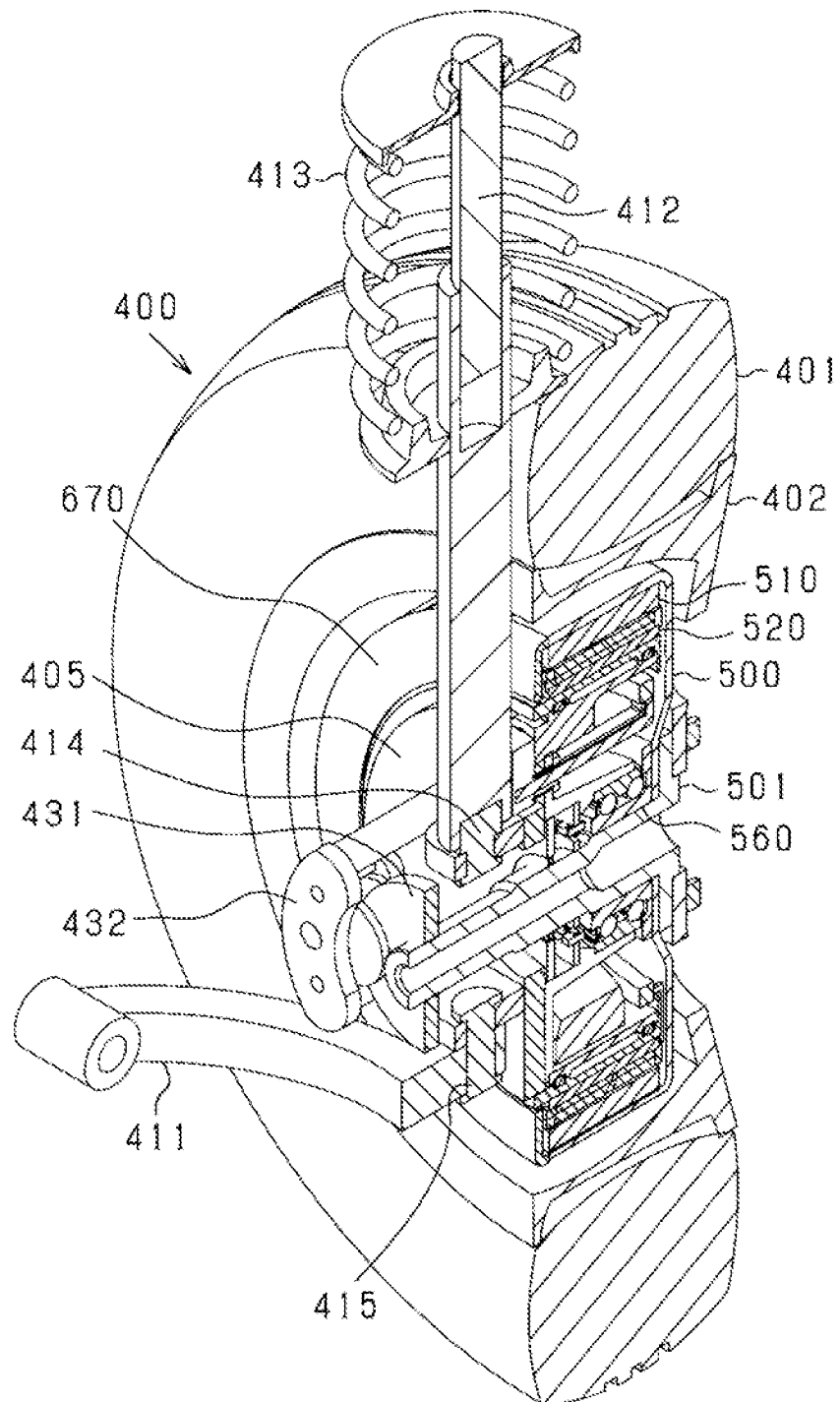


FIG. 46



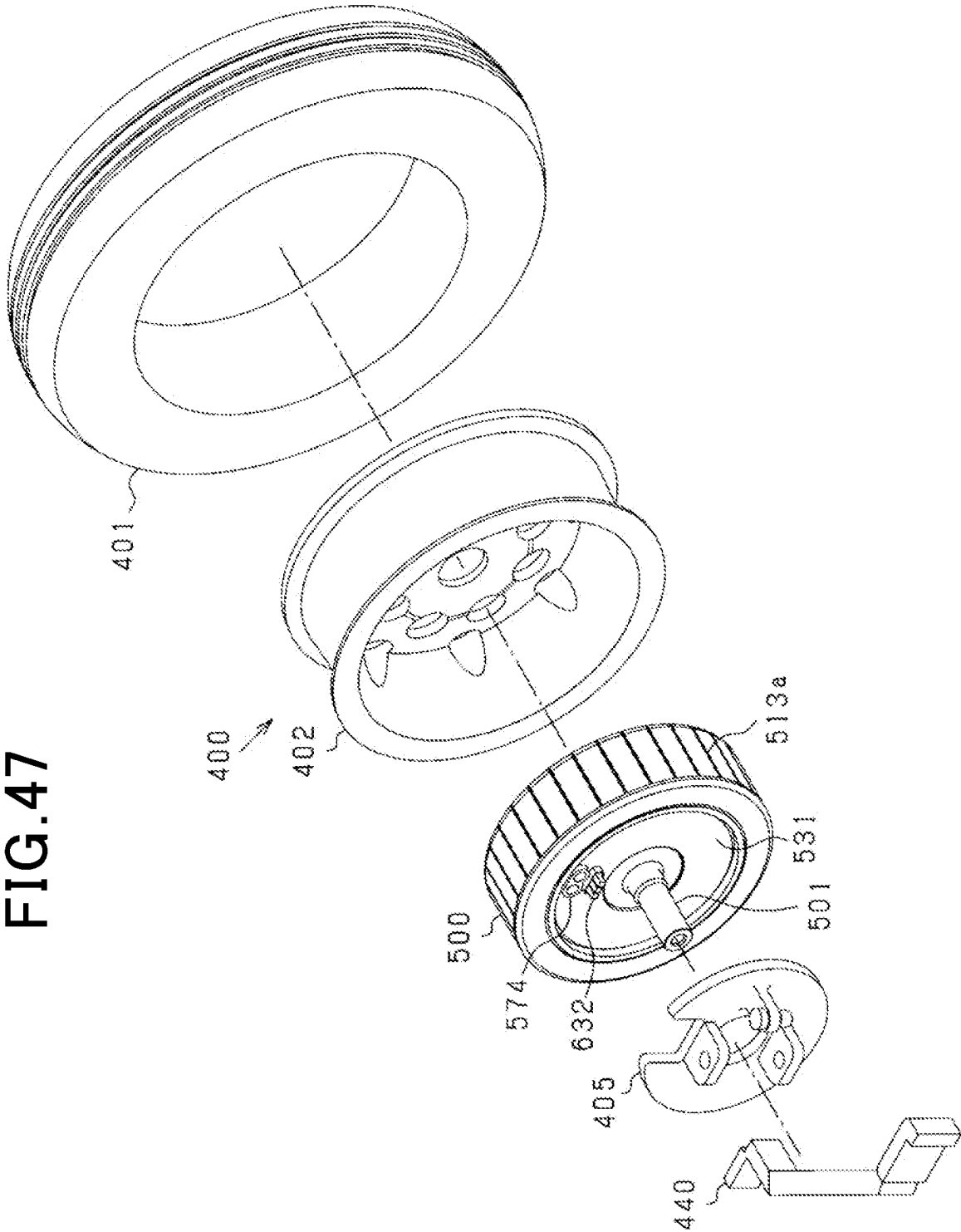


FIG. 48

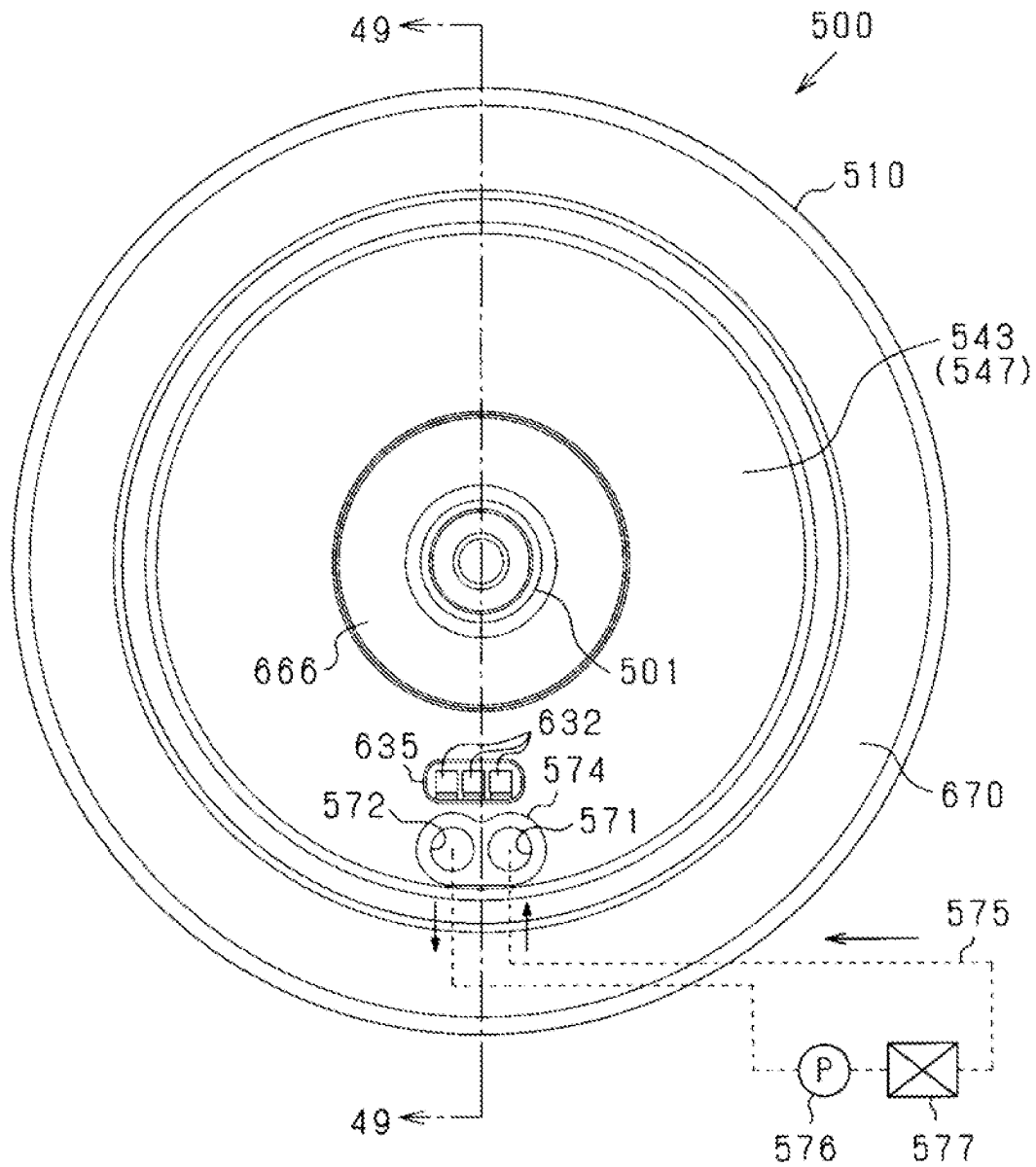


FIG. 49

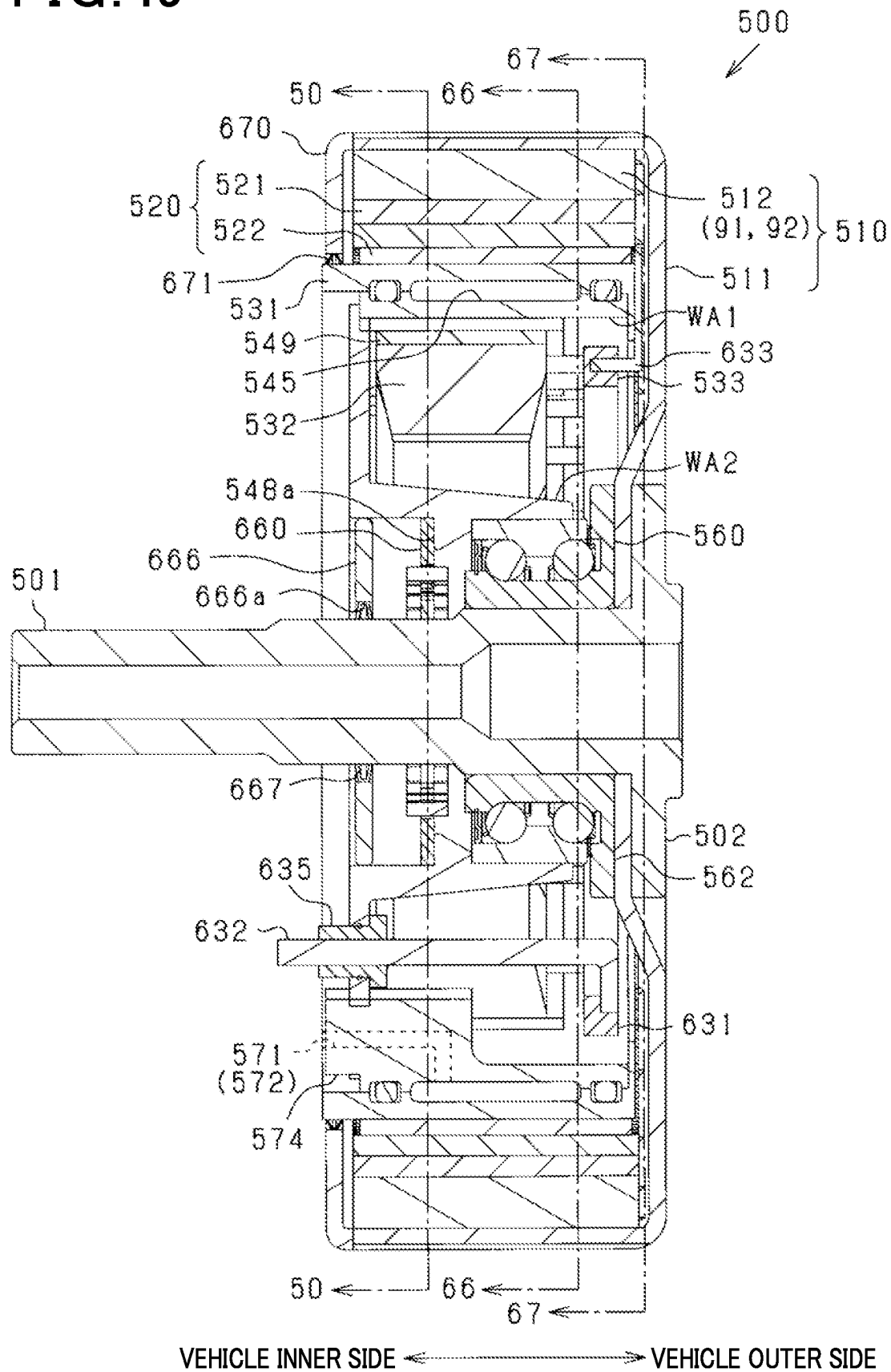


FIG. 50

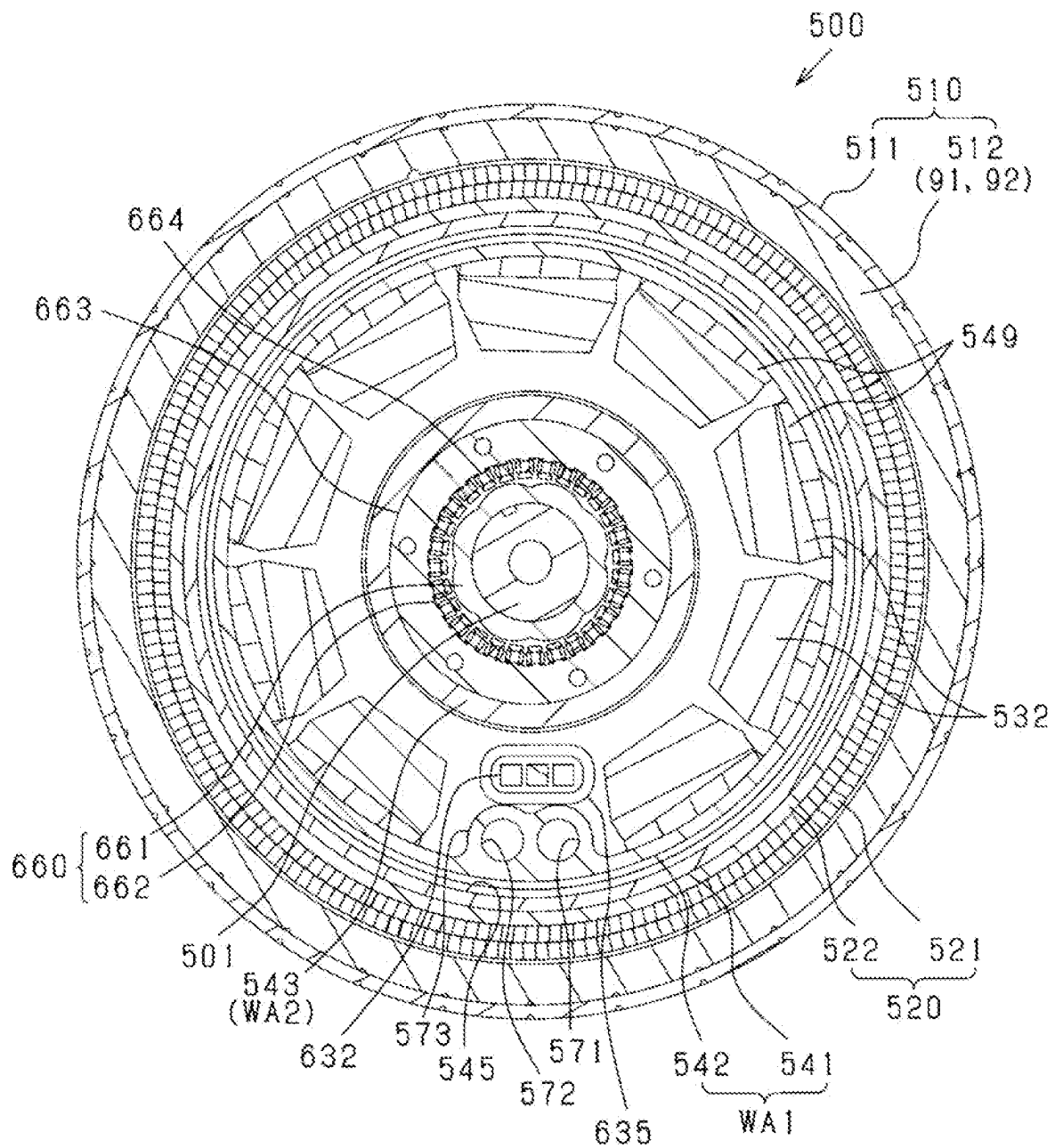


FIG. 51

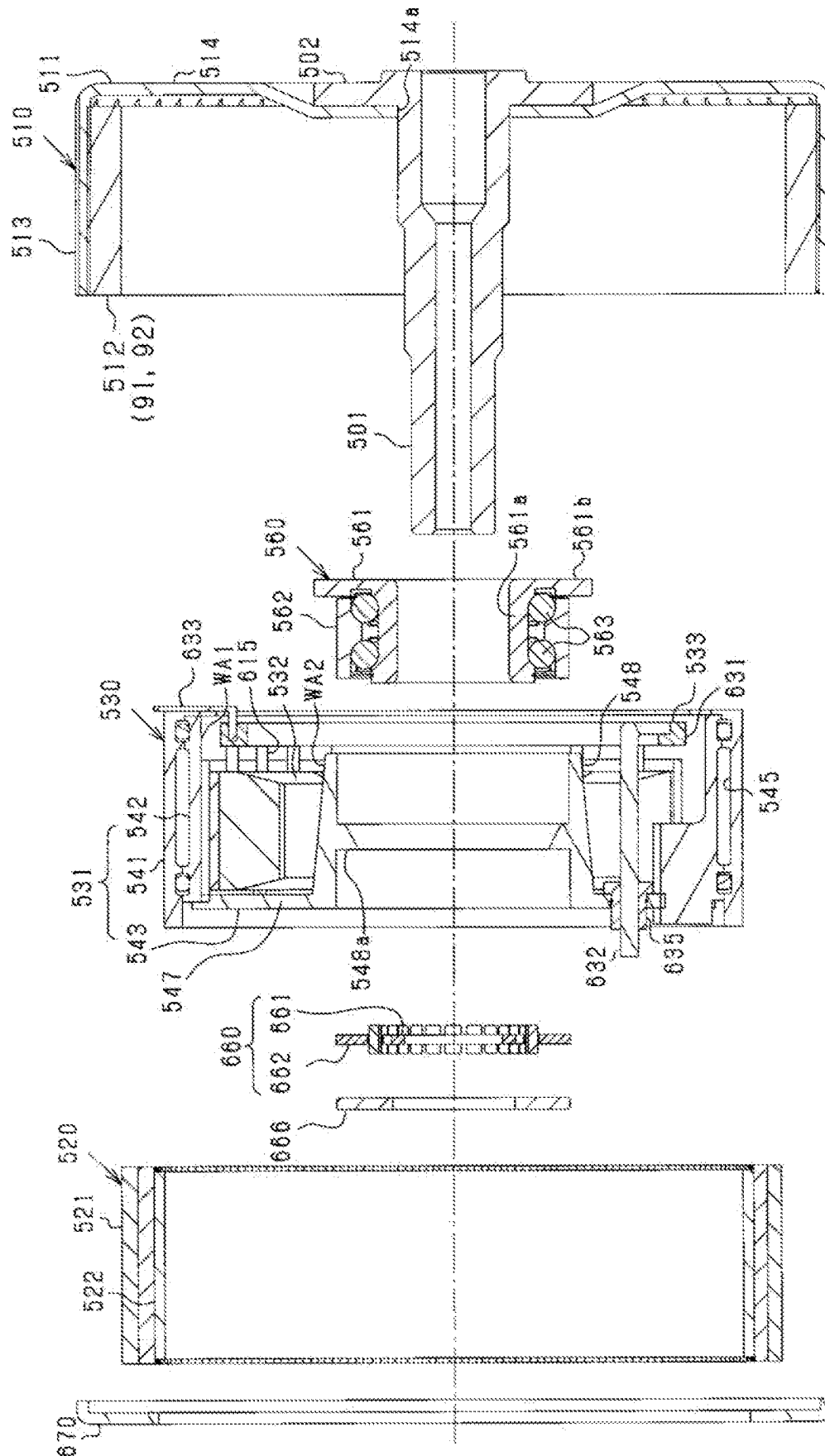


FIG. 52

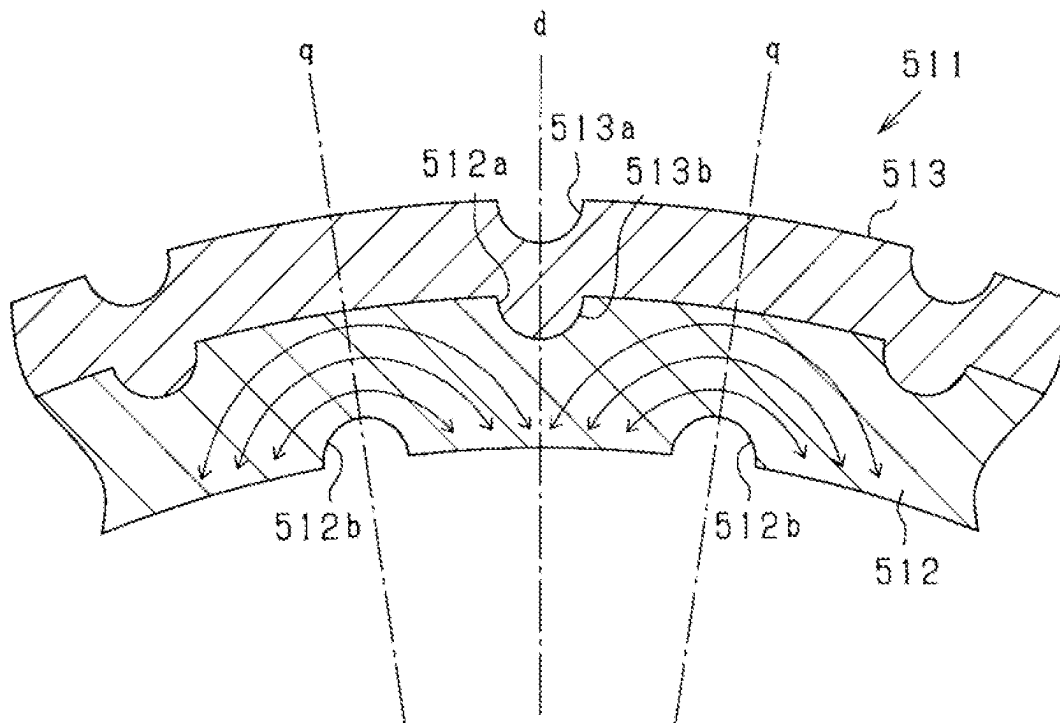


FIG. 53

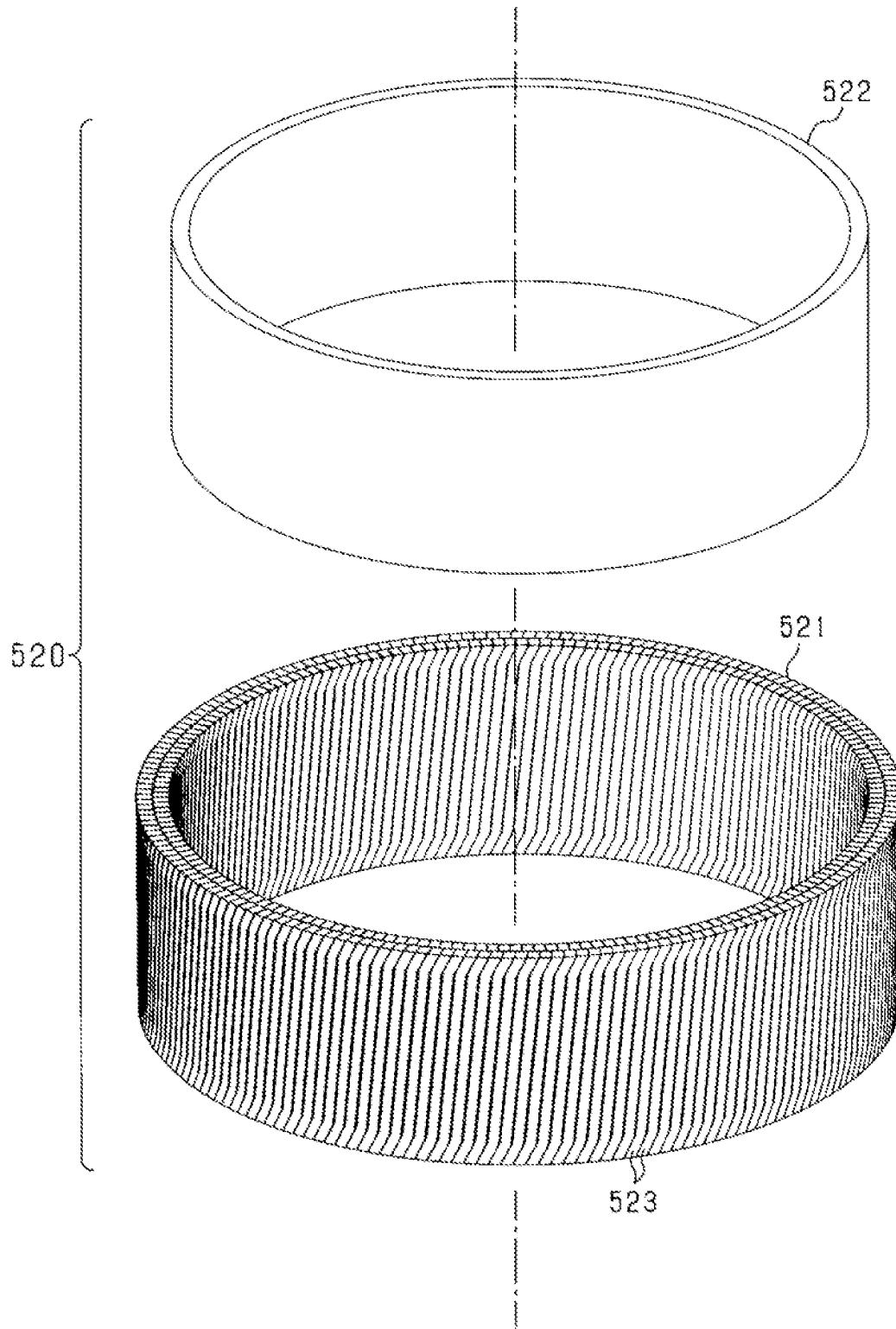


FIG.54

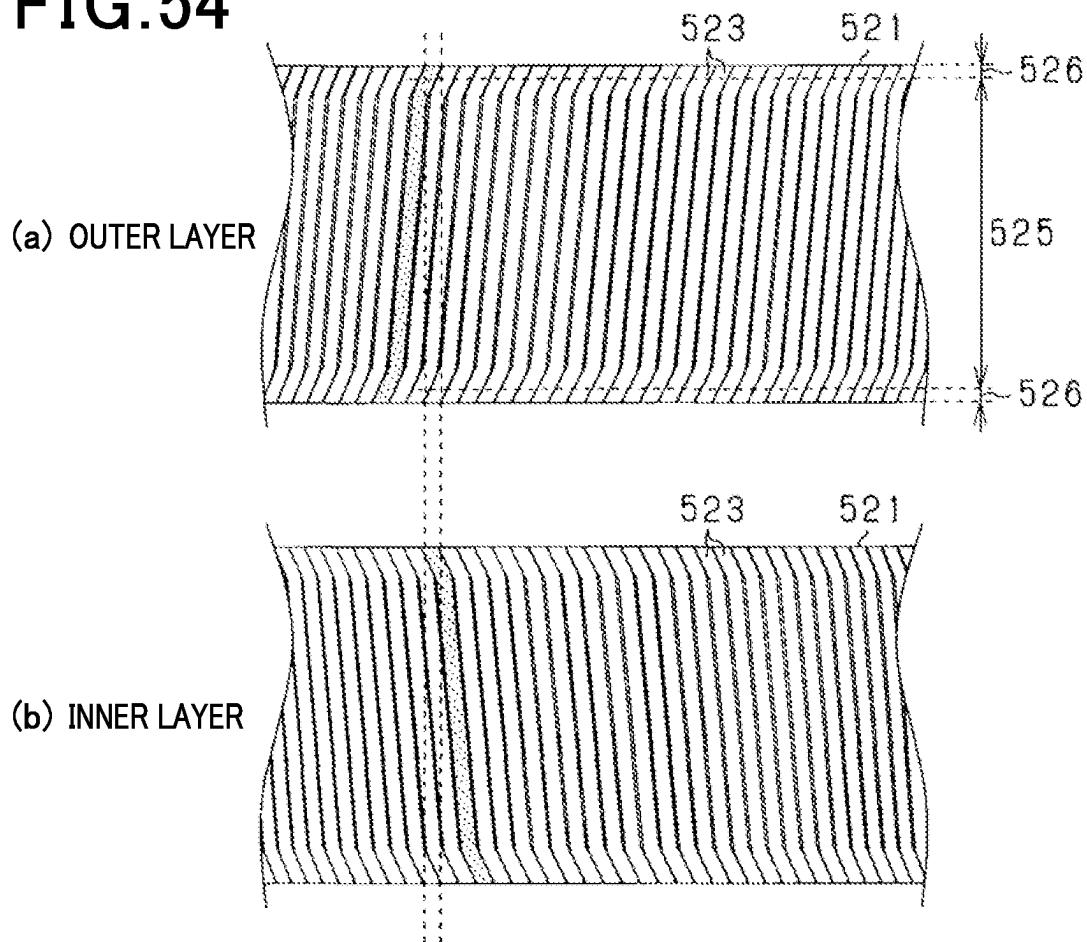


FIG.55

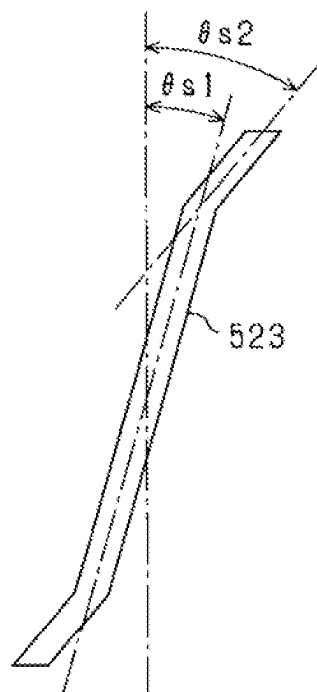


FIG. 56

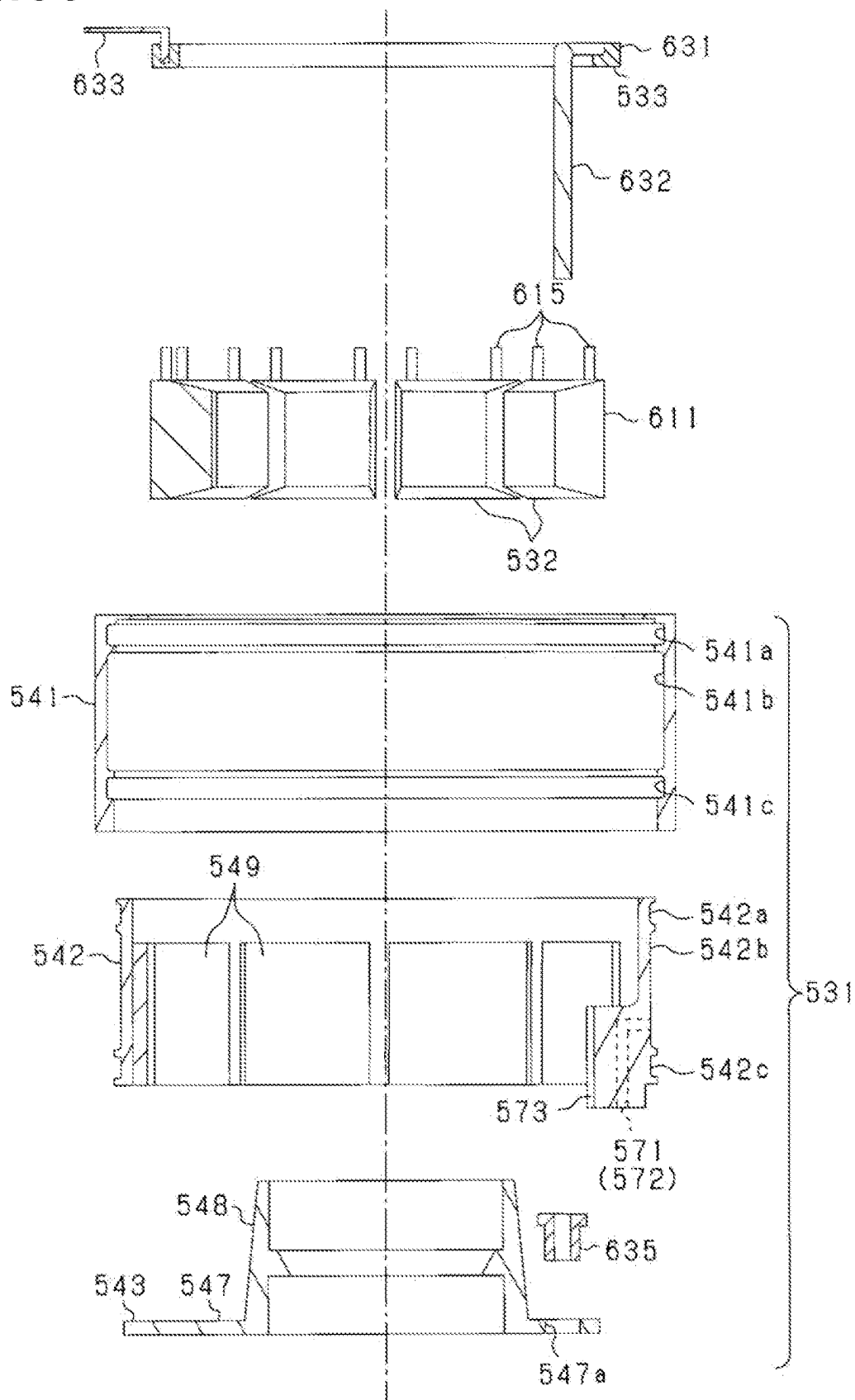


FIG. 57

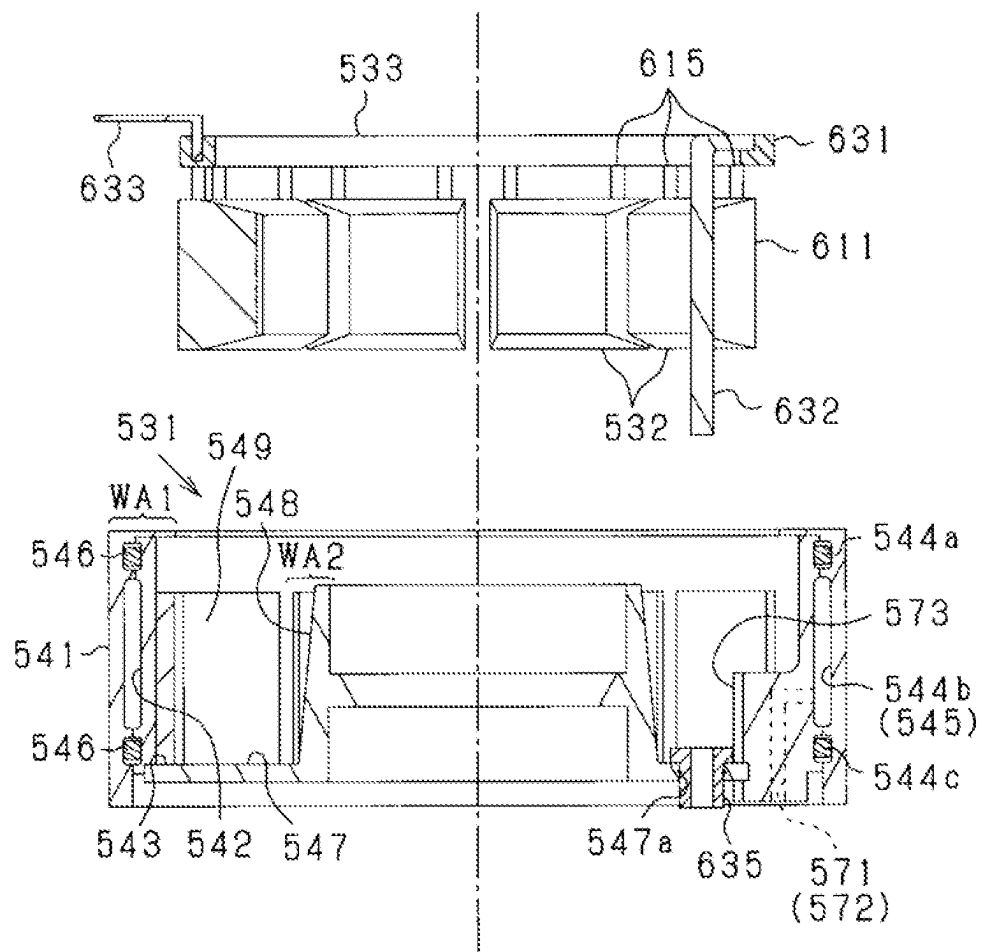


FIG. 58

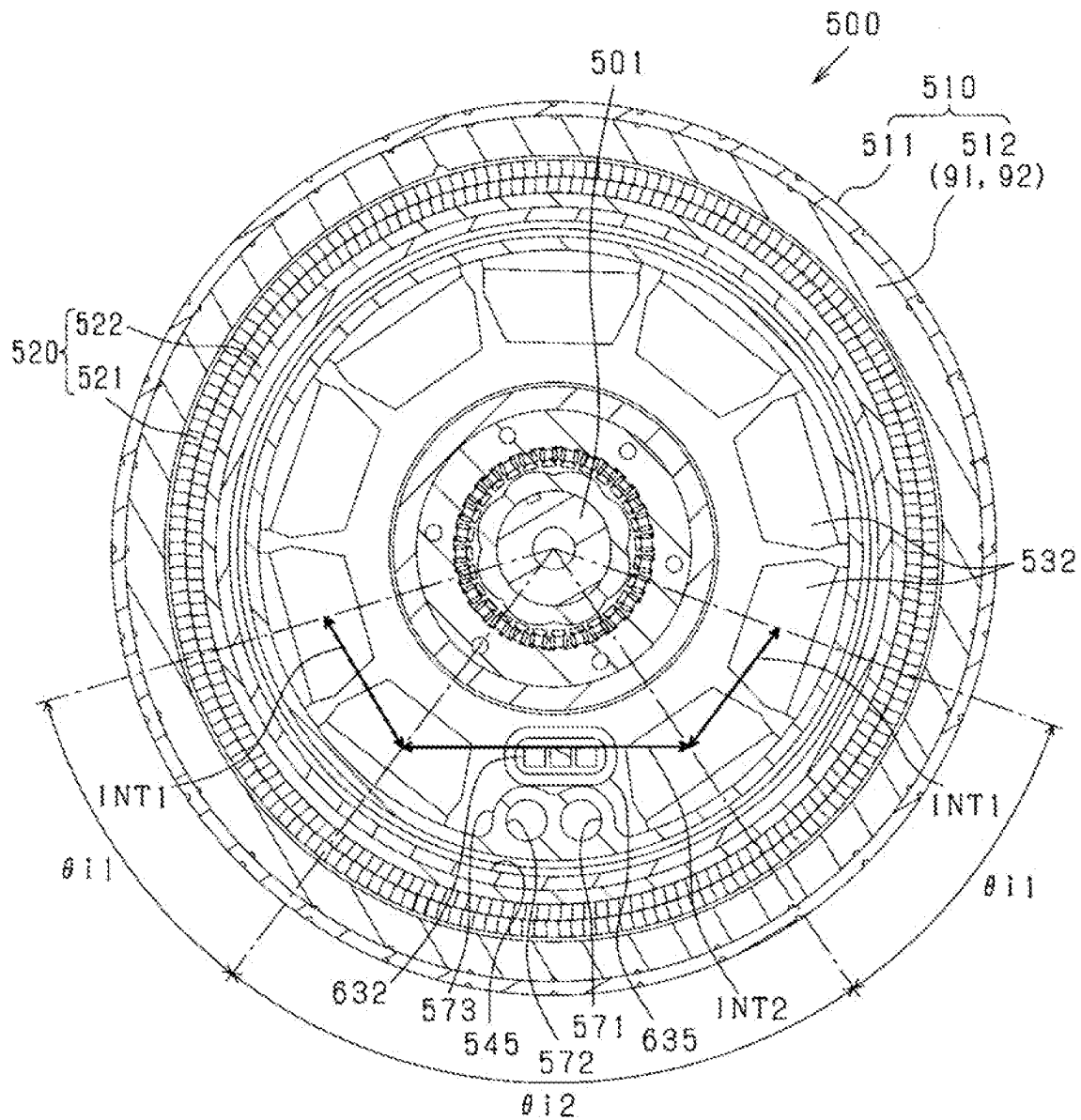


FIG. 59

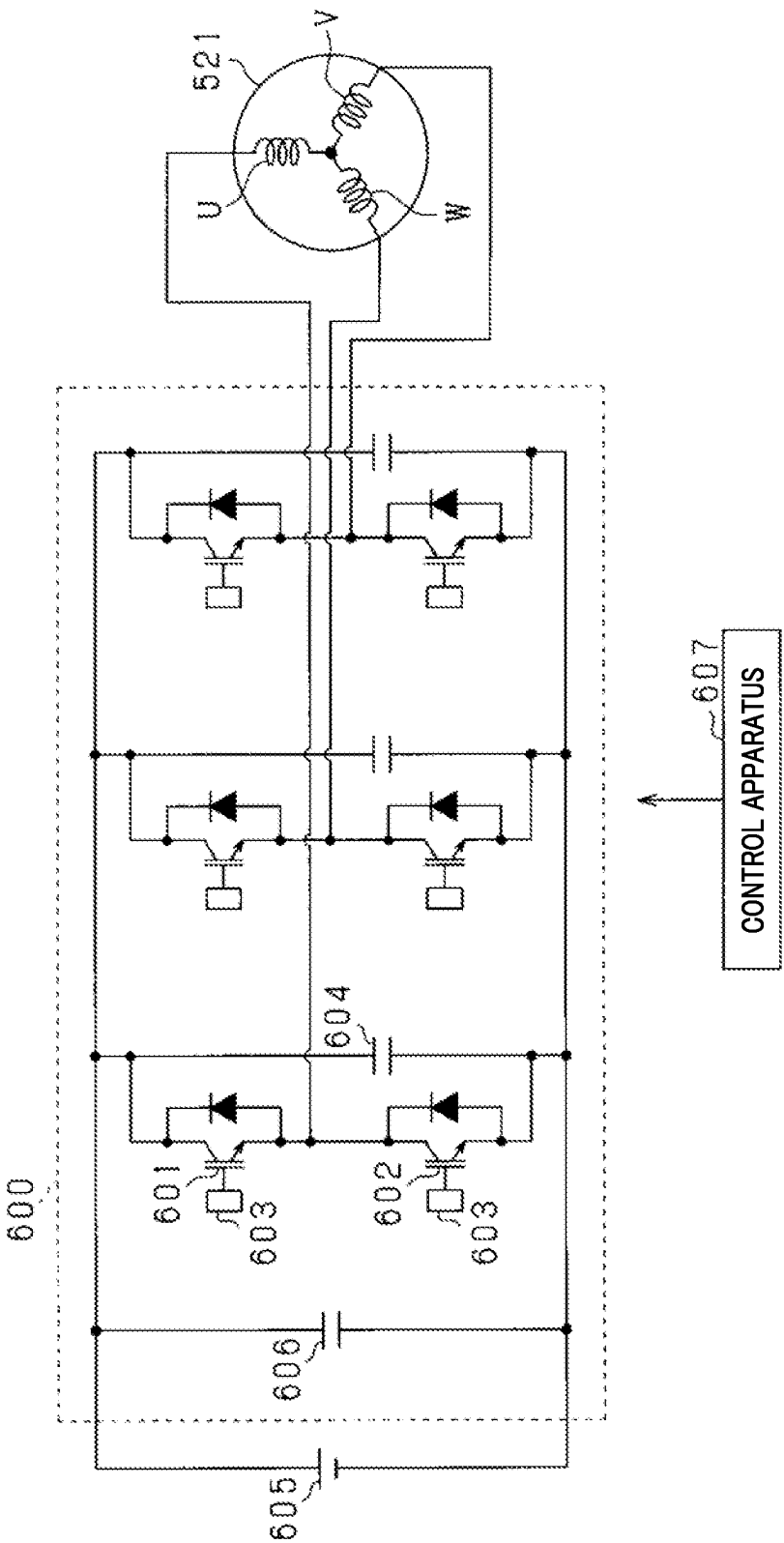


FIG. 60

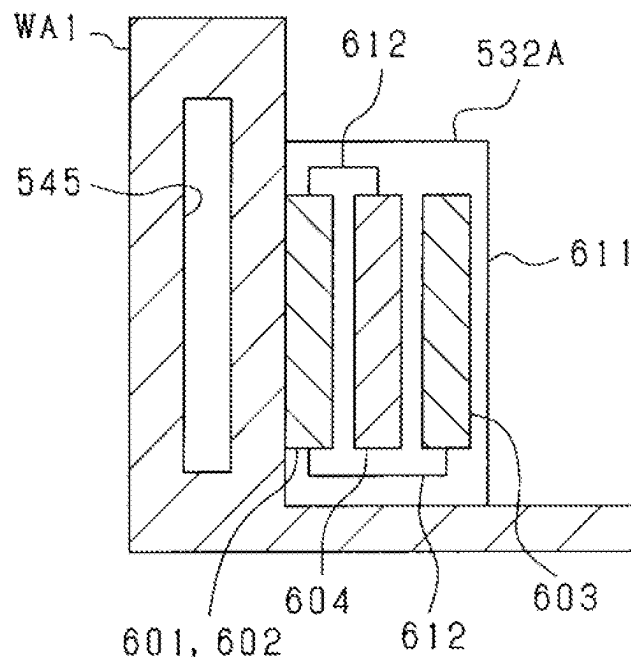


FIG. 61

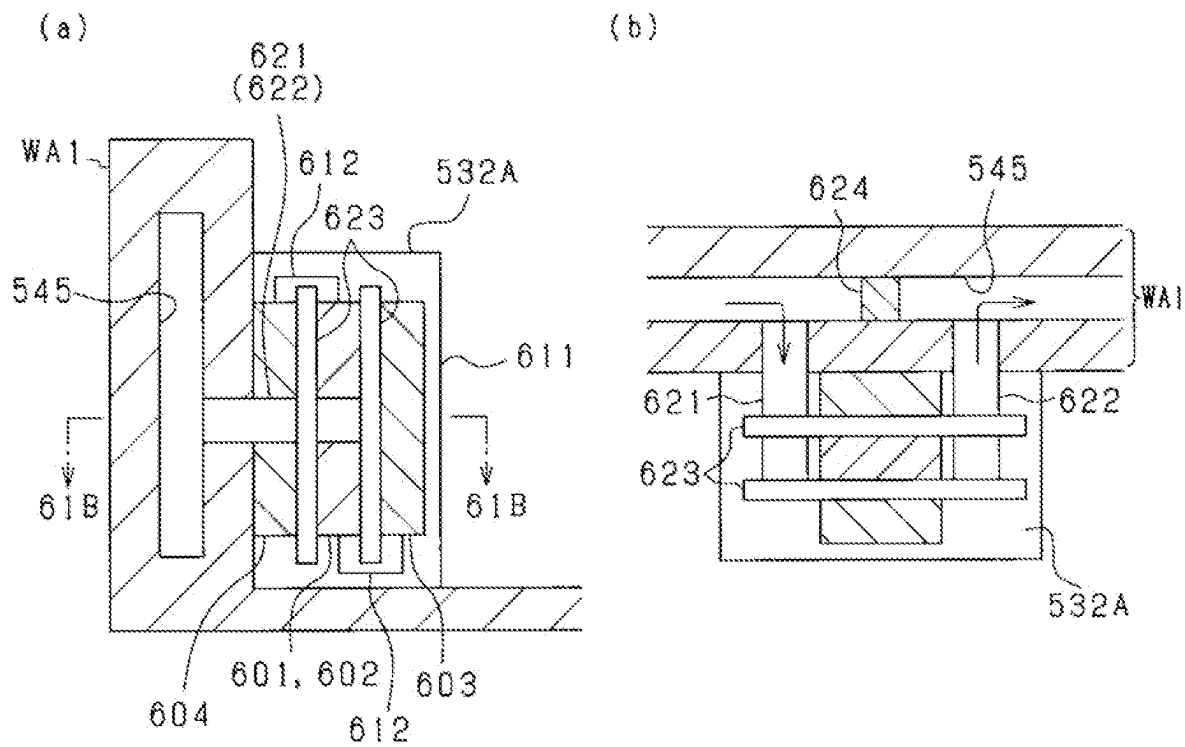


FIG. 63

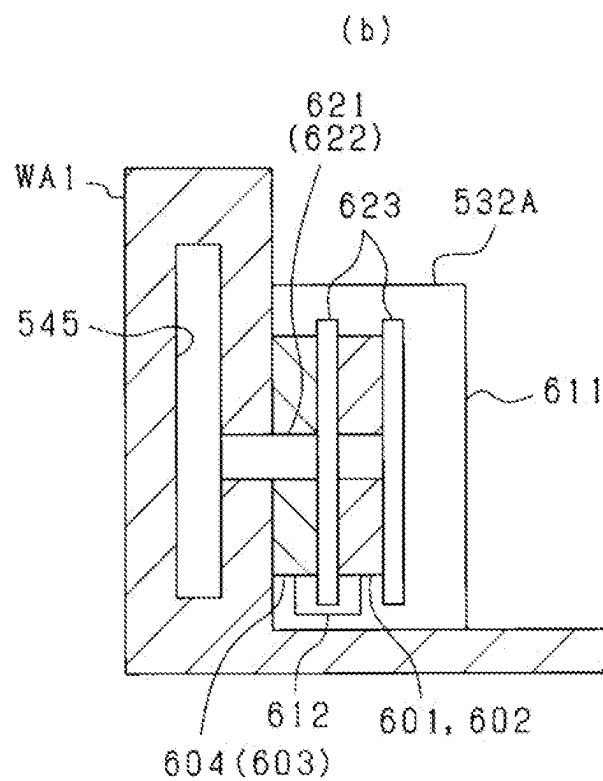
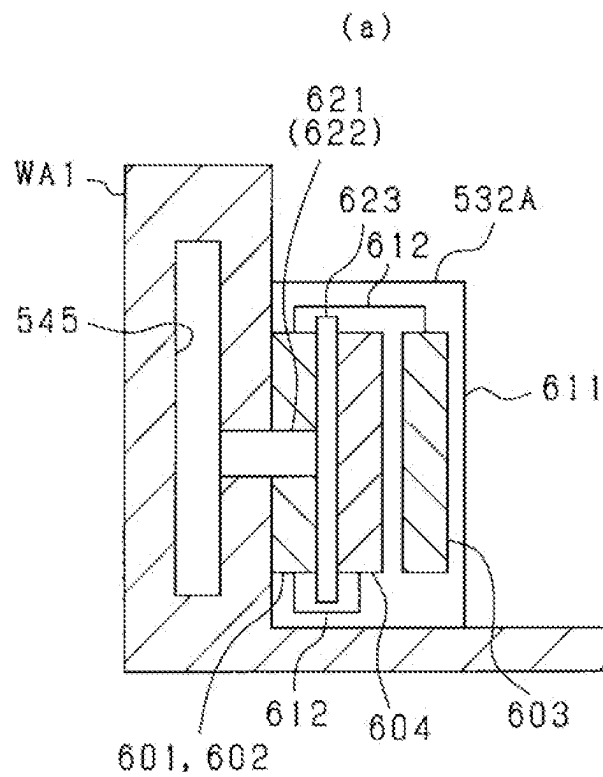


FIG. 64

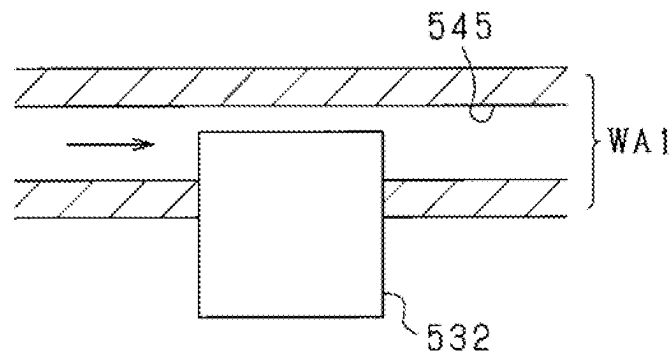


FIG. 65

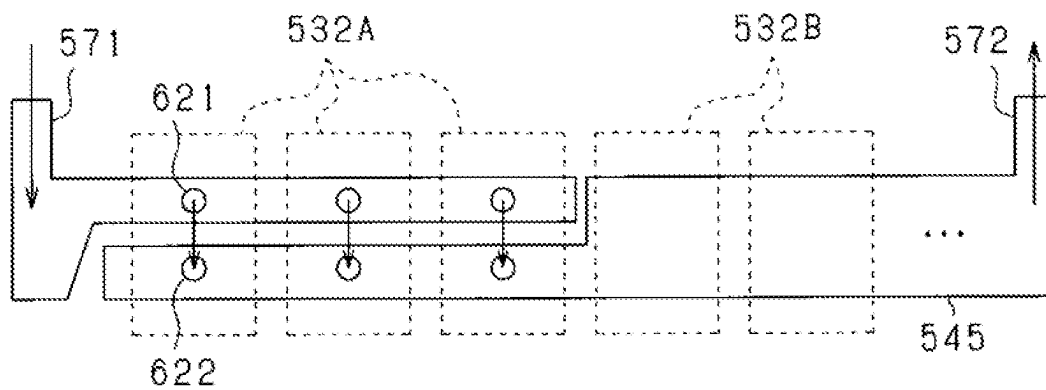


FIG. 66

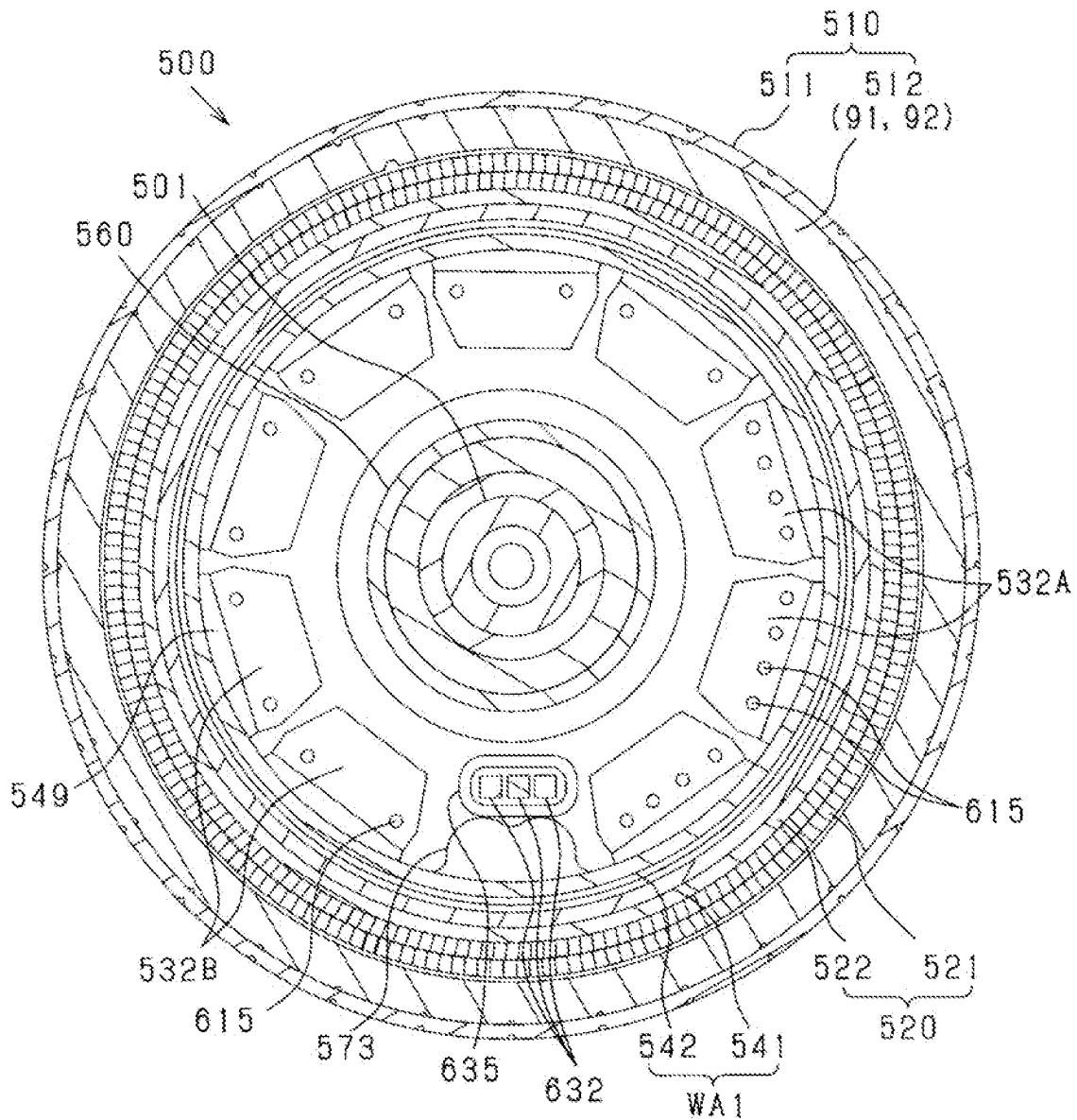


FIG. 67

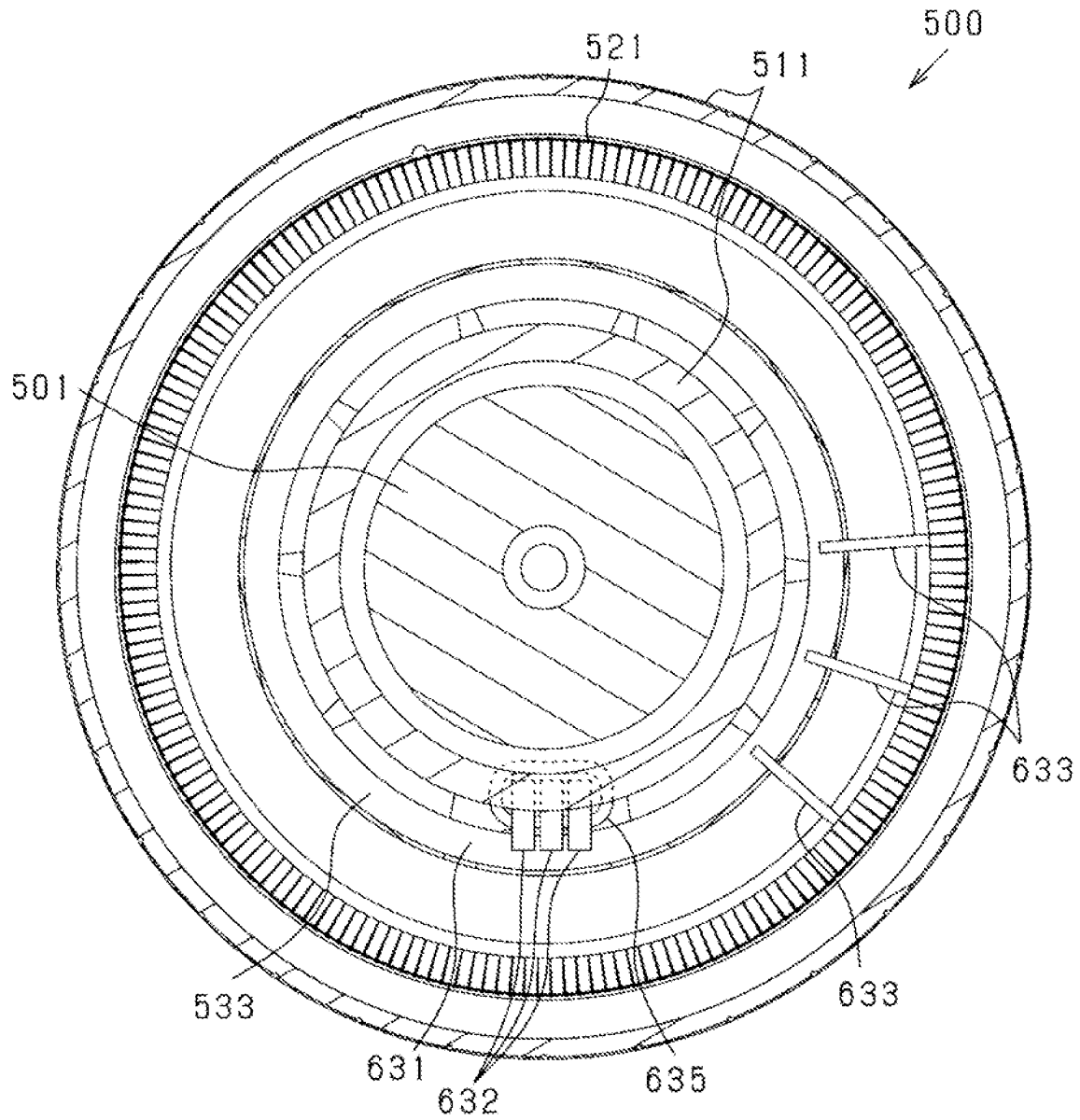


FIG. 68

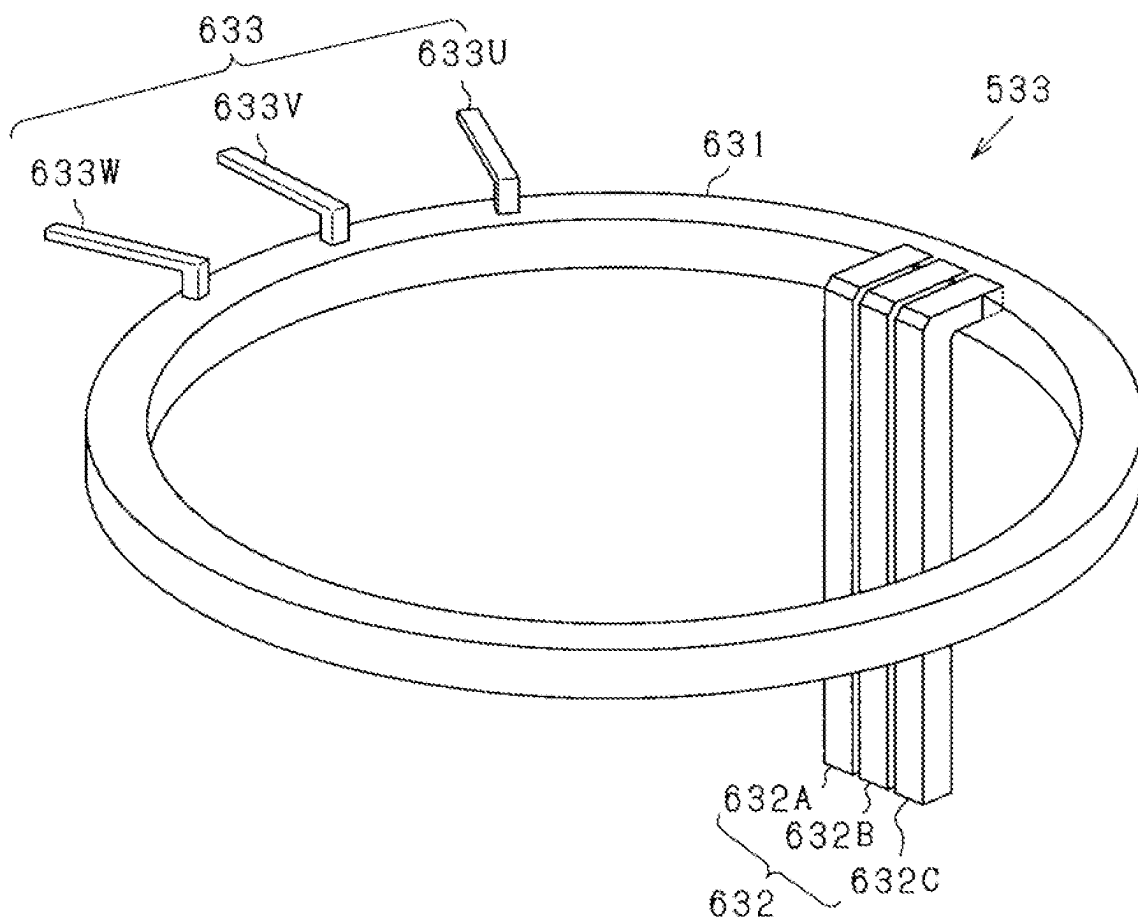


FIG. 69

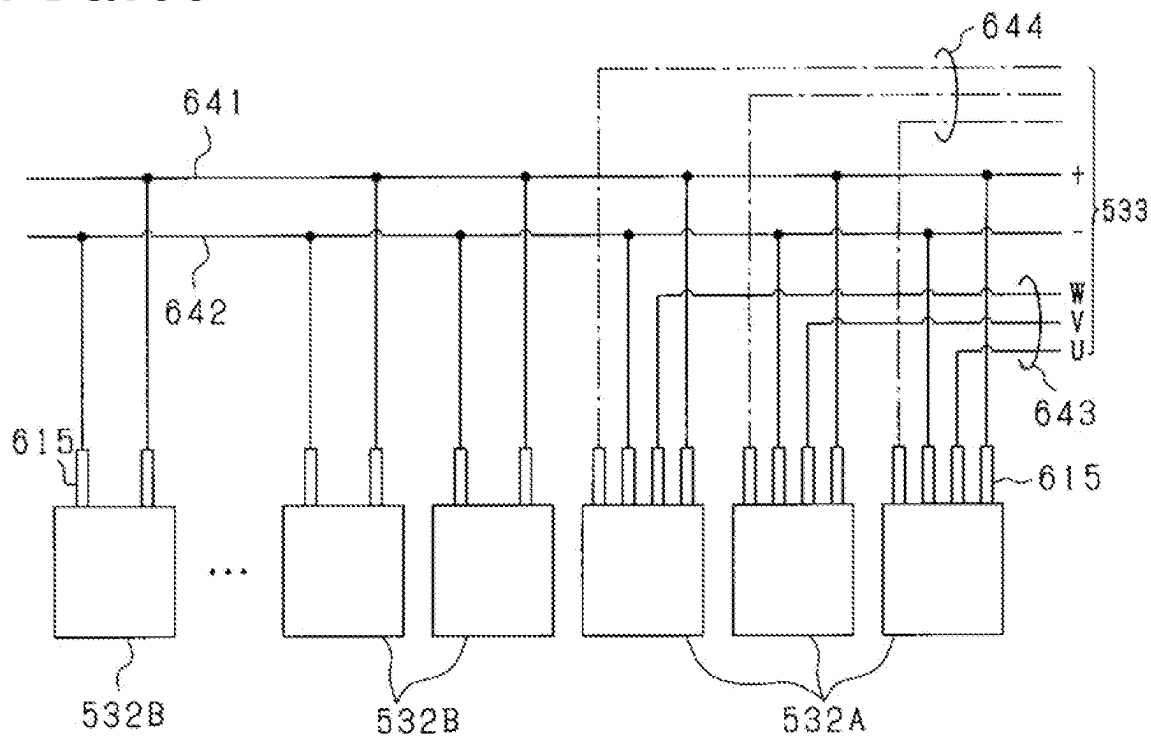


FIG. 70

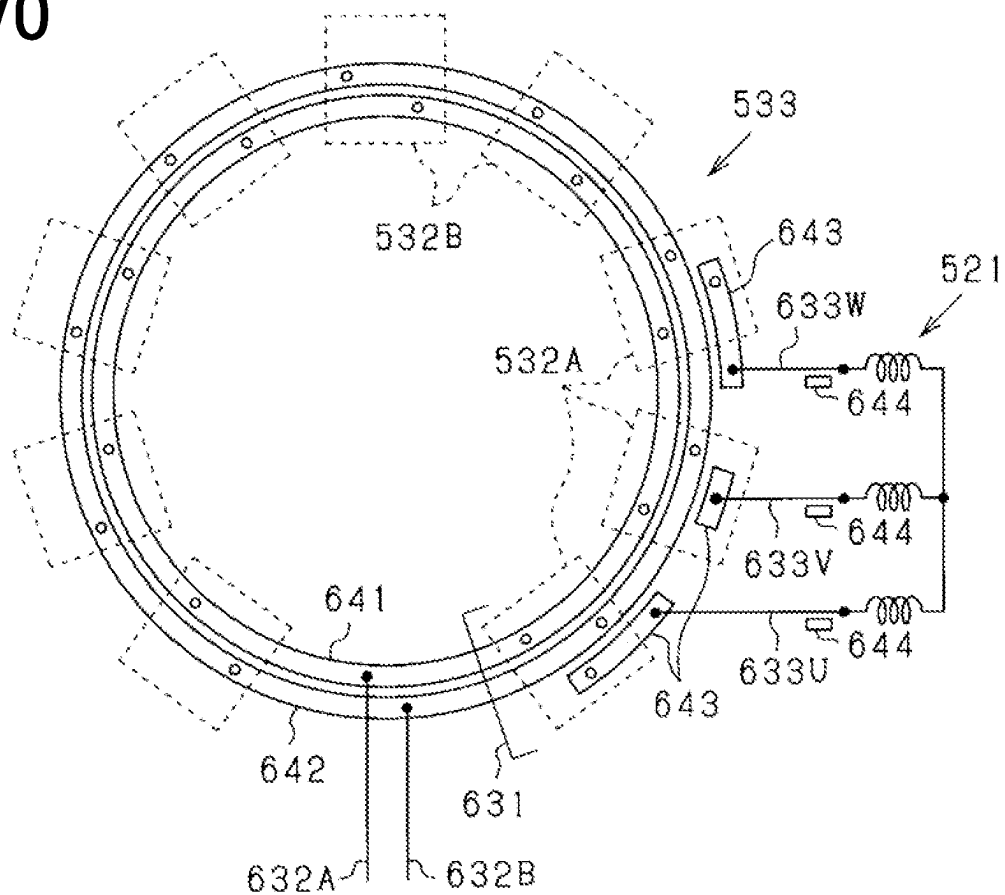


FIG. 71

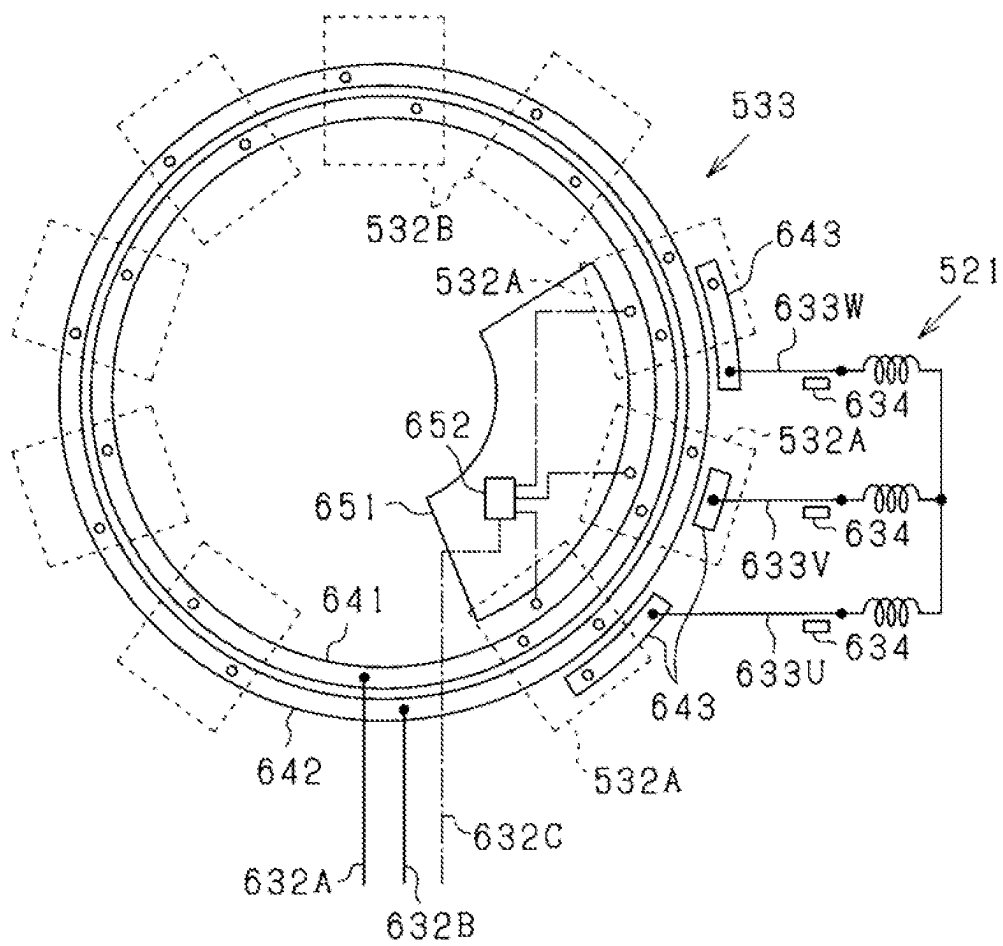


FIG. 72

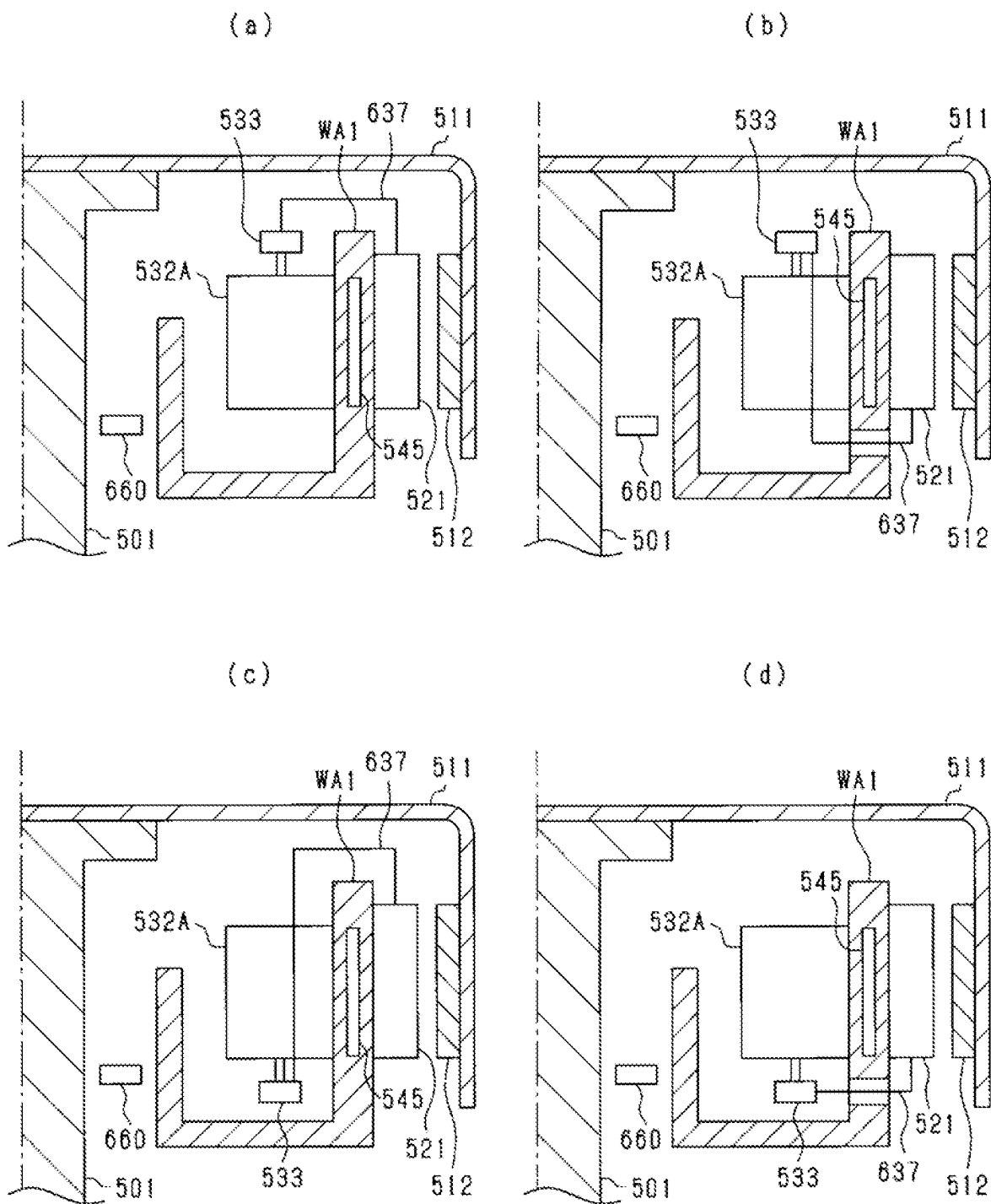
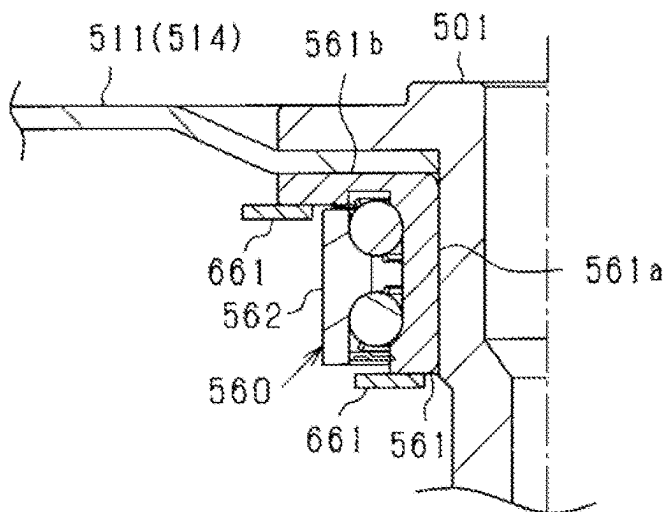
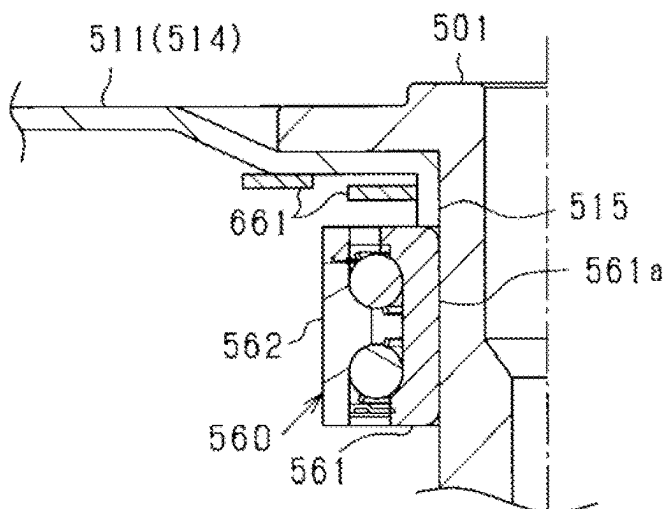


FIG. 73

(a)



(b)



(c)

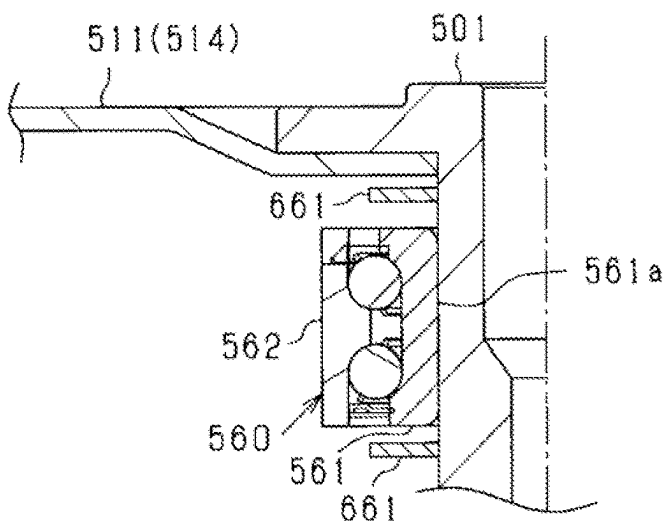
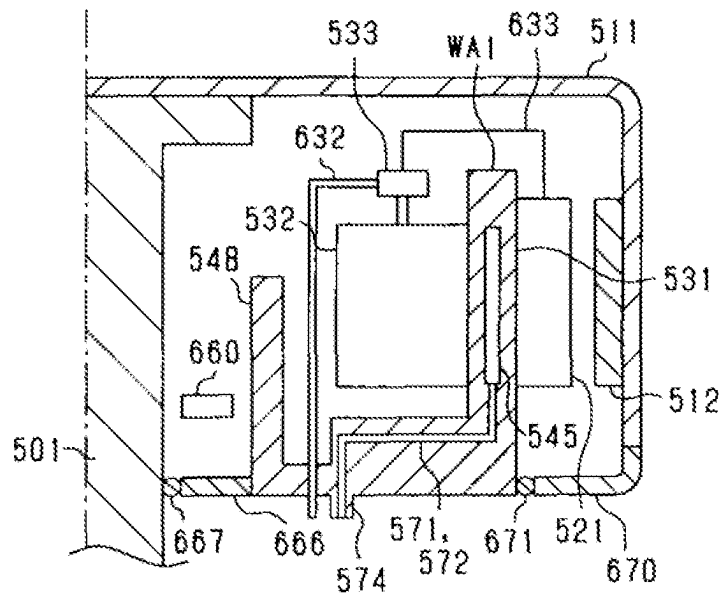


FIG. 74

(a)



(b)

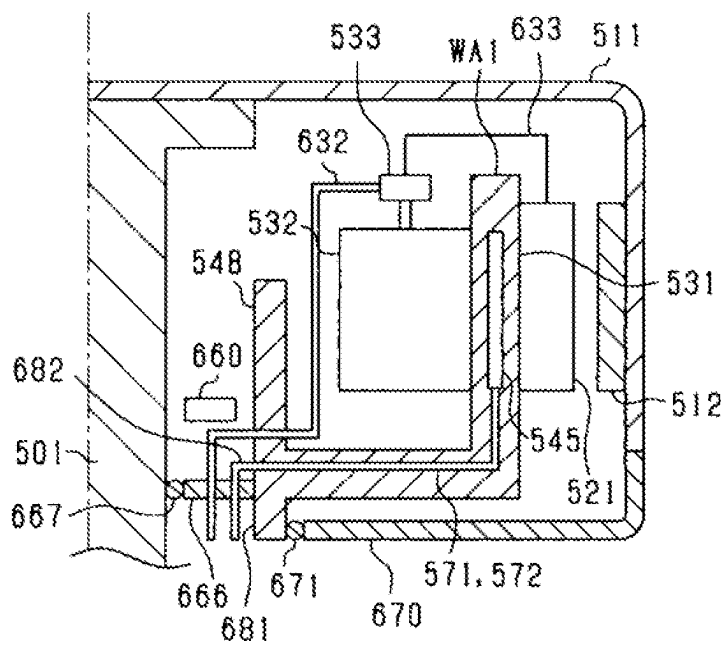


FIG. 75

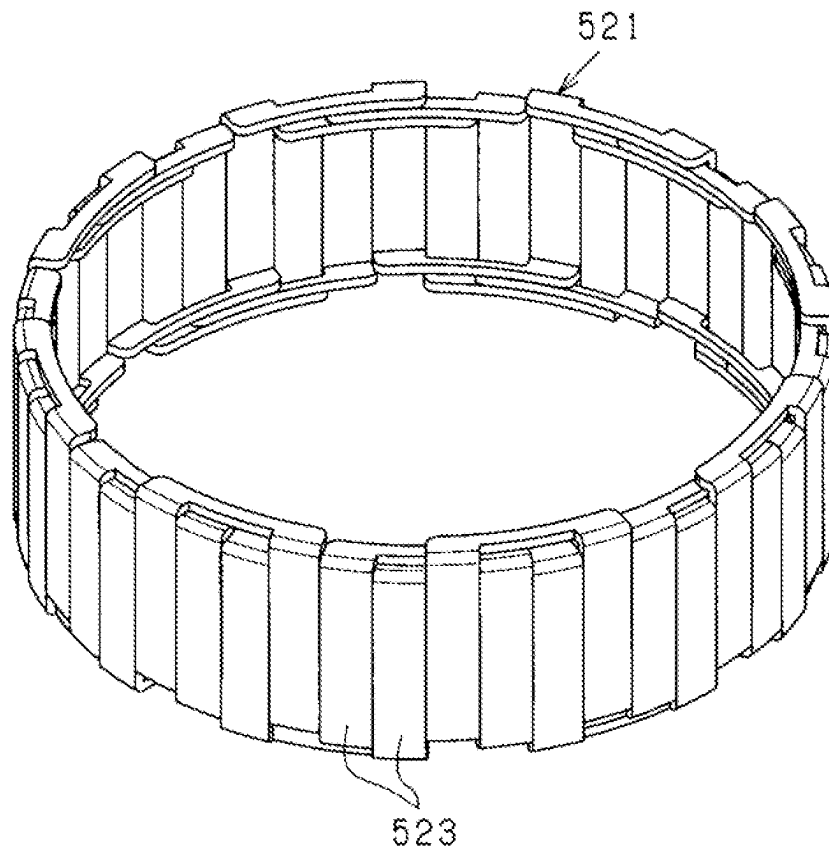


FIG. 79

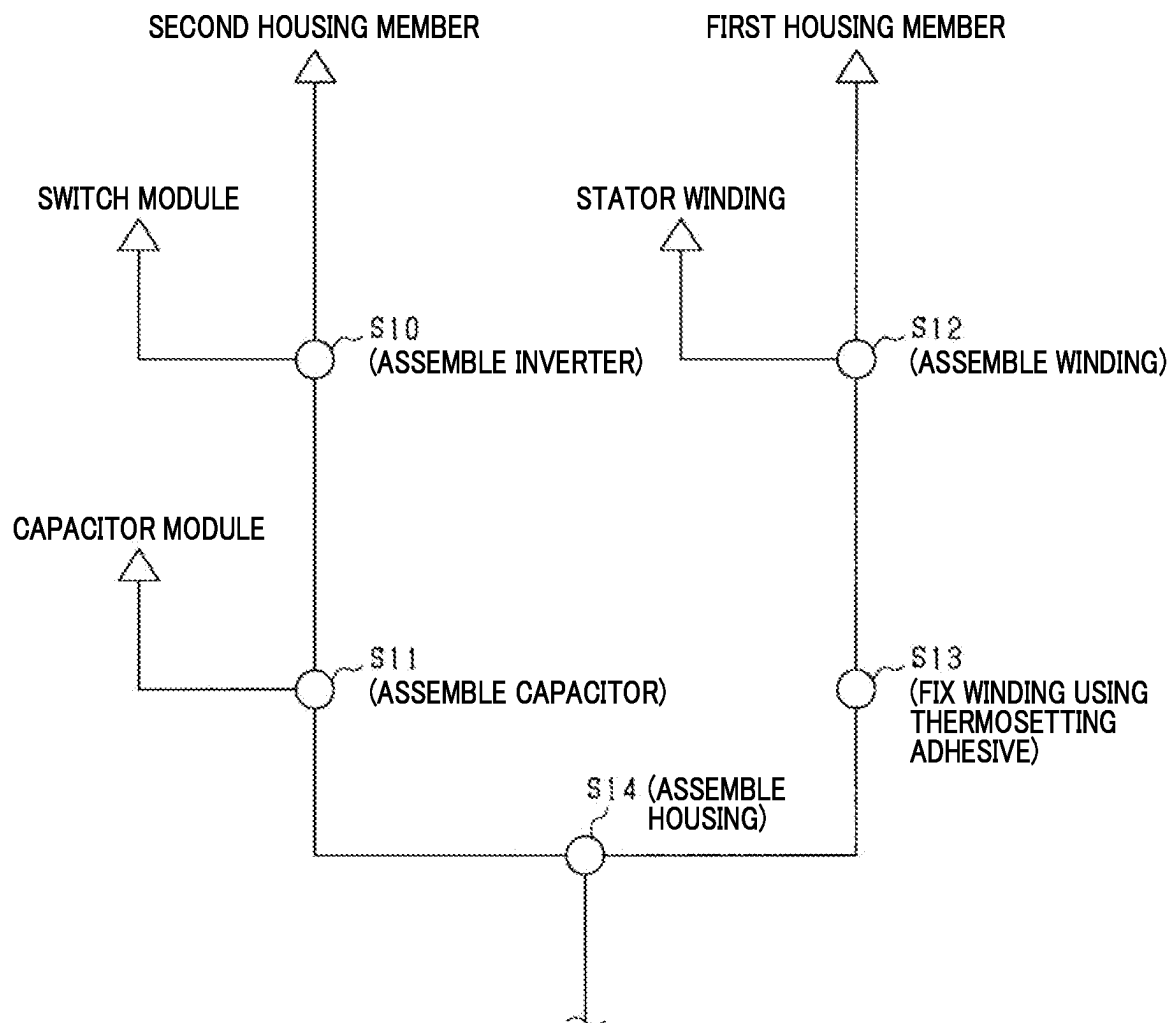


FIG. 81

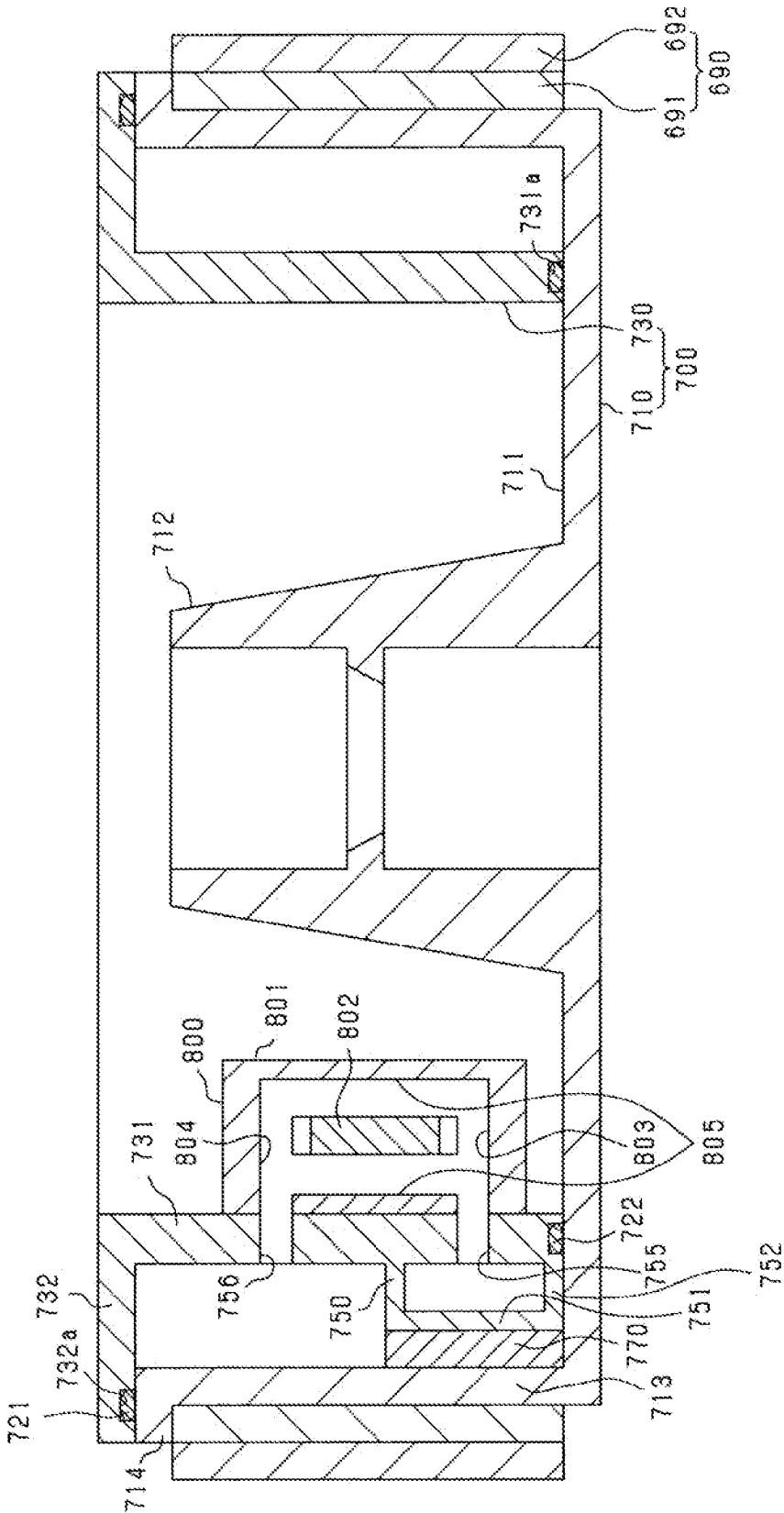


FIG. 82

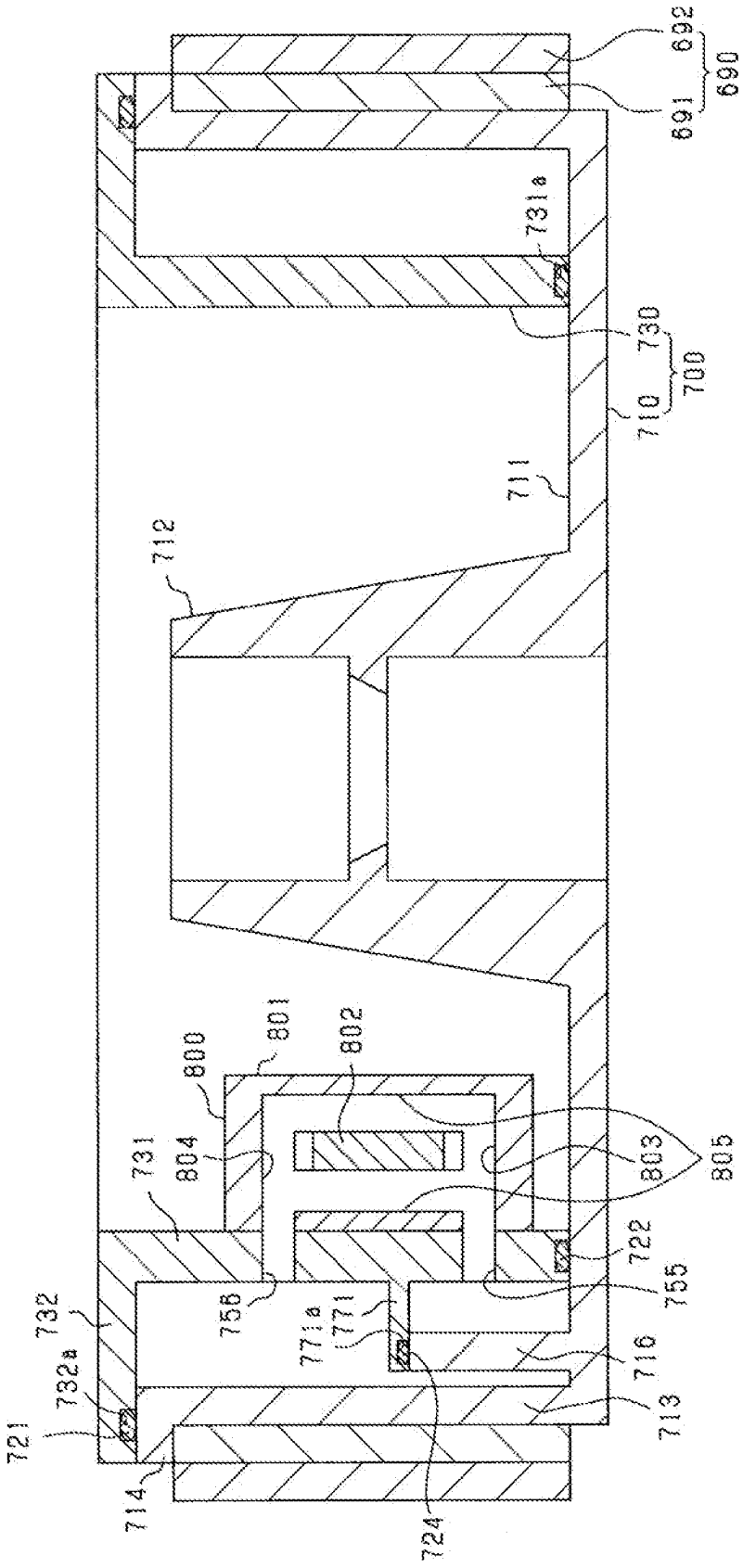


FIG. 83

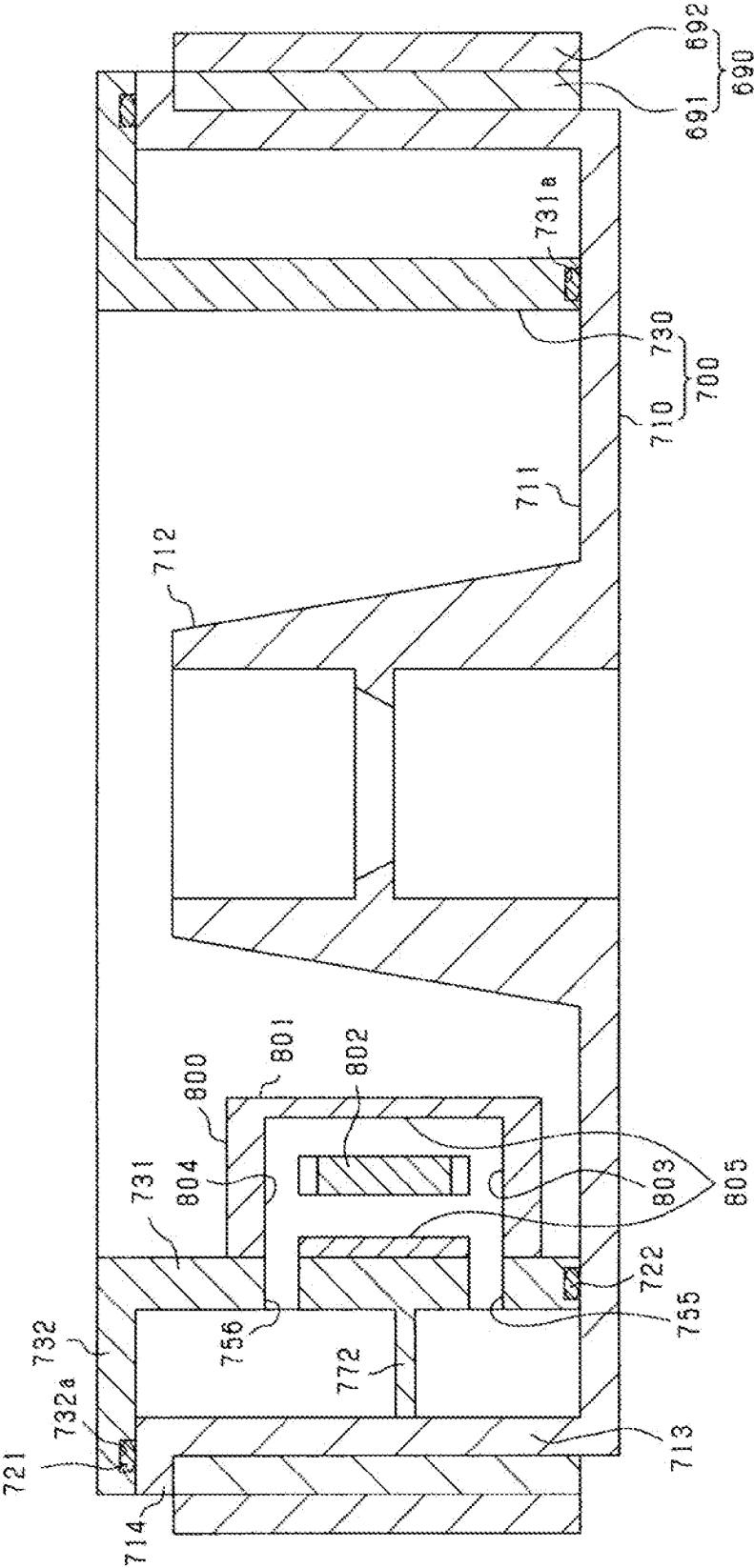


FIG. 84

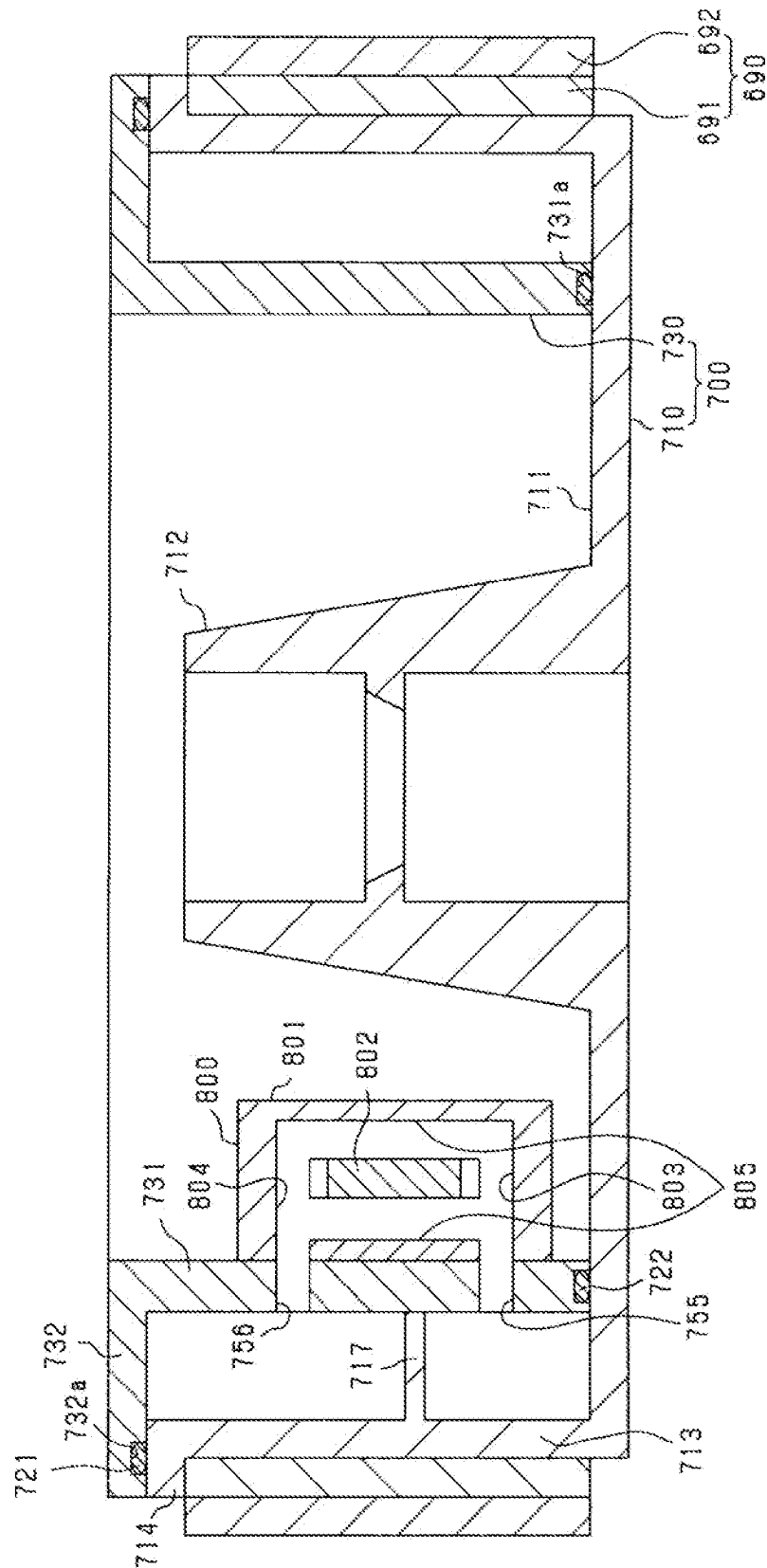


FIG. 85

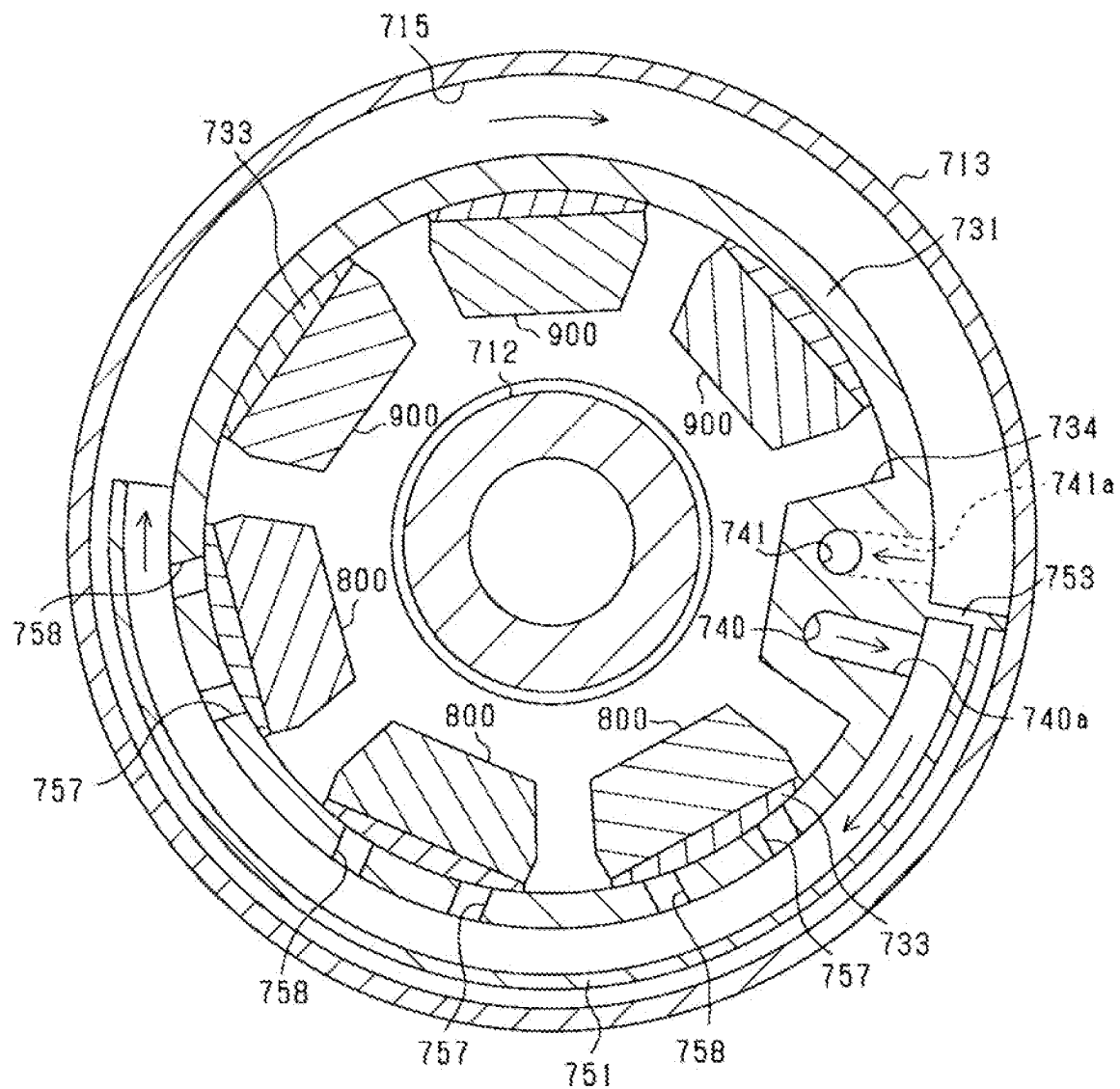


FIG. 86

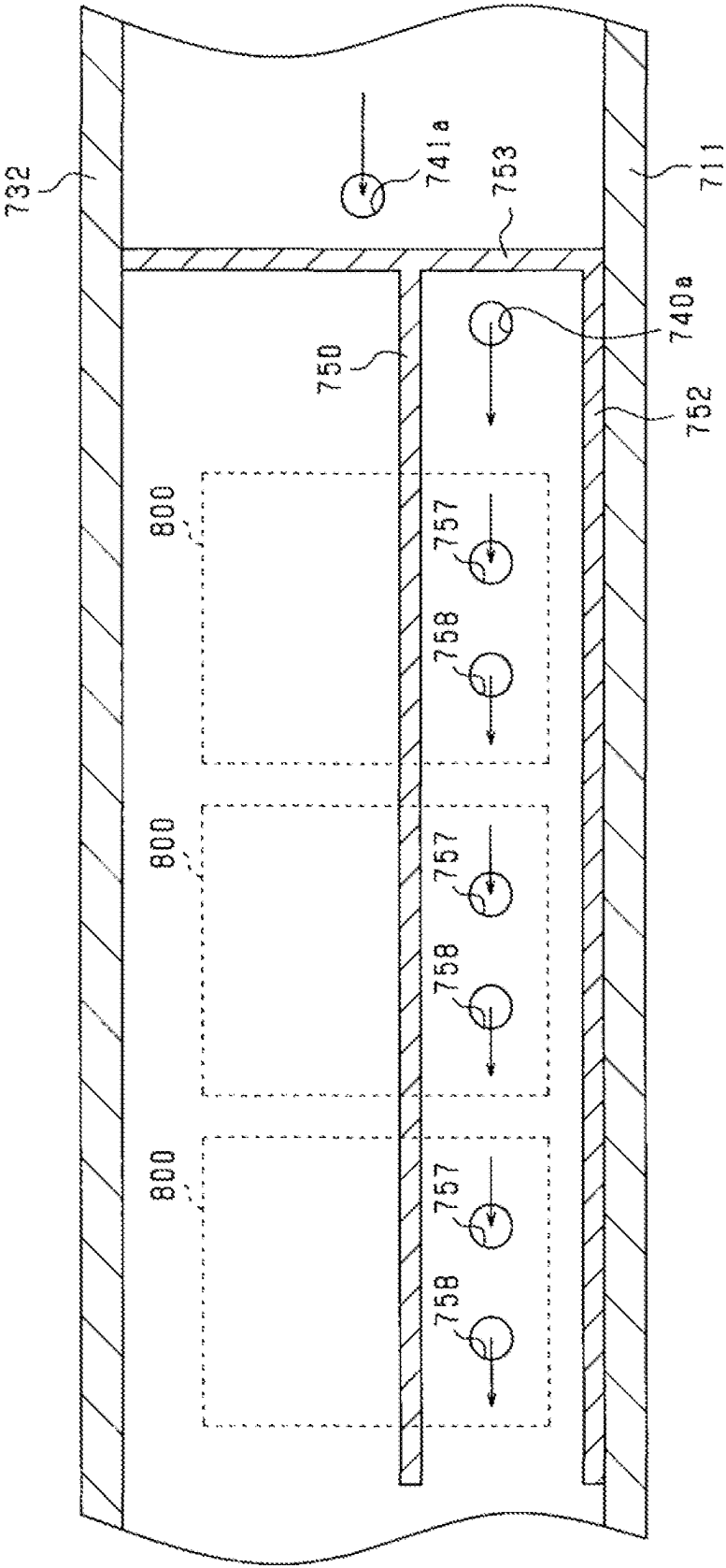
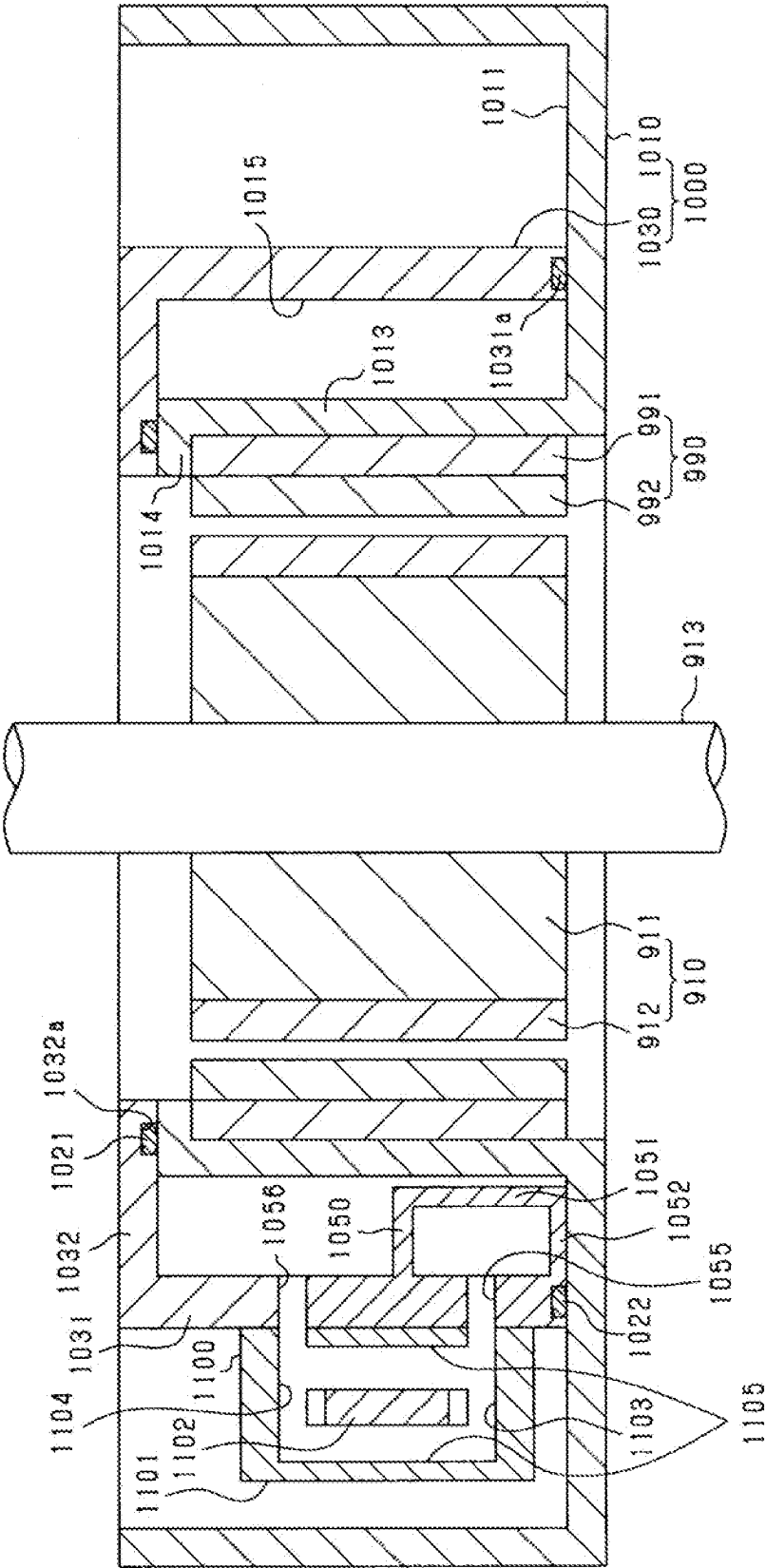


FIG. 87



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ROTATING ELECTRIC MACHINE AND MANUFACTURING METHOD FOR ROTATING ELECTRIC MACHINE

CROSS-REFERENCE TO RELATED APPLICATION

The present application is a continuation application of International Application No. PCT/JP2020/011837, filed on Mar. 17, 2020, which claims priority to Japanese Patent Application No. 2019-068789, filed on Mar. 29, 2019. The contents of these applications are incorporated herein by reference in their entirety.

BACKGROUND

Technical Field

A disclosure of this specification relates to a rotating electric machine and a manufacturing method for the rotating electric machine.

Related Art

A rotating electric machine that includes a field element (such as a rotor) that includes a magnet portion that has a plurality of magnetic poles of which polarities alternate in a circumferential direction, an armature (such as a stator) that has an armature winding of multiple phases, and a power converter that is electrically connected to the armature winding is known.

SUMMARY

One aspect of the present disclosure provides a rotating electric machine that includes: a field element that includes a magnet portion that has a plurality of magnetic poles of which polarities alternate in a circumferential direction; an armature that has an armature winding of multiple phases; and a power converter that has a first electrical component that generates heat as a result of energization, and a second electrical component that generates heat as a result of energization and has a smaller amount of heat generation than the first electrical component, and is electrically connected to the armature winding. Either of the field element and the armature is a rotor.

The rotating electric machine includes a housing member that includes a cylindrical portion, and in which the first electrical component and the second electrical component are arranged in the circumferential direction along the cylindrical portion on a counter-field element side in a radial direction of the cylindrical portion. The armature is assembled on the field element side in the radial direction of the cylindrical portion. A coolant passage that has an inlet portion into which a coolant flows and an outlet portion from which the coolant flows, and connects between the inlet portion and the outlet portion in one direction along the cylindrical portion is formed in the cylindrical portion.

The housing member includes a partition wall portion that partitions a portion of the coolant passage from the inlet portion to a specific position between the inlet portion and the outlet portion in the circumferential direction as an upstream-side path. The partition wall portion is separated in the radial direction from a wall portion of the cylindrical portion in which the armature is assembled. A downstream side of the upstream-side path in the coolant passage is a downstream-side path. On the counter-field element side in

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the radial direction of the cylindrical portion, the first electrical component is arranged on an inner side in the radial direction relative to the upstream-side path, and the second electrical component is arranged on the inner side in the radial direction relative to the downstream-side path.

BRIEF DESCRIPTION OF THE DRAWINGS

In the accompanying drawings:

FIG. 1 is a longitudinal cross-sectional perspective view of a rotating electric machine;

FIG. 2 is a longitudinal cross-sectional view of the rotating electric machine;

FIG. 3 is a cross-sectional view taken along line III-III in FIG. 2;

FIG. 4 is a cross-sectional view showing a portion of FIG. 3 in an enlarged manner;

FIG. 5 is an exploded view of the rotating electric machine;

FIG. 6 is an exploded view of an inverter unit;

FIG. 7 is a torque diagram of a relationship between ampere-turns of a stator winding and torque density;

FIG. 8 is a lateral cross-sectional view of a rotor and a stator;

FIG. 9 is a diagram showing a portion of FIG. 8 in an enlarged manner;

FIG. 10 is a lateral cross-sectional view of the stator;

FIG. 11 is a longitudinal cross-sectional view of the stator;

FIG. 12 is a perspective view of the stator winding;

FIG. 13 is a perspective view of a configuration of a conductor;

FIG. 14 is a schematic diagram of a configuration of a wire;

FIG. 15 illustrates, by (a) and (b), diagrams of an aspect of the conductors in an nth layer;

FIG. 16 is a side view of the conductors in the nth layer and an n+1th layer;

FIG. 17 is a diagram of a relationship between electrical angle and magnetic flux density in a magnet according to an embodiment;

FIG. 18 is a diagram of the relationship between electrical angle and magnetic flux density in a magnet of a comparative example;

FIG. 19 is an electric circuit diagram of a control system of the rotating electric machine;

FIG. 20 is a functional block diagram of a current feedback control process performed by a control apparatus;

FIG. 21 is a functional block diagram of a torque feedback control process performed by the control apparatus;

FIG. 22 is a lateral cross-sectional view of a rotor and a stator according to a second embodiment;

FIG. 23 is a diagram showing a portion of FIG. 22 in an enlarged manner;

FIG. 24 illustrates, by (a) and (b), detailed diagrams of a flow of magnetic flux in a magnet unit;

FIG. 25 is a cross-sectional view of the stator in a first modification;

FIG. 26 is a cross-sectional view of the stator in the first modification;

FIG. 27 is a cross-sectional view of the stator in a second modification;

FIG. 28 is a cross-sectional view of the stator in a third modification;

FIG. 29 is a cross-sectional view of the stator in a fourth modification;

FIG. 30 is a lateral cross-sectional view of the rotor and the stator in a seventh modification;

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FIG. 31 is a functional block diagram of a part of a process performed by an operating signal generating unit in an eighth modification;

FIG. 32 is a flowchart of the steps in a carrier frequency changing process;

FIG. 33 illustrates, by (a) to (c), diagrams of aspects of connection of conductors configuring a conductor group in a ninth modification;

FIG. 34 is a diagram of a configuration in which four pairs of conductors are arranged in a laminated manner in the ninth modification;

FIG. 35 is a lateral cross-sectional view of an inner-rotor-type rotor and stator in a tenth modification;

FIG. 36 is a diagram showing a portion of FIG. 35 in an enlarged manner;

FIG. 37 is a longitudinal cross-sectional view of an inner-rotor-type rotating electric machine;

FIG. 38 is a longitudinal cross-sectional view of a schematic configuration of the inner-rotor-type rotating electric machine;

FIG. 39 is a diagram of a configuration of a rotating electric machine having an inner-rotor structure in an eleventh modification;

FIG. 40 is a diagram of the configuration of the rotating electric machine having an inner-rotor structure in the eleventh modification;

FIG. 41 is a diagram of a configuration of a revolving-armature-type rotating electric machine in a twelfth modification;

FIG. 42 is a cross-sectional view of a configuration of a conductor in a fourteenth modification;

FIG. 43 is a diagram of a relationship among reluctance torque, magnet torque, and DM;

FIG. 44 is a diagram of teeth;

FIG. 45 is a perspective view of a vehicle wheel having an in-wheel-motor structure and a surrounding structure thereof;

FIG. 46 is a longitudinal cross-sectional view of the vehicle wheel and the surrounding structure thereof;

FIG. 47 is an exploded perspective view of the vehicle wheel;

FIG. 48 is a side view of a rotating electric machine viewed from a protruding side of a rotation shaft;

FIG. 49 is a cross-sectional view taken along line 49-49 in FIG. 48;

FIG. 50 is a cross-sectional view taken along line 50-50 in FIG. 49;

FIG. 51 is an exploded cross-sectional view of the rotating electric machine;

FIG. 52 is a partial cross-sectional view of a rotor;

FIG. 53 is a perspective view of a stator winding and a stator core;

FIG. 54 illustrates, by (a) and (b), front views of the stator winding in a planarly developed state;

FIG. 55 is a diagram of skew of a conductor;

FIG. 56 is an exploded cross-sectional view of an inverter unit;

FIG. 57 is an exploded cross-sectional view of the inverter unit;

FIG. 58 is a diagram of a state of arrangement of electrical modules in an inverter housing;

FIG. 59 is a circuit diagram of an electrical configuration of a power converter;

FIG. 60 is a diagram of an example of a cooling structure of a switch module;

FIG. 61 illustrates, by (a) and (b), diagrams of an example of the cooling structure of the switch module;

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FIG. 62 illustrates, by (a) to (c), diagrams of an example of the cooling structure of the switch module;

FIG. 63 illustrates, by (a) and (b), diagrams of an example of the cooling structure of the switch module;

FIG. 64 is a diagram of an example of the cooling structure of the switch module;

FIG. 65 is a diagram of an order of array of the electrical modules relative to a cooling water passage;

FIG. 66 is a cross-sectional view taken along line 66-66 in FIG. 49;

FIG. 67 is a cross-sectional view taken along line 67-67 in FIG. 49;

FIG. 68 is a perspective view of a bus bar module alone;

FIG. 69 is a diagram of a state of electrical connection between the electrical modules and the bus bar module;

FIG. 70 is a diagram of a state of electrical connection between the electrical modules and the bus bar module;

FIG. 71 is a diagram of a state of electrical connection between the electrical modules and the bus bar module;

FIG. 72 illustrates, by (a) to (d), configuration diagrams for explaining a first modification of an in-wheel motor;

FIG. 73 illustrates, by (a) to (c), configuration diagrams for explaining a second modification of the in-wheel motor;

FIG. 74 illustrates, by (a) and (b), configuration diagrams for explaining a third modification of the in-wheel motor;

FIG. 75 is a configuration diagram for explaining a fourth modification of the in-wheel motor;

FIG. 76 is a longitudinal cross-sectional view of the inverter housing in a fifth modification of the in-wheel motor and is a cross-sectional view taken along line 76-76 in FIG. 77;

FIG. 77 is a cross-sectional view taken along line 77-77 in FIG. 76;

FIG. 78 is a diagram in which a cross-sectional view taken along line 78-78 in FIG. 76 is developed in a circumferential direction;

FIG. 79 is a process control chart of a portion of a manufacturing process for the rotating electric machine;

FIG. 80 is a longitudinal cross-sectional view of the inverter housing in a sixth modification of the in-wheel motor;

FIG. 81 is a longitudinal cross-sectional view of the inverter housing in a seventh modification of the in-wheel motor;

FIG. 82 is a longitudinal cross-sectional view of the inverter housing in an eighth modification of the in-wheel motor;

FIG. 83 is a longitudinal cross-sectional view of the inverter housing in a ninth modification of the in-wheel motor;

FIG. 84 is a longitudinal cross-sectional view of the inverter housing in a tenth modification of the in-wheel motor;

FIG. 85 is a lateral cross-sectional view of the inverter housing in an eleventh modification of the in-wheel motor;

FIG. 86 is a configuration diagram for explaining a coolant passage; and

FIG. 87 is a longitudinal cross-sectional view of the rotating electric machine in a twelfth modification of the in-wheel motor.

DESCRIPTION OF THE EMBODIMENTS

As shown in JP-A-2017-070140, for example, a rotating electric machine that includes a field element (such as a rotor) that includes a magnet portion that has a plurality of magnetic poles of which polarities alternate in a circumfer-

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ential direction, an armature (such as a stator) that has an armature winding of multiple phases, and a power converter that is electrically connected to the armature winding is known.

The power converter includes a first electrical component that generates heat as a result of energization, and a second electrical component that generates heat as a result of energization and has a smaller amount of heat generation than the first electrical component. Therefore, the electrical components are required to be cooled.

Here, the rotating electric machine includes a housing member that has a cylindrical portion. The first electrical component and the second electrical components are arranged in the circumferential direction along the cylindrical portion on a counter-field element side in a radial direction of the cylindrical portion. In addition, a coolant passage that has an inlet portion into which a coolant flows and an outlet portion from which the coolant flows, and connects between the inlet portion and the outlet portion in one direction along the cylindrical portion is formed in the cylindrical portion. As a result of this configuration, the electrical components can be cooled by the coolant that flows through the coolant passage.

Here, the armature is assembled on the field element side in the radial direction of the cylindrical portion. When the armature winding that configures the armature is energized, the armature winding generates heat. In this case, the heat generated in the armature winding is transferred through the cylindrical portion to the coolant that flows through the coolant passage. As a result, a temperature of the coolant before the electrical components are cooled may increase and cooling performance regarding the electrical components may decrease. In particular, because the amount of heat generation of the first electrical component is greater than the amount of heat generation of the second electrical component, the first electrical component may not be able to be appropriately cooled.

It is thus desired to provide a rotating electric machine that is capable of appropriately cooling a first electrical component that configures a power converter, and a manufacturing method for the rotating electric machine.

A plurality of embodiments disclosed in this specification employ technical measures that differ from one another to achieve respective objects. Objects, features, and effects disclosed in this specification will be further clarified with reference to detailed descriptions that follow and accompanying drawings.

A first exemplary embodiment provides a rotating electric machine that includes: a field element that includes a magnet portion that has a plurality of magnetic poles of which polarities alternate in a circumferential direction; an armature that has an armature winding of multiple phases; and a power converter that has a first electrical component that generates heat as a result of energization, and a second electrical component that generates heat as a result of energization and has a smaller amount of heat generation than the first electrical component, and is electrically connected to the armature winding, in which either of the field element and the armature is a rotor.

The rotating electric machine includes a housing member that includes a cylindrical portion, and in which the first electrical component and the second electrical component are arranged in the circumferential direction along the cylindrical portion on a counter-field element side in a radial direction of the cylindrical portion. The armature is assembled on the field element side in the radial direction of the cylindrical portion. A coolant passage that has an inlet

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portion into which a coolant flows and an outlet portion from which the coolant flows, and connects between the inlet portion and the outlet portion in one direction along the cylindrical portion is formed in the cylindrical portion.

The housing member includes a partition wall portion that partitions a portion of the coolant passage from the inlet portion to a specific position between the inlet portion and the outlet portion in the circumferential direction as an upstream-side path. The partition wall portion is separated in the radial direction from a wall portion of the cylindrical portion in which the armature is assembled. A downstream side of the upstream-side path in the coolant passage is a downstream-side path. On the counter-field element side in the radial direction of the cylindrical portion, the first electrical component is arranged on an inner side in the radial direction relative to the upstream-side path, and the second electrical component is arranged on the inner side in the radial direction relative to the downstream-side path.

According to the first exemplary embodiment, of the coolant passage that is formed in the cylindrical portion, a portion from the inlet portion to the specific position between the inlet portion and the outlet portion is the upstream-side path that is partitioned by the partition wall portion. In addition, of the coolant passage, the downstream side of the upstream-side path is the downstream-side path. In this configuration, on the counter-field element side in the radial direction of the cylindrical portion, the first electrical component of which the amount of heat generation is relatively large is arranged on the inner side in the radial direction relative to the upstream-side path, and the second electrical component of which the amount of heat generation is relatively small is arranged on the inner side in the radial direction relative to the downstream-side path.

According to the first exemplary embodiment described above, of the first and second electrical components, the first component of which the amount of heat generation is relatively large can be preferentially cooled. In addition, according to the first exemplary embodiment, the partition wall portion is separated in the radial direction from the wall portion of the cylindrical portion in which the armature is assembled. Therefore, the coolant that flows through the upstream-side path is not easily affected by heat that is generated in the armature winding. Consequently, temperature increase in the coolant that flows through the upstream-side path can be suppressed and cooling performance regarding the first electrical component can be improved. In this manner, according to the first exemplary embodiment, the first electrical component can be appropriately cooled.

According to a second exemplary embodiment, in the first exemplary embodiment, the housing member includes: an end plate; an armature-side wall portion that extends in an axial direction from the end plate and has a circular cylindrical shape; a component-side wall portion that is provided on the counter-field element side in the radial direction relative to the armature-side wall portion, extends in the axial direction from the end plate, and has a circular cylindrical shape; and an overhang portion that is provided so as to be separated from the end plate in the axial direction, and extends from one end in the axial direction of the component-side wall portion to one end in the axial direction of the armature-side wall portion. The cylindrical portion is formed by the end plate, the armature-side wall portion, the overhang portion, and the component-side wall portion.

The partition wall portion includes: a first wall portion that extends from an intermediate position in the axial direction of the component-side wall portion to an intermediate position between the component-side wall portion and

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the armature-side wall portion in the radial direction, and is provided over an area from the inlet portion to the specific position in the circumferential direction; the second wall portion that extends from an end portion on the field element side in the radial direction of the first wall portion to the end plate in the axial direction, and a sealing wall portion that seals a downstream side of a passage that is partitioned by the end plate, the first wall portion, and the second wall portion.

A passage that is partitioned by the end plate, the component-side wall portion, the first wall portion, the second wall portion, and the sealing wall portion is the upstream-side path. On the counter-field element side in the radial direction of the component-side wall portion, a plurality of first electrical components are arranged on the inner side in the radial direction relative to the upstream-side path so as to be arrayed in the circumferential direction.

A coolant inflow port that passes through in the radial direction is formed in correspondence to each first electrical component in a portion of the component-side wall portion that is adjacent to the upstream-side path. A coolant outflow port that passes through in the radial direction is formed in correspondence to each first electrical component, over an area of the component-side wall portion from the inlet portion to the specific position in the circumferential direction in a portion that is further towards the overhang portion side in the axial direction than the first wall portion.

The rotating electric machine includes a communication path that is provided in correspondence to each first electrical component and guides the coolant that flows into the coolant inflow port to the coolant outflow port. Each first electrical component is cooled by the coolant that flows through the communication path.

According to the second exemplary embodiment, the housing member includes the end plate, the armature-side wall portion, the component-side wall portion, and the overhang portion. The partition wall portion includes the first wall portion, the second wall portion, and the sealing wall portion. A passage that is partitioned by the end plate, the component-side wall portion, the first wall portion, the second wall portion, and the sealing wall portion is the upstream-side path. In addition, the coolant inflow port and the coolant outflow port are formed in the component-side wall portion. The coolant that flows into the coolant inflow port is led to the coolant outflow port by the communication path. As a result of this configuration, the coolant that flows from the inlet portion flows to the downstream-side path by way of the upstream-side path, the coolant inflow port, the communication path, and the coolant outflow port.

According to the second exemplary embodiment, a plurality of first electrical components are arranged so as to be arrayed in the circumferential direction. The coolant inflow port, the communication path, and the coolant outflow port are provided for each first electrical component. Therefore, the coolant after cooling any first electrical component among the plurality of first electrical components is not used to cool other first electrical components. Consequently, performance in cooling the first electrical components can be improved. In addition, differences in cooling performance among the first electrical components can be reduced.

According to a third exemplary embodiment, in the second exemplary embodiment, the partition wall portion includes a third wall portion that extends in the radial direction from one end on the end plate side in the axial direction of the component-side wall portion to one end on the end plate side in the axial direction of the second wall portion. A passage that is partitioned by the end plate, the

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component-side wall portion, the first wall portion, the second wall portion, the third wall portion, and the sealing wall portion is the upstream-side path. The rotating electric machine includes: a first sealing member that seals between one end on the overhang portion side in the axial direction of the armature-side wall portion and the overhang portion, and has a circular annular shape; and a second sealing member that seals between one end on the end plate side in the axial direction of the component-side wall portion and the end plate, and has a circular annular shape.

According to the third exemplary embodiment, the configuration is such that the upstream-side path is partitioned by the third wall portion in addition to the end plate, the component-side wall portion, the first wall portion, and the second wall portion. As a result of this configuration, sealing between a space in which the electrical components are arranged in the component-side wall portion and the coolant passage can be performed, and coolant leakage between the upstream-side path and the downstream-side path through a path other than the communication path can be prevented, while a number of sealing members is reduced.

According to a fourth exemplary embodiment, in the first exemplary embodiment, the housing member includes: an end plate; an armature-side wall portion that extends in an axial direction from the end plate and has a circular cylindrical shape; a component-side wall portion that is provided on the counter-field element side in the radial direction relative to the armature-side wall portion, extends in the axial direction from the end plate, and has a circular cylindrical shape; and an overhang portion that is provided so as to be separated from the end plate in the axial direction, and extends from one end in the axial direction of the component-side wall portion to one end in the axial direction of the armature-side wall portion. The cylindrical portion is formed by the end plate, the armature-side wall portion, the overhang portion, and the component-side wall portion.

On the counter-field element side in the radial direction of the component-side wall portion, a plurality of first electrical components are arranged on the inner side in the radial direction relative to the upstream-side path so as to be arrayed in the circumferential direction. A coolant inflow port that passes through in the radial direction and a coolant outflow port that passes through in the radial direction are formed in correspondence to each first electrical component in a portion of the component-side wall portion that is adjacent to the upstream-side path. The rotating electric machine includes a communication path that is provided in correspondence to each first electrical component and guides the coolant that flows into the coolant inflow port to the coolant outflow port. Each first electrical component is cooled by the coolant that flows through the communication path.

According to the fourth exemplary embodiment, each of the first electrical components that are arrayed in the circumferential direction can be appropriately cooled.

According to a fifth exemplary embodiment, in the third or fourth exemplary embodiment, the first electrical component is an electrical module that is configured by a heat generating component, housed in a housing case, that generates heat as a result of energization. Each electrical module is provided with the communication path therein.

According to the fifth exemplary embodiment, cooling performance regarding the electrical modules can be further improved.

According to a sixth exemplary embodiment, in any one of the first to fifth exemplary embodiments, a passage area of the upstream-side path is smaller than a passage area of the downstream-side path.

A speed of flow of the coolant increases as the passage area over which the coolant flows decreases, and cooling performance improves. In light of this point, according to the sixth exemplary embodiment, the passage area of the upstream-side path is smaller than the passage area of the downstream-side path. Therefore, the speed of flow of the coolant that flows through the upstream-side path can be made higher than the speed of flow of the coolant that flows through the downstream-side path. Consequently, cooling performance regarding the first electrical components can be further improved.

According to a seventh exemplary embodiment, in any one of the first to sixth exemplary embodiments, the power converter includes a series-connection body of an upper arm switch and a lower arm switch that is provided for each phase of the armature winding, and a capacitor that is connected in parallel to the series-connection body. The first electrical component is a switch module that is configured by the upper arm switch and the lower arm switch that serve as heat generating components, housed in a housing case, that generate heat as a result of energization. The second electrical component is a capacitor module that is configured by the capacitor that serves as a heat generating component, housed in a housing case, that generates heat as a result of energization.

According to the seventh exemplary embodiment, the switch module can be appropriately cooled.

According to an eighth exemplary embodiment, in any one of the first to seventh exemplary embodiments, the rotor that serves as the field element is provided on an outer side in the radial direction of a stator that serves as the armature.

According to the eighth exemplary embodiment, the first electrical component can be appropriately cooled in an outer-rotor-type rotating electric machine.

A ninth exemplary embodiment provides a manufacturing method for a rotating electric machine. The rotating electric machine includes: a field element that includes a magnet portion that has a plurality of magnetic poles of which polarities alternate in a circumferential direction; an armature that has an armature winding of multiple phases; a power converter that has a first electrical component that generates heat as a result of energization, and a second electrical component that generates heat as a result of energization and has a smaller amount of heat generation than the first electrical component, and is electrically connected to the armature winding; a first housing member that includes an end plate, and an armature-side wall portion that extends in an axial direction from the end plate and has a circular cylindrical shape; and a second housing member that includes a component-side wall portion that is provided on the counter-field element side in the radial direction relative to the armature-side wall portion, extends in the axial direction from the end plate, and has a circular cylindrical shape, and an overhang portion that is provided so as to be separated from the end plate in the axial direction, and extends from one end in the axial direction of the armature-side wall portion to one end in the axial direction of the component-side wall portion.

Either of the field element and the armature is a rotor. The cylindrical portion is formed by the end plate, the armature-side wall portion, the overhang portion, and the component-side wall portion. A coolant passage that has an inlet portion into which a coolant flows and an outlet portion from which

the coolant flows, and connects between the inlet portion and the outlet portion in one direction along the cylindrical portion is formed in the cylindrical portion.

The manufacturing method for the rotating electric machine includes: a step of fixing the armature winding on the field element side in the radial direction of the armature-side wall portion by an adhesive that is cured by heating; a step of assembling, on a circumferential surface of a counter-field element in the radial direction of the component-side wall portion, the first electrical component on a circumferential surface from the inlet portion to a specific position between the inlet portion and the outlet portion in the circumferential direction, and the second electrical component on a circumferential surface from the specific position to the outlet portion in the circumferential direction; and a step of forming the coolant passage by combining the first housing member to which the armature winding is fixed, and the second housing member to which the first electrical component and the second electrical component are assembled.

In the rotating electric machine that is manufactured according to the ninth exemplary embodiment, the cylindrical portion is formed by the end plate and the armature-side wall portion that configure the first housing member, and the overhang portion and the component-side wall portion that configure the second housing member. In addition, a space that is partitioned by the end plate, the armature-side wall portion, the overhang portion, and the component-side wall portion is the coolant passage that connects between the inlet portion and the outlet portion in one direction along the cylindrical portion.

In this configuration, of the circumferential surface of the counter-field element in the radial direction of the component-side wall portion, the first electrical component of which the amount of heat generation is relatively large is assembled on the circumferential surface from the inlet portion to the above-described specific position in the circumferential direction, and the second electrical component of which the amount of heat generation is relatively small is assembled on the circumferential surface from the above-described specific position to the outlet portion in the circumferential direction.

As a result, the first electrical component is cooled by the coolant that flows through the upstream-side path of the coolant passage, and the second electrical component is cooled by the coolant that flows through the downstream-side path. Consequently, of the first and second electrical components, the first electrical component of which the amount of heat generation is relatively large can be preferentially cooled. In this manner, according to the ninth exemplary embodiment, the first electrical component can be appropriately cooled.

In addition, according to the ninth exemplary embodiment, in a manufacturing process for the rotating electric machine, before the first housing member and the second housing member are combined, the step of fixing the armature winding on the field element side in the radial direction of the armature-side wall portion by an adhesive that is cured by heating, and the step of assembling the first and second electrical components on the circumferential surface of the counter-field element in the radial direction of the component-side wall portion are performed.

Consequently, when the adhesive is heated, the electrical components that are assembled to the second housing member being heated can be prevented. Reliability of the electrical components can be prevented from decreasing during the manufacturing process.

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According to a tenth exemplary embodiment, in the ninth exemplary embodiment, the second housing member includes: a first wall portion that extends from an intermediate position in the axial direction of the component-side wall portion to an intermediate position between the component-side wall portion and the armature-side wall portion in the radial direction, and is provided over an area from the inlet portion to the specific position in the circumferential direction; the second wall portion that extends from an end portion on the field element side in the radial direction of the first wall portion to the end plate in the axial direction; and a sealing wall portion that seals a downmost-stream side of a passage that is partitioned by the end plate, the first wall portion, and the second wall portion. The step of forming the coolant passage is a step of forming an upstream-side path that is partitioned by the end plate, the component-side wall portion, the first wall portion, the second wall portion, and the sealing wall portion, and a downstream-side path that is further towards a downstream side than the upstream-side path as the coolant passage by combining the first housing member and the second housing member.

According to the tenth exemplary embodiment, the housing member that forms the coolant passage is divided into the first and second housing members. Thus, the upstream-side path that is a portion of the coolant passage and is provided to cool the first electrical component can be easily provided.

A plurality of embodiments will be described with reference to the drawings. According to the plurality of embodiments, sections that are functionally and/or structurally corresponding and/or related may be given the same reference numbers or reference numbers of which digits in the hundreds place and higher differ. Descriptions according to other embodiments can be referenced regarding the corresponding sections and/or related sections.

For example, a rotating electric machine according to a present embodiment is used as a vehicle power source. However, the rotating electric machine can be widely used for industrial use, in vehicles, household appliances, office automation (OA) equipment, and game machines, and the like. Here, sections according to the embodiments below that are identical or equivalent to each other are given the same reference numbers in the drawings. Descriptions of sections that have the same reference numbers are applicable therebetween.

First Embodiment

A rotating electric machine 10 according to a present embodiment is a synchronous-type multiphase alternating-current motor and has an outer-rotor structure (outer-revolution structure). An overview of the rotating electric machine 10 is shown in FIGS. 1 to 5.

FIG. 1 is a longitudinal cross-sectional perspective view of the rotating electric machine 10. FIG. 2 is a longitudinal cross-sectional view of the rotating electric machine 10 in a direction along a rotation shaft 11. FIG. 3 is a lateral cross-sectional view (cross-sectional view taken along line III-III in FIG. 2) of the rotating electric machine 10 in a direction orthogonal to the rotation shaft 11. FIG. 4 is a cross-sectional view showing a portion of FIG. 3 in an enlarged manner. FIG. 5 is an exploded view of the rotating electric machine 10.

Here, in FIG. 3, for the purpose of illustration, the rotation shaft 11 is omitted and hatching that indicates a cross-sectional plane is omitted. In the description below, a direction in which the rotation shaft 11 extends is an axial

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direction. A direction that radially extends from a center of the rotation shaft 11 is a radial direction. A direction that circumferentially extends with the rotation shaft 11 as a center is a circumferential direction.

The rotating electric machine 10 generally includes a bearing unit 20, a housing 30, a rotor 40, a stator 50, and an inverter unit 60. The rotating electric machine 10 is configured by all of these members being arranged coaxially with the rotation shaft 11 and assembled in the axial direction in a predetermined order. The rotating electric machine 10 according to the present embodiment is configured to include the rotor 40 that serves as a "field element", and the stator 50 that serves as an "armature". The rotating electric machine 10 is implemented as a revolving-field-type rotating electric machine.

The bearing unit 20 includes two bearings 21 and 22, and a holding member 23. The two bearings 21 and 22 are arranged so as to be separated from each other in the axial direction. The holding member 23 holds the bearings 21 and 22. For example, the bearings 21 and 22 are radial ball bearings. Each of the bearings 21 and 22 includes an outer ring 25, an inner ring 26, and a plurality of balls 27 that are arranged between the outer ring 25 and the inner ring 26. The holding member 23 has a circular cylindrical shape. The bearings 21 and 22 are assembled on a radially inner side of the holding member 23. In addition, the rotation shaft 11 and the rotor 40 are supported so as to freely rotate on a radially inner side of the bearings 21 and 22. The bearings 21 and 22 configure a set of bearings that rotatably support the rotation shaft 11.

In each of the bearings 21 and 22, the balls 27 are held by a retainer (not shown). In this state, a pitch between the balls is maintained. The bearings 21 and 22 have a sealing member in upper and lower portions in the axial direction of the retainer, and an interior thereof is filled with a non-conductive grease (such as a non-conductive urea-based grease). In addition, a position of the inner ring 26 is mechanically held by a spacer. A constant-pressure preload that projects in an up/down direction from an inner side is applied.

The housing 30 includes a peripheral wall 31 that forms a circular cylindrical shape. The peripheral wall 31 has a first end and a second end that are opposing in the axial direction thereof. The peripheral wall 31 has an end surface 32 in the first end and an opening 33 in the second end. The opening 33 is open over the overall second end. A circular hole 34 is formed in a center of the end surface 32. The bearing unit 20 is fixed by a fixing means, such as a screw or a rivet, in a state in which the bearing unit 20 is inserted into the hole 34. In addition, the rotor 40 that has a hollow circular cylindrical shape and the stator 50 that has a hollow circular cylindrical shape are housed inside the housing 30, that is, in an interior space that is demarcated by the peripheral wall 31 and the end surface 32.

According to the present embodiment, the rotating electric machine 10 is an outer-rotor type. Inside the housing 30, the stator 50 is arranged on a radially inner side of the rotor 40 that has the cylindrical shape. The rotor 40 is supported in a cantilevered manner by the rotation shaft 11 on the end surface 32 side in the axial direction.

The rotor 40 includes a magnet holder 41 that is formed into a hollow cylindrical shape and an annular magnet unit 42 that is provided on a radially inner side of the magnet holder 41. The magnet holder 41 has an approximately cup-like shape and functions as a magnet holding member. The magnet holder 41 includes a circular cylindrical portion

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43, a fixing portion (attachment) 44, and an intermediate portion 45. The circular cylindrical portion 43 has a circular cylindrical shape.

The fixing portion 14 also has a circular cylindrical shape and has a smaller diameter than the circular cylindrical portion 43. The intermediate portion 45 is a portion that connects the circular cylindrical portion 43 and the fixing portion 44. The magnet unit 42 is attached to an inner circumferential surface of the circular cylindrical portion 43.

Here, the magnet holder 41 is made of a cold-rolled steel sheet (steel plate cold commercial [SPCC]), a forging steel, a carbon fiber-reinforced plastic (CFRP), or the like that has sufficient mechanical strength.

The rotation shaft 11 is inserted into a through hole 44a in the fixing portion 44. The fixing portion 44 is fixed to the rotation shaft 11 that is arranged inside the through hole 44a. That is, the magnet holder 41 is fixed to the rotation shaft 11 by the fixing portion 44. Here, the fixing portion 44 may be fixed to the rotation shaft 11 by spline coupling or key coupling that uses recesses and protrusions, welding, crimping, or the like. As a result, the rotor 40 rotates integrally with the rotation shaft 11.

In addition, the bearings 21 and 22 of the bearing unit 20 are assembled on a radially outer side of the fixing portion 44. As described above, the bearing unit 20 is fixed to the end surface 32 of the housing 30. Therefore, the rotation shaft 11 and the rotor 40 are rotatably supported by the housing 30. As a result, the rotor 40 can freely rotate inside the housing 30.

The fixing portion 44 is provided in the rotor 40 in only one of two end portions that are opposing in the axial direction of the rotor 40. As a result, the rotor 40 is supported by the rotation shaft 11 in a cantilevered manner. Here, the fixing portion 44 of the rotor 40 is rotatably supported at two positions that differ in the axial direction, by the bearings 21 and 22 of the bearing unit 20.

That is, the rotor 40 is rotatably supported by the two bearings 21 and 22 that are separated in the axial direction of the rotor 40, in one of two end portions of the magnet holder 41 that are opposing in the axial direction of the magnet holder 41. Therefore, even in a structure in which the rotor 40 is supported by the rotation shaft 11 in a cantilevered manner, stable rotation of the rotor 40 is implemented. In this case, the rotor 40 is supported by the bearings 21 and 22 at positions that are shifted to one side relative to a center position in the axial direction of the rotor 40.

In addition, a dimension of a gap between the outer ring 25 and the inner ring 26, and the balls 27 differ between the bearing 22 of the bearing unit 20 that is closer to a center of the rotor 40 (lower side in the drawing) and the bearing 21 on a side opposite thereof (upper side in the drawing). For example, the gap dimension is greater in the bearing 22 that is closer to the center of the rotor 40 than in the bearing 21 on the side opposite thereof. In this case, even when shaking of the rotor 40 or vibration caused by imbalance attributed to component tolerance act on the bearing unit 20 on the side that is closer to the center of the rotor 40, effects of the shaking and the vibration are favorably absorbed. Specifically, a play dimension (gap dimension) is increased by a preload in the bearing 22 that is closer to the center of the rotor 40 (lower side in the drawing).

As a result, the vibration that occurs in the cantilevered-support structure is absorbed by the play portion. The preload may be either of a fixed-position preload and a constant-pressure preload. In the case of the fixed-position preload, the outer rings 25 of the bearing 21 and the bearing

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22 are both joined to the holding member 23 using a method such as press-fitting or bonding.

In addition, the inner rings 26 of the bearing 21 and the bearing 22 are both joined to the rotation shaft 11 using a method such as press-fitting or bonding. Here, the preload can be generated by the outer ring 25 of the bearing 21 being arranged in a position that differs in the axial direction from that of the inner ring 26 of the bearing 21. The preload can also be generated by the outer ring 25 of the bearing 22 being arranged in a position that differs in the axial direction from that of the inner ring 26 of the bearing 22.

Furthermore, in a case in which the constant-pressure preload is used, a preload spring, such as wave washer 24, is arranged in an area that is sandwiched between the bearing 22 and the bearing 21 so that the preload is generated in the axial direction from the same area that is sandwiched between the bearing 22 and the bearing 21, toward the outer ring 25 of the bearing 22. In this case as well, the inner rings 26 of the bearing 21 and the bearing 22 are both joined to the rotation shaft 11 using a method such as press-fitting or bonding. The outer ring 25 of the bearing 21 or the bearing 22 is arranged with a predetermined clearance between the outer ring 25 and the holding member 23.

As a result of a configuration such as this, a spring force of the preload spring acts on the outer ring 25 of the bearing 22 in a direction away from the bearing 21. In addition, as a result of this force being transmitted to the rotation shaft 11, a force that presses the inner ring 26 of the bearing 21 in the direction of the bearing 22 is applied. As a result, in both the bearings 21 and 22, the positions of the outer ring 25 and the inner ring 26 in the axial direction are shifted. The preload can be applied to the two bearings in a manner similar to the above-described fixed-position preload.

Here, when the constant-pressure preload is generated, the spring force is not necessarily required to be applied to the outer ring 25 of the bearing 22 as shown in FIG. 2. For example, the spring force may be applied to the outer ring 25 of the bearing 21. In addition, the inner ring 26 of either of the bearings 21 and 22 may be arranged with a predetermined clearance between the inner ring 26 and the rotation shaft 11. The outer rings 25 of the bearings 21 and 22 may be joined to the holding member 23 using a method such as press-fitting or bonding, and the preload may thereby be applied to the two bearings.

Furthermore, when force is applied such that the inner ring 26 of the bearing 21 separates from the bearing 22, force is preferably applied such that the inner ring 26 of the bearing 22 separates from the bearing 21 as well. Conversely, when force is applied such that the inner ring 26 of the bearing 21 approaches the bearing 22, force is preferably applied such that the inner ring 26 of the bearing 22 approaches the bearing 21 as well.

Here, when the present rotating electric machine 10 is applied to a vehicle for the purpose of a vehicle power source or the like, vibrations that have components in a direction which the preload is generated may be applied to a mechanism that generates the preload, or a direction of gravitational force that is applied to a target to which the preload is applied may change. Therefore, when the present rotating electric machine 10 is applied to a vehicle, the fixed-position preload is preferably used.

In addition, the intermediate portion 45 includes an annular inner shoulder portion 49a and an annular outer shoulder portion 49b. The outer shoulder portion 49b is positioned on an outer side of the inner shoulder portion 49a in the radial direction of the intermediate portion 45. The inner shoulder

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portion **49a** and the outer shoulder portion **49b** are separated from each other in the axial direction of the intermediate portion **45**.

As a result, the circular cylindrical portion **43** and the fixing portion **44** partially overlap in the radial direction of the intermediate portion **45**. That is, the circular cylindrical portion **43** protrudes further toward the outer side in the axial direction than a base end portion (a rear-side end portion on the lower side of the drawing) of the fixing portion **44**. In the present configuration, the rotor **40** can be supported to the rotation shaft **11** in a position that is closer to the center of gravity of the rotor **40**, compared to a case in which the intermediate portion **45** is provided in a planar shape without a step. Stable operation of the rotor **40** can be implemented.

In the above-described configuration of the intermediate portion **45**, a bearing-housing recess portion **46** that houses a portion of the bearing unit **20** is formed in the rotor **40** in an annular shape, in a position surrounding the fixing portion **44** in the radial direction and toward an inner side of the intermediate portion **45**. In addition, a coil-housing recess portion **47** that houses a coil end **54** of a stator winding **51** of the stator **50**, described hereafter, is formed in the rotor **40** in a position surrounding the bearing-housing recess portion **46** in the radial direction and toward an outer side of the intermediate portion **45**.

Furthermore, the housing recess portions **46** and **47** are arranged so as to be adjacent to each other on the inner side and the radially outer side. That is, a portion of the bearing unit **20** and the coil end **54** of the stator winding **51** are arranged so as to overlap on the inner side and the radially outer side. As a result, a length dimension in the axial direction of the rotating electric machine **10** can be shortened.

The intermediate portion **45** is provided so as to protrude toward the radially outer side from the rotation shaft **11** side. In addition, a contact preventing portion that extends in the axial direction and prevents contact with the coil end **54** of the stator winding **51** of the stator **50** is provided in the intermediate portion **45**. The intermediate portion **45** corresponds to a protruding portion.

An axial-direction dimension of the coil end **54** can be decreased and an axial length of the stator **50** can be shortened by the coil end **54** being bent toward the inner side or the radially outer side. The bending direction of the coil end **54** may be that which takes into consideration assembly with the rotor **40**.

When assembly of the stator **50** on the radially inner side of the rotor **40** is assumed, the coil end **54** may be bent toward the radially inner side on an insertion-end side relative to the rotor **40**. The bending direction of a coil end on a side opposite the coil end **54** may be arbitrary. However, in terms of manufacturing, a shape in which the coil end is bent toward the outer side that has spatial leeway is preferable.

In addition, the magnet unit **42** that serves as a magnet portion is configured by a plurality of permanent magnets that are arranged on the radially inner side of the circular cylindrical portion **43** such that polarities alternately change along the circumferential direction. As a result, the magnet unit **42** has a plurality of magnetic poles in the circumferential direction. However, details of the magnet unit **42** will be described hereafter.

The stator **50** is provided on the radially inner side of the rotor **40**. The stator **50** includes the stator winding **51** and a stator core **52**. The stator winding **51** is formed so as to be wound into an approximately cylindrical shape (annular shape). The stator core **52** is arranged on the radially inner

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side of the stator winding **51** and serves as a base member. The stator winding **51** is arranged so as to oppose the circular annular magnet unit **42** with a predetermined airgap therebetween. The stator winding **51** is made of a plurality of phase windings. Each of the phase windings is configured by a plurality of conductors that are arrayed in the circumferential direction being connected to one other at a predetermined pitch.

According to the present embodiment, a three-phase winding of a U-phase, a V-phase, and a W-phase and a three-phase winding of an X-phase, a Y-phase, and a Z-phase are used. Through use of two of these three-phase windings, the stator winding **51** is configured as a phase winding of six phases.

The stator core **52** has laminated steel sheets in which electromagnetic steel sheets are formed into a laminated circular annular shape. The electromagnetic steel sheet is a soft magnetic material. The stator core **52** is assembled on the radially inner side of the stator winding **51**. For example, the electromagnetic steel sheet is a silicon steel sheet in which about several % (such as 3%) silicon is added to iron. The stator winding **51** corresponds to an armature winding. The stator core **52** corresponds to an armature core.

The stator winding **51** includes a coil side portion **53** and coil ends **54** and **55**. The coil side portion **53** is a portion that overlaps the stator core **52** in the radial direction and is on the radially outer side of the stator core **52**. The coil ends **54** and **55** respectively protrude from one end side and another end side of the stator core **52** in the axial direction.

The coil side portion **53** opposes each of the stator core **52** and the magnet unit **42** of the rotor **40** in the radial direction. In a state in which the stator **50** is arranged on the inner side of the rotor **40**, of the coil ends **54** and **55** on both sides in the axial direction, the coil end **54** that is on the side of the bearing unit **20** (upper side in the drawing) is housed in the coil-housing recess portion **47** that is formed by the magnet holder **41** of the rotor **40**. However, details of the stator **50** will be described hereafter.

The inverter unit **60** includes a unit base **61** and a plurality of electrical components **62**. The unit base **61** is fixed to the housing **30** by a fastener such as a bolt. The plurality of electrical components **62** are assembled to the unit base **61**. For example, the unit base **61** may be made of a CFRP. The unit base **61** includes an end plate **63** and a casing **64**. The end plate **63** is fixed to an edge of the opening **33** of the housing **30**. The casing **64** is provided integrally with the end plate **63** and extends in the axial direction. The end plate **63** has a circular opening **65** in a center portion thereof. The casing **64** is formed so as to stand erect (protrude) from a circumferential edge portion of the opening **65**.

The stator **50** is assembled to an outer circumferential surface of the casing **64**. That is, an outer diameter dimension of the casing **64** is a dimension that is the same as an inner diameter dimension of the stator core **52** or slightly smaller than the inner diameter dimension of the stator core **52**. As a result of the stator core **52** being assembled on the outer side of the casing **64**, the stator **50** and the unit base **61** are integrated. In addition, because the unit base **61** is fixed to the housing **30**, in the state in which the stator core **52** is assembled to the casing **64**, the stator **50** is in a state of being integrated with the housing **30**.

Here, the stator core **52** may be assembled to the unit base **61** by bonding, shrink-fitting, press-fitting, or the like. As a result, positional shifting of the stator core **52** in the circumferential direction or the axial direction relative to the unit base **61** side is suppressed.

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In addition, a radially inner side of the casing **64** is a housing space for housing the electrical components **62**. The electrical components **62** are arranged in the housing space so as to surround the rotation shaft **11**. The casing **64** serves a role as a housing-space forming portion. The electrical components **62** are configured to actualize a semiconductor module **66** that configures an inverter circuit, a control board **67**, and a capacitor module **68**.

Here, the unit base **61** is provided on the radially inner side of the stator **50** and corresponds to a stator holder (armature holder) that holds the stator **50**. The housing **30** and the unit base **61** configure a motor housing of the rotating electric machine **10**. In the motor housing, the holding member **23** is fixed to the housing **30** on one side in the axial direction with the rotor **40** therebetween, and the housing **30** and the unit base **61** are coupled with each other on the other side. For example, in an electric vehicle that is an electric automobile or the like, the rotating electric machine **10** is mounted in the vehicle or the like by the motor housing being attached on the side of the vehicle or the like.

Here, the configuration of the inverter unit **60** will be further described with reference to FIG. **6**, in addition to above-described FIGS. **1** to **5**. FIG. **6** is an exploded view of the inverter unit **60**.

In the unit base **61**, the casing **64** includes a cylindrical portion **71** and an end surface **72** that is provided on one (an end portion on the bearing unit **20** side) of both ends that are opposing in the axial direction of the cylindrical portion **71**. A side opposite the end surface **72** of both end portions in the axial direction of the cylindrical portion **71** is completely open through the opening **65** of the end plate **63**.

A circular hole **73** is formed in a center of the end surface **72**. The rotation shaft **11** can be inserted into the hole **73**. A sealing member **171** that seals a gap between the end surface **72** and the outer circumferential surface of the rotation shaft **11** is provided in the hole **73**. For example, the sealing member **171** may be a sliding seal that is made of a resin material.

The cylindrical portion **71** of the casing **64** is a partitioning portion that partitions between the rotor **40** and the stator **50** that are arranged on a radially outer side thereof, and the electrical components **62** that are arranged on a radially inner side thereof. The rotor **40** and the stator **50**, and the electrical components **62** are respectively arranged so as to be arrayed on the inner side and the radially outer side with the cylindrical portion **71** therebetween.

In addition, the electrical component **62** is an electrical component that configures an inverter circuit. The electrical component **62** provides a power-running function for supplying a current to the phase windings of the stator winding **51** in a predetermined order and rotating the rotor **40**, and a power generation function for receiving input of a three-phase alternating-current current that flows through the stator winding **51** in accompaniment with the rotation of the rotation shaft **11** and outputting the three-phase alternating-current current outside as generated power.

Here, the electrical component **62** may only provide either of the power-running function and the power generation function. For example, when the rotating electric machine **10** is used as a vehicle power source, the power generation function is a regeneration function for outputting the three-phase alternating-current current outside as regenerative power.

As shown in FIG. **4**, as a specific configuration of the electrical components **62**, a capacitor module **68** that has a hollow circular cylindrical shape is provided around the rotation shaft **11**, and a plurality of semiconductor modules

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66 are arranged in an array in the circumferential direction on an outer circumferential surface of the capacitor module **68**. The capacitor module **68** includes a plurality of smoothing capacitors **68a** that are connected to one another in parallel.

Specifically, the capacitor **68a** is a laminated-type film capacitor that is made of a plurality of film capacitors being laminated. A lateral cross-section of the capacitor **68a** has a trapezoidal shape. The capacitor module **68** is configured by twelve capacitors **68a** being arranged so as to be annularly arrayed.

Here, for example, in a manufacturing process for the capacitor **68a**, a capacitor element is fabricated using an elongated film that has a predetermined width and is made of a plurality of films being laminated. The elongated film is cut into isosceles trapezoids such that a film-width direction serves as a trapezoid-height direction, and tops and bottoms of the trapezoids alternate. In addition, the capacitor **68a** is fabricated by electrodes and the like being attached to the capacitor element.

For example, the semiconductor module **66** has a semiconductor switching element, such as a metal-oxide-semiconductor field-effect transistor (MOSFET) or an insulated-gate bipolar transistor (IGBT), and is formed into an approximately plate-like shape.

According to the present embodiment, the rotating electric machine **10** includes two sets of three-phase windings. The inverter circuit is provided for each of the three-phase windings. Therefore, a semiconductor module group **66A** that is formed by a total of twelve semiconductor modules **66** being annularly arrayed is provided in the electrical components **62**.

The semiconductor module **66** is arranged so as to be sandwiched between the cylindrical portion **71** of the casing **64** and the capacitor module **68**. An outer circumferential surface of the semiconductor module group **66A** is in contact with an inner circumferential surface of the cylindrical portion **71**. An inner circumferential surface of the semiconductor module group **66A** is in contact with the outer circumferential surface of the capacitor module **68**. In this case, heat that is generated in the semiconductor module **66** is transmitted to the end plate **63** through the casing **64** and released from the end plate **63**.

The semiconductor module group **66A** may include a spacer **69** on the outer circumferential surface side, that is, between the semiconductor modules **66** and the cylindrical portion **71** in the radial direction. In this case, in the capacitor module **68**, a cross-sectional shape of a lateral cross-section that is orthogonal to the axial direction is a regular dodecagon. Meanwhile, a lateral cross-sectional shape of the inner circumferential surface of the cylindrical portion **71** is a circular shape.

Therefore, in the spacer **69**, an inner circumferential surface is a flat surface and an outer circumferential surface is a curved surface. The spacer **69** may be integrally provided on the radially outer side of the semiconductor module group **66A** so as to be continuous in a circular annular shape. The spacer **69** is a good heat conductor and, for example, may be made of a metal such as aluminum or a heat-radiation gel sheet. Here, the lateral cross-sectional shape of the inner circumferential surface of the cylindrical portion **71** can also be a dodecagon that is identical to the capacitor module **68**. In this case, both the inner circumferential surface and the outer circumferential surface of the spacer **69** may be flat surfaces.

In addition, according to the present embodiment, a cooling water passage **74** through which cooling water flows

is formed in the cylindrical portion **71** of the casing **64**. Heat that is generated in the semiconductor modules **66** is released to the cooling water that flows through the cooling water passage **74** as well. That is, the casing **64** includes a water-cooled mechanism.

As shown in FIGS. **3** and **4**, the cooling water passage **74** is formed into an annular shape so as to surround the electrical components **62** (the semiconductor modules **66** and the capacitor module **68**). The semiconductor modules **66** are arranged along the inner circumferential surface of the cylindrical portion **71**. The cooling water passage **74** is provided in a position that overlaps the semiconductor modules **66** on the inner side and the radially outer side.

The stator **50** is arranged on the outer side of the cylindrical portion **71** and the electrical components **62** are arranged on the inner side. Therefore, heat from the stator **50** is transmitted to the cylindrical portion **71** from the outer side thereof, and heat from the electrical components **62** (such as heat from the semiconductor modules **66**) is transmitted from the inner side. In this case, the stator **50** and the semiconductor modules **66** can be simultaneously cooled. Heat from heat generating components of the rotating electric machine **10** can be efficiently released.

Furthermore, at least a portion of the semiconductor modules **66** that configure a portion or an entirety of the inverter circuit that operates the rotating electric machine by performing energization of the stator winding **51** is arranged inside an area that is surrounded by the stator core **52** that is arranged on the radially outer side of the cylindrical portion **71** of the casing **64**. The entirety of a single semiconductor module **66** is preferably arranged inside the area that is surrounded by the stator core **52**. Furthermore, the entirety of all semiconductor modules **66** is preferably arranged inside the area that is surrounded by the stator core **52**.

In addition, at least a portion of the semiconductor modules **66** is arranged inside an area that is surrounded by the cooling water passage **74**. All of the semiconductor modules **66** is preferably arranged inside an area that is surrounded by a yoke **141**.

Moreover, the electrical components **62** include, in the axial direction, an insulating sheet **75** that is provided on one end surface of the capacitor module **68** and a wiring module **76** that is provided on another end surface. In this case, the capacitor module **68** includes two end surfaces that are opposing in the axial direction thereof, that is, a first end surface and a second end surface. The first end surface of the capacitor module **68** that is close to the bearing unit **20** opposes the end surface **72** of the casing **64** and overlaps the end surface **72** with the insulating sheet **75** sandwiched therebetween. In addition, the wiring module **76** is assembled to the second end surface of the capacitor module **68** that is close to the opening **65**.

The wiring module **76** includes a main body portion **76a** and a plurality of bus bars **76b** and **76c**. The main body portion **76a** is made of a synthetic resin material and has a circular plate shape. The plurality of bus bars **76b** and **76c** are embedded inside the main body portion **76a**. Electrical connection with the semiconductor modules **66** and the capacitor module **68** is achieved by the bus bars **76b** and **76c**.

Specifically, the semiconductor module **66** includes a connection pin **66a** that extends from an end surface in the axial direction thereof. The connection pin **66a** is connected to the bus bar **76b** on a radially outer side of the main body portion **76a**. In addition, the bus bar **76c** extends toward a side opposite the capacitor module **68** on the radially outer

side of the main body portion **76a**. The bus bar **76c** is connected to a wiring member **79** at a tip end portion thereof (see FIG. **2**).

As described above, the insulating sheet **75** is provided on the first end surface that is opposing in the axial direction of the capacitor module **68**, and the wiring module **76** is provided on the second surface of the capacitor module **68**. In this configuration, as a heat releasing path of the capacitor module **68**, a path from the first end surface and the second end surface of the capacitor module **68** to the end surface **72** and the cylindrical portion **71** is formed.

That is, a path from the first end surface to the end surface **72** and a path from the second end surface to the cylindrical portion **71** are formed. As a result, heat release from the end surface portions of the capacitor module **68** other than the outer circumferential surface on which the semiconductor modules **66** are provided can be performed. That is, heat release can be performed not only in the radial direction but also the axial direction.

In addition, the capacitor module **68** has a hollow circular cylindrical shape. The rotation shaft **11** is arranged in an inner circumferential portion thereof with a predetermined gap interposed therebetween. Therefore, heat from the capacitor module **68** can also be released from the hollow portion thereof. In this case, as a result of a flow of air being generated by the rotation of the rotation shaft **11**, the cooling effect thereof can be improved.

The circular plate-shaped control board **67** is attached to the wiring module **76**. The control board **67** includes a printed circuit board (PCB) on which a predetermined wiring pattern is formed. A control apparatus **77** that corresponds to a control unit that is made of various types of integrated circuits (IC), microcomputers, and the like is mounted on the board. The control board **67** is fixed to the wiring module **76** by a fixing means such as a screw. The control board **67** has an insertion hole **67a** through which the rotational shaft **11** is inserted in a center portion thereof.

Here, the wiring module **76** has a first surface and a second surface that oppose each other in the axial direction, that is, oppose each other in a thickness direction thereof. The first surface faces the capacitor module **68**. The wiring module **76** is provided with the control board **67** on the second surface thereof. The bus bar **76c** of the wiring module **76** extends from one side to the other side of both surfaces of the control board **67**. In this configuration, the control board **67** may be provided with a notch that prevents interference with the bus bar **76c**. For example, a portion of an outer edge portion of the control board **67** that has the circular shape may be notched.

As described above, the electrical components **62** are housed inside the space that is surrounded by the casing **64**, and the housing **30**, the rotor **40**, and the stator **50** are provided in layers on the outer side thereof. In this configuration, shielding from electromagnetic noise that is generated in the inverter circuit is suitably performed.

That is, in the inverter circuit, switching control being performed in each of the semiconductor modules **66** using pulse width modulation (PWM) control based on a predetermined carrier frequency and electromagnetic noise being generated as a result of the switching control can be considered. However, shielding from this electromagnetic noise can be suitably performed by the housing **30**, the rotor **40**, the stator **50**, and the like on the outer side of in the radial direction the electrical components **62**.

Furthermore, as a result of at least a portion of the semiconductor modules **66** being arranged inside the area that is surrounded by the stator core **52** that is arranged on

the radially outer side of the cylindrical portion 71 of the casing 64, compared to a configuration in which the semiconductor modules 66 and the stator winding 51 are arranged without the stator core 52 therebetween, even if magnetic flux is generated from the semiconductor modules 66, the stator winding 51 is not easily affected.

In addition, even if magnetic flux is generated from the stator winding 51, the semiconductor modules 66 are not easily affected. Here, it is even more effective to arrange the overall semiconductor modules 66 inside the area that is surrounded by the stator core 52 that is arranged on the radially outer side of the cylindrical portion 71 of the casing 64. In addition, when at least a portion of the semiconductor modules 66 is surrounded by the cooling water passage 74, an effect in which heat generated from the stator winding 51 and the magnet unit 42 does not easily reach the semiconductor modules 66 can be obtained.

A through hole 78 through which the wiring member 79 (see FIG. 2) is inserted is formed near the end plate 63 in the cylindrical portion 71. The wiring member 79 electrically connects the stator 50 on the outer side of the cylindrical portion 71 and the electrical components 62 on the inner side thereof.

As shown in FIG. 2, the wiring member 79 is connected to each of the end portion of the stator winding 51 and the bus bar 76c of the wiring module 76 by press-fitting, welding, or the like. For example, the wiring member 79 is preferably crushed to be flat. The through hole 78 may be provided in a single location or a plurality of locations.

According to the present embodiment, the through holes 78 are provided in two locations. In this configuration, winding terminals that extend from the two sets of three-phase windings can each easily be connected by the wiring member 79. This is suitable in terms of performing multi-phase connection.

As described above, as shown in FIG. 4, inside the housing 30, the rotor 40 and the stator 50 are provided in order from the radially outer side, and the inverter unit 60 is provided on the radially inner side of the stator 50. Here, when a radius of the inner circumferential surface of the housing 30 is d, the rotor 40 and the stator 50 are arranged further toward the radially outer side than a distance of $d \times 0.705$ from a rotational center of the rotor 40 is.

In this case, when an area on the radially inner side from an inner circumferential surface of the stator 50 (that is, an inner circumferential surface of the stator core 52) that is on the radially inner side, of the rotor 40 and the stator 50, is a first area X1 and an area from the inner circumferential surface of the stator 50 to the housing 30 in the radial direction is a second area X2, an area of a lateral cross-section which is the first area X1 is greater than an area of a lateral cross-section which is the second area X2.

In addition, in terms of an area over which the magnet unit 42 of the rotor 40 and stator winding 51 overlap in the radial direction, a volume of the first area X1 is greater than a volume of the second area X2.

Here, if the rotor 40 and the stator 50 are considered a magnetic circuit component assembly, inside the housing 30, the first area X1 that is on the radially inner side from an inner circumferential surface of the magnetic circuit component assembly has a greater volume than the second area X2 that is from the inner circumferential surface of the magnetic circuit component assembly to the housing 30 in the radial direction.

Next, the configurations of the rotor 40 and the stator 50 will be described in further detail.

As a configuration of a stator in a rotating electric machine, a configuration in which a plurality of slots are provided in the circumferential direction in a stator core that is made of laminated steel sheets and has a circular annular shape, and a stator winding is wound through the slots is generally known. Specifically, the stator core includes a plurality of teeth that extend in the radial direction from a yoke at predetermined intervals. The slots are formed between the teeth that are adjacent to each other in the circumferential direction. In addition, for example, a plurality of layers of conductors are housed inside the slots in the radial direction, and the stator winding is configured by these conductors.

However, in the above-described stator structure, during energization of the stator winding, magnetic saturation occurring in the teeth portion of the stator core in accompaniment with increase in magnetomotive force in the stator winding, and torque density of the rotating electric machine becoming limited as a result thereof can be considered. That is, in the stator core, magnetic saturation occurs as a result of a rotating magnetic flux that is generated by the energization of the stator winding being concentrated at the teeth.

In addition, as a configuration of an interior permanent magnet (IPM) rotor of a rotating electric machine, a configuration in which a permanent magnet is arranged on a d-axis and a rotor core is arranged on a q-axis of a d-q coordinate system is generally known. In such cases, as a result of the stator winding near the d-axis being excited, an excitation magnetic flux flows from the stator to the q-axis of the rotor as a result of Fleming's Rule. In addition, as a result, magnetic saturation over a wide area is thought to occur in a q-axis core portion of the rotor.

FIG. 7 is a torque diagram of a relationship between ampere-turns [AT] and torque density [Nm/L]. The ampere-turns indicates magnetomotive force in the stator winding. A broken line indicates characteristics of a typical IPM-rotor-type rotating electric machine. As shown in FIG. 7, in the typical rotating electric machine, as a result of the magnetomotive force being increased in the stator, magnetic saturation occurs in two locations that are the teeth portion between the slots and the q-axis core portion, and increase in torque becomes limited as a result. In this manner, in the typical rotating electric machine, an ampere-turns design value is limited by A1.

Here, according to the present embodiment, to eliminate limitations attributed to magnetic saturation, the rotating electric machine 10 is also provided with a configuration described below. That is, as a first modification, a slot-less structure is used in the stator 50 to eliminate magnetic saturation that occurs in the teeth of the stator core in the stator. In addition, a surface permanent magnet (SPM) rotor is used to eliminate magnetic saturation that occurs in the q-axis core portion of the IPM rotor.

As a result of the first modification, the above-described two locations in which magnetic saturation occurs can be eliminated. However, decrease in torque in a low-current region can be considered (refer to a single-dot chain line in FIG. 7). Therefore, as a second modification, a polar anisotropic structure in which a magnet magnetic path is extended and magnetic force is increased in the magnet unit 42 of the rotor 40 is used to recover the decrease in torque through magnetic flux enhancement in the SPM rotor.

In addition, as a third modification, recovery of the decrease in torque is achieved through use of a flattened conductor structure in which a thickness of the conductor in the radial direction of the stator 50 is reduced in the coil side portion 53 of the stator winding 51. Here, there is thought to

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be a potential risk of generation of larger eddy currents in the stator winding **51** that opposes the magnet unit **42**, as a result of the above-described polar anisotropic structure in which the magnetic force is increased.

However, as a result of the third modification, the generation of eddy currents in the radial direction in the stator winding **51** can be suppressed because of the flattened conductor structure that is thin in the radial direction. In this manner, as a result of these first to third configurations, even while significant improvement in torque characteristics can be expected through use of a magnet that has high magnetic force, as indicated by a solid line in FIG. **7**, concern regarding the generation of large eddy currents that may occur as a result of the magnet that has high magnetic force can be ameliorated as well.

Furthermore, as a fourth modification, a magnet unit that has a magnetic flux density distribution that is close to a sine wave is used through use of the polar anisotropic structure. As a result, a sine-wave matching ratio can be improved by pulse control, described hereafter, or the like and torque enhancement can be obtained. In addition, because changes in magnetic flux are more gradual compared to that of a radial magnet, eddy current loss (copper loss due to eddy currents) can also be further suppressed.

The sine-wave matching ratio will be described below. The sine-wave matching ratio can be determined based on a comparison between an actual measured waveform of a surface magnetic flux density distribution that is measured by a surface of a magnet being traced by a magnetic flux probe or the like, and a sine wave that has the same period and the same peak value. In addition, a proportion of an amplitude of a primary waveform that is a fundamental wave of the rotating electric machine relative to an amplitude of the actual measured waveform, that is, an amplitude obtained by another harmonic component being added to the fundamental wave corresponds to the sine-wave matching ratio.

As the sine-wave matching ratio increases, the waveform of the surface magnetic flux density distribution becomes closer to the sine-wave waveform. In addition, when a primary sine-wave current is supplied from an inverter to the rotating electric machine that includes a magnet that has an improved sine-wave matching ratio, because of this and the waveform of the surface magnetic flux density distribution of the magnet being close to the sine waveform as well, a large torque can be generated. Here, the surface magnetic flux density distribution may be estimated by a method other than actual measurement, such as by an electromagnetic field analysis using Maxwell's equations.

In addition, as a fifth modification, the stator winding **51** has a wire conductor body structure in which a plurality of wires are gathered together and bundled. As a result, because the wires are connected in parallel, a large current can be supplied. In addition, the generation of eddy currents that are generated in the conductors that are spread in the circumferential direction of the stator **50** as a result of the flattened conductor structure can be suppressed more effectively than when the conductors are made thinner in the radial direction as a result of the third modification, because a cross-sectional area of each wire is reduced. In addition, as a result of a configuration in which the plurality of wires are twisted together, regarding magnetomotive force from a conductor body, eddy currents from a magnetic flux that is generated based on a right-hand screw rule in a current conduction direction can be cancelled.

In this manner, as a result of the fourth modification and the fifth modification being further added, torque enhance-

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ment can be obtained while a magnet according to the second modification that has a high magnetic force that is used and, further, while the eddy current loss attributed to the high magnetic force is suppressed.

Descriptions of the above-described slot-less structure of the stator **50**, flattened conductor structure of the stator winding **51**, and polar anisotropic structure of the magnet unit **42** are separately added below. Here, first, the slot-less structure of the stator **50** and the flattened conductor structure of the stator winding **51** will be described.

FIG. **8** is a lateral cross-sectional view of the rotor **40** and the stator **50**. FIG. **9** is a diagram showing a portion of the rotor **40** and the stator **50** shown in FIG. **8** in an enlarged manner. FIG. **10** is a cross-sectional view showing a lateral cross-section of the stator **50** taken along line X-X in FIG. **11**. FIG. **11** is a cross-sectional view showing a longitudinal cross-section of the stator **50**. In addition, FIG. **12** is a perspective view of the stator winding **51**. Here, in FIGS. **8** and **9**, a magnetization direction of the magnets in the magnet unit **42** is indicated by an arrow.

As shown in FIGS. **8** to **11**, the stator core **52** is that in which a plurality of electromagnetic steel sheets are laminated in the axial direction. The stator core **52** has a circular cylindrical shape that has a predetermined thickness in the radial direction. The stator winding **51** is assembled on the radially outer side of the stator core **52** that is the rotor **42** side. In the stator core **52**, the outer circumferential surface on the rotor **40** side serves as a conductor setup portion (conductor body area). The outer circumferential surface of the stator core **52** has a curved surface shape that has substantially no unevenness.

A plurality of conductor groups **81** are arranged on the outer circumferential surface of the stator core **52** at predetermined intervals in the circumferential direction. The stator core **52** functions as a back yoke that serves as a portion of a magnetic circuit for rotating the rotor **40**. In this case, a tooth (that is, a core) that is made of a soft magnetic material is not provided between two conductor groups **81** that are adjacent to each other in the circumferential direction (that is, a slot-less structure).

According to the present embodiment, the structure is such that a resin material of a sealing member **57** enters a gap **56** between the conductor groups **81**. That is, in the stator **50**, an inter-conductor member that is provided between the conductor groups **81** in the circumferential direction is configured as the sealing member **57** that is a non-magnetic material. In terms of a state before sealing by the sealing member **57**, the conductor groups **81** are arranged on the radially outer side of the stator core **52**, at predetermined intervals in the circumferential direction so as to each be separated by the gap **56** that is a conductor-to-conductor area.

The stator **50** that has a slot-less structure is thereby constructed. In other words, each conductor group **81** is made of two conductors **82**, as described hereafter. Only a non-magnetic material occupies the area between two conductor groups **81** that are adjacent to each other in the circumferential direction of the stator **50**. The non-magnetic material may include a non-magnetic gas such as air, a non-magnetic liquid, and the like, in addition to the sealing member **57**. Hereafter, the sealing member **57** is also referred to as the inter-conductor member.

Here, the configuration in which the teeth are provided between the conductor groups **81** that are arrayed in the circumferential direction can be said to be a configuration in which, as a result of the teeth having a predetermined thickness in the radial direction and a predetermined width

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in the circumferential direction, a portion of the magnetic circuit, that is, a magnet magnetic path is formed between the conductor groups **81**. In this regard, the configuration in which the teeth are not provided between the conductor groups **81** can be said to be a configuration in which the above-described magnetic circuit is not formed.

As shown in FIG. **10**, the stator winding (that is, the armature winding) **51** is formed to have a predetermined thickness T2 (also referred to, hereafter, as a first dimension) and width W2 (also referred to, hereafter, as a second dimension). The thickness T2 is a shortest distance between the outer circumferential surface and the inner circumferential surface that oppose each other in the radial direction of the stator winding **51**. The width W2 is a length, in the circumferential direction of the stator winding **51**, of a portion of the stator winding **51** that functions as one of the multiple phases (in the example, three phases: three phases that are the U-phase, V-phase, and W-phase or three phases that are the X-phase, Y-phase, and Z-phase) of the stator winding **51**.

Specifically, in FIG. **10**, when the two conductor groups **81** that are adjacent to each other in the circumferential direction function as one of the three phases, such as the U-phase, the width W2 is from end to end of the two conductor groups **81** in the circumferential direction. In addition, the thickness T2 is less than the width W2.

Here, the thickness T2 is preferably less than a total width dimension of the two conductor groups **81** that are present within the width W2. In addition, if the cross-sectional shape of the stator winding **51** (more specifically, the conductors **82**) is perfectly circular, elliptical, or polygonal, of the cross-section of the conductors **82** along the radial direction of the stator **50**, a maximum length in the radial direction of the stator **50** on the cross-section may be W2 and a maximum length in the circumferential direction of the stator **50** on the same cross-section may be W2.

As shown in FIGS. **10** and **11**, the stator winding **51** is sealed by the sealing member **57** that is made of a synthetic resin material that serves as a sealing material (molding material). That is, the stator winding **51** is molded by the molding material, together with the stator core **52**. Here, the resin may be a non-magnetic body or an equivalent of a non-magnetic body in which Bs=0.

In terms of the lateral cross-section in FIG. **10**, the sealing member **57** is provided by the synthetic resin filling the area between the conductor groups **81**, that is, the gaps **56**. An insulation member is interposed between the conductor groups **81** as a result of the sealing member **57**. That is, the sealing member **57** functions as an insulation member in the gap **56**. The sealing member **57** is provided on the radially outer side of the stator core **52**, over an area that includes all of the conductor groups **81**, that is, over an area in which a thickness dimension in the radial direction is greater than the thickness dimension in the radial direction of each conductor group **81**.

In addition, in terms of the longitudinal cross-section in FIG. **11**, the sealing member **57** is provided over an area that includes a turn portion **84** of the stator winding **51**. The sealing member **57** is provided on the radially inner side of the stator winding **51**, over an area that includes at least a portion of an end surface of the stator core **52** that is opposing in the axial direction. In this case, the stator winding **51** is approximately entirely sealed by resin, excluding the end portion of the phase winding of each phase, that is, the connection terminals for the inverter circuit.

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The sealing member **57** is provided over an area that includes the end surface of the stator core **52**. In this configuration, the laminated steel sheets of the stator core **52** can be pressed toward the inner side in the axial direction by the sealing member **57**. As a result, the state of lamination of the steel sheets can be maintained using the sealing member **57**. Here, according to the present embodiment, the inner circumferential surface of the stator core **52** is not sealed by resin. However, instead, the overall stator core **52** including the inner circumferential surface of the stator core **52** may be sealed by resin.

When the rotating electric machine **10** is used as a vehicle power source, the sealing member **57** is preferably made of fluororesin that has high heat resistance, epoxy resin, polyphenylene sulfide (PPS) resin, polyether ether ketone (PEEK) resin, liquid crystal polymer (LCP) resin, silicone resin, polyamide-imide (PAI) resin, polyimide (PI) resin, or the like.

In addition, when a coefficient of linear expansion is considered from a perspective of suppressing cracks caused by differences in expansion, the sealing member **57** is preferably made of a material that is the same as that of an outer coating of the conductors of the stator winding **51**. That is, a silicone resin of which the coefficient of linear expansion is generally equal to or greater than twice that of other resins is preferably excluded.

Here, in electrical products that do not have an engine that uses combustion, like an electric vehicle, poly(p-phenylene oxide) (PPO) resin and phenolic resin that have heat resistance of about 180° C., and fiber-reinforced plastic (FRP) resin are also candidates. In fields in which ambient temperature of the rotating electric machine can be assumed to be less than 100° C., the materials are not limited to the foregoing.

The torque of the rotating electric machine **10** is proportional to the magnitude of the magnetic flux. Here, when the stator core has teeth, a maximum magnetic flux amount of the stator is dependent on and limited by the saturation magnetic flux density at the teeth. However, when the stator core does not have teeth, the maximum magnetic flux amount of the stator is not limited. Therefore, the configuration is advantageous in terms of increasing a conduction current to the stator winding **51** and achieving torque increase in the rotating electric machine **10**.

According to the present embodiment, inductance in the stator **50** decreases as a result of the structure (slot-less structure) in which the teeth are eliminated being used in the stator **50**. Specifically, whereas the inductance in a stator of a typical rotating electric machine in which conductors are housed in slots that are partitioned by a plurality of teeth is, for example, about 1 mH, the inductance is reduced to about 5 μH to 60 μH in the stator **50** according to the present embodiment.

According to the present embodiment, even with the rotating electric machine **10** that has the outer-rotor structure, a mechanical time constant Tm can be reduced through reduction of the inductance in the stator **50**. That is, the mechanical time constant Tm can be reduced while higher torque is obtained. Here, when inertia is J, inductance is L, a torque constant is Kt, and a counter electromotive force constant is Ke, the mechanical time constant Tm is calculated by a following expression.

$$Tm=(J \times L)/(Kt \times Ke)$$

In this case, it can be confirmed that the mechanical time constant Tm decreases as a result of decrease in the inductance L.

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The conductor groups **81** on the radially outer side of the stator core **52** are configured such that a plurality of conductors **82** of which a cross-section forms a flattened rectangular shape are arranged so as to be arrayed in the radial direction of the stator core **52**. The conductor **82** is arranged

to be oriented such that, on a lateral cross-section, radial direction dimension < circumferential direction dimension. As a result, thinness in the radial direction is achieved in each conductor group **81**. Furthermore, in addition to thinness in the radial direction being achieved, a conductor-body area extends in a planar manner to an area in which teeth were originally provided, and a flattened conductor area structure is formed. As a result, increase in a heat generation quantity of the conductors that becomes a concern as a result of the cross-sectional area becoming smaller as a result of being thinner is suppressed by the cross-sectional area of the conductor body being increased through flattening in the circumferential direction.

Here, even when the plurality of conductors are arrayed in the circumferential direction and connected in parallel, although decrease in a conductor-body cross-sectional area that amounts to the conductor coating occurs, effects based on the same reasoning can be obtained. Here, each of the conductor groups **81** and each of the conductors **82** may also be referred to as a conductive member, below.

Because slots are not provided, in the stator winding **51** according to the present embodiment, the conductor-body area that is occupied by the stator winding **51** in a single round in the circumferential direction can be designed to be greater than a conductor-body unoccupied area in which the stator winding **51** is not present.

Here, in a conventional rotating electric machine for a vehicle, the conductor-body area/conductor-body unoccupied area in a single round in the circumferential direction of the stator winding being equal to or less than 1 is a matter of course. Meanwhile, according to the present embodiment, the conductor groups **81** are provided such that the conductor-body area is equal to the conductor-body unoccupied area or the conductor-body area is greater than the conductor-body unoccupied area.

Here, as shown in FIG. 10, when a conductor area in which the conductors **82** (that is, a linear portions **83**, described hereafter) is arranged in the circumferential direction is a WA and an inter-conductor area between adjacent conductors **82** is WB, the conductor area WA is greater in the circumferential direction than the conductor area WB.

As the conductor group **81** in the stator winding **51**, a thickness dimension in the radial direction of the conductor group **81** is less than a width dimension in the circumferential direction corresponding to a single phase within a single magnetic pole. That is, the conductor group **81** is made of two layers of conductors **82** in the radial direction, and two conductor groups **81** are provided in the circumferential direction for a single phase within a single magnetic pole. In this configuration, a relationship expressed by $Tc \times 2 < Wc \times 2$ is established, where Tc is the thickness dimension in the radial direction of the conductor **82**, and Wc is the width dimension in the circumferential direction of the conductor **82**.

Here, as another configuration, the conductor group **81** may be made of two layers of conductors **82**, and a single conductor group **81** may be provided in the circumferential direction for a single phase within a single magnetic pole. In this configuration, a relationship expressed by $Tc \times 2 < Wc$ may be established. In short, the conductor portions (conductor groups **81**) that are arranged at predetermined intervals in the circumferential direction in the stator winding **51**

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are that in which the thickness dimension in the radial direction thereof is less than the width dimension in the circumferential direction corresponding to a single phase within a single magnetic pole.

In other words, each of the conductors **82** may be such that the thickness dimension Tc in the radial direction is less than the width dimension Wc in the circumferential direction. In addition, further, the thickness dimension (2Tc) in the radial direction of the conductor group **81** that is made of two layers of the conductors **82** in the radial direction, that is, the thickness dimension (2Tc) in the radial direction of the conductor group **81** may be less than the width dimension Wc in the circumferential direction.

The torque of the rotating electric machine **10** is approximately inversely proportional to the thickness in the radial direction of the stator core **52** of the conductor group **81**. In this regard, as a result of the thickness of the conductor group **81** being made thinner on the radially outer side of the stator core **52**, the configuration is advantageous in terms of achieving torque increase in the rotating electric machine **10**. A reason for this is that a distance from the magnet unit **42** of the rotor **40** to the stator core **52** (that is, a distance of a portion that includes no iron) can be reduced and magnetic resistance can be reduced. As a result, interlinkage flux in the stator core **52** by the permanent magnet can be increased and torque can be enhanced.

In addition, as a result of the thickness of the conductor group **81** being made thinner, even when magnetic flux leaks from the conductor group **81**, the magnetic flux can be easily recovered in the stator core **52**. The magnetic flux leaking outside and not being effectively used for torque improvement can be suppressed. That is, reduction in magnetic force as a result of magnetic flux leakage can be suppressed. The interlinkage flux in the stator core **52** by the permanent magnet can be increased, and torque can be enhanced.

The conductor **82** is made of a coated conductor in which a surface of a conductor body **82a** is covered by an insulation coating **82b**. Insulation is ensured between the conductors **82** that overlap each other in the radial direction and between the conductor **82** and the stator core **52**. When the wire **86**, described hereafter, is a self-fusing coated wire, the insulation coating **82b** is made of the coating of the wire **86**. Alternatively, the insulation coating **82b** may be made of an insulation member that is overlayed separately from the coating of the wire **86**.

Here, in each of the phase windings that are configured by the conductors **82**, insulation properties of the insulation coating **82b** are maintained, excluding an exposed portion for connection. For example, the exposed portion is an input/output terminal portion or a neutral point portion when a star connection is formed. In the conductor group **81**, the conductors **82** that are adjacent in the radial direction are mutually fixed using resin fixing or self-fusing coated wires. As a result, insulation breakdown, vibrations, and noise that occur as a result of the conductors **82** rubbing together are suppressed.

According to the present embodiment, the conductor body **82a** is configured as a bundle of a plurality of wires **86**. Specifically, as shown in FIG. 13, the conductor body **82a** is formed into a braided shape by the plurality of wires **86** being twisted. In addition, as shown in FIG. 14, the wire **86** is configured as a composite in which thin, fibrous conductive materials **87** are bundled.

For example, the wire **86** may be a composite of carbon nanotube (CNT) fibers. As the CNT fibers, fibers including boron-containing fine fibers in which at least a portion of carbon is replaced with boron may be used. As carbon-based

fine fibers, in addition to CNT fibers, vapor-grown carbon fibers (VGCF) and the like can be used. However, CNT fibers are preferably used. Here, the surface of the wire **86** is covered by a polymer insulation layer such as enamel. In addition, the surface of the wire **86** is preferably covered by a so-called enamel coating that is made of a coating of polyimide or a coating of amide-imide.

The conductors **82** configure the windings of n-phases in the stator winding **51**. In addition, the wires **86** of the conductor **82** (that is, the conductor body **82a**) are adjacent to each other in a state of contact. The conductor **82** is made of a wire bundle in which a winding conductor body has a portion that is formed by the plurality of wires **86** being twisted in one or more locations within a phase, and a resistance value between twisted wires **86** is greater than a resistance value of the wire **86** itself.

In other words, when two adjacent wires **86** have a first electrical resistivity in the direction in which the wires **86** are adjacent and each of the wires **86** has a second electrical resistivity in the length direction thereof, the first electrical resistivity is a greater value than the second electrical resistivity. Here, the conductor **82** may be a wire bundle that is formed by the plurality of wires **86**, and in which the plurality of wires **86** are covered by an insulation member that has a very high first electrical resistivity. In addition, the conductor body **82a** of the conductor **82** may be configured by the plurality of wires **86** that are twisted together.

In the above-described conductor body **82a**, because the plurality of wires **86** are twisted together, generation of eddy currents in the wires **86** can be suppressed and eddy currents in the conductor body **82a** can be decreased. In addition, as a result of the wires **86** being twisted, a section in which directions in which a magnetic field is applied are opposite each other is produced in a single wire **86**, and a counter electromotive voltage is canceled. Therefore, decrease in eddy currents can again be achieved. In addition, as a result of the wire **86** being made of the fibrous conductive materials **87**, thinning and significant increase in the number of twists can be obtained. Eddy currents can be more suitably reduced.

Here, an insulation method for the wires **86** herein is not limited to the above-described polymer insulation coating and may be a method in which flow of current is made difficult between the twisted wires **86** using contact resistance. That is, if a relationship is such that the resistance value between the twisted wires **86** is greater than the resistance value of the wire **86** itself, the above-described effects can be obtained as a result of a potential difference that is generated as a result of the difference in resistance values.

For example, as a result of a manufacturing facility for fabricating the wire **86** and a manufacturing facility for fabricating the stator **50** (armature) of the rotating electric machine **10** being used as separate discontinuous facilities, the wires **86** can become oxidized due to transportation time, work intervals, and the like. Contact resistance can be increased and is, therefore, favorable.

As described above, the conductor **82** has a cross-section that has a flattened rectangular shape. A plurality of conductors **82** are arranged so as to be arrayed in the radial direction. For example, the conductor **82** maintains the shape by a plurality of coated wires **86** that are the self-fusing coated wires that include a fusion layer and an insulation layer being bundled in a twisted state and the fusion layers being fused together.

Here, the conductor **82** may be formed by wires that do not have the fusion layer or wires that are the self-fusing

coated wires being hardened into a desired shape by a synthetic resin or the like in a twisted state. When the thickness of the insulation coating **82b** of the conductor **82** is, for example, 80 μm to 100 μm and thicker than a coating thickness (5 μm to 40 μm) of a conductor that is typically used, insulation between the conductor **82** and the stator core **52** can be ensured without an insulation paper or the like being interposed therebetween.

In addition, the insulation coating **82b** is preferably configured to have higher insulation properties than the insulation layer of the wire **86** and be capable of insulating between phases. For example, when the thickness of the polymer insulation layer of the wire **86** is about 5 μm , the thickness of the insulation coating **82b** of the conductor **82** is preferably about 80 μm to 100 μm , and made capable of suitably insulating between phases.

Furthermore, the conductor **82** may be configured such that the plurality of wires **86** are bundled without being twisted. That is, the conductor **82** may have any of a configuration in which the plurality of wires **86** are twisted over the overall length thereof, a configuration in which the plurality of wires **86** are twisted in a portion of the overall length, and a configuration in which the plurality of wires **86** are bundled without being twisted over the overall length. In summary, the conductor **82** that configures the conductor portion is a wire bundle in which the plurality of wires **86** are bundled, and the resistance value between the bundled wires is greater than the resistance value of the wire **86** itself.

The conductor **82** is formed by bending so as to be arranged in a predetermined arrangement pattern in the circumferential direction of the stator winding **51**. As a result, as the stator winding **51**, a phase winding is formed for each phase. As shown in FIG. 12, in the stator winding **51**, the coil side portion **53** is formed by the linear portion **83** of the conductor **82** that linearly extends in the axial direction, and the coil ends **54** and **55** are formed by the turn portions **84** that protrude further toward both outer sides than the coil side portion **53** in the axial direction.

As a result of the linear portion **83** and the turn portion **84** being alternately repeated, the conductors **82** are configured as a series of conductors in a wave-winding state. The linear portion **83** is arranged in a position that opposes the magnet unit **42** in the radial direction. The linear portions **83** of the same phase that are arranged with a predetermined interval therebetween in positions on the outer side in the axial direction of the magnet unit **42** are connected to each other by the turn portion **84**. Here, the linear portion **83** corresponds to a "magnet opposing portion".

According to the present embodiment, the stator winding **51** is formed by being wound into a circular annular shape by distributed winding. In this case, in the coil side portion **53**, the linear portions **83** are arranged in the circumferential direction at an interval that corresponds to a single pole pair of the magnet unit **42**, for each phase. In the coil ends **54** and **55**, the linear portions **83** of each phase are connected to each other by the turn portion **84** that is formed into a substantial V-shape.

In the linear portions **83** that form a pair in correspondence to a single pole pair, respective current directions are opposite each other. In addition, between one coil end **54** and the other coil end **55**, a combination of the pair of linear portions **83** that are connected by the turn portion **84** differs. As a result of the connections at the coil ends **54** and **55** being repeated in the circumferential direction, the stator winding **51** is formed into an approximately circular cylindrical shape.

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More specifically, the stator winding **51** is that in which the winding of each phase is configured using two pairs of conductors **82** for each phase, and one three-phase winding (U-phase, V-phase, and W-phase) and the other three-phase winding (X-phase, Y-phase, and Z-phase) of the stator winding **51** are provided in two layers that are on the inner side and the radially outer side. In this case, when the number of phases of the stator winding **51** is S (6 in the case of the example) and the number of conductors **82** per phase is m , $2 \times S \times m = 2Sm$ conductors **82** are formed for each pole pair. According to the present embodiment, the number of phases S is six and the number m is four, and the rotating electric machine has eight pole pairs (16 poles). Therefore, $6 \times 4 \times 8 = 192$ conductors **82** are arranged in the circumferential direction of the stator core **52**.

In the stator winding **51** shown in FIG. 12, in the coil side portion **53**, the linear portions **83** are arranged so as to overlap in two layers that are adjacent in the radial direction and, in the coil ends **54** and **55**, the turn portions **84** extend in the circumferential direction from the linear portions **83** that overlap in the radial direction, at directions that are opposite each other in the circumferential direction. That is, in the conductors **82** that are adjacent to each other in the radial direction, the directions of the turn portions **84** are opposite each other, excluding the end portions of the stator winding **51**.

Here, a winding structure of the conductors **82** in the stator winding **51** will be described in detail. According to the present embodiment, a plurality of conductors **82** that are formed by wave-winding are provided so as to overlap in a plurality of layers (such as two layers) that are adjacent in the radial direction. FIG. 15 illustrates, by (a) and (b), diagrams of an aspect of the conductors **82** in an n th layer.

FIG. 15 shows, by (a), the shape of the conductors **82** when viewed from a side of the stator winding **51**. FIG. 15 shows, by (b), the shape of the conductors **82** when viewed from one axial direction side of the stator winding **51**. Here, in FIG. 15 by (a) and (b), the positions in which the conductor groups **81** are arranged are respectively D1, D2, D3, . . . In addition, for convenience of description, only three conductors **82** are shown. The three conductors **82** are a first conductor **82_A**, a second conductor **82_B**, and a third conductor **82_C**.

In the conductors **82_A** to **82_C**, the linear portions **83** are all arranged in positions in the n th layer, that is, the same position in the radial direction. The linear portions **83** that are separated from each other by six positions (corresponding to $3 \times m$ pairs) in the circumferential direction are connected to each other by the turn portion **84**. In other words, in the conductors **82_A** to **82_C**, two of both ends of seven linear portions **83** that are arrayed in an adjacent manner in the circumferential direction of the stator winding **51** on the same circle of which a center is an axial center of the rotor **40** are connected to each other by a single turn portion **84**. For example, in the first conductor **82_A**, a pair of linear portions **83** are respectively arranged in D1 and D7, and the pair of linear portions **83** are connected to each other by the turn portion **84** that has an inverted V-shape.

In addition, the other conductors **82_B** and **82_C** are respectively arranged such that the positions in the circumferential direction are shifted by one position each in the same n th layer. In this case, because the conductors **82_A** to **82_C** are all arranged in the same layer, the turn portions **84** interfering with one another can be considered. Therefore, according to the present embodiment, an interference preventing portion in which a portion of each turn portion **84** is

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offset in the radial direction is formed in the turn portions **84** of the conductors **82_A** to **82_C**.

Specifically, the turn portion **84** of each of the conductors **82_A** to **82_C** includes a sloped portion **84a**, a peak portion **84b**, a sloped portion **84c**, and a return portion **84d**.

The sloped portion **84a** is a portion that extends in the circumferential direction on the same circle (first circle). The peak portion **84b** is shifted from the sloped portion **84a** further toward the radially inner side (upper side in FIG. 15 by (b)) than the same circle and reaches another circle (second circle). The sloped portion **84c** extends in the circumferential direction on the second circle. The return portion **84d** returns from the first circle to the second circle.

The peak portion **84b**, the sloped portion **84c**, and the return portion **84d** correspond to the interference preventing portion. Here, the sloped portion **84c** may be configured to shift toward the radially outer side relative to the sloped portion **84a**.

In other words, the turn portion **84** of each of the conductors **82_A** to **82_C** has the sloped portion **84a** on one side and the sloped portion **84c** on the other side, of both sides that sandwich the peak portion **84b** that is a center position in the circumferential direction. The positions in the radial direction of the sloped portions **84a** and **84c** (positions in a rearward direction on paper in FIG. 15 by (a) and positions in an up/down direction in FIG. 15 by (b)) differ from each other.

For example, the turn portion **84** of the first conductor **82_A** is configured to extend along the circumferential direction with a D1 position in the n th layer as a starting position, turn to the radial direction (such as toward the radially inner side) at the peak portion **84b** that is the center position in the circumferential direction, subsequently turn again to the circumferential direction, thereby extending again along the circumferential direction, and further, turn again to the radial direction (such as toward the radially outer side) at the returning portion **84d**, thereby reaching a D7 position in the n th layer that is a terminal position.

As a result of the above-described configuration, in the conductors **82_A** to **82_C**, the one sloped portions **84a** are arrayed from top to bottom in order from the first conductor **82_A**→second conductor **82_B**→third conductor **82_C**. In addition, the top to bottom order of the conductors **82_A** to **82_B** is interchanged at the peak portions **84b**, and the other sloped portions **84c** are arrayed from top to bottom in order from the third conductor **82_C**→second conductor **82_B**→first conductor **82_A**. Therefore, the conductors **82_A** to **82_C** can be arranged in the circumferential direction without interfering with one other.

Here, the conductor group **81** is formed by the plurality of conductors **82** being overlapped in the radial direction. In this configuration, the turn portion **84** that is connected to the linear portion **83** on the radially inner side, and the turn portion **84** that is connected to the linear portion **83** on the radially outer side, among the linear portions **83** of a plurality of layers, may be arranged so as to be further separated in the radial direction than the linear portions **84**.

In addition, when the conductors **82** of a plurality of layers are bent toward the same side in the radial direction at the end portions of the turn portions **84**, that is, near boundary portions with the linear portions **83**, insulation being compromised as a result of interference between the conductors **82** of adjacent layers may be prevented from occurring.

For example, in D7 to D9 in FIG. 15 by (a) and (b), the conductors **82** that overlap in the radial direction are each bent in the radial direction at the return portion **84d** of the

turn portion **84**. In this case, as shown in FIG. **16**, a radius of curvature of a bending portion may be made to differ between the conductor **82** of the n th layer and the conductor **82** of the $n+1$ th layer. Specifically, a radius of curvature $R1$ of the conductor **82** on the radially inner side (n th layer) is less than a radius of curvature $R2$ of the conductor **82** on the radially outer side ($n+1$ th layer).

In addition, an amount of shifting in the radial direction may be made to differ between the conductor **82** of the n th layer and the conductor **82** of the $n+1$ th layer. Specifically, a shift amount $S1$ of the conductor **82** on the radially inner side (n th layer) is less than a shift amount $S2$ of the conductor **82** on the radially outer side ($n+1$ th layer).

As a result of the above-described configuration, even when the conductors **82** that overlap in the radial direction are bent in the same direction, mutual interference between the conductors **82** can be suitably prevented. As a result, favorable insulation properties can be obtained.

Next, the structure of the magnet unit **42** in the rotor **40** will be described. According to the present embodiment, the magnet unit **42** is made of a permanent magnet. A permanent magnet of which a remanent flux density $Br=1.0$ [T] and intrinsic coercive force $Hcj=400$ [kA/m] or greater is assumed. In short, the permanent magnet that is used according to the present embodiment is a sintered magnet in which a granular magnetic material is sintered and solidified in a mold. The intrinsic coercive force Hcj on a J-H curve is equal to or greater than 400 [kA/m], and the remanent flux density Br is equal to or greater than 1.0 [T].

When 5000 to 10,000 [AT] is applied as a result of inter-phase excitation, if a permanent magnet of which a magnetic length of a single pole pair, that is, an N pole and an S pole, or in other words, a length of a path over which magnetic flux between the N pole and the S pole flows that passes through the magnet is 25 [mm] is used, $Hcj=10,000$ [A], indicating that demagnetization does not occur.

Still in other words, the magnet unit **42** is that in which saturation magnetic flux density J_s is equal to or greater than 1.2 [T], grain size is equal to or less than 10 [μ m], and when an orientation ratio is α , $J_s \times \alpha$ is equal to or greater than 1.0 [T].

A supplementary description is provided below, regarding the magnet unit **42**. The magnet unit **42** (magnet) is characteristic in that 2.15 [T] $\geq J_s \geq 1.2$ [T]. In other words, as the magnet that is used in the magnet unit **42**, NdFe11TiN, Nd2Fe14B, Sm2Fe17N3, an FeNi magnet that has L10-type crystals, and the like can be used.

Here, compositions such as SmCo5 (samarium-cobalt), FePt, Dy2Fe14B, and CoPt cannot be used. Also, 2.15 [T] $\geq J_s \geq 1.2$ [T] may be met even in magnets of the same type of compounds, such as Dy2Fe14B and Nd2Fe14B, in which dysprosium that is a heavy rare earth is typically used to impart the high coercive force of Dy, while only slightly losing the high J_s characteristics of neodymium. These magnets can be used in this case as well.

In such cases, for example, the magnet is referred to as ([Nd1-xDy]2Fe14B). Furthermore, 2.15 [T] $\geq J_s \geq 1.2$ [T] can be obtained even in two or more types of magnets that have differing compositions, such as magnets that are made of two or more types of materials, such as FeNi plus Sm2Fe17N3. For example, 2.15 [T] $\geq J_s \geq 1.2$ [T] can be obtained even in a mixed magnet in which coercive force is increased by a small amount of Dy2Fe14B, for example, of which $J_s < 1$ [T] being mixed with a Nd2Fe14B magnet of which $J_s = 1.6$ [T] and has leeway in terms of J_s .

In addition, in a rotating electric machine that operates at a temperature that is outside a range of human activity, such

as 60° C. or higher, that exceeds the temperatures of a desert, or such as for use in a vehicle motor in which an in-vehicle temperature approaches 80° C. when left stationary in the summer, the components of FeNi and Sm2Fe17N3 of which a coefficient of temperature dependence is particularly small are preferably included.

A reason for this is that, in motor operation ranging from a temperature state that is close to -40° C. in Northern Europe, which is within the range of human activity, to the aforementioned 60° C. or higher that exceeds the temperatures of a desert, or to heat resistance temperatures of about 180° C. to 240° C. of a coil enamel coating, motor characteristics significantly differ based on the coefficient of temperature dependence.

Therefore, optimal control and the like with the same motor driver becomes difficult. Through use of FeNi that has the L10-type crystals or Sm2Fe17N3, or the like, described above, because these magnets have a coefficient of temperature dependence that is equal to or less than half that of Nd2Fe14B, load placed on the motor driver can be suitably reduced.

In addition, the magnet unit **42** has a characteristic that, using the above-described magnet composition, a magnitude of particle size in a fine powder state before orientation is equal to or less than 10 μ m, and equal to or greater than a single magnetic-domain particle size. In a magnet, coercive force increases as a result of particles of a powder being micronized to the order of several hundred nm. Therefore, in recent years, powder that is as micronized as possible is used.

However, when the powder is too fine, the BH product of the magnet decreases as a result of oxidation and the like. Therefore, a particle size that is equal to or greater than the single magnet-domain particle size is preferable. When the particle size is up to the single magnet-domain particle size, it is known that coercive force increases as a result of micronization. Here, the magnitude of particle size described herein refers to the magnitude of particle size in a fine powder state in an orientation step, in terms of a manufacturing process of a magnet.

Furthermore, each of a first magnet **91** and a second magnet **92** of the magnet unit **42** is a sintered magnet that is formed by so-called sintering in which a magnetic powder is baked at a high temperature and hardened. This sintering is performed so that, when saturation magnetization J_s of the magnet unit **42** is equal to or greater than 1.2 T, the grain size of the first magnet **91** and the second magnet **92** is equal to or less than 10 μ m, and the orientation ratio is α , a condition that $J_s \times \alpha$ is equal to or greater than 1.0 T (tesla) is met.

In addition, the first magnet **91** and the second magnet **92** are each sintered to meet the following conditions. In addition, as a result of orientation being performed in the orientation step in the manufacturing process, unlike a definition of a magnetic force direction of an isotropic magnet as a result of a magnetizing step, the first magnet **91** and the second magnet **92** have a high orientation ratio. A high orientation ratio is set so that the saturation magnetization J_s of the magnet unit **42** according to the present embodiment is equal to or greater than 1.2 T, and the orientation ratio α of the first magnet **91** and the second magnet **92** is $J_r \geq J_s \times \alpha \geq 1.0$ [T].

Here, for example, the orientation ratio α referred to herein is, in each of the first magnet **91** or the second magnet **92**, $\alpha = 5/6$ when six easy axes of magnetization are present and, of the six easy axes of magnetization, five are oriented toward a direction A10 that is the same direction and the remaining one is oriented toward a direction B10 that is

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tilted at an angle of 90 degrees relative to the direction A10, and $\alpha=(5+0.707)/6$ when the remaining one is oriented toward a direction B10 that is tilted by 45 degrees relative to the direction A10, because the component of the remaining one that is oriented toward the direction A10 is $\cos 45^\circ=0.707$.

In the present example, the first magnet 91 and the second magnet 92 are formed by sintering. However, if the above-described conditions are met, the first magnet 91 and the second magnet 92 may be formed by other methods. For example, a method in which an MQ3 magnet or the like is formed can be used.

According to the present embodiment, because a permanent magnet of which the easy axis of magnetization is controlled by orientation is used, a magnetic circuit length inside the magnet can be made longer compared to the magnetic circuit length of a conventional linear orientation magnet that outputs 1.0 [T] or greater. That is, the magnetic circuit length for a single pole pair can be obtained using a smaller quantity of magnetic material.

In addition, compared to a design in which the conventional linear orientation magnet is used, even when the magnet is exposed to harsh high-temperature conditions, a reversible demagnetization range thereof can be maintained. In addition, the disclosers of the present application have found a configuration in which characteristics similar to those of a polar anisotropic magnet can be obtained even through use of a magnet of a conventional technology.

Here, the easy axis of magnetization refers to a crystal orientation at which magnetization is facilitated in a magnet. The orientation of the easy axis of magnetization in a magnet is a direction of which the orientation ratio that indicates an extent to which the directions of the easy axes of magnetization match is equal to or greater than 50% or a direction that is the average of the orientations of the magnet.

As shown in FIGS. 8 and 9, the magnet unit 42 is formed into a circular annular shape and is provided on the inner side of the magnet holder 41 (specifically, the radially inner side of the circular cylindrical portion 43). The magnet unit 42 includes the first magnet 91 and the second magnet 92 that are each a polar anisotropic magnet and of which the polarities differ from each other. The first magnet 91 and the second magnet 92 are alternately arranged in the circumferential direction. The first magnet 91 is a magnet that forms the N pole in a portion near the stator winding 51. The second magnet 92 is a magnet that forms the S pole in a portion near the stator winding 51. The first magnet 91 and the second magnet 92 are permanent magnets made of, for example, a rare earth magnet such as a neodymium magnet.

As shown in FIG. 9, in each of the magnets 91 and 92, the magnetization direction extends in a circular arc shape between a d-axis (direct axis) that is a magnetic pole center in a well known d-q coordinate system and a q-axis (quadrature axis) that is a magnetic pole boundary between the N pole and the S pole (in other words, the magnetic flux density is 0 tesla). In each of the magnets 91 and 92, on the d-axis side, the magnetization direction is the radial direction of the magnet unit 42 that has the circular annular shape. On the q-axis side, the magnetization direction of the magnet unit 42 that has the circular annular shape is the circumferential direction. This will be described in further detail, below.

As shown in FIG. 9, each of the magnets 91 and 92 includes a first portion 250 and two second portions 260 that are positioned on both sides of the first portion 250 in the circumferential direction of the magnet unit 42. In other words, the first portion 250 is closer to the d-axis than the

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second portion 260, and the second portion 260 is closer to the q-axis than the first portion 250.

In addition, the magnet unit 42 is configured such that the direction of an easy axis of magnetization 300 in the first portion 250 is more parallel to the d-axis than the direction of an easy axis of magnetization 310 in the second portion 260. In other words, the magnet unit 42 is configured such that an angle θ_{11} that the easy axis of magnetization 300 in the first portion 250 forms with the d-axis is smaller than an angle θ_{12} that the easy axis of magnetization 310 in the second portion 260 forms with the q-axis.

More specifically, the angle θ_{11} is an angle that is formed by the d-axis and the easy axis of magnetization 300 when a direction from the stator 50 (armature) toward the magnet unit 42 on the d-axis is forward. The angle θ_{12} is an angle that is formed by the q-axis and the easy axis of magnetization 310 when a direction from the stator 50 (armature) toward the magnet unit 42 on the q-axis is forward. Here, the angle θ_{11} and the angle θ_{12} are both equal to or less than 90° according to the present embodiment.

The easy axes of magnetization 300 and 310 herein are each based on a following definition. When, in respective portions of the magnets 91 and 92, one easy axis of magnetization is oriented toward a direction A11 and another easy axis of magnetization is oriented toward a direction B11, an absolute value ($|\cos \theta|$) of a cosine of an angle θ formed by the direction A11 and the direction B11 is the easy axis of magnetization 300 or the easy axis of magnetization 310.

That is, in each of the magnets 91 and 92, the orientation of the easy axis of magnetization differs between the d-axis side (the portion located closer to the d-axis) and the q-axis side (the portion located closer to the q-axis). On the d-axis side, the orientation of the easy axis of magnetization is an orientation that is close to a direction that is parallel to the d-axis. On the q-axis side, the orientation of the easy axis of magnetization is an orientation that is close to a direction that is orthogonal to the q-axis.

In addition, a magnet magnetic path that has a circular arc shape may be formed based on the orientations of the easy axes of magnetization. Here, in each of the magnets 91 and 92, the easy axis of magnetization on the d-axis side may have an orientation that is parallel to the d-axis and the easy axis of magnetization on the q-axis side may have an orientation that is orthogonal to the q-axis.

In addition, in the magnets 91 and 92, of the circumferential surface of each of the magnets 91 and 92, a stator-side outer surface that is on the stator 50 side (lower side in FIG. 9) and an end surface on the q-axis side in the circumferential direction serve as magnetic flux action surfaces that are inflow/outflow surfaces for the magnetic flux. The magnet magnetic path is formed so as to connect these magnetic flux action surfaces (the stator-side outer surface and the end surface on q-axis side).

In the magnet unit 42, as a result of the magnets 91 and 92, the magnetic flux flows between adjacent N and S poles in a circular arc shape. Therefore, for example, the magnet magnetic path is longer compared to that of a radial anisotropic magnet. Therefore, as shown in FIG. 17, the magnetic flux density distribution is close to a sine wave. As a result, unlike the magnetic flux density distribution of the radial anisotropic magnet shown as a comparative example in FIG. 18, the magnetic flux can be concentrated toward a center side of the magnetic pole. The torque of the rotating electric machine 10 can be increased.

In addition, a difference in the magnetic flux density distribution is present between the magnet unit 42 according

to the present embodiment and a conventional magnet that has a Halbach array. Here, in FIGS. 17 and 18, a horizontal axis indicates electrical angle and a vertical axis indicates magnetic flux density. In addition, in FIGS. 17 and 18, 90° on the horizontal axis indicates the d-axis (that is, the magnetic pole center), and 0° and 180° on the horizontal axis indicates the q-axis.

That is, as a result of the magnets 91 and 92 configured as described above, the magnet magnetic flux on the d-axis is strengthened and changes in the magnetic flux near the q-axis are suppressed. As a result, the magnets 91 and 92 of which the changes in surface magnetic flux from the q-axis to the d-axis are gradual at each magnetic pole can be suitably implemented.

For example, the sine-wave matching ratio of the magnetic flux density distribution may be a value that is equal to or greater than 40%. As a result, compared to a case in which a radial orientation magnet or a parallel orientation magnet of which the sine-wave matching ratio is about 30% is used, the amount of magnetic flux in a waveform center portion can be improved with certainty. In addition, when the sine-wave matching ratio is equal to or greater than 60%, the amount of magnetic flux in the waveform center portion can reliably be improved compared to that of a magnetic flux concentration array such as the Halbach array.

In the radial anisotropic magnet shown in FIG. 18, the magnetic density near the q-axis sharply changes. As the change in magnetic flux density becomes sharper, the eddy currents that are generated in the stator winding 51 increase. In addition, the change in magnetic flux on the stator winding 51 side also becomes sharp. In this regard, according to the present embodiment, the magnetic flux density distribution is a magnetic flux waveform that is close to a sine wave. Therefore, near the q-axis, the change in the magnetic flux density is smaller than the change in the magnetic flux density in the radial anisotropic magnet. As a result, the generation of eddy currents can be suppressed.

In the magnet unit 42, the magnetic flux is generated near the d-axis of each of the magnets 91 and 92 (that is, near the magnetic pole center) at an orientation that is orthogonal to the magnetic flux action surface 280 on the stator 50 side. The magnetic flux forms a circular arc shape that moves away from the d-axis as the magnetic flux moves away from the magnetic flux action surface 280 on the stator 50 side.

In addition, the magnetic flux becomes stronger as the magnetic flux becomes more orthogonal to the magnetic flux action surface. In this regard, in the rotating electric machine 10 according to the present embodiment, because the conductor groups 81 are thinner in the radial direction as described above, the center position in the radial direction of the conductor group 81 becomes close to the magnetic flux action surface of the magnet unit 42. A strong magnetic flux can be received in the stator 50 from the rotor 40.

In addition, the stator 50 is provided with the circular cylindrical stator core 52 on the radially inner side of the stator winding 51, that is, on the side opposite the rotor 40 with the stator winding 51 therebetween. Therefore, the magnetic flux that extends from the magnetic flux action surface of each magnet 91 and 92 is drawn to the stator core 52 and circles the stator core 52 using the stator core 52 as a portion of a magnetic path. In this case, the orientation and the path of the magnet magnetic flux can be optimized.

Hereafter, as a manufacturing method for the rotating electric machine 10, assembly steps for the bearing unit 20, the housing 30, the rotor 40, the stator 50, and the inverter unit 60 shown in FIG. 5 will be described. Here, the inverter unit 60 includes the unit base 61 and the electrical compo-

nents 62 as shown in FIG. 6. Work steps that include the assembly step for the unit base 61 and the electrical components 62 will be described. In the description below, an assembly that is made of the stator 50 and the inverter unit 60 is a first unit. An assembly that is made of the bearing unit 20, the housing 30, and the rotor 40 is a second unit.

The present manufacturing steps are: a first step of mounting the electrical components 62 on the radially inner side of the unit base 61; a second step of manufacturing the first unit by mounting the unit base 61 on the radially inner side of the stator 50; a third step of manufacturing the second unit by inserting the fixing portion 44 of the rotor 40 into the bearing unit 20 that is assembled to the housing 30; a fourth step of mounting the first unit on the radially inner side of the second unit; and a fifth step of fixing the housing 30 and the unit base 61 by fastening. An order of execution of these steps is the first step→second step→third step→fourth step→fifth step.

As a result of the above-described manufacturing method, after the bearing unit 20, the housing 30, the rotor 40, the stator 50, and the inverter unit 60 are assembled as a plurality of assemblies (sub-assemblies), these assemblies are assembled together. Therefore, ease of handling, completion of inspection for each unit, and the like can be implemented. Construction of a logical assembly line can be obtained. Therefore, multi-product production can also be easily accommodated.

At the first step, on at least either of the radially inner side of the unit base 61 and the outer portion in the radial direction of the electrical component 62, a good heat conductor that provides good heat conduction may be applied by coating, bonding, or the like, and in this state, the electrical component 62 may be mounted to the unit base 61. As a result, heat generation from the semiconductor module 66 can be efficiently transmitted to the unit base 61.

At the third step, an insertion operation of the rotor 40 may be performed while a coaxial state is maintained between the housing 30 and the rotor 40. Specifically, for example, a jig that prescribes the position of the outer circumferential surface of the rotor 40 (the outer circumferential surface of the magnet holder 41) or the inner circumferential surface of the rotor 40 (inner circumferential surface of the magnet unit 42) with reference to the inner circumferential surface of the housing 30 is used, and the housing 30 and the rotor 40 are assembled while either of the housing 30 and the rotor 40 is slid along the jig. As a result, heavy components can be assembled without an unbalanced load being applied to the bearing unit 20. Reliability of the bearing unit 20 is improved.

At the fourth step, the assembly of the first unit and the second unit may be performed while the coaxial state between the first unit and the second unit is maintained. Specifically, for example, a jig that prescribes the position of the inner circumferential surface of the unit base 61 with reference to the inner circumferential surface of the fixing portion 44 of the rotor 40 is used, and assembly of the units is performed while either of the first unit and the second unit is slid along the jig. As a result, because the rotor 40 and the stator 50 can be assembled while mutual interference at miniscule gaps between the rotor 40 and the stator 50 is prevented, defective products attributed to assembly, such as damage to the stator winding 51 and chipping of the permanent magnets, can be eliminated.

The order of the above-described steps can also be the second step→third step→fourth step→fifth step→first step. In this case, the delicate electrical components 62 are

assembled last. Stress applied to the electrical components 62 during the assembly step can be minimized.

Next, a configuration of a control system that controls the rotating electric machine 10 will be described. FIG. 19 is an electric circuit diagram of the control system of the rotating electric machine 10. FIG. 20 is a functional block diagram of a control process performed by the control apparatus 110.

In FIG. 19, two sets of three-phase windings 51a and 51b are shown as the stator winding 51. The three-phase winding 51a is made of the U-phase winding, the V-phase winding, and the W-phase winding. The three-phase winding 51b is made of the X-phase winding, the Y-phase winding, and the Z-phase winding. For the three-phase windings 51a and 51b, a first inverter 101 and a second inverter 102 that correspond to power converters are respectively provided.

The inverters 101 and 102 are configured by a full-bridge circuit that has the same number of upper and lower arms as the number of phases of the phase winding. Energization current is adjusted in each phase winding of the stator winding 51 by switching on/off of a switch (semiconductor switching element) that is provided in each arm.

A direct-current power supply 103 and a smoothing capacitor 104 are connected in parallel to the inverters 101 and 102. For example, the direct-current power supply 103 is configured by an assembled battery in which a plurality of unit batteries are connected in series. Here, each switch of the inverters 101 and 102 corresponds to the semiconductor module 66 shown in FIG. 1 and the like. The capacitor 104 corresponds to the capacitor module 68 shown in FIG. 1 and the like.

The control apparatus 110 includes a microcomputer that includes a central processing unit (CPU) and various memories. The control apparatus 110 performs energization control through switching on/off of the switches in the inverters 101 and 102 based on various types of detection information of the rotating electric machine 10, and requests for power-running drive and power generation. The control apparatus 110 corresponds to the control apparatus 77 shown in FIG. 6.

For example, the detection information of the rotating electric machine 10 includes a rotation angle (electrical angle information) of the rotor 40 that is detected by an angle detector such as resolver, a power-supply voltage (inverter input voltage) that is detected by a voltage sensor, and an energization current of each phase that is detected by a current sensor. The control apparatus 110 generates operating signals to operate the switches of the inverters 101 and 102, and outputs the operating signals. Here, for example, the request for power generation is a request for regenerative drive when the rotating electric machine 10 is used as a vehicle power source.

The first inverter 101 includes a serial-connection body of an upper arm switch Sp and a lower arm switch Sn for each of the three phases that are made of the U-phase, the V-phase, and the W-phase. A high-potential-side terminal of the upper arm switch Sp of each phase is connected to a positive electrode terminal of the direct-current power supply 103. A low-potential-side terminal of the lower arm switch Sn of each phase is connected to a negative electrode terminal (ground) of the direct-current power supply 103.

One end of each of the U-phase winding, the V-phase winding, and the W-phase winding is connected to an intermediate connection point between the upper arm switch Sp and the lower arm switch Sn of each phase. These phase windings are connected by a star connection (Y connection). Other ends of the phase windings are connected to one another at a neutral point.

The second inverter 102 has a configuration that is similar to that of the first inverter 101. The second inverter 102 includes a serial-connection body of an upper arm switch Sp and a lower arm switch Sn for each of the three phases that are made of the X-phase, the Y-phase, and the Z-phase. A high-potential-side terminal of the upper arm switch Sp of each phase is connected to the positive electrode terminal of the direct-current power supply 103. A low-potential-side terminal of the lower arm switch Sn of each phase is connected to the negative electrode terminal (ground) of the direct-current power supply 103.

One end of each of the X-phase winding, the Y-phase winding, and the Z-phase winding is connected to an intermediate connection point between the upper arm switch Sp and the lower arm switch Sn of each phase. These phase windings are connected by a star connection (Y connection). Other ends of the phase windings are connected to one another at a neutral point.

FIG. 20 shows a current feedback process for controlling the phase currents of the U-, V-, and W-phases, and a current feedback process for controlling the phase currents of the X-, Y-, and Z-phases. Here, first, the control process on the U-, V-, and W-phase side will be described.

In FIG. 20, a current command value setting unit 111 sets a d-axis current command value and a q-axis current command value based on a power-running torque command value or a power-generation torque command value for the rotating electric machine 10, and an electrical angular velocity ω obtained by time-differentiating the electrical angle θ , using a torque-dq map.

Here, the current command value setting unit 111 is provided to be shared between the U-, V-, and W-phase side and the X-, Y-, and Z-phase side. Here, for example, the power-generation torque command value is a regeneration-torque command value when the rotating electric machine 10 is used as a vehicle power source.

A dq converting unit 112 converts a current detection value (three phase current) from a current sensor that is provided for each phase to a d-axis current and a q-axis current that are components of an orthogonal two-dimensional rotating coordinate system in which a field direction (direction of an axis of a magnetic field or field direction) is the d-axis.

A d-axis current feedback control unit 113 calculates a d-axis command voltage as a manipulated variable for performing feedback control of the d-axis current to the d-axis current command value. In addition, a q-axis current feedback control unit 114 calculates a q-axis command voltage as a manipulated variable for performing feedback control of the q-axis current to the q-axis current command value. In the feedback control units 113 and 114, the command voltages are calculated using a proportional-integral (PI) feedback method based on deviation of the d-axis current and the q-axis current from the current command values.

A three-phase converting unit 115 converts the d-axis and q-axis command voltages to U-phase, V-phase, and W-phase command voltages. Here, the above-described units 111 to 115 are a feedback control unit that performs feedback control of a fundamental wave current based on dq transformation. The U-phase, V-phase, and W-phase command voltages are feedback control values.

In addition, an operating signal generating unit 116 generates an operating signal for the first inverter 101 based on the command voltages of the three phases using a known triangular-wave-carrier comparison method. Specifically, the operating signal generating unit 116 generates a switch

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operating signal (duty signal) for the upper and lower arms of each phase by PWM control based on a comparison of magnitude between a signal in which the command voltages of the three phases are standardized by the power supply voltage and a carrier signal such as a triangular wave signal.

Moreover, a similar configuration is provided on the X-, Y-, and Z-phase side as well. A dq converting unit **122** converts a current detection value (three phase currents) from a current sensor that is provided for each phase to a d-axis current and a q-axis current that are components of an orthogonal two-dimensional rotating coordinate system in which a field direction is the d-axis.

A d-axis current feedback control unit **123** calculates a d-axis command voltage and a q-axis current feedback control unit **124** calculates a q-axis command voltage. A three-phase converting unit **125** converts the d-axis and q-axis command voltages to X-phase, Y-phase, and Z-phase command voltages.

In addition, an operating signal generating unit **126** generates an operating signal for the second inverter **102** based on the command voltages of the three phases. Specifically, the operating signal generating unit **126** generates a switch operating signal (duty signal) for the upper and lower arms of each phase by PWM control based on a comparison of magnitude between a signal in which the command voltages of the three phases are standardized by the power supply voltage and a carrier signal such as a triangular wave signal.

A driver **117** turns on/off the switches Sp and Sn of each of the three phases in the inverters **101** and **102** based on the switch operating signals generated in the operating signal generating units **116** and **126**.

Next, a torque feedback control process will be described. For example, this process is mainly used for the purpose of increasing output and reducing loss in the rotating electric machine **10** under driving conditions in which the output voltages of the inverters **101** and **102** increase, such as in a high-rotation region and a high-output region. The control apparatus **110** selects either of the torque feedback control process and the current feedback control process based on the driving conditions of the rotating electric machine **10**, and performs the selected process.

FIG. **21** shows the torque feedback control process that corresponds to the U-, V-, and W-phases and the torque feedback control process that corresponds to the X-, Y-, and Z-phases. Here, in FIG. **21**, configurations that are identical to those in FIG. **20** are given the same reference numbers. Descriptions thereof are omitted. Here, first, the control process on the U-, V-, and W-phase side will be described.

A voltage amplitude calculating unit **127** calculates a voltage amplitude command that is a command value for a magnitude of a voltage vector, based on the power-running torque command value or the power-generation torque command value for the rotating electric machine **10**, and the electrical angular velocity ω obtained by time-differentiating the electrical angle θ .

A torque estimating unit **128a** calculates a torque estimation value that corresponds to the U-, V-, and W-phases based on the d-axis current and the q-axis current converted by the dq converting unit **112**. Here, the torque estimating unit **128a** may calculate the voltage amplitude command based on map information in which the d-axis current, the q-axis current, and the voltage amplitude command are associated.

A torque feedback control unit **129a** calculates a voltage phase command that is a command value for a phase of the voltage vector as a manipulated variable for performing feedback control of the torque estimation value to the

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power-running torque command value or the power-generation torque command value. In the torque feedback control unit **129a**, the voltage phase command is calculated using the PI feedback method, based on the deviation of the torque estimation value from the power-running torque command value or the power-generation torque command value.

An operating signal generating unit **130a** generates the operating signal of the first inverter **101** based on the voltage amplitude command, the voltage phase command, and the electrical angle θ . Specifically, the operating signal generating unit **130a** calculates the command voltages of the three phases based on the voltage amplitude command, the voltage phase command, and the electrical angle θ , and generates the switch operating signal for the upper and lower arms of each phase by PWM control based on a comparison of magnitude between a signal in which the calculated command voltages of the three phases are standardized by the power supply voltage and a carrier signal such as a triangular wave signal.

Here, the operating signal generating unit **130a** may generate the switch operating signal based on pulse pattern information that is map information in which the voltage amplitude command, the voltage phase command, the electrical angle θ , and the switch operating signal are associated, the voltage amplitude command, the voltage phase command, and the electrical angle θ .

Moreover, a similar configuration is provided on the X-, Y-, and Z-phase side as well. A torque estimating unit **128b** calculates a torque estimation value that corresponds to the X-, Y-, and Z-phases based on the d-axis current and the q-axis current converted by the dq converting unit **122**.

A torque feedback control unit **129b** calculates a voltage phase command as a manipulated variable for performing feedback control of the torque estimation value to the power-running torque command value or the power-generation torque command value. In the torque feedback control unit **129b**, the voltage phase command is calculated using the PI feedback method, based on the deviation of the torque estimation value from the power-running torque command value or the power-generation torque command value.

An operating signal generating unit **130b** generates the operating signal of the second inverter **102** based on the voltage amplitude command, the voltage phase command, and the electrical angle θ . Specifically, the operating signal generating unit **130b** calculates the command voltages of the three phases based on the voltage amplitude command, the voltage phase command, and the electrical angle θ , and generates the switch operating signal for the upper and lower arms of each phase by PWM control based on a comparison of magnitude between a signal in which the calculated command voltages of the three phases are standardized by the power supply voltage and a carrier signal such as a triangular wave signal. The driver **117** turns on/off the switches Sp and Sn of each of the three phases in the inverters **101** and **102** based on the switch operating signals generated in the operating signal generating units **130a** and **130b**.

Here, the operating signal generating unit **130b** may generate the switch operating signal based on pulse pattern information that is map information in which the voltage amplitude command, the voltage phase command, the electrical angle θ , and the switch operating signal are associated, the voltage amplitude command, the voltage phase command, and the electrical angle θ .

Here, in the rotating electric machine **10**, occurrence of electrical corrosion in the bearings **21** and **22** in accompaniment with generation of axial current is a concern. For

example, when energization of the stator winding **51** is switched by switching, distortion in the magnetic flux occurs as a result of a minute shift in switching timing (switching imbalance).

Electrical corrosion occurring as a result in the bearings **21** and **22** that support the rotation shaft **11** becomes a concern. The distortion in the magnetic flux occurs based on the inductance in the stator **50**. As a result of electromotive voltage in the axial direction that is generated by the distortion in the magnetic flux, insulation breakdown occurs inside the bearings **21** and **22**, and electrical corrosion progresses.

In this regard, according to the present embodiment, three measures that are described below are taken as electrical corrosion measures. A first electrical corrosion measure is an electrical corrosion suppression measure that is achieved by inductance being reduced in accompaniment with the stator **50** becoming coreless and the magnet magnetic flux of the magnet unit **42** being smoothed. A second electrical corrosion measure is an electrical corrosion suppression measure that is achieved by the rotation shaft having the cantilevered structure as a result of the bearings **21** and **22**. A third electrical corrosion measure is an electrical corrosion suppression measure that is achieved by the circular annular stator winding **51** being molded from a molding material together with the stator core **52**. Details of each of these measures will be separately described below.

First, in the first electrical corrosion measure, the stator **50** is configured to be toothless between the conductor groups **81** in the circumferential direction and provided with the sealing member **57** that is made of a non-magnetic material between the conductor groups **81**, instead of the teeth (core) (see FIG. 10).

As a result, inductance in the stator **50** can be reduced. As a result of reduction of inductance in the stator **50** being achieved, even if a shift in switching timing occurs during energization of the stator winding **51**, the occurrence of magnetic flux distortion attributed to the shift in switching timing can be suppressed and, furthermore, electrical corrosion suppression in the bearings **21** and **22** can be performed. Here, the inductance on the d-axis may be equal to or less than the inductance on the q-axis.

In addition, the magnets **91** and **92** are configured to be oriented such that, on the d-axis side, the orientation of the easy axis of magnetization is more parallel to the d-axis compared to the q-axis side (see FIG. 9). As a result, the magnetic flux on the d-axis is strengthened. The changes in surface magnetic flux (increase/decrease in magnetic flux) from the q-axis toward the d-axis at each magnetic pole becomes gradual. Therefore, sudden changes in voltage attributed to switching imbalance is suppressed. Moreover, a configuration that contributes to electrical corrosion suppression is achieved.

In the second electrical corrosion measure, in the rotating electric machine **10**, the bearings **21** and **22** are arranged so as to be concentrated on one side in the axial direction relative to a center in the axial direction of the rotor **40** (see FIG. 2). As a result, compared to a configuration in which a plurality of bearings are provided on both sides in the axial direction with a rotor therebetween, the effects of electrical corrosion can be reduced.

That is, the rotor is double-supported by the plurality of bearings. In this configuration, a closed circuit that passes through the rotor, the stator, and each of the bearings (that is, the bearings on both sides in the axial direction sandwiching the rotor) is formed in accompaniment with generation of a high-frequency magnetic flux. Electrical corro-

sion of the bearings as a result of an axial current becomes a concern. In contrast, the rotor **40** is cantilever-supported by the plurality of bearings **21** and **22**. In this configuration, the above-described closed circuit is not formed. Electrical corrosion of the bearings is suppressed.

In addition, the rotating electric machine **10** has a following configuration relative to the configuration for one-side arrangement of the bearings **21** and **22**. In the magnet holder **41**, the contact preventing portion that extends in the axial direction and prevents contact with the stator **50** is provided in the intermediate portion **45** that protrudes in the radial direction of the rotor **40** (see FIG. 2). In this case, in cases in which a closed circuit of the axial current is formed by way of the magnet holder **41**, a closed circuit length can be lengthened and circuit resistance thereof can be increased. As a result, electrical corrosion of the bearings **21** and **22** can be suppressed.

The holding member **23** of the bearing unit **20** is fixed to the housing on one side in the axial direction with the rotor **40** therebetween. In addition, on the other side, the housing **30** and the unit base **61** (stator holder) are coupled with each other (see FIG. 2). As a result of the present configuration, the configuration in which the bearings **21** and **22** are arranged in the axial direction of the rotation shaft **11** to be concentrated on one side in the axial direction can be suitably implemented.

In addition, in the present configuration, the unit base **61** is connected to the rotation shaft **11** via the housing **30**. Therefore, the unit base **61** can be arranged in a position that is electrically separated from the rotation shaft **11**. Here, if an insulation member such as resin is interposed between the unit base **61** and the housing **30**, a configuration in which the unit base **61** and the rotation shaft **11** are further electrically separated is achieved. As a result, electrical corrosion of the bearings **21** and **22** can be suitably suppressed.

In the rotating electric machine **10** according to the present embodiment, as a result of the one-sided arrangement of the bearings **21** and **22** and the like, axial voltage that acts on the bearings **21** and **22** is reduced. In addition, a potential difference between the rotor **40** and the stator **50** is reduced. Therefore, even when a conductive grease is not used in the bearings **21** and **22**, the potential difference acting on the bearings **21** and **22** can be reduced. The conductive grease is thought to generate noise because fine particles of carbon and the like are typically included.

In this regard, according to the present embodiment, a non-conductive grease is used in the bearings **21** and **22**. Therefore, a disadvantage in which noise is generated in the bearings **21** and **22** can be suppressed. For example, during application to an electric vehicle such as an electric automobile, measures against noise in the rotating electric machine **10** are considered to be required. This configuration can be suitably used as such a measure against noise.

In the third electrical corrosion measure, as a result of the stator winding **51** being molded from a molding material together with the stator core **52**, positional shifting of the stator winding **51** in the stator **50** is suppressed (see FIG. 11).

In particular, in the rotating electric machine **10** according to the present embodiment, because an inter-conductor member (teeth) is not provided between the conductor groups **81** in the circumferential direction in the stator winding **51**, concern that a positional shift may occur in the stator winding **51** can be considered. However, as a result of the stator winding **51** being molded together with the stator core **52**, shifting of the conductor position of the stator winding **51** is suppressed. Therefore, distortion in the magnetic flux as a result of a positional shift in the stator winding

51 and the occurrence of electrical corrosion in the bearings 21 and 22 as a result can be suppressed.

Here, the unit base 61 that serves as a housing member that fixes the stator core 51 is made of a CFRP. Therefore, for example, compared to a case in which the unit base 61 is made of aluminum or the like, electrical discharge to the unit base 61 is suppressed, and furthermore, a suitable electrical corrosion suppression measure can be obtained.

In addition, as an electrical corrosion suppression measure for the bearings 21 and 22, at least either of the outer ring 52 and the inner ring 26 can be made of a ceramic material. Alternatively, a configuration in which an insulation sleeve is provided on the outer side of the outer ring 25 or the like can also be used.

Hereafter, other embodiments will be described mainly focusing on differences with the first embodiment.

Second Embodiment

According to a present embodiment, the polar anisotropic structure of the magnet unit 42 in the rotor 40 is modified. This will be described in detail, below.

As shown in FIGS. 22 and 23, the magnet unit 42 is configured using a magnet array that is referred to as a Halbach array. That is, the magnet unit 42 includes a first magnet 131 of which a magnetization direction (orientation of a magnetization vector) is the radial direction and a second magnet 132 of which the magnetization direction (orientation of a magnetization vector) is the circumferential direction. The first magnets 131 are arranged at predetermined intervals in the circumferential direction. The second magnets 132 are arranged in positions between the first magnets 131 that are adjacent in the circumferential direction. For example, the first magnet 131 and the second magnet 132 are permanent magnets that are made of a rare earth magnet such as a neodymium magnet.

The first magnets 131 are arranged to be separated from each other in the circumferential direction, such that the poles on the side opposing the stator 50 (inner side in the radial direction) are alternately the N pole and the S pole. In addition, the second magnets 132 are arranged, such that the polarities alternate in the circumferential direction, adjacent to each of the first magnets 131.

The circular cylindrical portion 43 that is provided so as to surround these magnets 131 and 132 may be a soft magnetic body core that is made of a soft magnetic material and functions as a back core. Here, in the magnet unit 42 according to the second embodiment as well, the relationship of the easy axes of magnetization relative to the d-axis and the q-axis in the d-q coordinate system is the same as that according to the above-described first embodiment.

In addition, a magnetic body 133 that is made of a soft magnetic material is arranged on the radially outer side of the first magnet 131, that is, on the side of the circular cylindrical portion 43 of the magnet holder 41. For example, the magnetic body 133 may be made of an electromagnetic steel sheet, or a soft iron or a dust core material. In this case, a length in the circumferential direction of the magnetic body 133 is the same as the length in the circumferential direction of the first magnet 131 (in particular, the length in the circumferential direction of the outer circumferential portion of the first magnet 131).

In addition, a thickness in the radial direction of an integrated body in a state in which the first magnet 131 and the magnetic body 133 are integrated is the same as the thickness in the radial direction of the second magnet 132. In other words, the first magnet 131 has a thickness in the

radial direction that is thinner than the second magnet 132 by an amount corresponding to the magnetic body 133.

The magnets 131 and 132 and the magnetic body 133 are mutually fixed by an adhesive or the like. The radially outer side of the first magnet 131 in the magnet unit 42 is a side opposite the stator 50. The magnetic body 133 is provided on the side opposite the stator 50 (counter-stator side), of both sides of the first magnet 131 in the radial direction.

In the outer circumferential portion of the magnetic body 133, a key 134 that serves as a protruding portion that protrudes toward the radially outer side, that is, the circular cylindrical portion 43 side of the magnet holder 41 is formed. In addition, on the inner circumferential surface of the circular cylindrical portion 43, a key groove 135 that serves as a recess portion that houses the key 134 of the magnetic body 133 is formed. The protruding shape of the key 134 and the groove shape of the key groove 135 are identical. In correspondence to the keys 134 that are formed in the magnetic bodies 133, the same number of key grooves 135 as the keys 134 are formed.

As a result of engagement of the keys 134 and the key grooves 135, positional shifting of the first magnet 131, the second magnet 132, and the magnet holder 41 in the circumferential direction (rotation direction) is suppressed. Here, the circular cylinder portion 43 of the magnet holder 41 and the magnetic body 133 in which the key 134 and the key groove 135 are provided may be arbitrary. However, in a manner opposite to the description above, the key groove 135 can be provided in the outer circumferential portion of the magnetic body 133 and the key 134 can be provided in the inner circumferential portion of the circular cylindrical portion 43 of the magnet holder 41.

Here, in the magnet unit 42, as a result of the first magnets 131 and the second magnets 132 being alternately arrayed, the magnetic flux density at the first magnets 131 can be increased. Therefore, in the magnet unit 42, concentration of the magnetic flux on one surface can occur. Magnetic flux reinforcement on the side closer to the stator 50 can be obtained.

In addition, as a result of the magnetic body 133 being arranged on the radially outer side of the first magnet 131, that is, on the counter-stator side, partial magnetic saturation on the radially outer side of the first magnet 131 can be suppressed.

In addition, demagnetization of the first magnet 131 that occurs as a result of magnetic saturation can be suppressed. Consequently, magnetic force of the magnet unit 42 can be increased as a result. The magnet unit 42 according to the present embodiment has, so to speak, a configuration in which a portion of the first magnet 131 in which demagnetization easily occurs is replaced by the magnetic body 133.

FIG. 24 illustrates, by (a) and (b), diagrams that show a flow of magnetic flux in the magnet unit 42 in detail. FIG. 24 shows, by (a), a case in which a conventional configuration in which the magnetic body 133 is not provided in the magnet unit 42 is used. FIG. 24 shows, by (b), a case in which the configuration according to the present embodiment in which the magnetic body 133 is provided in the magnet unit 42 is used.

Here, FIG. 24 show, by (a) and (b), the circular cylindrical portion 43 and the magnet unit 42 of the magnet holder 41 in a linearly exploded state. A lower side of the drawings is the stator side and an upper side is the counter-stator side.

In FIG. 24 by (a), the magnetic flux action surface of the first magnet 131 and the side surface of the second magnet 132 are both in contact with the inner circumferential surface of the circular cylindrical portion 43. In addition, the

magnetic flux action surface of the second magnet **132** is in contact with the side surface of the first magnet **131**.

In this case, a composite magnetic flux is generated in the circular cylindrical portion **43**. The composite magnetic flux is made of a magnetic flux **F1** that passes through an outer-side path of the second magnet **132** and enters the contact surface with the first magnet **131**, and a magnetic flux that is approximately parallel to the circular cylindrical portion **43** and draws the magnetic flux **F2** of the second magnet **132**. Therefore, magnetic saturation partially occurring near the contact surface of the first magnet **131** and the second magnet **132** in the circular cylindrical portion **43** is a concern.

In this regard, in FIG. **24** by (b), the magnetic body **133** is provided between the magnetic flux action surface of the first magnet **131** and the inner circumferential surface of the circular cylindrical portion **43** on the side opposite the stator **50** of the first magnet **131**. Therefore, passage of magnetic flux is allowed by the magnetic body **133**. Consequently, magnetic saturation in the circular cylindrical portion **43** can be suppressed. Resistance against demagnetization is improved.

In addition, in FIG. **24** by (b), unlike in FIG. **24** by (a), magnetic flux **F2** that promotes magnetic saturation can be eliminated. As a result, permeance of the overall magnetic circuit can be effectively improved. As a result of a configuration such as this, the magnetic circuit characteristics thereof can be maintained even under harsh, high-temperature conditions.

Furthermore, compared to a radial magnet in a conventional SPM rotor, the magnet magnetic path that passes through the interior of the magnet is long. Therefore, magnet permeance increases. Magnetic force increases, and torque can be enhanced. Furthermore, because the magnetic flux is concentrated in the center of the d-axis, the sine-wave matching ratio can be increased. In particular, if a current waveform is a sine wave or a trapezoid wave by PWM control or a 120-degree energization switching integrated circuit (IC) be used, the torque can be more effectively enhanced.

Here, in cases in which the stator core **52** is made of electromagnetic steel sheets, the thickness in the radial direction of the stator core **52** may be $\frac{1}{2}$ of the thickness in the radial direction of the magnet unit **42** or greater than $\frac{1}{2}$. For example, the thickness in the radial direction of the stator core **52** may be equal to or greater than $\frac{1}{2}$ of the thickness in the radial direction of the first magnet **131** that is provided in a magnetic pole center of the magnet unit **42**.

In addition, the thickness in the radial direction of the stator core **52** may be less than the thickness in the radial direction of the magnet unit **42**. In this case, the magnet magnetic flux is about 1 [T] and the saturation magnetic flux density of the stator core **52** is 2 [T]. Therefore, as a result of the thickness in the radial direction of the stator core **52** being equal to or greater than $\frac{1}{2}$ of the thickness in the radial direction of the magnet unit **42**, magnetic flux leakage toward the inner circumferential side of the stator core **52** can be prevented.

In a magnet that has a Halbach structure or polar anisotropic structure, the magnetic path has a pseudo circular-arc shape. Therefore, the magnetic flux thereof can be increased in proportion to the thickness of the magnet that covers the magnetic flux in the circumferential direction.

In such a configuration, the magnetic flux that flows to the stator core **52** is thought to not exceed the magnetic flux in the circumferential direction. That is, when an iron-based metal that has a saturation magnetic flux density of 2 [T] is

used relative to a magnetic flux of 1 [T] of the magnet, if the thickness of the stator core **52** is equal to or greater than half the magnet thickness, a rotating electric machine that is compact and lightweight can be suitably kept high without the occurrence of magnetic saturation.

Here, because a diamagnetic field from the stator **50** acts on the magnet magnetic flux, the magnet magnetic flux typically becomes equal to or less than 0.9 [T]. Therefore, if the stator core has a thickness that is half that of the magnet, magnetic permeability thereof can be suitably kept high.

Modifications in which sections of the above-described configuration are modified will be described below.
(First Modification)

According to the above-described embodiment, the outer circumferential surface of the stator core **52** is a curved surface with substantially no unevenness, and a plurality of conductor groups **81** are arranged in an array at predetermined intervals on the outer circumferential surface thereof. However, this configuration may be modified. For example, as shown in FIG. **25**, the stator core **52** has a circular annular yoke **141** and a protruding portion **142**.

The yoke **141** is provided on the side opposite the rotor **40** (lower side in the drawing), of both sides in the radial direction of the stator winding **51**. The protruding portion **142** extends from the yoke **141** so as to protrude toward an area between the linear portions **83** that are adjacent to each other in the circumferential direction.

The protruding portion **142** is provided at predetermined intervals on the radially outer side of the yoke **141**, that is, on the rotor **40** side. The conductor groups **81** of the stator winding **51** engage with the protruding portions **142** in the circumferential direction and are arranged in an array in the circumferential direction while using the protruding portions **142** as positioning portions for the conductor groups **81**. Here, the protruding portion **142** corresponds to the "inter-conductor member".

The protruding portion **142** is configured such that a thickness dimension in the radial direction from the yoke **141**, or in other words, as shown in FIG. **25**, a distance **W** from an inner side surface **320** of the linear portion **83** that is adjacent to the yoke **141** to a peak of the protruding portion **142** in the radial direction of the yoke **141** is less than $\frac{1}{2}$ of a thickness dimension (**H1** in the drawing) in the radial direction of the linear portion **83** that is adjacent to the yoke **141** in the radial direction.

In other words, an area that is three-fourths of a dimension (thickness) **T1** of the conductor group **81** (conductive member) in the radial direction of the stator winding **51** (stator core **52**) (twice the thickness of the conductor **82**, or in other words, a minimum distance between the surface **320** of the conductor group **81** that is in contact with the stator core **52** and a surface **330** of the conductor group **81** that faces the rotor **40**) may be occupied by a non-magnetic member (sealing member **57**).

As a result of a thickness restriction of the protruding portion **142** such as this, the protruding portions **142** do not function as teeth between the conductor groups **81** (that is, the linear portions **83**) that are adjacent to each other in the circumferential direction, and formation of a magnetic path by the teeth does not occur.

The protruding portions **142** may not be provided between all of the conductor groups **81** that are arrayed in the circumferential direction. The protruding portion **142** is merely required to be provided between at least one set of conductor groups **81** that are adjacent in the circumferential direction. For example, the protruding portion **142** may be provided at equal intervals between every predetermined

number of conductor groups **81** in the circumferential direction. The shape of the protruding portion **142** may be an arbitrary shape, such as a rectangle or a circular arc.

In addition, the linear portions **83** may be provided in a single layer on the outer circumferential surface of the stator core **52**. Therefore, in a broad sense, all that is required is that the thickness dimension in the radial direction of the protruding portion **142** from the yoke **141** be less than $\frac{1}{2}$ of the thickness dimension in the radial direction of the linear portion **83**.

Here, when a virtual circle of which a center is the axial center of the rotation shaft **11** and that passes through a center position in the radial direction of the linear portion **83** that is adjacent to the yoke **141** in the radial direction is assumed, the protruding portion **142** may have a shape that protrudes from the yoke **141** within the range of the virtual circle, or in other words, a shape that does not protrude further toward the radially outer side (that is, the rotor **40** side) than the virtual circle.

As a result of the above-described configuration, the thickness dimension in the radial direction of the protruding portion **142** is limited. In addition, the protruding portion **142** does not function as the teeth between the linear portions **83** that are adjacent to each other in the circumferential direction. Therefore, compared to a case in which the teeth are provided between the linear portions **83**, the linear portions **83** that are adjacent to each other can be brought closer together. As a result, a cross-sectional area of the conductor body **82a** can be increased. Heat generation that occurs in accompaniment with the energization of the stator winding **51** can be reduced.

In this configuration, magnetic saturation can be alleviated as a result of the teeth not being provided. Energization current to the stator winding **51** can be increased. In this case, increase in the amount of heat generation in accompaniment with the increase in energization current can be suitably addressed. In addition, in the stator winding **51**, the turn portion **84** includes the interference preventing portion that is shifted in the radial direction and prevents interference with another turn portion **84**. Therefore, differing turn portions **84** can be arranged so as to be separated from each other in the radial direction. As a result, heat releasability can be improved even in the turn portions **84**. As a result of the foregoing, heat releasing performance in the stator **50** can be optimized.

In addition, if the yoke **141** of the stator core **52** and the magnet unit **42** of the rotor **40** (that is, the magnets **91** and **92**) are separated by a predetermined distance or more, the thickness dimension in the radial direction of the protruding portion **142** is not bound to $H1$ in FIG. **25**. Specifically, if the yoke **141** and the magnet unit **42** are separated by 2 mm or more, the thickness dimension in the radial direction of the protruding portion **142** may be equal to or greater than $H1$ in FIG. **25**.

For example, when the thickness dimension in the radial direction of the linear portion **83** exceeds 2 mm and the conductor group **81** is made of two layers of conductors **82** on the inner side and the radially outer side, the protruding portion **142** may be provided in a range up to a halfway position of the linear portion **83** that is not adjacent to the yoke **141**, that is, the conductor **82** in the second layer when counted from the yoke **141**. In this case, if the thickness dimension in the radial direction of the protruding portion **142** is up to $H1 \times 3/2$, as a result of the cross-sectional area of the conductors of the conductor group **81** being increased, the above-described effect can approximately be achieved.

In addition, the stator core **52** may be configured as shown in FIG. **26**. Here, in FIG. **26**, the sealing member **57** is omitted. However, the sealing member **57** may be provided. In FIG. **26**, the magnet unit **42** and the stator core **52** are shown in a linearly exploded state for convenience.

In FIG. **26**, the stator **50** includes the protruding portion **142** that serves as the inter-conductor member between the conductors **82** (that is, the linear portions **83**) that are adjacent in the circumferential direction. The stator **50** includes a portion **350** that, when the stator winding **51** is energized, magnetically functions together with one of the magnetic poles (the N pole or the S pole) of the magnet unit **42** and extends in the circumferential direction of the stator **50**.

When a length of this portion **350** in the circumferential direction of the stator **50** is W_n , when a total width (that is, a total dimension in the circumferential direction of the stator **50**) of the protruding portions **142** that are present in this length range W_n is W_t , the saturation magnetic flux density of the protruding portion **142** is B_s , the width dimension in the circumferential direction corresponding to a single pole of the magnet unit **42** is W_m , and the residual magnetic flux density of the magnet unit **42** is B_r , the protruding portion **142** is made of a magnetic material that satisfies a relationship expressed by:

$$W_t \times B_s \leq W_m \times B_r \quad (1).$$

Here, the range W_n is set to include a plurality of conductor groups **81** that are adjacent in the circumferential direction and of which an excitation period overlaps. At this time, a center of the gap **56** of the conductor groups **81** is preferably set as a reference (boundary) for setting the range W_n .

For example, in the case of the configuration shown as an example in FIG. **26**, the conductor groups **81** up to a fourth in order from the conductor group **81** of which the distance from the magnetic pole center of the N pole in the circumferential direction is the shortest corresponds to the foregoing plurality of conductor groups **81**. In addition, the range W_n is set to include the four conductor groups **81**. At this time, the ends of the range W_n (starting point and ending point) are the centers of the gaps **56**.

In FIG. **26**, because a half of the protruding portion **142** each is included in the two ends of the range W_n , the range W_n includes a total of four protruding portions **142**. Therefore, when a width of the protruding portion **142** (that is, the dimension of the protruding portion **142** in the circumferential direction of the stator **50**, or in other words, the interval between adjacent conductor groups **81**) is A , the total width of the protruding portions **142** that are included in the range is $W_t = \frac{1}{2}A + A + A + A + \frac{1}{2}A = 4A$.

Specifically, according to the present embodiment, the three-phase winding of the stator winding **51** is a distributed winding. In the stator winding **51**, relative to a single pole of the magnet unit **42**, the number of protrusions **142**, that is, the number of gaps **56** that are the areas between the conductor groups **81** is number of phases $\times Q$. Here, Q refers to the number of conductors **82** that are in contact with the stator core **52** among the conductors **82** of a single phase.

Here, when the conductor group **81** is that in which the conductors **82** are laminated in the radial direction of the rotor **40**, Q can also be considered the number of conductors **82** on the inner circumferential side of the conductor groups **81** of a single phase. In this case, when the three-phase winding of the stator winding **51** is energized in a predetermined order of the phases, the protruding portions **142** corresponding to two phases are excited within a single pole.

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Therefore, when the width dimension in the circumferential direction of the protruding portion 142 (that is, the gap 56) is A, the total width dimension Wt in the circumferential direction of the protruding portions 142 that are excited by the energization of the stator winding 51 within the range of a single pole of the magnet unit 42 is number of excited phases $\times Q \times A = 2 \times 2 \times A$.

In addition, with the total width dimension Wt prescribed in this manner, in the stator core 52, the protruding portion 142 is configured as a magnetic material that satisfies the relationship in (1), above. Here, the total width dimension Wt is also the circumferential-direction dimension of a portion within a single pole in which relative permeability may be greater than 1.

In addition, taking into consideration leeway, the total width dimension Wt may be the width dimension in the circumferential direction of the protruding portions 142 in a single magnetic pole. Specifically, because the number of protruding portions 142 relative to a single pole of the magnet unit 42 is number of phases $\times Q$, the width dimension (total width dimension Wt) in the circumferential direction of the protruding portions 142 in a single magnetic pole may be number of phases $\times Q \times A = 3 \times 2 \times A = 6A$.

Here, the distributed winding referred to herein is that in which a single pole pair of the stator winding 51 is present at a single pole-pair cycle (N pole and S pole) of the magnetic poles. The single pole pair of the stator winding 51 is made of the two linear portions 83 through which currents flow in opposite directions and that are electrically connected by the turn portion 84, and the turn portion 84. If the above-described condition is met, even a short pitch winding is considered an equivalent of a distributed winding of a full pitch winding.

Next, an example of a case of a concentrated winding will be described. The concentrated winding herein is that in which the width of a single pole pair of the magnetic poles and the width of a single pole pair of the stator winding 51 differ. As examples of the concentrated winding, those in which relationships in which the conductor groups 81 relative to a single magnetic pole pair is three, the conductor groups 81 relative to two magnetic pole pairs is three, the conductor groups 81 relative to four magnetic pole pairs is nine, and the conductor groups 81 relative to five magnetic pole pairs is nine are established can be given.

Here, in a case in which the stator winding 51 is a concentrated winding, when the three-phase winding of the stator winding 51 is energized in a predetermined order, the stator winding 51 corresponding to two phases is excited. As a result, the protruding portions 142 corresponding to two phases are excited. Therefore, the width dimension Wt in the circumferential direction of the protruding portions 142 that are excited by the energization of the stator winding 51 within the range of a single pole of the magnet unit 42 is $A \times 2$.

In addition, with the width dimension Wt prescribed in this manner, the protruding portion 142 is configured as a magnetic material that satisfies the relationship in (1), above. Here, in the case of the concentrated winding described above, a sum of the widths of the protruding portions 142 that are present in the circumferential direction of the stator 50 in the area surrounded by the conductor groups 81 of the same phase is A. In addition, Wm in the concentrated winding corresponds to a perimeter of a surface of the magnet unit 42 opposing an air gap $\times$ number of phases $\pm$ number of dispersions of the conductor group 81.

Here, in a magnet of which the BH product is equal to or greater than 20 [MG0e (kJ/m³)], such as a neodymium

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magnet, a samarium cobalt magnet, or a ferrite magnet, Bd is just over 1.0 [T]. In iron, Br is just over 2.0 [T]. Therefore, as a high output motor, in the stator 52, the protruding portion 142 is merely required to be made of a magnetic material that satisfies a relationship expressed by $Wt < \frac{1}{2} \times Wm$.

In addition, when the conductor 82 includes an outer-layer coating 182 as described hereafter, the conductors 82 may be arranged in the circumferential direction of the stator core 52 such that the outer-layer coatings 182 of the conductors 82 are in contact with each other. In this case, Wt can be considered to be 0 or the thickness of the outer-layer coatings 182 of both conductors 82 that are in contact.

In FIGS. 25 and 26, the inter-conductor member (protruding portion 142) that is disproportionately small relative to the magnet magnetic flux on the rotor 40 side is provided. Here, the rotor 40 is a flat surface-magnet-type rotor that has low inductance and does not have saliency in terms of magnetic resistance. In this configuration, inductance in the stator 50 can be reduced. The occurrence of magnetic flux distortion attributed to a shift in the switching timing of the stator winding 51 is suppressed. Furthermore, electrical corrosion of the bearings 21 and 22 is suppressed.

(Second Modification)

As the stator 50 that uses the inter-conductor member that satisfies the relationship in expression (1), above, a following configuration can also be used. In FIG. 27, a tooth-like portion 143 is provided as the inter-conductor member on the outer circumferential surface side (upper surface side in the drawing) of the stator core 52. The tooth-like portion 143 is provided at a predetermined interval in the circumferential direction so as to protrude from the yoke 141 and has a thickness dimension that is the same as that of the conductor group 81 in the radial direction. A side surface of the tooth-like portion 143 is connected to the conductors 82 of the conductor group 81. However, a gap may be provided between the tooth-like portion 143 and the conductors 82.

The tooth-like portion 143 is restricted regarding the width dimension in the circumferential direction and has a thin pole tooth (stator tooth) that is disproportionate to the amount of magnets. As a result of the configuration, the tooth-like portion 143 is saturated with certainty by the magnet magnetic flux at 1.8 T or greater, and inductance can be reduced by reduction in permeance.

Here, in the magnet unit 42, when a surface area for a single pole of the magnetic flux action surface on the stator side is Sm and the residual magnetic flux density of the magnet unit 42 is Br, the magnetic flux on the magnet unit side is, for example, $Sm \times Br$.

In addition, when the surface area on the rotor side of each tooth-like portion 143 is St, the number of conductors 82 for a single phase is m, and the tooth-like portions 143 corresponding to two phases are excited within a single pole by energization of the stator winding 51, the magnetic flux on the stator side is, for example, $St \times m \times 2 \times Bs$. In this case, inductance can be reduced as a result of the dimensions of the tooth-like portion 143 being restricted so as to satisfy a relationship expressed by:

$$St \times m \times 2 \times Bs < Sm \times Br \quad (2).$$

Here, in a case in which the dimensions of the magnet unit 42 and the tooth-like portion 143 in the axial direction are the same, when the width dimension in the circumferential direction corresponding to a single pole of the magnet unit 42 is Wm and a width dimension in the circumferential direction of the tooth-like portion 143 is Wst, expression (2) is replaced as in expression (3).

$$Wst \times m \times 2 \times Bs < Wm \times Br \quad (3)$$

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More specifically, for example, when an assumption is made that $B_s=2T$, $B_r=1T$, and $m=2$, expression (3), above, is a relationship expressed by $W_{st}<W_m/8$. In this case, reduction in induction is achieved as a result of the width dimension W_{st} of the tooth-like portion 143 being made less than $1/8$ of the width dimension W_m corresponding to a single pole of the magnet unit 42. Here, when m is 1, the width dimension W_{st} of the tooth-like portion 143 may be less than $1/4$ of the width dimension W_m corresponding to a single pole of the magnet unit 42.

Here, in expression (3), above, $W_{st}\times m\times 2$ corresponds to the width dimension in the circumferential direction of the tooth-like portion 143 that is excited by energization of the stator winding 51 within the range of a single pole of the magnet unit 42.

In FIG. 27, in a manner similar to the configurations in FIGS. 25 and 26, described above, the inter-conductor member (tooth-like portion 143) that is disproportionately small relative to the magnet magnetic flux on the rotor 40 side is provided. In this configuration, inductance in the stator 50 can be reduced. The occurrence of magnetic flux distortion attributed to a shift in the switching timing of the stator winding 51 is suppressed. Furthermore, electrical corrosion of the bearings 21 and 22 is suppressed.

(Third Modification)

According to the above-described embodiment, the sealing member 57 that covers the stator winding 51 is provided in a range that includes all of the conductor groups 81 on the outer side of the stator core 52 in the radial direction, that is, a range in which the thickness dimension in the radial direction becomes greater than the thickness dimension in the radial direction of the conductor group 81. However, this configuration may be modified.

For example, as shown in FIG. 28, the sealing member 57 is configured to be provided such that a portion of the conductor 82 protrudes outward. More specifically, the sealing member 57 is configured to be provided such that a portion of the conductor 82 on the outermost side in the radial direction of the conductor group 81 is exposed toward the radially outer side, that is, the stator 50 side. In this case, the thickness dimension in the radial direction of the sealing member 57 may be the same as the thickness dimension in the radial direction of the conductor group 81 or less than the thickness dimension.

(Fourth Modification)

As shown in FIG. 29, in the stator 50, the conductor groups 81 may not be sealed by the sealing member 57. That is, the sealing member 57 that covers the stator winding 51 may not be used. In this case, the inter-conductor member is not provided between the conductor groups 81 that are arrayed in the circumferential direction and gaps are formed. In short, the inter-conductor member is not provided between the conductor groups 81 that are arrayed in the circumferential direction. Here, air can be considered a non-magnetic body or an equivalent of a non-magnetic body in which $B_s=0$. Air may be provided in the gap.

(Fifth Modification)

When the inter-conductor member in the stator 50 is made of a non-magnetic material, a material other than resin can be used as the non-magnetic material. For example, a metal-based non-magnetic material can be used such as SUS304 that is an austenitic stainless steel being used.

(Sixth Modification)

The stator 50 may not include the stator core 52. In this case, the stator 50 is configured by the stator winding 51

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shown in FIG. 12. Here, in the stator 50 that does not include the stator core 52, the stator winding 51 may be sealed by a sealing material. Alternatively, the stator 50 may include a circular annular winding holding portion that is made of a non-magnetic material such as synthetic resin, instead of the stator core 52 that is made of a soft magnetic material. (Seventh Modification)

According to the above-described first embodiment, the plurality of magnets 91 and 92 that are arrayed in the circumferential direction are used as the magnet unit 42 of the rotor 40. However, this configuration may be modified. An annular magnet that is a circular annular permanent magnet may be used as the magnet unit 42.

Specifically, as shown in FIG. 30, an annular magnet 95 is fixed on the radially inner side of the circular cylindrical portion 43 of the magnet holder 41. A plurality of magnetic poles of which the polarities alternate in the circumferential direction are provided in the annular magnet 95. The magnet is integrally formed on both the d-axis and the q-axis. A circular-arc-shaped magnet magnetic path of which a direction of orientation on the d-axis of the magnetic pole is the radial direction and a direction of orientation on the q-axis between magnetic poles is the circumferential direction is formed in the annular magnet 95.

Here, in the annular magnet 95, the orientation is merely required to be such that a circular-arc-shaped magnet magnetic path in which the easy axis of magnetization is parallel to the d-axis or oriented to be close to parallel to the d-axis in a portion located close to the d-axis, and the easy axis of magnetization is orthogonal to the q-axis or oriented to be close to orthogonal to the q-axis in a portion located close to the q-axis is formed.

(Eighth Modification)

In a present modification, a part of a control method of the control apparatus 110 is modified. In the present modification, sections that differ from the configuration described according to the first embodiment will mainly be described.

First, processes within the operating signal generating units 116 and 126 shown in FIG. 20, and the operating signal generating unit 130a and 130b shown in FIG. 21 will be described with reference to FIG. 31. Here, the processes in the operating signal generating units 116, 126, 130a, and 130b are basically similar. Therefore, the process in the operating signal generating unit 116 will be described below as an example.

The operating signal generating unit 116 includes a carrier generating unit 116a and U-, V-, and W-phase comparators 116bU, 116bV, and 116bW. According to the present embodiment, the carrier generating unit 116a generates a triangular wave signal as a carrier signal SigC and outputs the carrier signal SigC.

The carrier signal SigC generated by the carrier generating unit 116a, and the U-, V-, and W-phase command voltages calculated by the three-phase converting unit 115 are inputted to the U-, V-, and W-phase comparators 116bU, 116bV, and 116bW. For example, the U-, V-, and W-phase command voltages are waveforms in the shape of sine waves, and phases are shifted from each other by 120° in electrical angle.

The U-, V-, and W-phase comparators 116bU, 116bV, and 116bW generate the operating signals for the switches Sp and Sn of the upper arms and the lower arms of the U-, V-, and W-phases in the first inverter 101 by PWM control based on a comparison of magnitude between the U-, V-, and W-phase command voltages and the carrier signal SigC.

Specifically, the operating signal generating unit 116 generates the operating signals for the switches Sp and Sn of

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the U-, V-, and W-phases by PWM control based on a comparison of magnitude between signals in which the U-, V-, and W-phase command voltages are standardized by the power supply voltage, and the carrier signal. The driver 117 turns on/off the switches Sp and Sn of the U-, V-, and W-phases in the first inverter 101 based on the operating signals generated by the operating signal generating unit 116.

The control apparatus 110 performs a process for changing the carrier frequency fc of the carrier signal SigC, that is, the switching frequency of the switches Sp and Sn. The carrier frequency fc is set to be high in a low-torque region or a high-rotation region of the rotating electric machine 10, and set to be low in a high-torque region of the rotating electric machine 10. This setting is performed to suppress decrease in controllability of the current that flows to each phase winding.

That is, inductance in the stator 50 can be reduced in accompaniment with the stator 50 being made coreless. Here, when the inductance decreases, the electrical time constant of the rotating electric machine 10 decreases. As a result, ripples in the current that flows to each phase winding may increase, controllability of the current that flows to the winding may decrease, and current control may diverge.

The effects of this decrease in controllability can become more pronounced when the current that flows to the winding (such as an effective value of the current) is in a low-current region than when the current is included in a high-current region. In response to this issue, in the present modification, the control apparatus 100 changes the carrier frequency fc.

A process for changing the carrier frequency fc will be described with reference to FIG. 32. For example, this process is repeatedly performed at a predetermined control cycle by the control apparatus 110 as a process of the operating signal generating unit 116.

At step S10, the control apparatus 110 determines whether the current that flows to the winding 51a of each phase is in the low-current region. This process is a process for determining that the current torque of the rotating electric machine 10 is in the low-torque region. As a method for determining whether the current is in the low-current region, for example, first and second methods below can be given.

<First Method>

The torque estimation value of the rotating electric machine 10 is calculated based on the d-axis current and the q-axis current that are converted by the dq converting unit 112. In addition, when the calculated torque estimation value is determined to be less than a torque threshold, the current flowing to the winding 51a is determined to be in the low-current region. When the torque estimation value is determined to be equal to or greater than the torque threshold, the current is determined to be in the high-current region. Here, for example, the torque threshold may be set to 1/2 of a starting torque (also referred to as a locked-rotor torque) of the rotating electric machine 10.

<Second Method>

When the rotation angle of the rotor 40 detected by the angle detector is determined to be equal to or greater than a speed threshold, the current that flows to the winding 51a is determined to be in the low-current region, that is, the high-rotation region. Here, for example, the speed threshold may be set to a rotation speed when a maximum torque of the rotating electric machine 10 is the torque threshold.

When a negative determination is made at step S10, the control apparatus 110 determines that the current is in the

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high-current region and proceeds to step S11. At step S11, the control apparatus 110 sets the carrier frequency fc as a first frequency fL.

When an affirmative determination is made at step S10, the control apparatus 110 proceeds to step S12 and sets the carrier frequency fc as a second frequency fH that is higher than the first frequency fL.

As a result of the present modification described above, the carrier frequency fc is set to be higher when the current that flows to each phase winding is in the low-current region than when the current is in the high-current region. Therefore, in the low-current region, the switching frequency of the switches Sp and Sn can be increased, and increase in current ripples can be suppressed. As a result, the decrease in current controllability can be suppressed.

Meanwhile, when the current that flows to each phase winding is in the high-current region, the carrier frequency fc is set to be lower than when the current is in the low frequency region. In the high-current region, the amplitude of the current that flows to the winding is greater than that in the low-current region. Therefore, the effect that the increase in current ripples that are attributed to the decrease in inductance has on current controllability is small. Consequently, in the high-current region, the carrier frequency fc can be set to be lower than that in the low-current region. Switching loss in the inverters 101 and 102 can be reduced.

In the present modification, modes described below are possible.

When the carrier frequency fc is set to the first frequency fL, when the affirmative determination is made at step S10 in FIG. 32, the carrier frequency fc may be gradually changed from the first frequency fL toward the second frequency fH.

In addition, when the carrier frequency fc is set to the second frequency fH, when the negative determination is made at step S10, the carrier frequency fc may be gradually changed from the second frequency fH toward the first frequency fL.

The operating signals of the switches may be generated by space vector modulation (SVM) control, instead of PWM control. In this case as well, the changes in the switching frequency described above can be applied.

(Ninth Modification)

According to the above-described embodiments, the conductors configuring the conductor group 81 that are in two pairs for each phase are connected in parallel as shown in FIG. 33 by (a). FIG. 33 illustrates, by (a), a diagram showing an electrical connection between first and second conductors 88a and 88b that are two pairs of conductors. Here, instead of the configuration shown in FIG. 33 by (a), the first and second conductors 88a and 88b may be connected in series as shown in FIG. 33 by (b).

In addition, a multiple layer conductor of three pairs or more may be arranged so as to be laminated in the radial direction. FIG. 34 shows a configuration in which first to fourth conductors 88a to 88d that are four pairs of conductors are arranged in a laminated manner. The first to fourth conductors 88a to 88d are arranged so as to be arrayed in the radial direction in order of first, second, third, and fourth conductors 88a, 88b, 88c, and 88d, from the conductor closest to the stator core 52.

Here, as shown in FIG. 33 by (c), the third and fourth conductors 88c and 88d may be connected in parallel. In addition, the first conductor 88a may be connected to one end of this parallel-connection body and the second conductor 88b may be connected to the other end. When the parallel connection is used, current density in the conductors

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that are connected in parallel can be reduced. Heat generation during energization can be suppressed.

Therefore, a cylindrical stator winding is assembled to a housing (unit base **61**) in which the cooling water passage **74** is formed. In this configuration, the first and second conductors **88a** and **88b** that are not connected in parallel are arranged on the stator core **52** side that is in contact with the unit base **61**, and the third and fourth conductors **88c** and **88d** that are connected in parallel are arranged on the counter-stator core side. As a result, the cooling performance of the conductors **88a** to **88d** in the multiple-layer conductor structure can be equalized.

Here, the thickness dimension in the radial direction of the conductor group **81** that is made of the first to fourth conductors **88a** to **88d** is merely required to be less than the width dimension in the circumferential direction corresponding to a single phase within a single magnetic pole. (Tenth Modification)

The rotating electric machine **10** may have an inner-rotor structure (internally revolving structure). In this case, for example, inside the housing **30**, the stator **50** may be provided on the radially outer side and the rotor **40** may be provided on the radially inner side thereof. In addition, the inverter unit **60** may be provided on one side or both sides of both ends in the axial direction of the stator **50** and the rotor **40**. FIG. **35** is a lateral cross-sectional view of the rotor **40** and the stator **50**. FIG. **36** is a diagram showing a portion of the rotor **40** and the stator **50** in an enlarged manner.

The configuration in FIGS. **35** and **36** in which the inner-rotor structure is presumed is a configuration that is similar to the configuration in FIGS. **8** and **9** in which the outer-rotor structure is presumed, aside from the rotor **40** and the stator **50** being reversed on the inner side and the radially outer side. In brief, the stator **50** includes the stator winding **51** that has a flattened conductor structure and the stator core **52** that does not have teeth. The stator winding **51** is assembled on the radially inner side of the stator core **52**. The stator core **52** has any of the configurations below, in a manner similar to that in the case of the outer-rotor structure.

- (A) In the stator **50**, the inter-conductor member is provided between the conductor portions in the circumferential direction, and when the width dimension in the circumferential direction of the inter-conductor member in a single magnetic pole is W_t , the saturation magnetic density of the inter-conductor member is B_s , the width dimension in the circumferential direction of the magnet unit in a single magnetic pole is W_m , and the residual magnetic flux density of the magnet unit is B_r , a magnetic material in which a relationship expressed by $W_t \times B_s < W_m \times B_r$ is satisfied is used as the inter-conductor member.
- (B) In the stator **50**, the inter-conductor member is provided between the conductor portions in the circumferential direction, and a non-magnetic material is used as the inter-conductor member.
- (C) In the stator **50**, the inter-conductor member is not provided between the conductor portions in the circumferential direction.

In addition, the foregoing similarly applies to the magnets **91** and **92** of the magnet unit **42**. That is, the magnet unit **42** is configured using the magnets **91** and **92** oriented such that, at locations near to the d-axis that is the magnetic pole center, the orientation of the easy axis of magnetization is parallel to the d-axis compared to at locations near to the q-axis that is the magnetic pole boundary. Details of the magnetization direction and the like of the magnets **91** and

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92 are as described above. The annular magnet **95** (see FIG. **30**) can be used in the magnet unit **42**.

FIG. **37** is a longitudinal cross-sectional view of the rotating electric machine **10** when the rotating electric machine **10** is the inner-rotor-type. FIG. **37** is a diagram that corresponds to FIG. **2** that has been described earlier. Differences with the configuration in FIG. **2** will briefly be described.

In FIG. **37**, the annular stator **50** is fixed on the inner side of the housing **30**, and the rotor **40** is rotatably provided on the inner side of the rotor **50** with a predetermined air gap therebetween. In a manner similar to that in FIG. **2**, the bearings **21** and **22** are arranged so as to be concentrated on one side in the axial direction relative to the center in the axial direction of the rotor **40**. As a result, the rotor **40** is cantilever-supported. In addition, the inverter unit **60** is provided on the inner side of the magnet holder **41** of the rotor **40**.

FIG. **38** shows another configuration of the rotating electric machine **10** that has the inner-rotor structure. In FIG. **38**, in the housing **30**, the rotation shaft **11** is rotatably supported by the bearings **21** and **22**, and the rotor **40** is fixed to the rotation shaft **11**. In a manner similar to the configuration shown in FIG. **2** and the like, the bearings **21** and **22** are arranged so as to be concentrated on one side in the axial direction relative to the center in the axial direction of the rotor **40**. The rotor **40** includes the magnet holder **41** and the magnet unit **42**.

In the rotating electric machine **10** in FIG. **38**, as a difference with the rotor **10** in FIG. **37**, the inverter unit **60** is not provided on the radially inner side of the rotor **40**. The magnet holder **41** is connected to the rotation shaft **11** in a position on the radially inner side of the magnet unit **42**. In addition, the stator **50** has the stator winding **51** and the stator core **52**, and is attached to the housing **30**. (Eleventh Modification)

Another configuration will be described as the rotating electric machine that has an inner-rotor structure. FIG. **39** is an exploded perspective view of a rotating electric machine **200**. FIG. **40** is a cross-sectional side view of the rotating electric machine **200**. Here, the up/down direction is indicated with reference to the state in FIGS. **39** and **40**.

As shown in FIGS. **39** and **40**, the rotating electric machine **200** includes a stator **203** and a rotor **204**. The stator **203** includes an annular stator core **201** and a multiple-phase stator winding **202**. The rotor **204** is arranged on the inner side of the stator core **201** so as to freely rotate. The stator **203** corresponds to an armature. The rotor **204** corresponds to a field element. The stator core **201** is configured by numerous silicon steel sheets being laminated. The stator winding **202** is attached to the stator core **201**. Although omitted in the drawings, the rotor **204** includes a rotor core and a plurality of permanent magnets that serve as a magnet unit.

A plurality of magnet insertion holes are provided in the rotor core at an even interval in the circular circumferential direction. The permanent magnets that are magnetized such that the magnetization directions alternately change for each adjacent magnetic pole are mounted in the magnet insertion holes. Here, the permanent magnet of the magnet unit may be that which has the Halbach array as described in FIG. **23** or a configuration similar thereto.

Alternatively, the permanent magnet of the magnet unit may be that which has the characteristics of polar anisotropy in which the orientation direction (magnetization direction) extends in a circular arc shape between the d-axis that is the

magnetic pole center and the q-axis that is the magnetic pole boundary, such as that described in FIGS. 9 and 30.

Here, the stator 203 may have any of the configurations below.

- (A) In the stator 203, the inter-conductor member is provided between the conductor portions in the circumferential direction, and when the width dimension in the circumferential direction of the inter-conductor member in a single magnetic pole is W_t , the saturation magnetic density of the inter-conductor member is B_s , the width dimension in the circumferential direction of the magnet unit in a single magnetic pole is W_m , and the residual magnetic flux density of the magnet unit is B_r , a magnetic material in which a relationship expressed by $W_t \times B_s \leq W_m \times B_r$ is satisfied is used as the inter-conductor member.
- (B) In the stator 203, the inter-conductor member is provided between the conductor portions in the circumferential direction, and a non-magnetic material is used as the inter-conductor member.
- (C) In the stator 203, the inter-conductor member is not provided between the conductor portions in the circumferential direction.

In addition, in the rotor 204, the magnet unit is configured using a plurality of magnets that are oriented such that, on the d-axis side that is the magnetic pole center, the orientation of the easy axis of magnetization is parallel to the d-axis compared to the side of the q-axis that is the magnetic pole boundary.

An annular inverter case 211 is provided on one end side in the axial direction of the rotating electric machine 200. The inverter case 211 is arranged such that a case lower surface is in contact with an upper surface of the stator core 201. A plurality of power modules 212 that configure an inverter circuit, a smoothing capacitor 213 that suppresses ripples in the voltage and the current that occur as a result of the switching operation of the semiconductor switching elements, the control board 214 that has a control unit, a current sensor 215 that detects a phase current, and a resolver stator 216 that is a rotation frequency sensor for the rotor 204 are provided inside the inverter case 211. The power modules 212 include IGBTs that are the semiconductor switching elements and diodes.

A power connector 217 and a signal connector 218 are provided on a peripheral edge of the inverter case 211. The power connector 217 is connected to a direct-current circuit of a battery that is mounted in the vehicle. The signal connector 218 is used to receive and transmit various signals between the rotating electric machine 200 side and a vehicle-side control apparatus. The inverter case 211 is covered by a top cover 219. Direct-current power from the onboard battery is inputted via the power connector 217, converted by the switching of the power modules 212, and supplied to the stator winding 202 of each phase.

A bearing unit 221 that rotatably holds the rotation shaft of the rotor 204 and an annular rear case 222 that houses the bearing unit 221 are provided on a side opposite the inverter case 211, of both sides in the axial direction of the stator core 201. For example, the bearing unit 221 includes two sets of bearings, and is arranged so as to be concentrated on one side in the axial direction relative to the center in the axial direction of the rotor 204. However, the plurality of bearings in the bearing unit 221 may be provided so as to be dispersed on both sides in the axial direction of the stator core 201, and the rotation shaft may be double-supported by the bearings. The rotating electric machine 200 is connected to the vehicle

side by the rear case 222 being fixed to an attachment portion of a gear case or a transmission of the vehicle by bolt-fastening.

A cooling passage 211a for allowing a coolant to flow is formed inside the inverter case 211. The cooling passage 211a is formed by a space that is provided in an annular recessing shape from a lower surface of the inverter case 211 being sealed by the upper surface of the stator core 201. The cooling passage 211a is formed so as to surround the coil end of the stator winding 202. A module case 212a for the power modules 212 is inserted inside the cooling passage 211a. A cooling passage 222a is also formed in the rear case 222 so as to surround the coil end of the stator winding 202. The cooling passage 222a is formed by a space that is provided in an annular recessing shape from an upper surface of the rear case 222 being sealed by a lower surface of the stator core 201.

(Twelfth Modification)

Up to this point, configurations that are implemented in a rotation-field-type rotating electric machine have been described. However, the configuration can be modified and implemented in a rotating-armature-type rotating electric machine. FIG. 41 shows a configuration of a rotating-armature-type rotating electric machine 230.

In the rotating electric machine 230 in FIG. 41, a bearing 232 is fixed to each of housings 231a and 231b, and a rotation shaft 233 is supported by the bearing 232 so as to freely rotate. For example, the bearing 232 is an oil-retaining bearing that includes a porous metal permeated with oil. A rotor 234 that serves as an armature is fixed to the rotation shaft 233. The rotor 234 includes a rotor core 235 and a multiple-phase rotor winding 236 that is fixed to an outer circumferential portion of the rotor core 235. In the rotor 234, the rotor core 235 has a slot-less structure. The rotor winding 236 has a flattened conductor structure. That is, the rotor winding 236 has a flattened structure in which an area for each phase is longer in the circumferential direction than the radial direction.

In addition, a stator 237 that serves as a field element is provided on the radially outer side of the rotor 234. The stator 237 includes the stator core 238 that is fixed to the housing 231a and a magnet unit 239 that is fixed to the inner circumferential side of the stator core 238. The magnet unit 239 is configured to include a plurality of magnetic poles of which the polarities alternate in the circumferential direction.

In a manner similar to the magnet unit 42 and the like described earlier, the magnet unit 239 is configured to be oriented such that, on the d-axis side that is the magnetic pole center, the orientation of the easy axis of magnetization is parallel to the d-axis compared to the side of the q-axis that is the magnetic pole boundary. The magnet unit 239 includes a sintered neodymium magnet that is oriented. The intrinsic coercive force thereof is equal to or greater than 400 [kA/m], and the remanent flux density B_r is equal to or greater than 1.0 [T].

The rotating electric machine 230 of the present example is a two-pole, three-coil brushed coreless motor. The rotor winding 236 is divided into three, and the magnet unit 239 has two poles. The number of poles and the number of coils of the brushed motor varies, such as 2:3, 4:10, or 4:21, depending on an intended use thereof.

A commutator 241 is fixed to the rotation shaft 233, and a plurality of brushes 242 are arranged on the radially outer side thereof. The commutator 241 is electrically connected to the rotor winding 236 via a conductor 243 that is embedded in the rotation shaft 233. A direct-current current

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flows in and out of the rotor winding **236** through the commutator **241**, the brushes **242**, and the conductor **243**. The commutator **241** is configured to be divided in the circumferential direction as appropriate, based on the number of phases of the rotor winding **236**. Here, the brushes **242** may be directly connected to a direct-current power supply such as a storage battery by electrical wiring, or may be connected to the direct-current power supply through a terminal block or the like.

A resin washer **244** that serves as a sealing member is provided in the rotation shaft **233**, between the bearing **232** and the commutator **241**. As a result of the resin washer **244**, oil that seeps out from the bearing **232** that is an oil-retaining bearing is suppressed from flowing out toward the commutator **241** side.

(Thirteenth Modification)

In the stator winding **51** of the rotating electric machine **10**, the conductors **82** may have a plurality of insulation coatings inside and outside thereof. For example, the conductor **82** may be configured by a plurality of conductors (wires) that have insulation coatings being bundled and the bundle being covered by an outer-layer coating.

In this case, the insulation coatings of the wires configure the insulation coatings on the inner side. The outer-layer coating configures the insulation coating on the outer side. In addition, in particular, insulation performance of the insulation coating on the outer side, among the plurality of insulation coatings of the conductor **82**, may be made higher than the insulation performance of the insulation coatings on the inner side. Specifically, a thickness of the insulation coating on the outer side is made thicker than a thickness of the insulation coatings on the inner side.

For example, the thickness of the insulation coating on the outer side may be 100 μm and the thickness of the insulation coating on the inner side may be 40 μm . Alternatively, a material that has a lower dielectric constant than the insulation coating on the inner side may be used as the insulation coating on the outer side. All that is required is that at least either of the foregoing is applied. Here, the wire may be configured as a bundle of a plurality of conductive materials.

As a result of insulation on the outermost layer of the conductor **82** being strengthened as described above, the conductor **82** becomes suitable for use in a high-voltage vehicle system. In addition, appropriate driving of the rotating electric machine **10** can be obtained even in elevated regions where air pressure is low.

(Fourteenth Modification)

In the conductor **82** that includes the plurality of insulation coatings inside and outside, at least either of a rate of linear expansion (coefficient of linear expansion) and bonding strength may differ between the insulation coating on the outer side and the insulation coating on the inner side. The configuration of the conductor **82** of the present modification is shown in FIG. **42**.

In FIG. **42**, the conductor **82** includes a plurality (four in the drawing) of wires **181**, an outer-layer coating **182** (outer insulation coating) that is made of resin, for example, and surrounds the plurality of wires **181**, and an intermediate layer **183** (intermediate insulation coating) that fills an area surrounding the wires **181** inside the outer layer coating **182**. The wire **181** includes a conductive portion **181a** that is made of a copper material and a conductor coating **181b** (inner insulation coating) that is made of an insulation material. In terms of the stator winding, insulation is provided between phases by the outer-layer coating **182**. Here, the wiring **181** may be configured as a bundle of a plurality of conductive materials.

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The intermediate layer **183** has a higher rate of linear expansion than the conductor coating **181b** of the wire **181** and a lower rate of linear expansion than the outer-layer coating **182**. That is, in the conductor **82**, the rate of linear expansion increases toward the outer side.

In general, in the outer-layer coating **182**, the coefficient of linear expansion is higher than that of the conductor coating **181b**. As a result of the intermediate layer **183** that has a rate of linear expansion that is midway between those of the outer-layer coating **182** and the conductor coating **181b**, the intermediate layer **183** functions as a cushion material and can prevent simultaneous breakage on the outer layer side and the inner layer side.

Furthermore, in the conductor **82**, the conductive portion **181a** and the conductor coating **181b** are bonded in the wire **181**. The conductor coating **181b** and the intermediate layer **183** are bonded together, and the intermediate layer **183** and the outer-layer coating **182** are bonded together. In these bonded portions, bonding strength weakens toward the outer side of the conductor **82**. That is, the bonding strength between the conductive portion **181a** and the conductor coating **181b** is weaker than the bonding strength between the conductor coating **181b** and the intermediate layer **183**, and also weaker than the bonding strength between the intermediate layer **183** and the outer-layer coating **182**.

In addition, when the bonding strength between the conductor coating **181** and the intermediate layer **183** and the bonding strength between the intermediate layer **183** and the outer-layer coating **182** are compared, the latter (on the outer side) may be weaker or equal. Here, for example, a magnitude of the bonding strength between coatings can be ascertained by tensile strength that is required when the two layers of coatings are peeled apart.

As a result of the bonding strength of the conductor **82** being set as described above, even if an inner/outer temperature difference occurs as a result of heat generation or cooling, breakage occurring on both the inner layer side and the outer layer side (co-breakage) can be suppressed.

Here, heat generation and temperature changes in the rotating electric machine mainly manifest as copper loss that is heat-generated from the conductive portion **181a** of the wire **181** and iron loss that is generated from within the core. However, these two types of losses are transmitted from the conductive portion **181a** inside the conductor **82** or from outside the conductor **82**. A heat generation source is not present in the intermediate layer **183**.

In this case, as a result of the intermediate layer **183** having bonding force that can serve as a cushion for both, simultaneous breakage thereof can be prevented. Therefore, favorable usage can be obtained even for use in fields that involve high voltage resistance or significant temperature changes, such as use in vehicles.

A supplementary description is provided below. For example, the wire **181** may be an enamel wire. In this case, the wire **181** includes a resin coating layer (conductor coating **181b**) made of polyamide (PA), PI, PAI, or the like. In addition, the outer-layer coating **182** on the outer side of the wire **181** is preferably made of a similar PA, PI, PAI, or the like and thick in terms of thickness. As a result, breakage of the coating due to a difference in linear expansion is suppressed.

Here, as the outer-layer coating **182**, in addition to that in which measures are taken by the material, such as PA, PI, or PAI being made thick, use of that in which the dielectric constant is smaller than that of PI or PAI, such as PPS, PEEK, fluororesin, polycarbonate, silicon resin, epoxy, polyethylene naphthalate, or liquid crystal polymer (LCP), is

also preferred in terms of increasing conductor density in the rotating electric machine. As a result of these resins, even when the coating is thinner than a PI or PAI coating that is equivalent to the conductor coating **181b** or of equal thickness to the conductor coating **181b**, the insulation performance thereof can be increased. As a result, an occupancy ratio of the conductive portion can be increased.

In general, the above-described resin provides insulation in which the dielectric constant is more favorable than that of the insulation coating of the enamel wire. Of course, there are examples in which the dielectric constant is made deteriorated due to a state of molding or adulteration. Among the foregoing, PPS and PEEK generally have a greater coefficient of linear expansion than an enamel coating. However, because the coefficient of linear expansion thereof is less than that of other resins, PPS and PEEK are suitable as the outer-layer coating in the second layer.

In addition, the bonding strength between the two types of coatings (intermediate insulation coating and outer-layer insulation coating) on the outer side of the wire **181** and the enamel coating of the wire **181** is preferably weaker than the bonding strength between the copper wire in the wire **181** and the enamel coating. As a result, a phenomenon in which the enamel coating and the two types of coatings break simultaneously is suppressed.

When a water-cooled structure, a liquid-cooled structure, or an air-cooled structure is added to the stator, thermal stress and impact stress are thought to basically be applied from the outer-layer coating **182** and beyond. However, even in cases in which the insulation layer of the wire **181** and the above-described two types of coatings are made of differing resins, as a result of a portion in which the coatings are not bonded being provided, the thermal stress and impact stress can be reduced.

That is, the insulation structure is formed by a space being provided between the two types of coatings and the wire (enamel wire), and fluororesin, polycarbonate, silicon resin, epoxy, polyethylene naphthalate, or LCP being used. In this case, the outer-layer coating and the inner-layer coating are preferably bonded using an adhesive material that has a low dielectric constant and a low coefficient of linear expansion, such as epoxy.

As a result, in addition to mechanical strength, coating breakage as a result of friction caused by shaking due to vibrations in the conductive portion and the like, or breakage of the outer-layer coating as a result of the difference in coefficient of linear expansion can be suppressed.

As an outermost-layer fixing that is generally a final step for the periphery of the stator winding and imparts mechanical strength, fixing, and the like, relative to the conductor **82** that is configured as described above, a resin, such as epoxy, PPS, PEEK, or LCP, of which moldability is favorable and properties such as the dielectric constant and the coefficient of linear expansion are similar to the properties of the enamel coating is preferred.

In general, resin potting using urethane or silicon is commonly performed. However, in the above-described resin, the coefficient of linear expansion thereof differs by almost two-fold compared to other resins, and thermal stress that may shear the resin is generated. Therefore, the resin is unsuitable for use at 60 V or higher for which strict insulation regulations are internationally applied. In this regard, as a result of a final insulation step that is easily fabricated by injection molding or the like using epoxy, PPS, PEEK, LCP, or the like, the requirements described above can be achieved.

Modifications other than those described above are listed below.

A distance DM in the radial direction between a surface on the armature side in the radial direction of the magnet unit **42** and the axial center of the rotor may be equal to or greater than 50 mm. Specifically, for example, the distance DM in the radial direction between the surface on the radially inner side of the magnet unit **42** (specifically, the first and second magnets **91** and **92**) shown in FIG. 4 and the axial center of the rotor **40** may be equal to or greater than 50 mm.

As the rotating electric machine that has a slot-less structure, a small-scale rotating electric machine that is used for models of which output ranges from several tens to several hundred watts and the like is known. In addition, the disclosers of the present application have not ascertained examples in which the slot-less structure is used in a large-scale rotating electric machine for industrial use that typically exceeds 10 kW. The disclosers of the present application have examined reasons therefor.

The rotating electric machines that have become mainstream in recent years are largely classified into the following four types. These rotating electric machines are a brushed motor, a squirrel-cage-type induction motor, a permanent-magnet-type synchronous motor, and a reluctance motor.

In the brushed motor, excitation current is supplied via a brush. Therefore, in the case of a large-scale brushed motor, the brush may become large and maintenance may become complicated. As a result, there is a history in that, in accompaniment with the remarkable advancements in semiconductor technology, the brushed motors have been replaced by brushless motors such as induction motors. Meanwhile, in the field of compact motors, many coreless motors are also being supplied across the world because of advantages in terms of low inertia and economic efficiency.

In the squirrel-cage-type induction motor, the principle is that torque is generated by a magnetic field that is generated by a stator winding on a primary side being received by a core of a rotor on a secondary side, an induction current being sent in a concentrated manner to a squirrel-cage-type conductor, and a reaction magnetic field being formed. Therefore, from the perspective of compactness and higher efficiency of an apparatus, eliminating the core from both the stator side and the rotor side cannot necessarily be said to be expedient.

The reluctance motor is a motor that simply uses changes in reluctance in the core. In principle, eliminating the core is not preferable.

In the permanent-magnet-type synchronous motor, the IPM (that is, an embedded magnet-type rotor) has become mainstream in recent years. Unless there are special circumstances, large-scale machines in particular are often IPMs.

The IPM has a characteristic of having both magnet torque and reluctance torque. The IPM is operated while proportions of these torques are adjusted as appropriate by inverter control. Therefore, the IPM is a compact motor that has excellent controllability.

When, based on analysis by the disclosers of the present application, the torques on the rotor surface that generates the magnet torque and the reluctance torque are drawn with the distance DM in the radial direction between the surface on the armature side in the radial direction of the magnet unit and the axial center of the rotor, that is, a radius of the stator core of a typical inner rotor is taken on a horizontal axis, the torques are as shown in FIG. 43.

As shown in expression (eq1), below, whereas a potential of the magnet torque is determined by magnetic field

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strength generated by the permanent magnet, a potential of the reluctance torque is determined by inductance, particularly a magnitude of a q-axis inductance, as shown in expression (eq2), below.

$$\text{Magnet torque} = k \cdot \Psi \cdot I_q \quad (\text{eq1})$$

$$\text{Reluctance torque} = k \cdot (L_q - L_d) \cdot I_q \cdot I_d \quad (\text{eq2})$$

Here, the magnetic field strength of the permanent magnet and the magnitude of the inductance in the winding are compared based on DM. The magnetic field strength generated by the permanent magnet, that is, a magnetic flux amount Ψ is proportional to a total area of the permanent magnet on a surface that opposes the stator. In the case of a circular cylindrical rotor, the total area is the surface area of a circular cylinder.

Strictly speaking, because the N pole and the S pole are present, the magnet field strength is proportional to an occupied area that is half the circular cylindrical surface. The surface area of the circular cylinder is proportional to a radius of the circular cylinder and a circular cylinder length. That is, if the circular cylinder length is fixed, the surface area is proportional to the radius of the circular cylinder.

Meanwhile, although an inductance L_q of the winding is dependent on core shape, sensitivity is low. Rather, because the inductance L_q is proportional to a square of the number of windings of the stator winding, dependence on the number of windings is high. Here, when μ is the magnetic permeability of the magnetic circuit, N is the number of windings, S is the cross-sectional area of the magnetic circuit, and δ is an effective length of the magnetic circuit, inductance $L = \mu \cdot N^2 \cdot S / \delta$.

The number of windings of the winding is dependent on a size of a winding space. Therefore, in the case of a circular cylindrical motor, the number of windings is dependent on the winding space of the stator, that is, the slot area. As shown in FIG. 44, the slot area is proportional to a product $a \times b$ of a length dimension a in the circumferential direction and a length dimension b in the radial direction, because the shape of the slot is approximately a quadrangle.

The length dimension in the circumferential direction of the slot increases as the diameter of the circular cylinder increases. Therefore, the length dimension in the circumferential direction of the slot is proportional to the diameter of the circular cylinder. The length dimension in the radial direction of the slot is simply proportional to the diameter of the circular cylinder. That is, the slot area is proportional to a square of the diameter of the circular cylinder.

In addition, as is clear from expression (eq2), above, the reluctance torque is proportional to a square of the stator current. Therefore, the performance of the rotating electric machine is determined by the manner in which a large current can be supplied. The performance is dependent on the slot area of the stator. From the foregoing, if the length of the circular cylinder is fixed, the reluctance torque is proportional to the square of the diameter of the circular cylinder. With this in mind, a diagram in which a relationship between the magnetic torque, the reluctance torque, and DM is plotted is FIG. 43.

As shown in FIG. 43, the magnet torque linearly increases relative to DM. The reluctance torque quadratically increases relative to DM. It is clear that, when DM is relatively small, the magnet torque is dominant. The reluctance torque becomes dominant as the stator core radius increases.

The disclosers of the present application have reached a conclusion that, under predetermined conditions, an inter-

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section between the magnet torque and the reluctance torque in FIG. 43 is near a stator core radius of about 50 mm. That is, in a 10 kW-class motor in which the stator core radius sufficiently exceeds 50 mm, because use of reluctance torque is currently mainstream, eliminating the core is difficult. This is presumed to be one reason for which the slot-less structure is not used in the field of large-scale machinery.

In the case of a rotating electric machine in which a core is used in the stator, magnetic saturation of the core is an issue at all times. In particular, in a radial-gap-type rotating electric machine, the longitudinal cross-sectional shape of the rotation shaft is fan-shaped for each magnetic pole. A magnetic path width becomes narrower toward the inner circumferential side of the apparatus, and a dimension on the inner circumferential side of a teeth portion that forms the slots determines a performance limit of the rotating electric machine.

Regardless of how high performance the permanent magnet that is used is, if magnetic saturation occurs in this section, the performance of the permanent magnet cannot be sufficiently obtained. To prevent magnetic saturation from occurring in this section, the inner circumference is designed to be large, thereby resulting in a larger apparatus.

For example, in a distributed-winding rotating electric machine, if the winding is a three-phase winding, magnetic flux is supplied so as to be distributed among three to six teeth per magnetic pole. However, because the magnetic flux tends to become concentrated at the teeth toward the front in the circumferential direction, the magnetic flux does not flow evenly to the three to six teeth. In this case, while the magnetic flux flows in a concentrated manner to a portion (such as one or two) of the teeth, the teeth that are magnetically saturated also move in the circumferential direction in accompaniment with the rotation of the rotation shaft. This is also a factor in the generation of slot ripples.

From the foregoing, in the rotating electric machine that has a slot-less structure and of which DM is equal to or greater than 50 mm, the teeth are preferably eliminated to resolve magnetic saturation. However, when the teeth are eliminated, magnetic resistance in the magnetic circuit in the rotor and the stator increases, and the torque of the rotating electric machine decreases. A reason for the increase in magnetic resistance is, for example, the air gap between the rotor and the stator becoming larger.

Therefore, in the rotating electric machine that has the slot-less structure in which DM is equal to or greater than 50 mm, described above, there is room for improvement regarding the enhancement of torque. Therefore, there is significant merit in applying the above-described configuration that enables torque to be enhanced, to the rotating electric machine that has the slot-less structure and in which the DM is equal to or greater than 50 mm, described above.

Here, the distance DM in the radial direction between the surface on the armature side in the radial direction of the magnet unit and the axial center of the rotor may be equal to or greater than 50 mm in not only the rotating electric machine that has the outer-rotor structure, but also the rotating electric machine that has the inner rotor structure as well.

The stator winding 51 of the rotating electric machine 10 may be configured such that the linear portions 83 of the conductors 82 are provided in a single layer in the radial direction. In addition, when the linear portions 83 are arranged in a plurality of layers on the inner side and the radially outer side, the number of layers may be arbitrary. The linear portions 83 may be provided in three layers, four layers, five layers, six layers, or the like.

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For example, in FIG. 2, the rotation shaft 11 is provided so as to protrude toward both one end side and the other end side of the rotating electric machine 10 in the axial direction. However, this configuration may be modified. The rotation shaft 11 may be configured to protrude toward only one end side.

In this case, with a portion that is cantilever-supported by the bearing unit 20 as an end portion, the rotation shaft 11 may be provided so as to extend toward the outer side in the axial direction thereof.

In the present configuration, because the rotation shaft 11 does not protrude inside the inverter unit 60, an internal space of the inverter unit 60, or specifically, the internal space of the cylindrical portion 71 can be more widely used.

In the rotating electric machine 10 configured as described above, a non-conductive grease is used in the bearings 21 and 22. However, this configuration may be modified. A conductive grease may be used in the bearings 21 and 22. For example, a conductive grease that includes metal particles, carbon particles, or the like is used.

As a configuration in which the rotation shaft 11 is supported so as to rotate freely, the bearings may be provided in two locations, on one end side and the other end side in the axial direction of the rotor 40. In this case, in terms of the configuration in FIG. 1, the bearings may be provided in two locations, on one end side and the other end side with the inverter unit 60 therebetween.

In the rotating electric machine 10 configured as described above, the intermediate portion 45 of the magnet holder 41 in the rotor 40 includes the inner shoulder portion 49a and the annular outer shoulder portion 49b. However, these shoulder portions 49a and 49b may be eliminated, and the intermediate portion 45 may be configured to have a flat surface.

In the rotating electric machine 10 configured as described above, the conductor body 82a is a bundle of a plurality of wires 86 in the conductor 82 of the stator winding 51. However, this configuration may be modified. A square conductor that has a rectangular cross-section may be used as the conductor 82. In addition, a circular conductor that has a circular cross-sectional shape or an elliptical cross-sectional shape may be used as the conductor 82.

In the rotating electric machine 10 configured as described above, the inverter unit 60 is provided on the radially inner side of the stator 50. However, instead, the inverter unit 60 may not be provided on the radially inner side of the stator 50. In this case, an internal area that is the radially inner side of the stator 50 may be left as an empty space. In addition, a component other than the inverter unit 60 can be arranged in the internal area.

In the rotating electric machine 10 configured as described above, the housing 30 may not be provided. In this case, for example, the rotor 40, the stator 50, and the like may be held in a portion of the wheel or another vehicle component.

(Embodiment as an In-Wheel Motor for a Vehicle)

Next, an embodiment in which the rotating electric machine is provided integrally with a vehicle wheel of a vehicle as an in-wheel motor will be described.

FIG. 45 is a perspective view of a vehicle wheel 400 that has an in-wheel motor structure and a surrounding structure thereof. FIG. 46 is a longitudinal cross-sectional view of the vehicle wheel 400 and the surrounding structure thereof. FIG. 47 is an exploded perspective view of the vehicle wheel 400. Each of these drawings is a perspective view in which the vehicle wheel 400 is viewed from inside the vehicle.

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Here, in the vehicle, the in-wheel motor structure according to the present embodiment can be applied in various modes. For example, in a vehicle that has two wheels each in the front and rear of the vehicle, the in-wheel motor according to the present embodiment can be applied to the two wheels on the front side of the vehicle, the two wheels on the rear side of the vehicle, or the four wheels in the front and rear of the vehicle. However, the in-wheel motor according to the present embodiment can also be applied to a vehicle in which at least either of the front and rear of the vehicle has a single wheel. Here, the in-wheel motor is an application example of a drive unit for a vehicle.

As shown in FIGS. 45 to 47, for example, the vehicle wheel 400 includes a tire 401 that is a known tire that is filled with air, a wheel 402 that is fixed to an inner circumferential side of the tire 401, and a rotating electric machine 500 that is fixed to an inner circumferential side of the wheel 402. The rotating electric machine 500 includes a fixed portion that is a portion that includes a stator and a rotating portion that is a portion that includes a rotor. The fixed portion is fixed to a vehicle body side.

In addition, the rotating portion is fixed to the wheel 402. The tire 401 and the wheel 402 rotate as a result of the rotation of the rotating unit. Here, in the rotating electric machine 500, a detailed configuration including the fixed portion and the rotating portion will be described hereafter.

In addition, in the vehicle wheel 400, as peripheral apparatuses, a suspension apparatus that holds the vehicle wheel 400 to a vehicle body (not shown), a steering apparatus that enables an orientation of the vehicle wheel 400 to be changed, and a brake apparatus that performs braking of the vehicle wheel 400 are attached.

The suspension apparatus is an independent-suspension-type suspension. For example, application of an arbitrary type, such as a trailing arm type, a strut type, a wishbone type, or a multilink type, is possible. According to the present embodiment, as the suspension apparatus, a lower arm 411 is provided so as to be oriented to extend toward the vehicle-body center side, and a suspension arm 412 and a spring 413 are provided so as to be oriented to extend in the vertical direction.

For example, the suspension arm 412 may be configured as a shock absorber. However, a detailed illustration thereof is omitted. The lower arm 411 and the suspension arm 412 are each connected to the vehicle body side and connected to a circular-disk-shaped base plate 405 that is fixed to the fixed portion of the rotating electric machine 500. As shown in FIG. 46, on the rotating electric machine 500 side (base plate 405 side), the lower arm 411 and the suspension arm 412 are supported by support axes 414 and 415 so as to be in a coaxial state with each other.

In addition, as the steering apparatus, for example, application of a rack-and-pinion type structure or a ball-and-nut type structure, or application of a hydraulic power steering system or an electric power steering system is possible. According to the present embodiment, a rack apparatus 421 and a tie rod 422 are provided as the steering apparatus. The rack apparatus 421 is connected to the base plate 405 on the rotating electric machine 500 side by the tie rod 422.

In this case, when the rack apparatus 421 is operated in accompaniment with the rotation of a steering shaft (not shown), the tie rod 422 moves in a left/right direction of the vehicle. As a result, the vehicle wheel 400 rotates around the support shafts 414 and 415 of the lower arm 411 and the suspension arm 412 and a vehicle-wheel direction is changed.

As the brake apparatus, application of a disk brake or a drum brake is suitable. According to the present embodiment, as the brake apparatus, a disk rotor **431** that is fixed to the rotation shaft **501** of the rotating electric machine **500** and a brake caliper **432** that is fixed to the base plate **405** on the rotating electric machine **500** side are provided. In the brake caliper **432**, a brake pad is operated by hydraulic pressure or the like. As a result of the brake pad being pressed against the disk rotor **431**, braking force caused by friction is generated and rotation of the vehicle wheel **400** is stopped.

In addition, a housing duct **440** that houses electrical wiring **H1** and a cooling pipe **H2** that extend from the rotating electric machine **500** is attached to the vehicle wheel **400**. The housing duct **440** is provided so as to extend from an end portion on the fixed portion side of the rotating electric machine **500**, along an end surface of the rotating electric machine **500**, and avoid the suspension arm **412**. The housing duct **440** is fixed to the suspension arm **412** in this state.

As a result, a connection portion to the housing duct **440** of the suspension arm **412** has a fixed positional relationship with the base plate **405**. Therefore, stress that is generated in the electrical wiring **H1** and the cooling pipe **H2** as a result of vibrations in the vehicle and the like can be suppressed. Here, the electrical wiring **H1** is connected to an onboard power supply unit and an onboard electronic control unit (ECU) (not shown). The cooling pipe **H2** is connected to a radiator (not shown).

Next, a configuration of the rotating electric machine **500** that is used as the in-wheel motor will be described in detail. According to the present embodiment, an example in which the rotating electric machine **500** is applied to the in-wheel motor is given. The rotating electric machine **500** has superior efficiency and output compared to a motor of a vehicle drive unit that has a speed reducer as in conventional technology.

That is, if the rotating electric machine **500** is used for a purpose that enables actualization of more practical pricing (lower pricing), compared to conventional technology, through cost reduction, the rotating electric machine **500** may also be used as a motor for purposes other than the vehicle drive unit. In such cases as well, in a manner similar to that when the rotating electric machine **500** is applied to the in-wheel motor, superior performance is exhibited. Here, operation efficiency refers to an index that is used during testing in traveling mode to derive fuel efficiency of a vehicle.

An overview of the rotating electric machine **500** is shown in FIGS. **48** to **51**. FIG. **48** is a side view of the rotating electric machine **500** viewed from a protruding side of the rotation shaft **501** (inner side of the vehicle).

FIG. **49** is a longitudinal cross-sectional view of the rotating electric machine **500** (a cross-sectional view taken along line **49-49** in FIG. **48**). FIG. **50** is a lateral cross-sectional view of the rotating electric machine **500** (a cross-sectional view taken along line **50-50** in FIG. **49**). FIG. **51** is an exploded cross-sectional view in which constituent elements of the rotating electric machine **500** are in an exploded state. In the description below, a direction in which the rotation shaft **501** extends in an outer-side direction of the vehicle body in FIG. **51** is an axial direction. A direction that radially extends from the rotation shaft **501** is a radial direction.

In FIG. **48**, on a center line that is drawn to form a cross-section **49** that passes through a center of the rotation shaft **501**, that is, a rotational center of a rotating portion,

each of two directions that extend in a circumferential manner from an arbitrary point excluding the rotational center of the rotation portion is a circumferential direction. In other words, the circumferential direction may be either of a clockwise direction and a counter-clockwise direction with an arbitrary point on the cross-section **49** as a starting point.

In addition, in terms of a vehicle-mounted state, a right side in FIG. **49** is a vehicle outer side and a left side is a vehicle inner side. In other words, in terms of the vehicle-mounted state, a rotor **510** described hereafter is arranged further toward the outer-side direction of the vehicle body than a rotor cover **670**.

The rotating electric machine **500** according to the present embodiment is an outer-rotor-type, surface-magnet-type rotating electric machine. The rotating electric machine **500** generally includes the rotor **510**, a stator **520**, an inverter unit **530**, a bearing **560**, and the rotor cover **670**. The rotating electric machine **10** is configured by all of these components being arranged coaxially with the rotation shaft **501** that is provided integrally with the rotor **510** and assembled in the axial direction in a predetermined order.

In the rotating electric machine **500**, the rotor **510** and the stator **520** each have a circular cylindrical shape and are arranged so as to oppose each other with an airgap therebetween. As a result of the rotor **510** integrally rotating with the rotation shaft **501**, the rotor **510** rotates on the radially outer side of the stator **520**. The rotor **510** corresponds to a "field element". The stator **520** corresponds to an "armature".

The rotor **510** includes an approximately circular cylindrical rotor carrier **511** and an annular magnet unit **512** that is fixed to the rotor carrier **511**. The rotation shaft **501** is fixed to the rotor carrier **511**.

The rotor carrier **511** includes a circular cylindrical portion **513**. The magnet unit **512** is fixed to an inner circumferential surface of the inner cylindrical portion **513**. That is, the magnet unit **512** is provided so as to be surrounded by the circular cylindrical portion **513** of the rotor carrier **511** from the radially outer side.

In addition, the circular cylindrical portion **513** includes a first end and a second end that are opposing in the axial direction thereof. The first end is positioned in a direction on the outer side of the vehicle body. The second end is positioned in a direction in which the base plate **405** is present. In the rotor carrier **511**, the first end of the circular cylindrical portion **513** is provided so as to be continuous with an end plate **514**.

That is, the circular cylindrical portion **513** and the end plate **514** are an integrated structure. The second end of the circular cylindrical portion **513** is open. For example, the rotor carrier **511** is formed by a cold-rolled steel sheet (SPCC or SPHC that has a thicker plate thickness than SPCC), a forging steel, a CFRP, or the like that has sufficient mechanical strength.

An axial length of the rotation shaft **501** is longer than a dimension in the axial direction of the rotor carrier **511**. In other words, the rotation shaft **501** protrudes toward the open end side (vehicle inner-side direction) of the rotor carrier **511**, and the above-described brake apparatus and the like are attached to the end portion on the protruding side.

A through hole **514a** is formed in a center portion of the end plate **514** of the rotor carrier **511**. The rotation shaft **501** is fixed to the rotor carrier **511** in a state in which the rotation shaft **501** is inserted into the through hole **514a** of the end plate **514**. The rotation shaft **501** has a flange **502** that extends so as to be oriented to intersect (be orthogonal to) the axial direction in a portion in which the rotor carrier **511**

is fixed. The rotation shaft **501** is fixed to the rotor carrier **511** in a state in which the flange and the surface on the vehicle outer side of the end plate **514** are surface-joined. Here, in the vehicle wheel **400**, the wheel **402** is fixed using a fastener such as a bolt that is erected in the direction of the vehicle outer side, from the flange **502** of the rotation shaft **501**.

In addition, the magnet unit **512** is configured by a plurality of permanent magnets that are arranged such that the polarities alternately change along the circumferential direction of the rotor **510**. As a result, the magnet unit **512** has a plurality of magnetic poles in the circumferential direction.

For example, the permanent magnet is fixed to the rotation carrier **511** by bonding. The magnet unit **512** has the configuration that is described as the magnet unit **42** in FIGS. **8** and **9** according to the first embodiment. As the permanent magnet, a sintered neodymium magnet of which the intrinsic coercive force is equal to or greater than 400 [kA/m], and the remanent flux density Br is equal to or greater than 1.0 [T] is used.

In a manner similar to the magnet unit **42** in FIG. **9** and the like, the magnet unit **512** includes the first magnet **91** and the second magnet **92** that are polar anisotropic magnets and of which the polarities differ from each other.

As described in FIGS. **8** and **9**, in each of the magnets **91** and **92**, the orientation of the easy axis of magnetization differs between the d-axis side (the portion located closer to the d-axis) and the q-axis side (the portion located closer to the q-axis). On the d-axis side, the orientation of the easy axis of magnetization is an orientation that is close to a direction that is parallel to the d-axis. On the q-axis side, the orientation of the easy axis of magnetization is an orientation that is close to a direction that is orthogonal to the q-axis. In addition, a magnet magnetic path that has a circular arc shape is formed as a result of orientation based on the orientations of the easy axes of magnetization.

Here, in each of the magnets **91** and **92**, the easy axis of magnetization on the d-axis side may have an orientation that is parallel to the d-axis and the easy axis of magnetization on the q-axis side may have an orientation that is orthogonal to the q-axis. In short, the magnet unit **239** is configured to be oriented such that, on the d-axis side that is the magnetic pole center, the orientation of the easy axis of magnetization is parallel to the d-axis compared to the side of the q-axis that is the magnetic pole boundary.

As a result of the magnets **91** and **92**, the magnet magnetic flux on the d-axis is strengthened and changes in the magnetic flux near the q-axis are suppressed. As a result, the magnets **91** and **92** of which the changes in surface magnetic flux from the q-axis to the d-axis is gradual at each magnetic pole can be suitably implemented. As the magnet unit **512**, the configuration of the magnet unit **42** shown in FIGS. **22** and **23**, or the configuration of the magnet unit **42** shown in FIG. **30** can also be used.

Here, the magnet unit **512** may have a stator core (back yoke) that includes a plurality of electromagnetic steel sheets being laminated in the axial direction on the side of the circular cylindrical portion **513** of the rotor carrier **511**, that is, the outer circumferential surface side. That is, the rotor core may be provided on the radially inner side of the circular cylindrical portion **513** of the rotor carrier **511**, and the permanent magnet (magnets **91** and **92**) is provided on the radially inner side of the rotor core.

As shown in FIG. **47**, recess portions **513a** are formed in a direction that extends in the axial direction at predetermined intervals in the circumferential direction in the cir-

cular cylindrical portion **513** of the rotor carrier **511**. For example, the recess portions **513a** are formed by press machining. As shown in FIG. **52**, a protruding portion **513b** is formed on the inner circumferential surface side of the circular cylindrical portion **513**, in a position that is on a back side of the recess portion **513a**. Meanwhile, on the outer circumferential surface side of the magnet unit **512**, the recess portion **512a** is formed to match the protruding portion **513b** of the circular cylindrical portion **513b**.

As a result of the protruding portion **513b** of the circular cylindrical portion **513** entering the recess portion **512a**, positional shifting in the circumferential direction of the magnet unit **512** is suppressed. That is, the protruding portion **513** on the rotor carrier **511** side functions as a rotation stopping portion of the magnet unit **512**. Here, a method for forming the protruding portion **513b** is arbitrary and may be other than press machining.

In FIG. **52**, the direction of the magnet magnetic path in the magnet unit **512** is indicated by an arrow. The magnet magnetic path extends in a circular arc shape so as to straddle the q-axis that is the magnetic pole boundary. In addition, on the d-axis that is the magnetic pole center, the magnet magnetic path is oriented to be parallel or close to parallel to the d-axis. In the magnet unit **512**, the recess portion **512b** is formed for each position corresponding to the q-axis on the inner circumferential surface side.

In this case, in the magnet unit **512**, the length of the magnet magnetic path differs between that on a side close to the stator **520** (lower side in the drawing) and that on a side away from the stator **520** (upper side in the drawing). The length of the magnet magnetic path is shorter on the side closer to the stator **520**. The recess portion **512b** is formed in a position at which the length of the magnet magnetic path is the shortest.

That is, in the magnet unit **512**, taking into consideration the difficulty in generating sufficient magnet magnetic flux in a location in which the length of the magnet magnetic path is short, the magnet is eliminated in the location at which the magnet magnetic flux is weak.

Here, an effective magnetic flux density Bd of a magnet increases as a length of a magnetic circuit passing through the interior of the magnet becomes longer. In addition, a permeance coefficient Pc and the effective magnetic flux density Bd of the magnet have a relationship in which when one increases, the other increases. In FIG. **52**, described above, the amount of magnets can be reduced while decrease in the permeance coefficient Pc that is an indicator of the magnitude of the effective magnetic flux density Bd of the magnet is suppressed.

Here, in B-H coordinates, an intersecting point between a permeance straight line and a demagnetization curved line based on the shape of the magnet is an operation point. The magnetic flux density at the operation point is the effective magnetic flux density Bd of the magnet. In the rotating electric machine **500** according to the present embodiment, an amount of iron in the stator **520** is reduced. In this configuration, the approach in which the magnetic circuit straddles the q-axis is very effective.

In addition, the recess portion **512b** of the magnet unit **512** can be used as an air passage that extends in the axial direction. Therefore, air cooling performance can also be improved.

Next, the configuration of the stator **520** will be described. The stator **520** includes a stator winding **521** and a stator core **522**. FIG. **53** is a perspective view of the stator winding **521** and the stator core **522** in an exploded state.

The stator winding **521** is made of a plurality of phase windings that are formed so as to be wound into an approximately cylindrical shape (annular shape). The stator core **522** that serves as a base member is assembled to the radially inner side of the stator winding **521**. According to the present embodiment, as a result of phase windings of the U-phase, V-phase, and W-phase being used, the stator winding **521** is configured as phase windings of three phases. Each phase winding is configured by two layers of conductors **523** on the inner side and the radially outer side. In a manner similar to the stator **50** described earlier, the stator **520** is characterized by having a slot-less structure and a flattened conductor structure in the stator winding **521**. The stator **520** has a configuration that is similar to or like the stator **50** shown in FIGS. **8** to **16**.

The configuration of the stator core **522** will be described. In a manner similar to the stator core **52** described earlier, the stator core **522** is that in which a plurality of electromagnetic steel sheets are laminated in the axial direction and has a circular cylindrical shape that has a predetermined thickness in the radial direction. The stator winding **521** is assembled to the stator core **522** on the radially outer side that is the rotor **510** side. The outer circumferential surface of the stator core **522** has a curved surface shape that has substantially no unevenness. In a state in which the stator winding **521** is assembled thereto, the conductors **523** that configure the stator winding **521** are arranged so as to be arrayed in the circumferential direction on the outer circumferential surface of the stator core **522**. The stator core **522** functions as a back core.

The stator **520** may be that which uses any of (A) to (C), below.

- (A) In the stator **520**, an inter-conductor member is provided between the conductors **523** in the circumferential direction, and when the width dimension in the circumferential direction of the inter-conductor member in a single magnetic pole is W_t , the saturation magnetic density of the inter-conductor member is B_s , the width dimension in the circumferential direction of the magnet unit **512** in a single magnetic pole is W_m , and the residual magnetic flux density of the magnet unit **512** is B_r , a magnetic material in which a relationship expressed by $W_t \times B_s \leq W_m \times B_r$ is satisfied is used as the inter-conductor member.
- (B) In the stator **520**, the inter-conductor member is provided between the conductors **523** in the circumferential direction, and a non-magnetic material is used as the inter-conductor member.
- (C) In the stator **520**, the inter-conductor member is not provided between the conductors **523** in the circumferential direction.

As a result of the configuration of the stator **520** such as this, inductance is reduced compared to a rotating electric machine that has a typical teeth structure in which teeth (core) for establishing a magnetic path is provided between the conductor portions that serve as the stator winding. Specifically, the inductance can be made $1/10$ or less. In this case, because impedance decreases in accompaniment with the decrease in inductance, output power relative to input power of the rotating electric machine **500** is increased.

Furthermore, this configuration can contribute to increase in torque. In addition, compared to a rotating electric machine that uses an embedded-magnet-type rotor in which torque output is performed using a voltage of an impedance component (in other words, using reluctance torque), a high-output rotating electric machine can be provided.

According to the present embodiment, the stator winding **521** is configured to be integrally molded from a molding material (insulation member) that is made of resin or the like, together with the stator core **522**. The mold material is interposed between the conductors **523** that are arrayed in the circumferential direction. Based on this structure, the stator **520** according to the present embodiment corresponds to configuration (B), among (A) to (C), described above.

In addition, the conductors **523** that are adjacent to each other in the circumferential direction are such that end surfaces in the circumferential direction are in contact with each other or are closely arranged with a minute gap therebetween. Based on this configuration, the stator **520** may have configuration (C), described above. Here, when configuration (A), described above, is used, a protruding portion may be provided on the outer circumferential surface of the stator core **522** to match an orientation of the conductors **523** in the axial direction, that is, for example, to match a skew angle if the stator winding **521** has a skewed structure.

Next, the configuration of the stator winding **521** will be described with reference to FIG. **54** by (a) and (b). FIG. **54** illustrates, by (a) and (b), front views in which the stator winding **521** is developed in a planar manner. FIG. **54** shows, by (a), each conductor **523** that is positioned on the outer layer in the radial direction. FIG. **54** shows, by (b), each conductor **523** that is positioned in the inner layer in the radial direction.

The stator winding **521** is formed by being wound into a circular annular shape by distributed winding. In the stator winding **521**, a conductor material is wound in two layers on the inner side and the radially outer side. In addition, skewing is applied in differing directions between the conductors **523** on the inner layer side and the outer layer side (see FIG. **54** by (a) and (b)). The conductors **523** are mutually insulated. The conductor **523** may be configured as a bundle of a plurality of wires **86** (see FIG. **13**).

In addition, for example, the conductors **523** that are of the same phase and that have the same energization direction are provided so as to be arrayed two at a time in the circumferential direction. In the stator winding **521**, a single conductor portion of the same phase is configured by the conductors **523** that are in two layers in the radial direction and two conductors in the circumferential direction (that is, a total of four conductors). The conductor portion is provided one each within a single magnetic pole.

In the conductor portion, a thickness dimension in the radial direction thereof is preferably smaller than a width dimension in the circumferential direction corresponding to a single phase within a single magnetic pole. The stator winding **521** preferably has a flattened conductor structure, as a result. Specifically, for example, in the stator winding **521**, a single conductor portion of the same phase may be configured by the conductors **523** that are in two layers in the radial direction and four conductors in the circumferential direction (that is, a total of eight conductors).

Alternatively, on a conductor cross-section of the stator winding **521** shown in FIG. **50**, the width dimension in the circumferential direction may be greater than the thickness dimension in the radial direction. The stator winding **51** shown in FIG. **12** can also be used as the stator winding **521**. However, in this case, a space for housing the coil end of the stator winding is required to be secured inside the rotor carrier **511**.

In the stator winding **521**, the conductors **523** are arranged in an array in the circumferential direction so as to be tilted at a predetermined angle relative to the stator core **522**, in

coil sides 525 that overlap on the inner side and the radially outer side. In addition, the conductors 523 are reversed (doubled back) toward the inner side in the axial direction at coil ends 526 on both sides that are further on the outer side in the axial direction than the stator core 522, and continuously connected.

In FIG. 54 by (a), an area that serves as the coil side 525 and an area that serves as the coil end 526 are each shown. The conductor 523 on the inner layer side and the conductor 523 on the outer layer side are connected to each other at the coil end 526. As a result, each time the conductor 523 is reversed (each time the conductor 523 is doubled back) in the axial direction at the coil end 526, the conductor 523 alternately switches between the inner layer side and the outer layer side. In other words, the stator winding 521 is configured such that, in the conductors 523 that are continuous in the circumferential direction, switching between inner and outer layers is performed to match a reversal of a direction of a current.

In addition, in the stator winding 521, two types of skewing of which skew angles differ between that of end portion areas that are both ends in the axial direction and that of a center area that is sandwiched between the end portion areas are applied.

That is, as shown in FIG. 55, in the conductor 523, a skew angle $\theta s1$ of the center area and a skew angle $\theta s2$ of the end portion area differ. The skew angle $\theta s1$ is smaller than the skew angle $\theta s2$. The end portion area is prescribed as an area that includes the coil side 525 in the axial direction. The skew angle $\theta s1$ and the skew angle $\theta s2$ are tilt angles at which the conductors 523 are tilted relative to the axial direction. The skew angle $\theta s1$ of the center area may be prescribed to be an angle range that is appropriate for eliminating harmonic components of the magnetic flux that are generated as a result of energization of the stator winding 521.

As a result of the skew angles of the conductor 523 in the stator winding 521 differing between that of the center area and that of the end portion areas, and the skew angle $\theta s1$ of the center area being smaller than the skew angle $\theta s2$ of the end portion areas, a winding factor of the stator winding 521 can be increased while reduction of the coil end 526 is achieved.

In other words, a length of the coil end 526, that is, a conductor length of the portion that projects out from the stator core 522 in the axial direction can be shortened while a desired winding factor is ensured. As a result, torque enhancement can be implemented while size reduction of the rotating electric machine 50 is implemented.

Here, an appropriate range of the skew angle $\theta s1$ of the center area will be described. When an X-number of conductors 523 are arranged within a single magnetic pole in the stator winding 521, an X-order harmonic component being generated as a result of the energization of the stator winding 521 can be considered. When the number of phases is S and the number of pairs is m, $X=2 \times S \times m$.

The disclosers of the present application have focused on the following. That is, because the X-order harmonic component is a component that composes a composite wave of an X-1-order harmonic component and an X+1-order harmonic component, the X-order harmonic component can be reduced as a result of at least either of the X-1-order harmonic component and the X+1-order harmonic component being reduced. In light of this focus, the disclosers of the present application have found that the X-order harmonic component can be reduced as a result of the skew

angle $\theta s1$ being set within an angle range of $360^\circ/(X+1)$ to $360^\circ/(X-1)$ in electrical angles.

For example, when $S=3$ and $m=2$, to reduce the harmonic component of X=12th order, the skew angle $\theta s1$ is set within an angle range of $360^\circ/13$ to $360^\circ/11$. That is, the skew angle $\theta s1$ may be set to an angle within a range of 27.7° to 32.7° .

As a result of the skew angle $\theta s1$ of the center area being set as described above, in the center area, the NS-alternating magnet magnetic flux can be actively interlinked. The winding factor of the stator winding 521 can be increased.

The skew angle $\theta s2$ of the end portion area is an angle that is greater than the skew angle $\theta s1$ of the center area, described above. In this case, the angle range of the skew angle $\theta s2$ is $\theta s1 < \theta s2 < 90^\circ$.

In addition, in the stator winding 521, the conductor 523 on the inner layer side and the conductor 523 on the outer layer side may be connected by welding or bonding of the end portions of the conductors 523. Alternatively, the conductor 523 on the inner layer side and the conductor 523 on the outer layer side may be connected by bending them. In the stator winding 521, the end portion of each phase winding is electrically connected to a power converter (inverter) by a bus bar or the like in the coil end 526 on one side (that is, one end side in the axial direction), of the coil ends 526 on both sides in the axial direction. Therefore, here, a configuration in which the conductors are connected to each other in the coil end 526 will be described, while differentiation is made between the coil end 526 on the bus-bar connection side and the coil end 526 on an opposite side thereof.

As a first configuration, the conductors 523 are connected by welding in the coil ends 526 on the bus-bar connection side, and the conductors 523 are connected by a means other than welding in the coil ends 526 on the opposite side thereof.

For example, as a means other than welding, connection by bending of the conductor material can be considered. In the coil end 526 on the bus-bar connection side, the bus bar being welded to the end portions of the phase windings can be assumed. Therefore, as a result of the configuration in which the conductors 523 are connected by welding in the same coil end 526 thereof, the welding portion can be performed in a series of steps and work efficiency can be improved.

As a second configuration, the conductors 523 are connected by a means other than welding in the coil ends 526 on the bus-bar connection side, and the conductors 523 are connected by welding in the coil ends 526 on the opposite side thereof.

In this case, if the conductors 523 are connected by welding in the coil ends 526 on the bus-bar connection side, a need to keep sufficient separation distance between the bus bar and the coil ends 526 to prevent contact between the welding portion and the bus bar arises. However, as a result of the present configuration, the separation distance between the bus bar and the coil ends 526 can be reduced. As a result, restrictions related to the length of the stator winding 521 in the axial direction or the bus bar can be relaxed.

As a third configuration, the conductors 523 are connected by welding in the coil ends 526 on both sides in the axial direction. In this case, all of the conductor materials that are prepared before welding can be of a short wire length. Work efficiency can be improved through elimination of a bending step.

As a fourth configuration, the conductors 523 are connected by a means other than welding in the coil ends 526 on both sides in the axial direction. In this case, sections in

which welding is performed can be minimized in the stator winding **521**. Concern regarding insulation peeling occurring at a welding step can be reduced.

In addition, at a step of fabricating the circular annular stator winding **521**, a strip-shaped winding that is aligned in a planar shape may be fabricated, and the strip-shaped winding may subsequently be formed into an annular shape. In this case, in a state in which the stator winding is in the form of the planar, strip-shaped winding, welding of the conductors at the coil ends **526** may be performed as required.

When the planar, strip-shaped winding is formed into the annular shape, the strip-shaped winding may be formed into an annular shape using a circular columnar tool that has the same diameter as the stator core **522**, by the winding being wrapped around the circular columnar tool. Alternatively, the strip-shaped winding may be directly wrapped around the stator core **522**.

Here, the configuration of the stator winding **521** can also be modified in the following manner.

For example, in the stator winding **521** shown in FIG. **54** by (a) and (b), the skew angles of the center area and the end portion area may be the same.

In addition, in the stator winding **521** shown in FIG. **54** by (a) and (b), the end portions of the conductors **523** of the same phase that are adjacent to each other in the circumferential direction may be connected to each other by a crossover wire that extends in a direction that is orthogonal to the axial direction.

The number of layers of the stator winding **521** is merely required to be $2 \times n$ layers (n being a natural number). The stator winding **521** can have four layers, six layers, or the like, instead of two layers.

Next, the inverter unit **530** that is a power conversion unit will be described. Here, a configuration of the inverter unit **530** will be described with reference to FIGS. **56** and **57** that are exploded cross-sectional views of the inverter unit **530**. Here, FIG. **57** shows components shown in FIG. **56** as two subassemblies.

The inverter unit **530** includes an inverter housing **531**, a plurality of electrical modules **532** that are assembled to the inverter housing **531**, and a bus bar module **533** that electrically connects the electrical modules **532**.

The inverter housing **531** includes an outer wall member **541**, an inner wall member **542**, and a boss formation member **543**. The outer wall member **541** has a circular cylindrical shape. The inner wall member **542** has a circular cylindrical shape of which an outer circumference diameter is smaller than a diameter of the outer wall member **541**, and is arranged on the radially inner side of the outer wall member **541**. The boss formation member **543** is fixed to one end side in the axial direction of the inner wall member **542**.

The members **541** to **543** are preferably made of a conductive material, and for example, is made of a CFRP. The inverter housing **531** is configured by the outer wall member **541** and the inner wall member **542** being assembled so as to be overlapped on the inner side and the radially outer side, and the boss formation member **543** being assembled to one end side in the axial direction of the inner wall member **542**. This assembled state is the state shown in FIG. **57**.

The stator core **522** is fixed to the radially outer side of the outer wall member **541** of the inverter housing **531**. As a result, the stator **520** and the inverter unit **530** are integrated.

As shown in FIG. **56**, a plurality of recess portions **541a**, **541b**, and **541c** are formed on an inner circumferential surface of the outer wall member **541**. In addition, a plurality

of recess portions **542a**, **542b**, and **542c** are formed on an outer circumferential surface of the inner wall member **542**. Furthermore, as a result of the outer wall member **541** and the inner wall member **542** being assembled together, three hollow portions **544a**, **544b**, and **544c** are formed between the outer wall member **541** and the inner wall member **542** (see FIG. **57**).

Among the hollow portions **544a**, **544b**, and **544c**, the hollow portion **544b** in the center is used as a cooling water passage **545** through which cooling water that serves as a coolant flows. In addition, a sealing member **546** is housed in the hollow portions **544a** and **544c** on both sides sandwiching the hollow portion **544b** (cooling water passage **545**). The hollow portion **544b** (cooling water passage **545**) is sealed as a result of the sealing member **546**. The cooling water passage **545** will be described in detail hereafter.

In addition, in the boss formation member **543**, an end plate **547** that has a circular-disk ring shape, and a boss portion **548** that protrudes from the end plate **547** toward a housing interior are provided. The boss portion **548** is provided in a hollow cylindrical shape.

For example, as shown in FIG. **51**, of a first end of the inner wall member **542** in the axial direction and a second end on the protruding side (that is, the vehicle inner side) of the rotation shaft **501** that opposes the first end, the boss formation member **543** is fixed to the second end. Here, in the vehicle wheel **400** shown in FIGS. **45** to **47**, the base plate **405** is fixed to the inverter housing **531** (more specifically, the end plate **547** of the boss formation member **543**).

The inverter housing **531** is configured to have a double layer of peripheral walls in the radial direction with the axial center as a center. The peripheral wall on the outer side of the double layer of peripheral walls is formed by the outer wall member **541** and the inner wall member **542**. The peripheral wall on the inner side is formed by the boss portion **548**.

Here, in the description below, the peripheral wall on the outer side that is formed by the outer wall member **541** and the inner wall member **542** is also referred to as an "outer peripheral wall WA1", and the peripheral wall on the inner side that is formed by the boss portion **548** is also referred to as an "inner peripheral wall WA2".

An annular space is formed between the outer peripheral wall WA1 and the inner peripheral wall WA2 in the inverter housing **531**. The plurality of electrical modules **532** are arranged so as to be arrayed in the circumferential direction inside the annular space. The electrical module **532** is fixed to the inner circumferential surface of the inner wall member **542** by bonding, screw-fastening, or the like. According to the present embodiment, the inverter housing **531** corresponds to a "housing member". The electrical module **532** corresponds to an "electrical component".

The bearing **560** is housed on the inner side of the inner peripheral wall WA2 (boss portion **548**). The rotation shaft **501** is supported by the bearing **560** so as to freely rotate. The bearing **560** is a hub bearing that rotatably supports the vehicle wheel **400** in a vehicle-wheel center portion. The bearing **560** is provided in a position that overlaps the rotor **510**, the stator **520**, and the inverter unit **530** in the axial direction.

In the rotating electric machine **500** according to the present embodiment, as a result of the magnet unit **512** being able to be made thinner in accompaniment with the orientation in the rotor **510**, and the slot-less structure and the flattened conductor structure being used in the stator **520**, the thickness dimension in the radial direction of the magnetic

circuit portion can be reduced and the hollow space further toward the radially inner side than the magnetic circuit portion is can be expanded.

As a result, arrangement of the magnetic circuit portion, the inverter unit **530**, and the bearing **560** in a state in which the magnetic circuit portion, the inverter unit **530**, and the bearing **560** are laminated in the radial direction becomes possible. The boss portion **548** serves as a bearing holding portion that holds the bearing **560** on the inner side thereof.

For example, the bearing **560** is a radial ball bearing. The bearing **560** includes an inner ring **561**, an outer ring **562**, and a plurality of balls **563**. The inner ring **561** forms a cylindrical shape. The outer ring **562** forms a cylindrical shape that has a larger diameter than the inner ring and is arranged on the radially outer side of the inner ring **561**. The plurality of balls **563** are arranged between the inner ring **561** and the outer ring **562**. The bearing **560** is fixed to the inverter housing **531** by the outer ring **562** being assembled to the boss formation member **543**, and the inner ring **561** is fixed to the rotation shaft **501**. This inner ring **561**, outer ring **562**, and balls **563** are all made of a metal material such as carbon steel.

In addition, the inner ring **561** of the bearing **560** has a cylindrical portion **561a** that houses the rotation shaft **501** and a flange **561b** that extends in a direction that intersects (is orthogonal to) the axial direction from one end portion in the axial direction of the cylindrical portion **561a**. The flange **561b** is a portion that is in contact with the end plate **514** of the rotor carrier **511** from the inner side.

In a state in which the bearing **560** is assembled to the rotation shaft **501**, the rotor carrier **511** is held so as to be sandwiched between the flange **502** of the rotation shaft **501** and the flange **561b** of the inner ring **561**. In this case, the flange **502** of the rotation shaft **501** and the flange **561b** of the inner ring have the same angle of intersection relative to the axial direction as each other (according to the present embodiment, both are right angles). The rotor carrier **511** is held so as to be sandwiched between these flanges **502** and **561b**.

The rotor carrier **511** is supported from the inner side by the inner ring **561** of the bearing **560**. In this configuration, an angle of the rotor carrier **511** relative to the rotation shaft **501** can be held at an appropriate angle. Furthermore, a degree of parallelism of the magnet unit **512** relative to the rotation shaft **501** can be favorably maintained. As a result, even when the rotor carrier **511** is expanded in the radial direction, resistance against vibration and the like can be improved.

Next, the electrical modules **532** that are housed in the inverter housing **531** will be described.

The plurality of electrical modules **532** are that in which electrical components such as the semiconductor switching element that configures the power converter and the smoothing capacitor are divided into a plurality of groups and individually modularized. The electrical modules **532** include a switch module **532A** that includes the semiconductor switching element that is a power element, and a capacitor module **532B** that includes the smoothing capacitor.

As shown in FIGS. **49** and **50**, a plurality of spacers **549** that have flat surfaces for attaching the electrical modules **532** are fixed to the inner circumferential surface of the inner wall member **542**. The electrical module **532** is attached to the spacer **549**. That is, whereas the inner circumferential surface of the inner wall member **542** is a curved surface, an attachment surface of the electrical module **532** is a flat surface. Therefore, a flat surface is formed on the inner

circumferential surface side of the inner wall member **542** by the spacer **549**, and the electrical module **532** is fixed to the flat surface.

Here, the configuration in which the spacer **549** is interposed between the inner wall member **542** and the electrical module **532** is not a requisite. The electrical module **532** can also be directly attached to the inner wall member **542** by the inner circumferential surface of the inner wall member **542** being a flat surface or the attachment surface of the electrical module **532** being a curved surface.

In addition, the electrical module **532** can also be fixed to the inverter housing **531** in a state in which the electrical module **532** is not in contact with the inner circumferential surface of the inner wall member **542**. For example, the electrical module **532** is fixed to the end plate **547** of the boss formation member **543**. The switch module **532A** can be fixed in a state of contact with the inner circumferential surface of the inner wall member **542**, and the capacitor module **532B** can be fixed in a state of non-contact with the inner circumferential surface of the inner wall member **542**.

Here, when the spacer **549** is provided on the inner circumferential surface of the inner wall member **542**, the outer peripheral wall **WA1** and the spacer **549** correspond to a "cylindrical portion". In addition, when the spacer **549** is not used, the outer peripheral wall **WA1** corresponds to the "cylindrical portion".

As described above, the cooling water passage **545** through which the cooling water that serves as a coolant flows is formed in the outer peripheral wall **WA1** of the inverter housing **531**. Each electrical module **532** is cooled by the cooling water that flows through the cooling water passage **545**.

Here, as the coolant, a cooling oil can also be used instead of the cooling water. The cooling water passage **545** is provided in an annular shape along the outer peripheral wall **WA1**. The cooling water that flows through the cooling water passage **545** flows from an upstream side to a downstream side via each electrical module **532**. According to the present embodiment, the cooling water passage **545** is provided in an annular shape so as to overlap each electrical module **532** on the inner side and the radially outer side and surround each electrical module **532**.

The inner wall member **542** is provided with an inlet passage **571** through which the cooling water flows into the cooling water passage **545**, and an outlet passage **572** through which the cooling water flows out from the cooling water passage **545**. The plurality of electrical modules **532** are fixed to the inner circumferential surface of the inner wall member **542** as described above.

In this configuration, a space between the electrical modules that are adjacent in the circumferential direction is more expanded in a single location than other spaces. A protruding portion **573** in which a portion of the inner wall member **542** protrudes toward the radially inner side is formed in the expanded portion. In addition, the inlet passage **571** and the outlet passage **572** are provided so as to be laterally arrayed along the radial direction in the protruding portion **573**.

A state of the arrangement of the electrical modules **532** in the inverter housing **531** is shown in FIG. **58**. Here, FIG. **58** is the same longitudinal cross-sectional view as FIG. **50**.

As shown in FIG. **58**, the electrical modules **532** are arranged so as to be arrayed in the circumferential direction with an interval between the electrical modules in the circumferential direction being a first interval INT1 or a second interval INT2. The second interval INT2 is an interval that is wider than the first interval INT1. For example, each of the intervals INT1 and INT2 is a distance

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between center positions of two electrical modules **532** that are adjacent in the circumferential direction.

In this case, the interval between the electrical modules that are adjacent in the circumferential direction without the protruding portion **573** therebetween is the first interval INT1. The interval between the electrical modules that are adjacent in the circumferential direction with the protruding portion **573** therebetween is the second interval INT2. That is, the interval between the electrical modules that are adjacent in the circumferential direction is widened in a portion thereof. The protruding portion **573** is provided, for example, in a portion that is a center of the widened interval (second interval INT2).

The intervals INT1 and INT2 may be a circular arc distance between the center positions of the two electrical modules **532** that are adjacent in the circumferential direction, on the same circle around the rotation shaft **51**. Alternatively, the interval between the electrical modules in the circumferential direction may be defined by angle intervals $\theta i1$ and $\theta i2$ with the rotation shaft **501** as a center ($\theta i1 < \theta i2$).

Here, in FIG. **58**, the electrical modules **532** that are arrayed at the first interval INT1 are arranged in a state in which the electrical modules **532** are separated from each other in the circumferential direction (state of non-contact). However, instead of this configuration, the electrical modules **532** may be arranged in a state in which the electrical modules **532** are in contact with each other in the circumferential direction.

As shown in FIG. **48**, a water-flow port **574** in which passage end portions of the inlet passage **571** and the outlet passage **572** are formed is provided in the end plate **547** of the boss formation member **543**. A circulation path **575** that circulates the cooling water is connected to the inlet passage **571** and the outlet passage **572**. The circulation path **575** is made of a cooling water pipe. A pump **576** and a heat releasing apparatus **577** are provided on the circulation path **575**. The cooling water circulates through the cooling water passage **545** and the circulation path **575** in accompaniment with driving of the pump **576**. The pump **576** is an electric pump. For example, the heat releasing apparatus **577** is a radiator that releases heat from the cooling water into the atmosphere.

As shown in FIG. **50**, the stator **520** is arranged on the outer side of the outer peripheral wall WA1 and the electrical modules **532** are arranged on the inner side. Therefore, heat from the stator **520** is transmitted to the outer peripheral wall WA1 from the outer side. In addition, heat from the electrical modules **532** is transmitted to the outer peripheral wall WA1 from the inner side.

In this case, the stator **50** and the electrical modules **532** can be simultaneously cooled by the cooling water that flows through the cooling water passage **545**. Heat from heat generating components of the rotating electric machine **500** can be efficiently released.

Here, an electrical configuration of the power converter will be described with reference to FIG. **59**.

As shown in FIG. **59**, the stator winding **521** is made of the U-phase winding, the V-phase winding, and the W-phase winding. An inverter **600** is connected to the stator winding **521**. The inverter **600** is configured by a full-bridge circuit that includes the same number of upper and lower arms as the number of phases. The inverter **600** is provided with a serial-connection body that is made of an upper arm switch **601** and a lower arm switch **602**, for each phase. The switches **601** and **602** are each turned on/off by a drive circuit **603**. The winding of each phase is energized based on the on/off of the switches **601** and **602**.

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For example, each of the switches **601** and **602** is made of a semiconductor switching element, such as a MOSFET or an IGBT. In addition, a charge-supplying capacitor **604** that supplies the switches **601** and **602** with electric charge that is required during switching is connected in parallel to the serial-connection body of the switches **601** and **602** in the upper and lower arms of each phase.

A control apparatus **607** includes a microcomputer that includes a CPU and various memories. The control apparatus **607** performs energization control through switching on/off of the switches **601** and **602** based on various types of detection information of the rotating electric machine **500**, and requests for power-running drive and power generation.

For example, the control apparatus **607** performs on/off control of the switches **601** and **602** by PWM control at a predetermined switching frequency (carrier frequency) or rectangular wave control. The control apparatus **607** may be an internal control apparatus that is provided inside the rotating electric machine **500** or may be an external control apparatus that is provided outside the rotating electric machine **500**.

Here, in the rotating electric machine **500** according to the present embodiment, the electrical time constant decreases as a result of decrease in the inductance in the stator **520**. Under such circumstances in which the electrical time constant is small, the switching frequency (carrier frequency) is preferably increased and switching speed is preferably increased. In this regard, wiring inductance decreases as a result of the charge-supplying capacitor **604** being connected in parallel to the serial-connection body of the switches **601** and **602** of each phase. Appropriate surge measures can be taken even when the switching speed is increased.

A high-potential-side terminal of the inverter **600** is connected to a positive electrode terminal of a direct-current power supply **605**, and a low-potential-side terminal is connected to a negative electrode terminal (ground) of the direct-current power supply **605**. In addition, a smoothing capacitor **606** is connected to the high-potential-side terminal and the low-potential-side terminal of the inverter **600**, in parallel with the direct-current power supply **605**.

The switch module **532A** includes the switches **601** and **602** (semiconductor switching elements), the drive circuit **603** (specifically, an electrical element that configures the drive circuit **603**), and the charge-supplying capacitor **604** as heat generating components. In addition, the capacitor module **532B** includes the smoothing capacitor **606** as the heat generating component. A specific configuration example of the switch module **532A** is shown in FIG. **60**.

As shown in FIG. **60**, the switch module **532A** includes a module case **611** that serves as a housing case. In addition, the switch module **532A** includes the switches **601** and **602** that amount to a single phase, the drive circuit **603**, and the charge-supplying capacitor **604** that are housed inside the module case **611**. Here, the drive circuit **603** is configured as a dedicated IC or a circuit board, and is provided in the switch module **532A**.

For example, the module case **611** is made of an insulation material such as resin. The module case **611** is fixed to the outer peripheral wall WA1 in a state in which a side surface thereof is in contact with the inner circumferential surface of the inner wall member **542** of the inverter unit **530**.

An interior of the module case **611** is filled with a molding material such as resin. Inside the module case **611**, the switches **601** and **602** and the drive circuit **603**, and the switches **601** and **602** and the capacitor **604** are each electrically connected by wiring **612**. Here, specifically, the

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switch module **532A** is attached to the outer peripheral wall **WA1** with the spacer **549** therebetween. However, illustration of the spacer **549** is omitted.

In a state in which the switch module **532A** is fixed to the outer peripheral wall **WA1**, cooling performance is higher toward a side closer to the outer peripheral wall **WA1** in the switch module **532A**, that is, toward a side closer to the cooling water passage **545**. Therefore, an order of array of the switches **601** and **602**, the drive circuit **603**, and the capacitor **604** is prescribed based on the cooling performance.

Specifically, when amounts of heat generation are compared, the order from the greatest is the switches **601** and **602**, the capacitor **604**, and the drive circuit **603**. Therefore, the switches **601** and **602**, the capacitor **604**, and the drive circuit **603** are arranged in this order from the side closer to the outer peripheral wall **WA1** to match the order of magnitude of the amounts of heat generation. Here, a contact surface of the switch module **532A** may be smaller than a contactable surface of the inner circumferential surface of the inner wall member **542**.

Here, a detailed illustration of the capacitor module **532B** is omitted. However, the capacitor module **532B** is configured such that the capacitor **606** is housed inside a module case that has the same shape and size as the switch module **532A**. In a manner similar to the switch module **532A**, the capacitor module **532B** is fixed to the outer peripheral wall **WA1** in a state in which the side surface of the module case **611** is in contact with the inner circumferential surface of the inner wall member **542** of the inverter housing **531**.

The switch module **532A** and the capacitor module **532B** are not necessarily required to be concentrically arrayed on the radially inner side of the outer peripheral wall **WA1** of the inverter housing **531**. For example, the switch module **532A** may be arranged further toward the radially inner side than the capacitor module **532B** is. Alternatively, the switch module **532A** and the capacitor module **532B** may be arranged in reverse of the foregoing configuration.

During driving of the rotating electric machine **500**, heat exchange is performed between the switch module **532A** and the capacitor module **532B**, and the cooling water passage **545** via the inner wall member **542** of the outer peripheral wall **WA1**. As a result, cooling of the switch module **532A** and the capacitor module **532B** is performed.

The electrical module **532** may each be configured such that the cooling water is drawn into the interior thereof, and cooling by the cooling water is performed in the module interior. Here, a water-cooled structure of the switch module **532A** will be described with reference to FIG. **61** by (a) and (b). FIG. **61** shows, by (a), a longitudinal cross-sectional view of a cross-sectional structure of the switch module **532A** in a direction crossing the outer peripheral wall **WA1**. FIG. **61** shows, by (c), a cross-sectional view taken along line **61B-61B** in FIG. **61** by (a).

As shown in FIG. **61** by (a) and (b), in addition to including the module case **611**, the switches **601** and **602** corresponding to a single phase, the drive circuit **603**, and the capacitor **604** in a manner similar to that in FIG. **60**, the switch module **532A** includes a cooling apparatus that includes a pair of pipe portions **621** and **622**, and a cooler **623**.

In the cooling apparatus, the pair of pipe portions **621** and **622** are made of an inflow-side pipe portion **621** through which the cooling water flows into the cooler **623** from the cooling water passage **545** of the outer peripheral wall **WA1**, and an outflow-side pipe portion **622** from which the cooling

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water flows into the cooling water passage **545** from the cooler **623**. The cooler **623** is provided based on a cooling target.

In the cooling apparatus, a single stage or a plurality of stages of coolers **623** is used. In FIG. **61** by (a) and (b), two stages of coolers **623** are provided so as to be separated from each other in a direction away from the cooling water passage **545**, that is, the radial direction of the inverter unit **530**. The cooling water is supplied to each of the coolers **623** via the pair of pipe portions **621** and **622**. For example, the cooler **623** has an interior that is a hollow cavity. However, the interior of the cooler **623** may be provided with an inner fin.

In the configuration that includes the two stages of coolers **623**, each of (1) the outer peripheral wall **WA1** side of the first-stage cooler **623**, (2) between the first-stage and second-stage coolers **623**, and (3) the counter-outer peripheral wall side of the second-stage cooler **623** is a location in which an electrical component to be cooled is arranged.

These locations are (2), (1), (3) in order from that with the highest cooling performance. That is, the location that is sandwiched between the two coolers **623** has the highest cooling performance. In the locations that are adjacent to either one of the coolers **623**, the location closer to the outer peripheral wall **WA1** (cooling water passage **545**) has a higher cooling performance.

Taking this into consideration, as shown in FIG. **61** by (a) and (b), the switches **601** and **602** are arranged (2) between the first-stage and second-stage coolers **623**, the capacitor **604** is arranged on (1) the outer peripheral wall **WA1** side of the first-stage cooler **623**, and the drive circuit **603** is arranged on (3) the counter-outer peripheral wall side of the second-stage cooler **623**. Here, although not shown, the drive circuit **603** and the capacitor **604** may be arranged in reverse.

In any case, the switches **601** and **602** and the drive circuit **603**, and the switches **601** and **602** and the capacitor **604** are respectively connected by the wirings **612** inside the module case **611**. In addition, because the switches **601** and **602** are positioned between the drive circuit **603** and the capacitor **604**, the wiring **612** that extends toward the drive circuit **603** from the switches **601** and **602** and the wiring **612** that extends toward the capacitor **604** from the switches **601** and **602** have a relationship in which the wirings **612** extend in directions that are opposite each other.

As shown in FIG. **61** by (b), the pair of pipe portions **621** and **622** are arranged so as to be arrayed in the circumferential direction, that is, on an upstream side and a downstream side of the cooling water passage **545**. The cooling water flows from the inflow-side pipe portion **621** that is positioned on the upstream side into the cooler **623** and subsequently, the cooling water flows from the outflow-side pipe portion **622** that is positioned on the downstream side.

Here, to promote inflow of the cooling water into the cooling apparatus, the cooling water passage **545** may be provided with a regulating unit **624** that regulates the flow of cooling water, in a position between the inflow-side pipe portion **621** and the outflow-side pipe portion **622** when viewed in the circumferential direction. The restricting portion **624** may be a blocking portion that blocks the cooling water passage **545** or a narrowing portion that reduces a passage area of the cooling water passage **545**.

FIG. **62** shows, by (a) to (c), another cooling structure of the switch module **532A**. FIG. **62** shows, by (a), a longitudinal cross-sectional view of the cross-sectional structure of the switch module **532A** in a direction crossing the outer

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peripheral wall WA1. FIG. 62 shows, by (b), a cross-sectional view taken along line 62B-62B in FIG. 62 by (a).

In FIG. 62 by (a) and (b), as a difference with the configuration in FIG. 61 by (a) and (b), described above, the arrangement of the pair of pipe portions 621 and 622 in the cooling apparatus differs. The pair of pipe portions 621 and 622 are arranged so as to be arrayed in the axial direction.

In addition, as shown in FIG. 62 by (c), in the cooling water passage 545, a passage portion that communicates with the inflow-side pipe portion 621 and a passage portion that communicates with the outflow-side pipe portion 622 are provided so as to be separated in the axial direction. These passage portions communicate through the pipe portions 621 and 622 and the coolers 623.

In addition, a following configuration can also be used as the switch module 532A.

In a configuration shown in FIG. 63 by (a), compared to the configuration in FIG. 61 by (a), the cooler 623 is changed from two stages to one stage. In this case, the location that has the highest cooling performance inside the module case 611 differs from that in FIG. 61 by (a). The location on the outer peripheral wall WA1 side, of both sides in the radial direction of the cooler 623 (both sides in the left/right direction in the drawing), has the highest cooling performance.

Next, the cooling performance decreases in the order of a location on the counter-outer peripheral wall side of the cooler 623 and a location away from the cooler 623. Taking this into consideration, as shown in FIG. 63 by (a), the switches 601 and 602 are arranged in the location on the outer peripheral wall WA1 side, of both sides in the radial direction of the cooler 623 (both sides in the left/right direction in the drawing). The capacitor 604 is arranged in the location on the counter-outer peripheral wall side of the cooler 623. The drive circuit 603 is arranged in a location away from the cooler 623.

In addition, in the switch module 532A, the configuration in which the switches 601 and 602 corresponding to a single phase, the drive circuit 603, and the capacitor 604 are housed inside the module case 611 can be modified. For example, the switches 601 and 602 corresponding to a single phase and either of the drive circuit 603 and the capacitor 604 may be housed inside the module case 611.

In FIG. 63 by (b), inside the module case 611, in addition to the pair of pipe portions 621 and 622 and the two stages of coolers 623 being provided, the switches 601 and 602 are arranged between the first-stage and second-stage coolers 623, and the capacitor 604 or the drive circuit 603 is arranged on the outer peripheral wall WA1 side of the first-stage cooler 623. In addition, the switches 601 and 602 and the drive circuit 603 may be integrated into a semiconductor module, and the semiconductor module and the capacitor 604 may be housed inside the module case 611.

Here, in FIG. 63 by (b), in the switch module 532A, a capacitor may be arranged on a side opposite the switches 601 and 602 in at least either of the coolers 623 that are arranged on both sides sandwiching the switches 601 and 602. That is, the capacitor 604 may be arranged on only either of the outer peripheral wall WA1 side of the first-stage cooler 623 and the counter-peripheral wall side of the second-stage cooler 623. Alternatively, the capacitor 604 may be arranged on both sides.

According to the present embodiment, the cooling water is drawn into the module interior from the cooling water passage 545 for only the switch module 532A, of the switch module 532A and the capacitor module 532B. However, the configuration may be modified.

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The cooling water may be drawn into both modules 532A and 532B from the cooling water passage 545.

In addition, the cooling water may come into direct contact with the outer surface of each electrical module 532 and may cool each electrical module 532. For example, as shown in FIG. 64, the cooling water is in contact with the outer surface of the electrical module 532 due to the electrical module 532 being embedded in the outer peripheral wall WA1.

In this case, a configuration in which a portion of the electrical module 532 is immersed inside the cooling water passage 545, or a configuration in which the cooling water passage 545 is further expanded in the radial direction than that in the configuration in FIG. 58 and the like, and the overall electrical module 532 is immersed inside the cooling water passage 545 can be considered. When the electrical module 532 is immersed inside the cooling water passage 545, if a fin is provided in the immersed module case 611 (an immersed portion of the module case 611), cooling performance can be further improved.

In addition, the electrical modules 532 include the switch module 532A and the capacitor module 532B. When both are compared, there is a difference in the amount of heat generation. Taking this into consideration, the arrangement of the electrical modules 532 in the inverter housing 531 can be modified as well.

For example, as shown in FIG. 65, a plurality of switch modules 532A are arrayed in the circumferential direction without being dispersed and are arranged on the upstream side of the cooling water passage 545, that is, the side close to the inlet passage 571. In this case, the cooling water that flows in from the inlet passage 571 is first used to cool the three switch modules 532A and subsequently used to cool the capacitor modules 532B.

Here, in FIG. 65, the pair of pipe portions 621 and 622 are arranged so as to be arrayed in the axial direction as in FIG. 62 by (a) and (b), above. However, the arrangement is not limited thereto. The pair of pipe portions 621 and 622 may be arranged so as to be arrayed in the circumferential direction as in FIG. 61 by (a) and (b), above.

Next, a configuration related to the electrical connection of the electrical modules 532 and the bus bar module 533 will be described. FIG. 66 is a cross-sectional view taken along line 66-66 in FIG. 49. FIG. 67 is a cross-sectional view taken along line 67-67 in FIG. 49. FIG. 68 is a perspective view showing a bus bar module 533 alone. Here, the configuration related to the electrical connection between the electrical modules 532 and the bus bar module 533 will be described with reference to these drawings.

As shown in FIG. 66, in the inverter housing 531, three switch modules 532A are arranged so as to be arrayed in the circumferential direction in a position adjacent in the circumferential direction to the protruding portion 573 that is provided in the inner wall member 542 (that is, the protruding portion 573 in which the inlet passage 571 and the outlet passage 572 that communicate with the cooling water passage 545 are provided), and six capacitor modules 532B are arranged so as to be arrayed in the circumferential direction, further adjacent thereto.

As an overview of the foregoing, in the inverter housing 531, the inner side of the outer peripheral wall WA1 is evenly divided into ten areas (that is, the number of modules+1) in the circumferential direction. Of the ten areas, the electrical modules 532 are arranged one each in nine areas. The protruding portion 573 is provided in the remaining one area. The three switch modules 532A are a U-phase module, a V-phase module, and a W-phase module.

As shown in FIG. 66, and above-described FIGS. 56 and 57, and the like, each electrical module 532 (switch module 532A and capacitor module 532B) includes a plurality of module terminals 615 that extend from the module case 611. The module terminal 615 is a module input/output terminal that enables electrical input and output to be performed in the electrical module 532. The module terminal 615 is provided so as to be oriented to extend in the axial direction. More specifically, the module terminal 615 is provided so as to extend from the module case 611 toward a rear side (vehicle outer side) of the rotor carrier 511 (see FIG. 51).

Each module terminal 615 of the electrical module 532 is connected to the bus bar module 533. The number of module terminals 615 differs between the switch module 532A and the capacitor module 532B. Four module terminals 615 are provided in the switch module 532A and two module terminals 615 are provided in the capacitor module 532B.

In addition, as shown in FIG. 68, the bus bar module 533 includes an annular portion 631 that forms a circular annular shape, three external connection terminals 632 that extend from the annular portion 631 and enable connection to an external apparatus, such as a power supply apparatus or an ECU, and a winding connection terminal 633 that is connected to a winding end portion of each phase in the stator winding 521. The bus bar module 533 corresponds to a “terminal module”.

The annular portion 631 is arranged in a position that is on the radially inner side of the outer peripheral wall WA1 in the inverter housing 531 and on one side in the axial direction of the electrical modules 532.

For example, the annular portion 631 has a circular annular main body portion that is formed by an insulation member that is made of resin or the like, and a plurality of bus bars that are embedded inside main body portion. The plurality of bus bars are connected to the module terminals 615 of each electrical module 532, each external connection terminal 632, and each phase winding of the stator winding 521. Details thereof are described hereafter.

The external connection terminal 632 is made of a high-potential-side power terminal 632A and a low-potential-side power terminal 632B that are connected to the power supply apparatus, and a single signal terminal 632C that is connected to an external ECU. These external connection terminals 632 (632A to 632C) are provided so as to be arrayed in a single row in the circumferential direction and extend in the axial direction on the radially inner side of the annular portion 631.

As shown in FIG. 51, in a state in which the bus bar module 533 is assembled to the inverter housing 531 together with the electrical modules 532, one end of the external connection terminal 632 protrudes from the end plate 547 of the boss formation member 543.

Specifically, as shown in FIGS. 56 and 57, an insertion hole 547a is provided in the end plate 547 of the boss formation member 543. A circular cylindrical grommet 635 is attached to the insertion hole 547a, and the external connection terminal 632 is provided so as to be inserted through the grommet 635. The grommet 635 also functions as a connector seal.

The winding connection terminal 633 is a terminal that is connected to the winding end portion of each phase of the stator winding 521 and is provided so as to extend from the annular portion 631 toward the radially outer side. The winding connection terminal 633 includes a winding connection terminal 633U that is connected to the end portion of the U-phase winding of the stator winding 521, a winding connection terminal 633V that is connected to the end

portion of the V-phase winding, and a winding connection terminal 633W that is connected to the end portion of the W-phase winding.

A current sensor 634 that detects a current (U-phase current, V-phase current, and W-phase current) that flows to each of these winding connection terminals 633 and each phase winding may be provided (see FIG. 70).

Here, the current sensor 634 may be arranged outside the electrical module 532 in the periphery of each winding connection terminal 633. Alternatively, the current sensor 634 may be arranged inside the electrical module 532.

Here, the connection between the electrical modules 532 and the bus bar module 533 will be described in detail with reference to FIGS. 69 and 70.

FIG. 69 shows the electrical modules 532 developed in plan view, and schematically shows a state of electrical connection between the electrical modules 532 and the bus bar module 533. FIG. 70 is a diagram that schematically shows the connection between the electrical modules 532 and the bus bar modules 533 in a state in which the electrical modules 532 are arranged in a circular annular shape. Here, in FIG. 69, a path for power transmission is indicated by a solid line and a path for signal transmission is indicated by a single-dot chain line. Only the path for power transmission is shown in FIG. 70.

The bus bar module 533 includes a first bus bar 41, a second bus bar 42, and a third bus bar 43 as bus bars for power transmission. Of the bus bars, the first bus bar 641 is connected to the power terminal 632A on the high potential side and the second bus bar 642 is connected to the power terminal 632B on the low potential side. In addition, three third bus bars 643 are respectively connected to the U-phase winding connection terminal 633U, the V-phase winding connection terminal 633V, and the W-phase winding connection terminal 633W.

Moreover, the winding connection terminals 633 and the third bus bars 643 are sections that tend to generate heat as a result of operation of the rotating electric machine 10. Therefore, a terminal block (not shown) may be interposed between the winding connection terminals 633 and the third bus bars 643.

In addition, the terminal block may be placed in contact with the inverter housing 531 that includes the cooling water passage 545. Alternatively, as a result of the winding connection terminals 633 and the third bus bars 643 being bent into a crank-like shape, the winding connection terminals 633 and the third bus bars 643 may be placed in contact with the inverter housing 531 that includes the cooling water passage 545.

As a result of a configuration such as this, the heat that is generated in the winding connection terminals 633 and the third bus bars 643 can be released to the cooling water inside the cooling water passage 545.

Here, in FIG. 70, the first bus bar 641 and the second bus bar 642 are shown as bus bars that form a circular annular shape. However, these bus bars 641 and 642 are not necessarily required to be connected in a circular annular shape and may form an approximately C-like shape in which a portion in the circumferential direction is discontinuous.

In addition, because the winding connection terminals 633U, 633V, and 633W are merely required to be individually connected to the switching modules 532A that correspond to the respective phases, the winding connection terminals 633U, 633V, and 633W may be directly connected to the switch modules 532A (in actuality, the module terminals 615) without the bus bar modules 533 therebetween.

Meanwhile, each switch module **532A** includes four module terminals **615** that are made of a positive-electrode-side terminal, a negative-electrode-side terminal, a winding terminal, and a signal terminal. Of the module terminals **615**, the positive-electrode-side terminal is connected to the first bus bar **641**, the negative-electrode-side terminal is connected to the second bus bar **642**, and the winding terminal is connected to the third bus bar **643**.

In addition, the bus bar module **533** includes a fourth bus bar **644** that serves as a bus bar for the signal transmission system. The signal terminal of each switch module **532A** is connected to the fourth bus bar **644**, and the fourth bus bar **644** is connected to the signal terminal **632C**.

According to the present embodiment, a control signal for each switch module **532A** is inputted from the external ECU via the signal terminal **632C**. That is, the switches **601** and **602** in the switch module **532A** are turned on/off by the control signal that is inputted via the signal terminal **632C**.

Therefore, the switch module **532A** is configured to be connected to the signal terminal **632C** without going through a control apparatus that is provided inside the rotating electric machine, midway. However, this configuration may be modified. A control apparatus may be provided inside the rotating electric machine and a control signal from the control apparatus may be inputted to the switch module **532A**. This configuration is shown in FIG. **71**.

The configuration in FIG. **71** includes a control board **651** on which a control apparatus **652** is mounted. The control apparatus **652** is connected to each switch module **532A**. In addition, the signal terminal **632C** is connected to the control apparatus **652**. In this case, for example, the control apparatus **652** receives input of a command signal that is related to power-running or power generation from the external ECU that is a higher-order control apparatus, and turns on/off the switches **601** and **602** of each switch module **532A** as appropriate, based on the command signal.

In the inverter unit **530**, the control board **651** may be arranged further toward the vehicle outer side (rear side of the rotor carrier **511**) than the bus bar module **533** is. Alternatively, the control board **651** may be arranged between the electrical modules **532** and the end plate **547** of the boss formation member **543**. The control board **651** may be arranged such that at least a portion thereof overlaps the electrical modules **532** in the axial direction.

In addition, the capacitor module **532B** includes two module terminals **615** that are made of a positive-electrode-side terminal and a negative-electrode-side terminal. The positive-electrode-side terminal is connected to the first bus bar **641** and the negative-electrode-side terminal is connected to the second bus bar **642**.

As shown in FIGS. **49** and **50**, inside the inverter housing **531**, the protruding portion **573** that includes the inlet passage **571** and the outlet passage **572** for the cooling water is provided inside the inverter housing **531** in a position that is arrayed with the electrical modules **532** in the circumferential direction. In addition, the external connection terminal **632** is provided so as to be adjacent in the radial direction to the protruding portion **573**. In other words, the protruding portion **573** and the external connection terminal **632** are provided in the same angular position in the circumferential direction.

According to the present embodiment, the external connection terminal **632** is provided in a position on the radially inner side of the protruding portion **573**. In addition, when viewed from the vehicle inner side of the inverter housing **531**, the water-flow port **574** and the external connection

terminal **632** are provided so as to be arrayed in the radial direction on the end plate **547** of the boss formation member **543** (see FIG. **48**).

In this case, as a result of the protruding portion **573** and the external connection terminal **632** being arranged so as to be arrayed in the circumferential direction together with the plurality of electrical modules **532**, size reduction as the inverter unit **530**, and further, size reduction as the rotating electric machine **500** can be implemented.

With reference to FIGS. **45** and **47** that show the structure of the vehicle wheel **400**, the cooling pipe H2 is connected to the water-flow port **574** and the electrical wiring H1 is connected to the external connection terminal **632**. In this state, the electrical wiring H1 and the cooling pipe H2 are housed in the housing duct **440**.

Here, in the above-described configuration, three switch modules **532A** are arranged in an array in the circumferential direction adjacent to the external connection terminal **632** inside the inverter housing **531**, and the six capacitor modules **532B** are arranged in an array in the circumferential direction further adjacent thereto. However, the configuration may be modified.

For example, the three switch modules **532A** may be arranged so as to be arrayed in a position farthest from the external connection terminal **632**, that is, a position on a side opposite the external connection terminal **632** with the rotation shaft **501** therebetween. In addition, the switch modules **532A** can be distributively arranged such that the capacitor modules **532B** are arranged on both sides of the switch modules **532A**.

As a result of the configuration in which the switch modules **532A** are arranged in the position farthest from the external connection terminal **632**, that is, in the position on the side opposite the external connection terminal **632** with the rotation shaft **501** therebetween, malfunction attributed to mutual inductance between the external connection terminal **632** and the switch modules **532A**, and the like can be suppressed.

Next, a configuration related to a resolver **660** that is provided as a rotation angle sensor will be described.

As shown in FIGS. **49** to **51**, the resolver **660** that detects the electrical angle θ of the rotating electric machine **500** is provided in the inverter housing **531**. The resolver **660** is an electromagnetic-induction-type sensor. The resolver **660** includes a resolver rotor **661** that is fixed to the rotation shaft **501** and a resolver stator **662** that is arranged in an opposing manner on the radially outer side of the resolver **661**.

The resolver rotor **661** has a circular-disk ring shape and is provided coaxially with the rotation shaft **501** in a state in which the rotation shaft **501** is inserted into the resolver rotor **661**. The resolver stator **662** includes a stator core **663** that has a circular annular shape and a stator coil **664** that is wound around a plurality of teeth that are formed in the stator core **663**. An excitation coil of a single phase and output coils of two phases are included in the stator coil **664**.

The excitation coil of the stator coil **664** is excited by a sine-wave excitation signal. A magnetic flux that is generated in the excitation coil by the excitation signal interlinks the pair of output coils. At this time, a relative arrangement relationship between the excitation coil and the pair of output coils periodically changes based on a rotation angle of the resolver rotor **661** (that is, a rotation angle of the rotation shaft **501**). Therefore, the number of magnetic fluxes (number of flux interlinkage) that interlink the pair of output coils periodically changes.

According to the present embodiment, the pair of output coils and the excitation coil are arranged such that phases of

voltages that are respectively generated in the pair of output coils are offset from each other by $\pi/2$. As a result, respective output voltages of the pair of output coils are modulated waves obtained by the excitation signal being respectively modulated by modulation waves $\sin \theta$ and $\cos \theta$. More specifically, when the excitation signal is $\sin \Omega t$, the modulation waves are respectively $\sin \theta \times \sin \Omega t$ and $\cos \theta \times \sin \Omega t$.

The resolver **660** includes a resolver digital converter. The resolver digital converter calculates the electrical angle θ by detection based on the generated modulated waves and the excitation signal.

For example, the resolver **660** is connected to the signal terminal **632C** and the calculation result of the resolver digital converter is outputted to an external apparatus via the signal terminal **632C**. In addition, when the control apparatus is provided inside the rotating electric machine **500**, the calculation result of the resolver digital converter is inputted to the control apparatus.

Here, an assembly structure of the resolver **660** in the inverter housing **531** will be described.

As shown in FIGS. **49** and **51**, the boss portion **548** of the boss formation member **543** that configures the inverter housing **531** has a hollow cylindrical shape. A protruding portion **548a** that extends in a direction that is orthogonal to the axial direction is formed on an inner circumferential side of the boss portion **548**.

In addition, the resolver stator **662** is fixed by a screw or the like in a state in which the resolver stator **662** is in contact with the protruding portion **548a** in the axial direction. Inside the boss portion **548**, the bearing **560** is provided on one side in the axial direction with the protruding portion **548a** therebetween. In addition, the resolver **660** is coaxially provided on the other side.

Furthermore, in the hollow portion of the boss portion **548**, the protruding portion **548a** is provided on one side of the resolver **660** in the axial direction and a circular-disk ring-shaped housing cover **666** that closes a housing space of the resolver **660** is attached on the other side.

The housing cover **666** is made of a conductive material such as a CFRP. A hole **666a** into which the rotation shaft **501** is inserted is formed in a center portion of the housing cover **666**. A sealing member **667** that seals a space between the housing cover **666** and the outer circumferential surface of the rotation shaft **501** is provided in the hole **666a**. A resolver housing space is sealed by the sealing material **667**. For example, the sealing material **667** may be a sliding seal that is made of a resin material.

The space in which the resolver **660** is housed is a space that is surrounded by the boss portion **548** that has a circular annular shape in the boss formation member **543**, and sandwiched between the bearing **560** and the housing cover **666** in the axial direction. The surrounding of the resolver **660** is surrounded by a conductive material. As a result, the effects of electromagnetic noise on the resolver **660** can be suppressed.

In addition, as described above, the inverter housing **531** includes the outer peripheral wall **WA1** and the inner peripheral wall **WA2** that form two layers (see FIG. **57**). The stator **520** is arranged on the outer side of the peripheral walls that form the two layers (the outer side of the outer peripheral wall **WA1**), the electrical modules **532** are arranged between the two layers of peripheral walls (between **WA1** and **WA2**), and the resolver **660** is arranged on the inner side of the two layers of peripheral walls (the inner side of the inner peripheral wall **WA2**). The inverter housing **531** is a conductive member.

Therefore, the stator **520** and the resolver **660** are arranged so as to be separated by a conductive partition wall (in particular, two layers of conductive partition walls according to the present embodiment). Occurrence of mutual magnetic interference on the stator **520** side (magnetic circuit side) and the resolver **660** can be suitably suppressed.

Next, a rotor cover **670** that is provided on a side of an open end portion of the rotor carrier **511** will be described.

As shown in FIGS. **49** and **51**, one side of the rotor carrier **511** in the axial direction is open. An approximately circular-disk ring-shaped rotor cover **670** is attached to the open end portion. The rotor cover **670** may be fixed to the rotor carrier **511** by an arbitrary joining method such as welding, bonding, or screw fastening. The rotor cover **670** preferably has a portion in which a dimension is set so as to be smaller than an inner circumference of the rotor carrier **511** such that movement in the axial direction of the magnet unit **512** can be suppressed.

An outer diameter dimension of the rotor cover **670** coincides with an outer diameter dimension of the rotor carrier **511** and an inner diameter dimension is a dimension that is slightly larger than an outer diameter dimension of the inverter housing **531**. Here, the outer diameter dimension of the inverter housing **531** and the inner diameter dimension of the stator **520** are the same.

As described above, the stator **520** is fixed on the radially outer side of the inverter housing **531**. In a joining portion in which the stator **520** and the inverter housing **531** are joined to each other, the inverter housing **531** protrudes in the axial direction relative to the stator **520**. In addition, the rotor cover **670** is attached so as to surround the protruding portion of the inverter housing **531**.

In this case, a sealing member **671** that seals a space between an end surface on the inner circumferential side of the rotor cover **670** and an outer circumferential surface of the inverter housing **531** is provided therebetween. A housing space of the magnet unit **512** and the stator **520** is sealed by the sealing member **671**. For example, the sealing member **671** may be a sliding seal that is made of a resin material.

According to the present embodiment described in detail above, the following excellent effects are achieved.

In the rotating electric machine **500**, the outer peripheral wall **WA1** of the inverter housing **531** is arranged on the radially inner side of the magnetic circuit portion that is made of the magnet unit **512** and the stator winding **521**. The cooling water passage **545** is formed in the outer peripheral wall **WA1**. In addition, the plurality of electrical modules **532** are arranged on the radially inner side of the outer peripheral wall **WA1** in the circumferential direction along the outer peripheral wall **WA1**.

As a result, the magnetic circuit portion, the cooling water passage **545**, and the power converter can be arranged so as to be laminated in the radial direction of the rotating electric machine **500**. Efficient component arrangement can be obtained while reduction in dimension in the axial direction is achieved. In addition, efficient cooling can be performed in the plurality of electrical modules **532** that configure the power converter. As a result, in the rotating electric machine **500**, high efficiency and size reduction can be implemented.

The electrical modules **532** (switch module **532A** and capacitor module **532B**) that have heat generating components such as the semiconductor switching element and the capacitor are provided so as to be in contact with the inner circumferential surface of the outer peripheral wall **WA1**. As a result, the heat from the electrical module **532** is transmitted to the outer peripheral wall **WA1** and the electrical

module **532** is suitably cooled as a result of heat exchange in the outer peripheral wall **WA1**.

In the switch module **532A**, the coolers **623** are arranged on both sides sandwiching the switches **601** and **602**, and the capacitor **604** is arranged on a side opposite the switches **601** and **602** in at least either of the coolers **623** on both sides of the switches **601** and **602**. As a result, cooling performance regarding the switches **601** and **602** can be improved. In addition, cooling performance regarding the capacitor **604** can be improved.

In the switch module **532A**, the coolers **623** are arranged on both sides sandwiching the switches **601** and **602**, the drive circuit **603** is arranged on a side opposite the switches **601** and **602** in at least either of the coolers **623** on both sides of the switches **601** and **602**, and the capacitor **604** is arranged on the side opposite the switches **601** and **602** in the other cooler **623**. As a result, the cooling performance regarding the switches **601** and **602** can be improved. In addition, cooling performance regarding the drive circuit **603** and the capacitor **604** can also be improved.

For example, in the switch module **532A**, the cooling water is supplied from the cooling water passage **545** into the module interior, and the semiconductor switching elements and the like are cooled by the cooling water. In this case, the switch module **532A** is cooled by heat exchange by the cooling water in the module interior in addition to heat exchange by the cooling water in the outer peripheral wall **WA1**. As a result, the cooling effect of the switch module **532A** can be improved.

In the cooling system in which the cooling water is supplied into the cooling water passage **545** from the external circulation path **575**, the switch module **532A** is arranged on an upstream side close to the inlet passage **571** of the cooling water passage **545** and the capacitor module **532B** is arranged further toward the downstream side than the switch module **532A** is. In this case, under an assumption that the cooling water that flows through the cooling water passage **545** is at a lower temperature toward the upstream side, a configuration that preferentially cools the switch module **532A** can be implemented.

A portion of the gaps between electrical modules that are adjacent to each other in the circumferential direction is widened, and the protruding portion **573** that includes the inlet passage **571** and the outlet passage **572** is provided in the portion that is the widened gap (second interval **INT2**). As a result, the inlet passage **571** and the outlet passage **572** of the cooling water passage **545** can be suitably formed in a portion that is on the radially inner side of the outer peripheral wall **WA1**.

That is, a flow amount of coolant is required to be ensured to improve cooling performance. Therefore, increasing opening areas of the inlet passage **571** and the outlet passage **572** can be considered. In this regard, as a result of a portion of the gaps between the electrical modules being widened and the protruding portion **573** being provided as described above, the inlet passage **571** and the outlet passage **572** that are of the desired size can be suitably formed.

The external connection terminal **632** of the bus bar module **533** is arranged in a position that is arrayed with the protruding portion **573** in the radial direction on the radially inner side of the outer peripheral wall **WA1**. That is, the external connection terminal **632** is arranged together with the protruding portion **573** in the portion in which the gap between electrical modules that are adjacent to each other in the circumferential direction is widened (the portion corresponding to the second interval **INT2**). As a result, the

external connection terminal **632** can be suitably arranged while interference with the electrical modules **532** is avoided.

In the outer-rotor-type rotating electric machine **500**, the stator **520** is fixed on the radially outer side of the outer peripheral wall **WA1** and the plurality of electrical modules **532** are arranged on the radially inner side thereof.

As a result, the heat from the stator **520** is transmitted to the outer peripheral wall **WA1** from the radially outer side thereof and the heat from the electrical modules **532** is transmitted from the radially inner side. In this case, the stator **520** and the electrical modules **532** can be simultaneously cooled by the cooling water that flows through the cooling water passage **545**. Heat from the heat generating components of the rotating electric machine **500** can be efficiently released.

The electrical module **532** on the radially inner side and the stator winding **521** on the radially outer side with the outer peripheral wall **WA1** therebetween are electrically connected by the winding connection terminal **633** of the bus bar module **533**. In addition, in this case, the winding connection terminal **633** is provided in a position away from the cooling water passage **545** in the axial direction.

As a result, even when the cooling water passage **545** is formed in an annular shape in the outer peripheral wall **WA1**, that is, a configuration in which the inner side and the outer side of the outer peripheral wall **WA1** are divided by the cooling water passage **545**, the electrical module **532** and the stator winding **521** can be suitably connected.

In the rotating electric machine **500** according to the present embodiment, as a result of the teeth (core) between the conductors **523** that are arrayed in the circumferential direction in the stator **520** being made smaller or eliminated, torque restrictions attributed to magnetic saturation that occurs between the conductors **523** are suppressed and torque decrease is suppressed by the conductor **523** being a thin, flat type.

In this case, even if outer diameter dimensions of the rotating electric machine **500** are the same, as a result of the stator **520** being made thinner, the area on the radially inner side of the magnetic circuit portion can be expanded. The outer peripheral wall **WA1** that includes the cooling water passage **545** and the plurality of electrical modules **532** that are provided on the radially inner side of the outer peripheral wall **WA1** can be suitably arranged using the inner area.

In the rotating electric machine **500** according to the present embodiment, the magnet magnetic flux on the d-axis is reinforced by the magnet magnetic flux being concentrated on the d-axis side in the magnet unit **512**. Torque enhancement that accompanies the reinforcement of the magnetic flux can be obtained.

In this case, in accompaniment with a thickness dimension in the radial direction of the magnet unit **512** being able to be made smaller (thinner), the area on the radially inner side of the magnetic circuit portion can be expanded. The outer peripheral wall **WA1** that includes the cooling water passage **545** and the plurality of electrical modules **532** that are provided on the radially inner side of the outer peripheral wall **WA1** can be suitably arranged using the inner area.

In addition, the bearing **560** and the resolver **660** can also be similarly suitably arranged in the radial direction, in addition to the magnetic circuit portion, the outer peripheral wall **WA1**, and the plurality of electrical modules **532**.

The vehicle wheel **400** in which the rotating electric machine **500** is used as the in-wheel motor is mounted in the vehicle body by the base plate **405** that is fixed to the inverter housing **531** and a mounting mechanism such as a suspen-

sion apparatus. Here, because size reduction is implemented in the rotating electric machine 500, space saving can be obtained even when assembly to a vehicle body is assumed. Therefore, a configuration that is advantageous in terms of expansion of an installation area for a power supply apparatus, such as a battery, or expansion of a vehicle cabin space in the vehicle can be implemented.

Modifications related to the in-wheel motor will be described below.

(First Modification of the In-Wheel Motor)

In the rotating electric machine 500, the electrical module 532 and the bus bar module 533 are arranged on the radially inner side of the outer peripheral wall WA1 of the inverter unit 530. In addition, the electrical module 532 and the bus bar module 533 are arranged on the inner side, and the stator winding 521 is arranged on the radially outer side with the outer peripheral wall WA1 therebetween.

In this configuration, the position of the bus bar module 533 relative to the electrical module 532 can be arbitrarily set. In addition, in a case in which the phase windings of the stator winding 521 and the bus bar module 533 are connected so as to cross the outer peripheral wall WA1 in the radial direction, a position in which a winding connection line (such as the winding connection terminal 633) used for the connection is guided can be arbitrarily set.

That is, as the position of the bus bar module 533 relative to the electrical module 532, (a1) a configuration in which the bus bar module 533 is further toward the vehicle outer side than the electrical module 532 in the axial direction, that is, toward the rear side on the rotor carrier 511 side, and (a2) a configuration in which the bus bar module 533 is further toward the vehicle inner side than the electrical module 533 in the axial direction, that is, toward the front side on the rotor carrier 511 side, can be considered.

In addition, as the position in which the winding connection line is guided, (131) a configuration in which the winding connection line is guided on the vehicle outer side in the axial direction, that is, on the rear side on the rotor carrier 511 side, and (132) a configuration in which the winding connection line is guided on the vehicle inner side in the axial direction, that is, on the front side on the rotor carrier 511 side, can be considered.

Hereafter, four configuration examples related to an arrangement of the electrical modules 532, the bus bar module 533, and the winding connection line will be described with reference to FIG. 72 by (a) to (d).

FIG. 72 shows, by (a) to (d), longitudinal cross-sectional views showing the configuration of the rotating electric machine 500 in a simplified manner. In FIG. 72 by (a) to (d), configurations that are already described are given the same reference numbers. A winding connection line 637 is electrical wiring that connects the phase windings of the stator winding 521 and the bus bar module 533. For example, the above-described winding connection terminal 633 corresponds to the winding connection line 637.

In the configuration in FIG. 72 by (a), the above-described ($\alpha 1$) is used as the position of the bus bar module 533 relative to the electrical module 532, and the above-described ($\beta 1$) is used as the position for guiding the winding connection line 637. That is, the electrical module 532 and the bus bar module 533, and the stator winding 521 and the bus bar module 533 are both connected on the vehicle outer side (rear side of the rotor carrier 511). Here, this configuration corresponds to the configuration shown in FIG. 49.

As a result of the present configuration, the cooling water passage 545 can be provided in the outer peripheral wall WA1 without concern regarding interference with the wind-

ing connection line 637. In addition, the winding connection line 637 that connects the stator winding 521 and the bus bar module 533 can be easily implemented.

In FIG. 72 by (b), the above-described ($\alpha 1$) is used as the position of the bus bar module 533 relative to the electrical module 532, and the above-described ($\beta 2$) is used as the position for guiding the winding connection line 637. That is, the electrical module 532 and the bus bar module 533 are connected on the vehicle outer side (rear side of the rotor carrier 511), and the stator winding 521 and the bus bar module 533 are connected on the vehicle inner side (front side of the rotor carrier 511).

As a result of the present configuration, the cooling water passage 545 can be provided in the outer peripheral wall WA1 without concern regarding interference with the winding connection line 637.

In FIG. 72 by (c), the above-described ($\alpha 2$) is used as the position of the bus bar module 533 relative to the electrical module 532, and the above-described ($\beta 1$) is used as the position for guiding the winding connection line 637. That is, the electrical module 532 and the bus bar module 533 are connected on the vehicle inner side (front side of the rotor carrier 511), and the stator winding 521 and the bus bar module 533 are connected on the vehicle outer side (rear side of the rotor carrier 511).

In FIG. 72 by (d), the above-described ($\alpha 2$) is used as the position of the bus bar module 533 relative to the electrical module 532, and the above-described ($\beta 2$) is used as the position for guiding the winding connection line 637. That is, the electrical module 532 and the bus bar module 533, and the stator winding 521 and the bus bar module 533 are both connected on the vehicle inner side (front side of the rotor carrier 511).

According to the configurations in FIG. 72 by (c) and (d), because the bus bar module 533 is arranged on the vehicle inner side (front side of the rotor carrier 511), if an electrical component such as a fan motor is added, wiring thereof is thought to be facilitated. In addition, the bus bar module 533 can be brought closer to the resolver 660 that is arranged further toward the vehicle inner side than the bearing is. Wiring of the resolver 660 is thought to be facilitated.

(Second Modification of the In-Wheel Motor)

Modifications of an attachment structure of the resolver rotor 661 will be described below. That is, the rotation shaft 501, the rotor carrier 511, and the inner ring 561 of the bearing 560 are a rotating body that integrally rotates. Modifications of an attachment structure of the resolver rotor 661 relative to the rotation body will be described below.

FIG. 73 shows, by (a) to (c), configuration diagrams of examples of the attachment structure of the resolver rotor 611 relative to the above-described rotation body. In all of the configurations, the resolver 660 is surrounded by the rotor carrier 511, the inverter housing 531, and the like, and is provided in a sealed space that is protected from exposure to moisture, dirt, and the like from outside. In FIG. 73 by (a) among (a) to (c), the bearing 560 has the same configuration as that in FIG. 49.

In addition, in FIG. 73 by (b) and (c), the bearing 560 has a configuration differing from that in FIG. 49, and is arranged in a position away from the end plate 514 of the rotor carrier 511. Two locations are shown as examples of an attachment location of the resolver 611 in the drawings. Here, the resolver stator 662 is not shown. However, the boss portion 548 of the boss formation member 543 may be extended to the outer circumferential side of the resolver

rotor **661** or the vicinity thereof, and the resolver stator **662** may be fixed to the boss portion **548**.

In the configuration in FIG. **73** by (a), the resolver rotor **661** is attached to the inner ring **561** of the bearing **560**. Specifically, the resolver rotor **661** is provided on the end surface in the axial direction of the flange **561b** of the inner ring **561**. Alternatively, the resolver rotor **661** is provided on the end surface in the axial direction of the cylindrical portion **561a** of the inner ring **561**.

In FIG. **73** by (b), the resolver rotor **661** is attached to the rotor carrier **511**. Specifically, the resolver rotor **661** is provided on the inner surface of the end plate **514** of the rotor carrier **511**. Alternatively, the rotor carrier **511** includes a cylindrical portion **515** that extends from an inner circumferential edge portion of the end plate **514** along the rotation shaft **501**. In this configuration, the resolver rotor **661** is provided on an outer circumferential surface of the cylindrical portion **515** of the rotor carrier **511**. In the latter case, the resolver rotor **661** is arranged between the end plate **514** of the rotor carrier **511** and the bearing **560**.

In FIG. **73** by (c), the resolver rotor **661** is attached to the rotation shaft **501**. Specifically, the resolver rotor **661** is provided between the end plate **514** of the rotor carrier **511** and the bearing **560** in the rotation shaft **501**. Alternatively, the resolver rotor **661** may be arranged in the rotation shaft **501** on the side opposite the rotor carrier **511** with the bearing **560** therebetween.

(Third Modification of the In-Wheel Motor)

Modifications of the inverter housing **531** and the rotor cover **670** will be described with reference to Fig. by (a) to (b). FIG. **74** shows, by (a) and (b), longitudinal cross-sectional views showing the configuration of the rotating electric machine **500** in a simplified manner. In FIG. **74** by (a) and (b), configurations that are already described are given the same reference numbers. Here, a configuration shown in FIG. **74** by (a) essentially corresponds to the configuration described with reference to FIG. **49** and the like. A configuration shown in FIG. **74** by (b) corresponds to a configuration in which a portion of the configuration in FIG. **74** by (a) is modified.

As shown in FIG. **74** by (a), the rotor cover **670** that is fixed to the open end portion of the rotor carrier **511** is provided so as to surround the outer peripheral wall **WA1** of the inverter housing **531**. That is, the end surface on the inner diameter side of the rotor cover **670** opposes the outer circumferential surface of the outer peripheral wall **WA1**, and the sealing member **671** is provided therebetween.

In addition, the housing cover **666** is attached in the hollow portion of the boss portion **548** of the inverter housing **531**, and the sealing member **667** is provided between the housing cover **666** and the rotation shaft **501**. The external connection terminal **632** that configures the bus bar module **533** passes through the inverter housing **531** and extends toward the vehicle inner side (lower side in the drawings).

In addition, in the inverter housing **531**, the inlet passage **571** and the outlet passage **572** that communicate with the cooling water passage **545** are formed, and the water-flow port **574** that includes the passage end portions of the inlet passage **571** and the outlet passage **572** is formed.

In contrast, as shown in FIG. **74** by (b), an annular protruding portion **81** that extends toward the protruding side (vehicle inner side) of the rotation shaft **501** is formed in the inverter housing **531** (specifically, the boss formation member **543**). The rotor cover **670** is provided so as to surround the protruding portion **81** of the inverter housing **531**. That is, the end surface on the inner diameter side of the

rotor cover **670** opposes an outer circumferential surface of the protruding portion **81**, and the sealing member **671** is provided therebetween.

In addition, the external connection terminal **632** that configures the bus bar module **533** passes through the boss portion **548** of the inverter housing **531** and extends to the hollow area of the boss portion **548**. In addition, the external connection terminal **632** passes through the housing cover **666** and extends toward the vehicle inner side (lower side in the drawing).

Furthermore, in the inverter housing **531**, the inlet passage **571** and the outlet passage **572** that communicate with the cooling water passage **545** are formed. The inlet passage **571** and the outlet passage **572** extend to the hollow area of the boss portion **548** and extend further toward the vehicle inner side (lower side in the drawing) than the housing cover **666** by a relay pipe **682**. In the present configuration, the pipe portion that extends from the housing cover **666** toward the vehicle inner side is the water-flow port **574**.

According to the configurations in FIG. **74** by (a) and (b), the rotor carrier **511** and the rotor cover **670** can be suitably rotated relative to the inverter housing **531** while sealability of the interior space of the rotor carrier **511** and the rotor cover **670** is maintained.

In addition, in particular, according to the configuration in FIG. **74** by (b), the inner diameter of the rotor cover **670** is smaller compared to that in the configuration in FIG. **74** by (a). Therefore, the inverter housing **531** and the rotor cover **670** can be provided in two layers in the axial direction in a position that is further toward the vehicle inner side than the electrical module **532** is. Issues caused by electromagnetic noise that are a concern in the electrical module **532** can be suppressed. In addition, a sliding diameter of the sealing member **671** is decreased as a result of the decrease in the inner diameter of the rotor cover **670**. Mechanical loss in a rotation sliding portion can be suppressed.

(Fourth Modification of the In-Wheel Motor)

A modification of the stator winding **521** will be described below. FIG. **75** shows a modification related to the stator winding **521**.

As shown in FIG. **75**, the stator winding **521** is wound by wave winding using a conductor material of which the lateral cross-section forms a rectangular shape, such that a long side of the conductor material is oriented to extend in the circumferential direction.

In this case, the conductors **523** of each phase that serve as the coil side in the stator winding **521** are arranged at predetermined pitch intervals for each phase and are connected to each other at the coil end. The conductors **523** that are adjacent to each other in the circumferential direction in the coil side are in contact with each other at the end surfaces in the circumferential direction or are closely arranged with a minute gap therebetween.

In addition, in the stator winding **521**, the conductor material is bent in the radial direction for each phase at the coil end. More specifically, the stator winding **521** (conductor material) is bent toward the radially inner side in a position that differs for each phase in the axial direction. As a result, interference among the phase windings of the U-phase, V-phase, and W-phase is prevented.

In the configuration in the drawing, the phase windings are made to differ only by an amount corresponding to the thickness of the conductor material, and the conductor material is bent at a right angle toward the radially inner side for each phase. The length dimensions between both ends in the axial direction of the conductors **523** that are arrayed in the circumferential direction may be the same.

Here, when the stator core **522** is assembled to the stator winding **521** and the stator **520** is fabricated, a portion of the circular annular shape of the stator winding **521** may be detached so as to be disconnected (that is, the stator winding **521** becomes approximately C-shaped), and after the stator core **522** is assembled to the inner circumferential side of the stator winding **521**, the detached portions may be connected to each other and the stator winding **521** may be formed into the circular annular shape.

In addition to the foregoing, the stator core **522** can be divided into a plurality of pieces (such as three or more pieces) in the circumferential direction. The core pieces that are divided into a plurality of pieces can be assembled to the inner circumferential side of the stator winding **521** that is formed into the circular annular shape.

(Fifth Modification of the In-Wheel Motor)

According to the present embodiment, the cooling structure inside the inverter housing is modified. Hereafter, because the cooling structure is the main modification point, descriptions of sections of the rotating electric machine other than the cooling structure are omitted as appropriate.

The rotating electric machine according to the present embodiment is an outer-rotor-type, surface-magnet-type rotating electric machine. The rotating electric machine generally includes a rotor, a stator, an inverter unit, a bearing, and a rotor cover. The rotating electric machine is configured by all of these components being arranged coaxially with a rotation shaft that is provided integrally with the rotor, and assembled in the axial direction in a predetermined order.

As shown in FIG. **76**, a stator **690** includes a stator core **691** and a stator winding **692**. The stator winding **692** is composed of a plurality of phase windings that are formed into a substantially cylindrical shape (annular shape) by winding. The stator core **691** is assembled on the inner side in the radial direction of the stator winding **692**. The stator **690** is characterized by having a slot-less structure in a manner similar to the stator **50** described earlier.

In a manner similar to the stator core **52** described earlier, the stator core **691** is that in which a plurality of electromagnetic steel sheets are laminated in the axial direction and has a circular cylindrical shape that has a predetermined thickness in the radial direction. The stator winding **692** is assembled to the stator core **691** on the outer side in the radial direction that is the rotor side. The outer circumferential surface of the stator core **691** has a curved surface shape that has no unevenness. In a state in which the stator winding **692** is assembled thereto, conductors that configure the stator winding **692** are arranged so as to be arrayed in the circumferential direction on the outer circumferential surface of the stator core **691**. The stator core **691** functions as a back core.

Next, an inverter housing **700** (corresponding to the “housing member”) that configures the inverter unit, and a plurality of electrical modules that are assembled to the inverter housing **700** will be described with reference to FIGS. **76** to **78**. Here, descriptions of a bus bar module that configures the inverter unit and the like are omitted.

The inverter housing **700** includes a first housing member **710** and a second housing member **730**. For example, the housing members **710** and **730** are cast components that are composed of aluminum.

The first housing member **710** includes an end plate **711** that has a circular-disk ring shape, and a boss portion **712** that protrudes from an end portion on the inner side in the radial direction of the end plate **711** towards a housing interior. The boss portion **712** is provided in a hollow

cylindrical shape and an inner side thereof serves as a bearing holding portion that holds a bearing. Here, the boss portion **712** is similar to the configuration described above. Therefore, a description thereof is omitted.

The first housing member **710** includes an outer wall portion **713** (corresponding to an “armature-side wall portion”). The outer wall portion **713** extends in the axial direction from one end on the outer side in the radial direction of the end plate **711** and forms a circular cylindrical shape. One end in the axial direction of the outer wall portion **713** bends towards the outer side in the radial direction and forms an annular end portion **714** that has a circular annular shape. The stator core **691** is fixed to an outer circumferential surface of the outer wall portion **713**. One end in the axial direction of the stator core **691** is in contact with the annular end portion **714**. The annular end portion **714** provides a positioning function for the stator core **691** in the axial direction. An outer diameter dimension of the stator core **691** and an outer diameter dimension of the annular end portion **714** are identical or substantially identical.

The second housing member **730** includes an inner wall portion **731** and an overhang portion **732**. The inner wall portion **731** is formed into a circular cylindrical shape of which an outer diameter dimension is smaller than that of the outer wall portion **713**, and is arranged on the inner side in the radial direction of the outer wall portion **713**. The overhang portion **732** extends towards the outer side in the radial direction from a first end in the axial direction of the inner wall portion **731** and forms a circular annular shape. The overhang portion **732** extends to a position that overlaps the annular end portion **714** in the radial direction. In a portion of the overhang portion **732** that opposes the annular end portion **714**, a first recessed portion **732a** that has a circular annular shape is formed. A first sealing member **721** (such as an O-ring) that has a circular annular shape is housed in the first recessed portion **732a**. A portion of the annular end portion **714** that opposes the first recessed portion **732a** is a flat sealing surface.

In a second end in the axial direction of the inner wall portion **731**, a second recessed portion **731a** that has a circular annular shape is formed. A second sealing member **722** that has a circular annular shape is housed in the second recessed portion **731a**. A portion of the end plate **711** that opposes the second recessed portion **731a** is a flat sealing surface. As a result of the sealing members **721** and **722**, a cooling water passage **715** (coolant passage) that is partitioned by the overhang portion **732**, the inner wall portion **731**, the end plate **711**, and the outer wall portion **713** is sealed.

An annular space is formed between the inner wall portion **731** and the boss portion **712** in the inverter housing **700**. A plurality of electrical modules are arranged so as to be arrayed in the circumferential direction inside the annular space. The electrical modules are fixed to an inner circumferential surface of the inner wall portion **731** so as to be in contact with the inner circumferential surface by bonding, screw-fastening, or the like.

Next, the electrical modules that are housed in the inverter housing **700** will be described.

As described above, the plurality of electrical modules are those in which electrical components such as the semiconductor switching element that configures the power converter and the smoothing capacitor are divided into a plurality of groups and individually modularized. The electrical modules include a switch module **800** (corresponding to a “first electrical component”) that includes the semiconduc-

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tor switching element that is a power element, and a capacitor module 900 (corresponding to a “second electrical component”) that includes the smoothing capacitor.

In a manner similar to the switch module 532A described above, the switch module 800 includes a module case 801 that serves as a housing case. In addition, the switch module 800 includes upper- and lower-arm switches 802 that amount to a single phase that is housed in the module case 801, and a drive circuit (not shown). According to the present embodiment, the switch module 800 has a greater amount of heat generation due to energization than the capacitor module 900. Here, the capacitor module 900 has a configuration similar to that of the capacitor module 532B described above.

As shown in FIG. 77, a plurality of spacers 733 that have flat surfaces for attaching the electrical modules are fixed to the inner circumferential surface of the inner wall portion 731. The electrical module is attached to the spacer 733. That is, whereas the inner circumferential surface of the inner wall portion 731 is a curved surface, an attachment surface of the electrical module is a flat surface. Therefore, the configuration is such that a flat surface is formed on the inner circumferential surface side of the inner wall portion 731 by the spacer 733, and the electrical module is fixed to the flat surface. In FIG. 77, details of the interior of the electrical module and the stator 690 are omitted.

Here, the configuration in which the spacer 733 is interposed between the inner wall portion 731 and the electrical module is not a requisite. The electrical module can also be directly attached to the inner wall portion 731 by the inner circumferential surface of the inner wall portion 731 being a flat surface or the attachment surface of the electrical module being a curved surface. The spacer 733 is omitted in FIG. 76.

Here, when the spacers 733 are provided on the inner circumferential surface of the inner wall portion 731, the end plate 711, the outer wall portion 713, the overhang portion 732, the inner wall portion 731, and the spacers 733 correspond to a “cylindrical portion”. The inner wall portion 731 and the spacers 733 correspond to a “component-side wall portion”. Meanwhile, when the spacers 733 are not provided, the end plate 711, the outer wall portion 713, the overhang portion 732, and the inner wall portion 731 correspond to the “cylindrical portion”. The inner wall portion 731 corresponds to the “component-side wall portion”.

A space that is partitioned by the end plate 711, the outer wall portion 713, the overhang portion 732, and the inner wall portion 731 is a C-shaped cooling water passage 715.

According to the present embodiment, as shown in FIG. 77, the modules 800 and 900 are attached on the inner circumferential surface of the inner wall portion 731 at even intervals θ_1 in the circumferential direction. The inner wall portion 731 is provided with an inlet passage 740 through which cooling water that serves as a coolant (liquid coolant) flows into the cooling water passage 715, and an outlet passage 741 through which the cooling water flows out from the cooling water passage 715. A portion between the switch module 800 and the capacitor module 900 of the inner wall portion 731 is a protruding portion 734 that protrudes towards the inner side in the radial direction.

In addition, the inlet passage 740 and the outlet passage 741 are provided so as to be laterally arrayed along the radial direction in the protruding portion 734. Here, the above-described circulation path for circulating the cooling water is connected to the inlet passage 740 and the outlet passage 741. An inlet portion 740 that connects to the inlet passage 740 and an outlet portion 741a that connects to the outlet

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passage 741 are formed on an outer circumferential surface of the inner wall portion 731 in positions that are shifted in the circumferential direction.

A portion of the cooling water passage 715 from the inlet portion 740a to a specific position between the inlet portion 740a and the outlet portion 741a in the circumferential direction is partitioned and set as an upstream-side path, and of the cooling water passage 15. A downstream side of the cooling water passage 715 from the upstream-side path is a downstream-side path. With reference to FIG. 77, the above-described specific position is a position between the switch module 800 and the capacitor module 900 in the circumferential direction. The upstream- and downstream-side paths will be described in detail hereafter.

The second housing member 730 includes a first wall portion 750, a second wall portion 751, a third wall portion 752, a first sealing wall portion 753, and a second sealing wall portion 754. The first wall portion 750 is a portion that extends towards the outer side in the radial direction from an intermediate position in the axial direction of the inner wall portion 731. The first wall portion 750 is provided over an area from the inlet portion 740a to the above-described specific position in the circumferential direction. According to the present embodiment, the first wall portion 750, the second wall portion 751, the third wall portion 752, the first sealing wall portion 753, and the second sealing wall portion 754 correspond to a “partition wall portion”.

The second wall portion 751 is a portion that extends in the axial direction from one end on the outer side in the radial direction of the first wall portion 750 to the end plate 711. The second wall portion 751 is separated from the outer wall portion 713 in the radial direction. The third wall portion 752 is a portion that extends in the radial direction from one end on the end plate 711 side in the axial direction of the inner wall portion 731 to one end on the end plate 711 side in the axial direction of the second wall portion 751.

The first sealing wall portion 753 is a portion that extends in the radial direction from the inner wall portion 731 to the inner circumferential surface of the outer wall portion 713 between the inlet portion 740a and the outlet portion 741a in the circumferential direction, and extends in the axial direction from the end plate 711 to the overhang portion 732. The cooling water passage 715 connects between the inlet portion 740a and the outlet portion 741a in one direction along the inner wall portion 731 as a result of the first sealing wall portion 753.

Consequently, the cooling water that cools the modules 800 and 900 is not used again to cool the modules 800 and 900 before being cooled by the heat releasing apparatus 577 described above. Here, for example, each of the end plate 711, the overhang portion 732, and the outer wall portion 713, and the first sealing wall portion 753 may be bonded by an adhesive (such as an anaerobic adhesive), or a sealing member may be provided between each of the end plate 711, the overhang portion 732, and the outer wall portion 713, and the first sealing wall portion 753.

The second sealing wall portion 754 is a portion that extends from one end in the circumferential direction of the first wall portion 750 to the third wall portion 752 in the axial direction, and extends from the second wall portion 751 to the outer circumferential surface of the inner wall portion 731 in the radial direction. A downstream-side of a passage that is partitioned by the first wall portion 750, the second wall portion 751, the third wall portion 752, and the inner wall portion 731 is sealed by the second sealing wall portion 754. According to the present embodiment, of the cooling water passage 715, a passage that is partitioned by

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the first wall portion 750, the second wall portion 751, the third wall portion 752, the inner wall portion 731, the first sealing wall portion 750, and the second sealing wall portion 754 is the upstream-side path. In addition, a portion of the cooling water passage 715 other than the upstream-side path is the downstream-side path.

A coolant inflow port 755 that passes through in the radial direction is formed in a portion of the inner wall portion 731 that opposes the second wall portion 751 in the radial direction. The coolant inflow ports 755 are individually formed in correspondence to the switch modules 800. The coolant inflow ports 755 are formed in an array in the circumferential direction.

Coolant outflow ports 756 that pass through in the radial direction are individually formed in correspondence to the switch modules 800 in a portion of the inner wall portion 731 on a side opposite the coolant inflow ports 755 with the first wall portion 750 therebetween. The coolant outflow ports 756 are formed in an array in the circumferential direction.

As shown in FIG. 76, the switch module 800 is configured such that the cooling water is drawn inside thereof, and cooling by the cooling water is performed in the module interior. The switch module 800 includes first to third pipe portions 803 to 805 (corresponding to "communication paths"). The first pipe portion 803 is connected to the coolant inflow ports 755. The second pipe portion 804 is connected to the coolant outflow ports 756. The third pipe portion 805 is provided in a pair so as to pass over both surface sides of the switch 802, and connects the first pipe portion 803 and the second pipe portion 804.

As a result of the above-described configuration, the cooling water that flows in from the inlet portion 740a flows through the upstream-side path, the coolant inflow port 755, the pipe portions 803 to 805, the coolant outflow port 756, and the downstream-side path, to the outlet portion 741a.

According to the present embodiment, as shown in FIG. 76, a passage area of the upstream-side path is smaller than a passage area of the downstream-side path. Therefore, a speed of flow of the cooling water that flows through the upstream-side path is faster than a speed of flow of the cooling water that flows through the downstream-side path.

Next, a manufacturing process for the rotating electric machine will be described with reference to FIG. 79. FIG. 79 is a process flow (process flow chart) that shows a portion of the manufacturing process. In the flow, as is well-known, a triangle mark indicates work or a component. A circle mark indicates a process such as assembly or machining.

In a process S10, the switch modules 800 are assembled to the second housing member 730. Specifically, the switch modules 800 are assembled to the inner circumferential surface of the inner wall portion 731 that configures the second housing member 730.

In a process S11, the capacitor modules 900 are assembled to the second housing member 730. Specifically, the capacitor modules 900 are assembled to the inner circumferential surface of the inner wall portion 731 that configures the second housing member 730.

In a process S12 that is of a separate flow from the processes S10 and S11, the stator 690 that includes the stator winding 692 is assembled to the first housing member 710. Specifically, the stator core 691 is assembled to the outer circumferential surface of the outer wall portion 713 that configures the first housing member 710, and the stator winding 692 is assembled to the outer circumferential surface of the stator core 691.

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In a process S13, the stator winding 692 is fixed to the outer circumferential surface of the stator core 691 by a wet-type adhesive that is cured by heating. Specifically, in a state in which the adhesive is interposed between the stator core 691 and the stator winding 692, the outer side in the radial direction of the first housing member 710 to which the stator 690 is assembled is heated until a temperature of the adhesive reaches a curing temperature thereof. According to the present embodiment, the curing temperature of the adhesive is higher than an allowable upper-limit temperature (such as 170° C.) of the modules 800 and 900. Here, as the adhesive, for example, an epoxy resin can be used. Specifically, Epiform (registered trademark) can be used.

Here, in the process S11 or S12, the stator winding 692 may be assembled after the outer circumferential surface of the stator core 691 is coated with the adhesive, or the stator winding 692 that is coated with the adhesive may be assembled to the stator core 691.

In a process S14, the second housing member 730 after completion of the process S11 and the first housing member 710 after completion of the process S13 are combined, and the inverter housing 700 is formed. As a result, the cooling water passage 15 that is configured by the upstream- and downstream-side paths is formed.

According to the present embodiment described above, the following effects can be obtained.

Of the cooling water passage 715, the passage that is partitioned by the first wall portion 750, the second wall portion 751, the third wall portion 752, the inner wall portion 731, the first sealing wall portion 753, and the second sealing wall portion 754 is the upstream-side path and the portion other than the upstream-side path is the downstream-side path. In this configuration, the switch modules 800 are cooled by the cooling water that flows through the upstream-side path. The capacitor modules 900 are cooled through the inner wall portion 731 by the cooling water that flows through the downstream-side path. As a result, of the modules 800 and 900, the switch modules 800 that have a relatively large amount of heat generation can be preferentially cooled.

In addition, the second wall portion 751 is provided so as to be separated in the radial direction from the outer wall portion 713 to which the stator 690 is assembled. Therefore, the cooling water that flows through the upstream-side path is not easily affected by the heat that is generated in the stator winding 692. As a result, temperature increase in the cooling water that flows through the upstream-side path can be suppressed. Cooling performance regarding the switch modules 800 can be improved.

A passage by which the cooling water that flows through the upstream-side path directly flows to the downstream-side path is not provided. The cooling water that flows through the upstream-side path flows to the downstream-side path by way of the coolant inflow port 755, the pipe portions 803 to 805, and the coolant outflow port 756. Therefore, the coolant after cooling any switch module among the plurality of switch modules 800 is not used to cool other switch modules. As a result, cooling performance regarding the switch modules 800 can be improved. Differences in cooling performance among the switches 800 can be reduced.

The upstream-side path is formed by the second wall portion 751, in addition to the first wall portion 750, the second wall portion 751, the inner wall portion 731, the first sealing wall portion 753, and the second sealing wall portion 754 that configure the second housing member 731. Therefore, as a result of the two sealing members that are the first sealing member 721 and the second sealing member 722,

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sealing between the annular space further towards the inner diameter side than the inner wall portion 731 and the cooling water passage 15 can be performed. In addition, cooling water leakage between the upstream-side path and the downstream-side path through a path other than the coolant inflow port 755, the pipe portions 803 to 805, and the coolant outflow port 756 can be prevented.

The pipe portions 803 to 805 are provided inside the switch modules 800. As a result, cooling performance regarding the switch modules 800 can be further improved.

The passage area of the upstream-side path is smaller than the passage area of the downstream-side path. As a result, cooling performance regarding the switching modules 800 can be further improved.

In the manufacturing process for the rotating electric machine, before the process S14 in which the first housing member 710 and the second housing member 730 are combined, the process S13 in which the stator winding 692 is fixed by an adhesive that is cured by heating, and the processes S10 and S11 in which the modules 800 and 900 are assembled to the second housing member 730 are performed. Therefore, the modules 800 and 900 that are assembled to the second housing member 730 being heated when the adhesive is heated can be prevented. Reliability of the modules 800 and 900 decreasing during the manufacturing process can be prevented.

In addition, the processes S10 and S11 and the processes S12 and S13 are processes of separate flows. Therefore, a degree of freedom in the manufacturing process for the rotating electric machine can be improved.

The first wall portion 750, the second wall portion 751, the inner wall portion 731, the first sealing wall portion 753, and the second sealing wall portion 754 for forming the upstream-side path are provided in the second housing member 730. As a result of the second housing member 730 and the first housing member 710 being combined, the cooling water passage 15 that is configured by the upstream-side path and the downstream-side path is formed. Consequently, the upstream-side path for preferentially cooling the switch modules 800 can be easily provided.

(Sixth Modification of the In-Wheel Motor)

According to the present embodiment, as shown in FIG. 80, the third wall portion 752 is not provided in the second housing member 730. In FIG. 80, configurations that are identical or correspond to the configurations shown in FIG. 76 above are given the same reference numbers for convenience.

The second housing member 730 includes a first wall portion 760 and a second wall portion 761, in addition to the first sealing wall portion 753 and the second sealing wall portion 754. The first wall portion 760 is a portion that extends towards the outer side in the radial direction from the intermediate position in the axial direction of the inner wall portion 731, and is provided over an area from the inlet portion 740a to the above-described specific position in the circumferential direction. The second wall portion 761 is a portion that extends in the axial direction from one end on the outer side in the radial direction of the first wall portion to the end plate 711. The second wall portion 761 is separated from the outer wall portion 713 in the radial direction. According to the present embodiment, the first sealing wall portion 753, the second sealing wall portion 754, the first wall portion 760, and the second wall portion 761 correspond to the "partition wall portion".

According to the present embodiment, of the cooling water passage 715, a passage that is partitioned by the first wall portion 760, the second wall portion 761, the end plate

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711, the inner wall portion 731, the first sealing wall portion 753, and the second sealing wall portion 754 is the upstream-side path. In addition, the portion of the cooling water passage 715 other than the upstream-side path is the downstream-side path.

A third recessed portion 761a is formed on respective one ends in the axial direction of the second wall portion 761, the first sealing wall portion 753, and the second sealing wall portion 754. A third sealing member 723 is housed in the third recessed portion 761a. Cooling water leakage between the upstream-side path and the downstream-side path is prevented by the third sealing member 723.

(Seventh Modification of the In-Wheel Motor)

According to the present embodiment, as shown in FIG. 81, a heat insulating material 770 is provided between the first wall portion 750 and the outer wall portion 713. In FIG. 81, configurations that are identical or correspond to the configurations shown in FIG. 76 above are given the same reference numbers for convenience.

As a result of the heat insulating material 770, heat that is generated in the stator winding 692 is not easily transferred to the cooling water that flows through the upstream-side path. Consequently, temperature increase in the cooling water that flows through the upstream-side path can be suppressed.

(Eighth Modification of the In-Wheel Motor)

According to the present embodiment, as shown in FIG. 82, the configuration of the inverter housing 700 is modified. In FIG. 82, configurations that are identical or correspond to the configurations shown in FIG. 76 above are given the same reference numbers for convenience.

The second housing member 730 includes a first wall portion 771 that extends towards the outer side in the radial direction from the intermediate position in the axial direction of the inner wall portion 731. The first wall portion 771 is a configuration that corresponds to the first wall portion 750 in FIG. 76.

The first housing member 710 includes a second wall portion 716 that extends from the end plate 711 to the first wall portion 771 in the axial direction. The second wall portion 716 is a configuration that corresponds to the second wall portion 751 in FIG. 76. According to the present embodiment, the first wall portion 771, the second wall portion 716, the first sealing wall portion 753, and the second sealing wall portion 754 correspond to the "partition wall portion".

A third recessed portion 771a is formed in a portion of the first wall portion 771 that opposes the second wall portion 716. A third sealing member 724 is housed in the third recessed portion 771a. Here, for example, the sealing wall portions 753 and 754 may be portions that configure the first housing member 710. In this case, for example, sealing members may be provided between the sealing wall portions 753 and 754, and the first wall portion 771 as well. Alternatively, the sealing wall portions 753 and 754, and the first wall portion 771 may be bonded by an anaerobic adhesive.

According to the present embodiment described above, of the cooling water passage 715, a passage that is partitioned by the first wall portion 771, the second wall portion 716, the end plate 711, the inner wall portion 731, the first sealing wall portion 753, and the second sealing wall portion 754 is the upstream-side path.

(Ninth Modification of the In-Wheel Motor)

According to the present embodiment, as shown in FIG. 83, the configuration of the inverter housing 700 is modified. In FIG. 83, configurations that are identical or correspond to

the configurations shown in FIG. 76 above are given the same reference numbers for convenience.

The second housing member 730 includes an intermediate wall portion 772 that extends towards the outer side in the radial direction from the intermediate position in the axial direction of the inner wall portion 731 to the inner circumferential surface of the outer wall portion 713. The intermediate wall portion 772 is provided over an area from the inlet portion 740a to the above-described specific position in the circumferential direction. For example, the intermediate wall portion 772 and the outer wall portion 713 are bonded by an anaerobic adhesive. According to the present embodiment, the intermediate wall portion 772, the first sealing wall portion 753, and the second sealing wall portion 754 correspond to the "partition wall portion".

Here, for example, the sealing wall portion 753 and 754 may be portions that configure the second housing member 730. In this case, for example, sealing members may be provided between the sealing wall portions 753 and 754, and the end plate 711. In addition, the sealing wall portions 753 and 754, and the outer wall portion 713 may be bonded by the anaerobic adhesive.

According to the present embodiment described above, of the cooling water passage 715, a passage that is partitioned by the intermediate wall portion 772, the outer wall portion 73, the end plate 711, the inner wall portion 731, the first sealing wall portion 753, and the second sealing wall portion 754 is the upstream-side path. In addition, according to the present embodiment as well, the passage area of the upstream-side path is smaller than the passage area of the downstream-side path. Consequently, cooling performance regarding the switch modules 800 can be improved.

(Tenth Modification of the In-Wheel Motor)
According to the present embodiment, as shown in FIG. 84, the configuration of the inverter housing 700 is modified. In FIG. 84, configurations that are identical or correspond to the configurations shown in FIG. 83 above are given the same reference numbers for convenience.

The first housing member 710 includes an intermediate wall portion 717 that extends towards the inner side in the radial direction from an intermediate position in the axial direction of the outer wall portion 713 to the outer circumferential surface of the inner wall portion 731. The intermediate wall portion 717 is a configuration that corresponds to the intermediate wall portion 772 in FIG. 83. For example, the intermediate wall portion 717 and the inner wall portion 731 are bonded by an anaerobic adhesive.

Here, for example, the sealing wall portions 753 and 754 may be portions that configure the first housing member 710. In this case, for example, sealing members may be provided between the sealing wall portions 753 and 754, and the end plate 711. In addition, the sealing wall portions 753 and 754, and the inner wall portion 731 may be bonded by an anaerobic adhesive.

According to the present embodiment described above, of the cooling water passage 715, a passage that is partitioned by the intermediate wall portion 717, the outer wall portion 713, the end plate 711, the inner wall portion 731, the first sealing wall portion 753, and the second sealing wall portion 754 is the upstream-side path. In addition, according to the present embodiment as well, the passage area of the upstream-side path is smaller than the passage area of the downstream-side path. Consequently, cooling performance regarding the switch modules 800 can be improved.

(Eleventh Modification of the In-Wheel Motor)

According to the present embodiment, as shown in FIGS. 85 and 86, the configurations of the inverter housing 700 and

the switch modules 800 are modified. In FIGS. 85 and 86, configurations that are identical or correspond to the configurations shown in FIGS. 76 and 77 above are given the same reference numbers for convenience.

According to the present embodiment, the second sealing wall portion 754 is not provided. In addition, according to the present embodiment, coolant inflow ports 757 and coolant outflow ports 758 that correspond to the switch modules 800 are formed so as to be arrayed in the circumferential direction in a portion of the inner wall portion 731 that opposes the second wall portion 751 in the radial direction.

According to the present embodiment, the cooling water passage inside each switch module 800 is configured in manner similar to the configuration shown in FIG. 61 by (b) above. Here, a restricting portion such as that shown in FIG. 61 by (b) above may be provided between the coolant inflow port 757 and the coolant outflow port 758 in the circumferential direction, of the upstream-side path that is partitioned by the first wall portion 750, the second wall portion 751, the third wall portion 752, and the first sealing wall portion 753. Specifically, the restricting portion is a wall portion that extends in the axial direction from the first wall portion 750 to the third wall portion 752, over the radial direction from the second wall portion 751 to the inner wall portion 731. As a result of the restricting portion, the cooling water can be appropriately sent into the cooling inflow port 757.

(Twelfth Modification of the In-Wheel Motor)

The cooling mechanism described in the fifth modification of the in-wheel motor can also be applied to the inner-rotor-type rotating electric machine, as shown in FIG. 87.

The rotating electric machine according to the present embodiment generally includes a rotor 910, a stator 990, an inverter unit, and a bearing. The rotating electric machine is configured by all of these components being arranged coaxially with a rotation shaft 913 that is provided integrally with the rotor 910 and assembled in the axial direction in a predetermined order. The rotor 910 includes a rotor core 911 and a magnet portion 912.

The stator 990 includes a stator core 991 and a stator winding 992. The stator winding 992 is configured by a plurality of phase windings that are formed into a substantially cylindrical shape by winding. The stator core 991 is assembled on the outer side in the radial direction of the stator winding 992. The stator 990 is characterized by having a slot-less structure and the like. The stator core 991 has a circular cylindrical shape. The stator winding 992 is assembled to the stator core 991 on the inner side in the radial direction that is the rotor side.

An inverter housing 700 includes a first housing member 1010 and a second housing member 1030. For example, the housing members 1010 and 1030 are cast components that are composed of aluminum.

The first housing member 1010 includes an end plate 1011 that has a circular-disk ring shape, and an inner wall portion 1013 (corresponding to the "armature-side wall portion"). The inner wall portion 1013 extends in the axial direction from one end on the inner side in the radial direction of the end plate 711 and forms a circular cylindrical shape. One end in the axial direction of the inner wall portion 1013 bends towards the inner side in the radial direction and forms an annular end portion 1014 that has a circular annular shape. The stator core 991 is fixed to an inner circumferential surface of the inner wall portion 1013. One end in the axial direction of the stator core 991 is in contact with the annular end portion 1014. The annular end portion 1014 provides a positioning function for the stator core 991 in the

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axial direction. An outer diameter dimension of the stator core **991** and an outer diameter dimension of the annular end portion **1014** are identical or substantially identical.

The second housing member **1030** includes an outer wall portion **1031** (corresponding to the “component-side wall portion”) and an overhang portion **1032**. The outer wall portion **1031** is formed into a circular cylindrical shape of which an outer diameter dimension is greater than that of the inner wall portion **1013**, and is arranged on the outer side in the radial direction of the inner wall portion **1013**. The overhang portion **1032** extends towards the inner side in the radial direction from a first end in the axial direction of the outer wall portion **1031** and forms a circular annular shape. The overhang portion **1032** extends to a position that overlaps the annular end portion **1014** in the radial direction. In a portion of the overhang portion **1032** that opposes the annular end portion **1014**, a first recessed portion **1032a** that has a circular annular shape is formed. A first sealing member **1021** (such as an O-ring) that has a circular annular shape is housed in the first recessed portion **1032a**. A portion of the annular end portion **1014** that opposes the first recessed portion **1032a** is a flat sealing surface.

In a second end in the axial direction of the outer wall portion **1031**, a second recessed portion **1031a** that has a circular annular shape is formed. A second sealing member **1022** that has a circular annular shape is housed in the second recessed portion **1031a**. A portion of the end plate **1011** that opposes the second recessed portion **1031a** is a flat sealing surface. As a result of the sealing members **1021** and **1022**, a C-shaped cooling water passage **1015** that is partitioned by the overhang portion **1032**, the outer wall portion **1031**, the end plate **1011**, and the inner wall portion **1013** is sealed.

In a manner similar to the configuration shown in FIG. 77, a plurality of electrical modules are assembled on an outer circumferential surface of the outer wall portion **1031** so as to be arrayed in the circumferential direction. FIG. 87 shows a switch module **1100** of the electrical modules. In a manner similar to the switch module **800** described above, the switch module **1100** includes a module case **1101**, upper- and lower-arm switches **1102** that amount to a single phase, and first to third pipe portions **1103** to **1105**.

In a manner similar to the configuration shown in FIG. 76, the second housing member **1030** includes a first wall portion **1050**, a second wall portion **1051**, a third wall portion **1052**, and a first sealing wall portion and a second sealing wall portion (not shown). The first wall portion **1050** is a portion that extends towards the inner side in the radial direction from an intermediate position in the axial direction of the outer wall portion **1031**. The second wall portion **1051** is a portion that extends in the axial direction from one end on the inner side in the radial direction of the first wall portion **1050** to the end plate **1011**. The second wall portion **1051** is separated from the inner wall portion **1013** in the radial direction. The third wall portion **1052** is a portion that extends in the radial direction from one end on the end plate **1011** side in the axial direction of the outer wall portion **1031** to one end on the end plate **1011** side in the axial direction of the second wall portion **1051**.

A coolant inflow port **1055** that passes through in the radial direction is formed in a portion of the inner wall portion **1031** that opposes the second wall portion **1051** in the radial direction. The coolant inflow ports **1055** are individually formed in correspondence to the switch modules **1100**. The coolant inflow ports **1055** are formed in an array in the circumferential direction.

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Coolant outflow ports **1056** that pass through in the radial direction are individually formed in correspondence to the switch modules **1100** in a portion of the outer wall portion **1031** on a side opposite the coolant inflow ports **1055** with the first wall portion **1050** therebetween. The coolant outflow ports **1056** are formed in an array in the circumferential direction. The first pipe portion **1103** of the switch module **110** is connected to the coolant inflow port **1055**. The second pipe portion **1104** is connected to the coolant outflow port **1056**.

According to the present embodiment described above, in the inner-rotor-type rotating electric machine, effects similar to the effects in the fifth modification of the in-wheel motor can be obtained. Here, according to the present embodiment as well, the rotating electric machine can be manufactured using a manufacturing process similar to the manufacturing process described in fifth modification of the in-wheel motor.

(Other Modifications)

In the configuration shown in FIG. 76, the passage through which the cooling water flows may not be provided inside the switch module **800**. In this case, for example, a pipe portion that connects the coolant inflow port **755** and the coolant outflow port **756** may be provided in the rotating electric machine, and the switch module **800** may be in contact with the pipe portion.

In the configuration shown in FIG. 76, the switch module **800** may be embedded in the inner wall portion **731** (see FIG. 64).

In the configurations shown in FIGS. 85 and 86, the passage through which the cooling water flows may not be provided inside the switch module **800**. Even in this case, because the switch module **800** is in contact with the inner circumferential surface of the inner wall portion **1013** that is adjacent to the upstream-side path, the switch module **800** can be appropriately cooled.

Of the switch module and the capacitor module, the capacitor module may have a greater amount of heat generation due to energization. In this case, for example, in the configuration shown in FIG. 76, arrangement positions of the capacitor modules and the switch modules may be interchanged. In addition, in this case, a passage through which the cooling water flows may be provided inside the capacitor modules.

For example, as shown in FIG. 50, the inlet passage **571** and the outlet passage **572** of the cooling water passage **545** are provided so as to be collected in a single location in the rotating electric machine **500**. However, this configuration may be modified such that the inlet passage **571** and the outlet passage **572** are each provided in positions that differ in the circumferential direction.

For example, the inlet passage **571** and the outlet passage **572** may be provided in positions that differ by 180 degrees in the circumferential direction. Alternatively, a plurality of at least either of the inlet passage **571** and the outlet passage **572** may be provided.

In the vehicle wheel **400** according to the above-described embodiment, the rotation shaft **501** protrudes toward one side in the axial direction of the rotating electric machine **500**. However, the configuration may be modified. The rotation shaft **501** may protrude toward both sides in the axial direction. As a result, for example, a suitable configuration can be implemented in a vehicle in which at least either of the front and the rear of the vehicle has a single wheel.

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An inner-rotor-type rotating electric machine can also be used as the rotating electric machine **500** that is used in the vehicle wheel **400**.

The rotating electric machine is not limited to that which has a star connection and may be that which has a Δ connection.

The disclosure of the present specification is not limited to the embodiments given as examples. The disclosure includes the embodiments given as examples, as well as modifications by a person skilled in the art based on the embodiments. For example, the disclosure is not limited to the combinations of components and/or elements described according to the embodiments. The disclosure can be carried out using various combinations. The disclosure may have additional sections that can be added to the embodiments. The disclosure includes that in which a component and/or element according to an embodiment has been omitted. The disclosure includes replacements and combinations of components and/or elements between one embodiment and another embodiment. The technical scope that is disclosed is not limited to the descriptions according to the embodiments. Several technical scopes that are disclosed are cited in the scope of claims. Furthermore, the technical scopes should be understood to include all modifications within the meaning and scope of equivalency of the scope of claims.

While the present disclosure has been described with reference to embodiments thereof, it is to be understood that the disclosure is not limited to the embodiments and constructions. The present disclosure is intended to cover various modification examples and modifications within the range of equivalency. In addition, various combinations and configurations, and further, other combinations and configurations including more, less, or only a single element thereof are also within the spirit and scope of the present disclosure.

What is claimed is:

1. A rotating electric machine comprising:

a field element that includes a magnet portion that has a plurality of magnetic poles of which polarities alternate in a circumferential direction;

an armature that has an armature winding of multiple phases, either of the field element and the armature being a rotor;

a power converter that has a first electrical component that generates heat as a result of energization, and a second electrical component that generates heat as a result of energization and has a smaller amount of heat generation than the first electrical component, and is electrically connected to the armature winding;

a housing member that includes a cylindrical portion, the first electrical component and the second electrical component being arranged in the circumferential direction along the cylindrical portion on a counter-field element side in a radial direction of the cylindrical portion, wherein

the armature is assembled on the field element side in the radial direction of the cylindrical portion,

a coolant passage that has an inlet portion into which a coolant flows and an outlet portion from which the coolant flows, and connects between the inlet portion and the outlet portion in one direction along the cylindrical portion is formed in the cylindrical portion,

the housing member includes a partition wall portion that partitions a portion of the coolant passage from the inlet portion to a specific position between the inlet portion and the outlet portion in the circumferential direction as an upstream-side path,

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the partition wall portion is separated in the radial direction from a wall portion of the cylindrical portion in which the armature is assembled,

a downstream side of the upstream-side path in the coolant passage is a downstream-side path, and

on the counter-field element side in the radial direction of the cylindrical portion, the first electrical component is arranged on an inner side in the radial direction relative to the upstream-side path, and the second electrical component is arranged on the inner side in the radial direction relative to the downstream-side path.

2. The rotating electric machine according to claim 1, wherein:

the housing member further includes

an end plate,

an armature-side wall portion that extends in an axial direction from the end plate and has a circular cylindrical shape,

a component-side wall portion that is provided on the counter-field element side in the radial direction relative to the armature-side wall portion, extends in the axial direction from the end plate, and has a circular cylindrical shape, and

an overhang portion that is provided so as to be separated from the end plate in the axial direction, and extends from one end in the axial direction of the component-side wall portion to one end in the axial direction of the armature-side wall portion;

the cylindrical portion is formed by the end plate, the armature-side wall portion, the overhang portion, and the component-side wall portion;

the partition wall portion includes

a first wall portion that extends from an intermediate position in the axial direction of the component-side wall portion to an intermediate position between the component-side wall portion and the armature-side wall portion in the radial direction, and is provided over an area from the inlet portion to the specific position in the circumferential direction,

a second wall portion that extends from an end portion on the field element side in the radial direction of the first wall portion to the end plate in the axial direction, and

a sealing wall portion that seals a downstream side of a passage that is partitioned by the end plate, the first wall portion, and the second wall portion;

a passage that is partitioned by the end plate, the component-side wall portion, the first wall portion, the second wall portion, and the sealing wall portion is the upstream-side path;

on the counter-field element side in the radial direction of the component-side wall portion, a plurality of first electrical components are arranged on the inner side in the radial direction relative to the upstream-side path so as to be arrayed in the circumferential direction;

a coolant inflow port that passes through in the radial direction is formed in correspondence to each first electrical component in a portion of the component-side wall portion that is adjacent to the upstream-side path;

a coolant outflow port that passes through in the radial direction is formed in correspondence to each first electrical component, over an area of the component-side wall portion from the inlet portion to the specific position in the circumferential direction in a portion that is further towards the overhang portion side in the axial direction than the first wall portion;

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the rotating electric machine includes a communication path that is provided in correspondence to each first electrical component and guides the coolant that flows into the coolant inflow port to the coolant outflow port; and

each first electrical component is cooled by the coolant that flows through the communication path.

3. The rotating electric machine according to claim 2, wherein:

the partition wall portion includes a third wall portion that extends in the radial direction from one end on the end plate side in the axial direction of the component-side wall portion to one end on the end plate side in the axial direction of the second wall portion;

a passage that is partitioned by the end plate, the component-side wall portion, the first wall portion, the second wall portion, the third wall portion, and the sealing wall portion is the upstream-side path; and the rotating electric machine further includes

a first sealing member that seals between one end on the overhang portion side in the axial direction of the armature-side wall portion and the overhang portion, and has a circular annular shape, and

a second sealing member that seals between one end on the end plate side in the axial direction of the component-side wall portion and the end plate, and has a circular annular shape.

4. The rotating electric machine according to claim 2, wherein:

the first electrical component is an electrical module that is configured by a heat generating component, housed in a housing case, that generates heat as a result of energization; and

each electrical module is provided with the communication path therein.

5. The rotating electric machine according to claim 1, wherein:

the housing member further includes

an end plate,

an armature-side wall portion that extends in an axial direction from the end plate and has a circular cylindrical shape,

a component-side wall portion that is provided on the counter-field element side in the radial direction relative to the armature-side wall portion, extends in the axial direction from the end plate, and has a circular cylindrical shape, and

an overhang portion that is provided so as to be separated from the end plate in the axial direction, and extends from one end in the axial direction of the component-side wall portion to one end in the axial direction of the armature-side wall portion;

the cylindrical portion is formed by the end plate, the armature-side wall portion, the overhang portion, and the component-side wall portion;

on the counter-field element side in the radial direction of the component-side wall portion, a plurality of first electrical components are arranged on the inner side in the radial direction relative to the upstream-side path so as to be arrayed in the circumferential direction;

a coolant inflow port that passes through in the radial direction and a coolant outflow port that passes through in the radial direction are formed in correspondence to each first electrical component in a portion of the component-side wall portion that is adjacent to the upstream-side path;

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the rotating electric machine includes a communication path that is provided in correspondence to each first electrical component and guides the coolant that flows into the coolant inflow port to the coolant outflow port; and

each first electrical component is cooled by the coolant that flows through the communication path.

6. The rotating electric machine according to claim 5, wherein:

the first electrical component is an electrical module that is configured by a heat generating component, housed in a housing case, that generates heat as a result of energization; and

each electrical module is provided with the communication path therein.

7. The rotating electric machine according to claim 6, wherein:

a passage area of the upstream-side path is smaller than a passage area of the downstream-side path.

8. The rotating electric machine according to claim 7, wherein:

the power converter includes a series-connection body of an upper arm switch and a lower arm switch that is provided for each phase of the armature winding, and a capacitor that is connected in parallel to the series-connection body;

the first electrical component is a switch module that is configured by the upper arm switch and the lower arm switch that serve as heat generating components, housed in a housing case, that generate heat as a result of energization; and

the second electrical component is a capacitor module that is configured by the capacitor that serves as a heat generating component, housed in a housing case, that generates heat as a result of energization.

9. The rotating electric machine according to claim 8, wherein:

the rotor that serves as the field element is provided on an outer side in the radial direction of a stator that serves as the armature.

10. The rotating electric machine according to claim 1, wherein:

a passage area of the upstream-side path is smaller than a passage area of the downstream-side path.

11. The rotating electric machine according to claim 1, wherein:

the power converter includes a series-connection body of an upper arm switch and a lower arm switch that is provided for each phase of the armature winding, and a capacitor that is connected in parallel to the series-connection body;

the first electrical component is a switch module that is configured by the upper arm switch and the lower arm switch that serve as heat generating components, housed in a housing case, that generate heat as a result of energization; and

the second electrical component is a capacitor module that is configured by the capacitor that serves as a heat generating component, housed in a housing case, that generates heat as a result of energization.

12. The rotating electric machine according to claim 1, wherein:

the rotor that serves as the field element is provided on an outer side in the radial direction of a stator that serves as the armature.

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